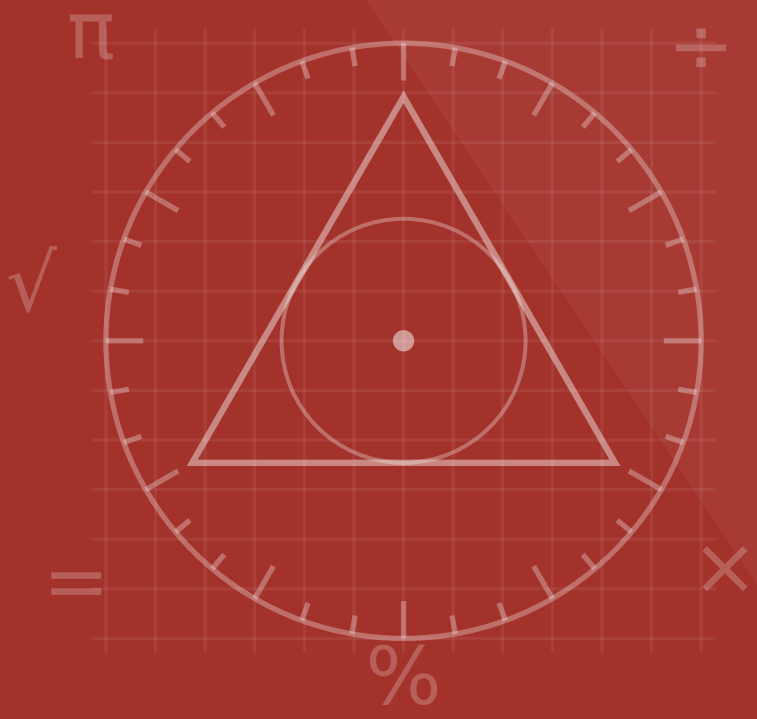
VENVLEARNING

TEACHER EDITION



Mathematics

GRADE 7

Guided lessons, pacing, and complete answer keys

9 units

VENVLEARNING

Grade 7
Mathematics

TEACHER EDITION

Unit 1 Proportional Relationships
Unit 2 Percents
Unit 3 Adding and Subtracting Integers
Unit 4 Adding and Subtracting Rational Numbers
Unit 5 Multiplying and Dividing Rational Numbers
Unit 6 Expressions, Equations, and Inequalities
Unit 7 Statistics and Probability
Unit 8 Scale Drawings and Geometric Figures
Unit 9 Angles, Circles, Area, and Volume

A VenvLearning family homeschool curriculum

How to use this Teacher Edition

This book holds everything needed to teach the course: unit overviews, a plan for every lesson with what to say and do, the student lesson with Try It answers, and answer keys for every workbook page, weekly check and unit test.

- Daily routine (about 45–55 minutes): Warm-Up 5 · Teach 15 · Guided Practice 10 · Independent Practice 15 · Exit Ticket 5.
- Say: lines are a suggested script. Use your own words.
- Four lessons a week, then the Weekly Check (about 20 minutes), then corrections or a project.
- Each plan's Pacing row and each unit's pacing guide give the week and lesson order. Set the days to fit your own calendar.

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UNIT 1 · TEACHER OVERVIEW

Proportional Relationships

Ratios, unit rates, the constant of proportionality, and proportional tables, graphs, and equations.

Many real-world quantities grow together at a constant rate: pay and hours, miles and gallons, cost and pounds. In this unit you will find unit rates, recognize proportional relationships, and represent them with tables, graphs, and equations.

Big ideas

- Compute unit rates, including with fractions.
- Decide whether two quantities are proportional.
- Find the constant of proportionality and write $y = kx$.
- Graph and interpret proportional relationships.
- Solve multistep ratio and rate problems.

Standards

Code	Standard
7.RP.1	Compute unit rates associated with ratios of fractions.
7.RP.2	Recognize and represent proportional relationships: tables, graphs, constant of proportionality, equations, interpreting points.
7.RP.3	Use proportional relationships to solve multistep ratio and percent problems.

What students should already know

- Ratios and rates (Grade 6).
- Fraction and decimal operations.
- Plotting points in the first quadrant.

Materials for the unit

- Graph paper
- Ruler

- Calculator for checking
- Store ads or receipts

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	1.1	Ratios and Rates
Week 1	Lesson 2 of 4	1.2	Unit Rates
Week 1	Lesson 3 of 4	1.3	Unit Rates with Fractions
Week 1	Lesson 4 of 4	1.4	Comparing Unit Rates
Week 1	End of week	Check	Weekly Check 1: Ratios and Unit Rates
Week 2	Lesson 1 of 4	1.5	What Is a Proportional Relationship?
Week 2	Lesson 2 of 4	1.6	The Constant of Proportionality
Week 2	Lesson 3 of 4	1.7	Equations for Proportional Relationships
Week 2	Lesson 4 of 4	1.8	Is It Proportional? Tables and Equations
Week 2	End of week	Check	Weekly Check 2: Proportional Tables and Equations
Week 3	Lesson 1 of 4	1.9	Graphing Proportional Relationships
Week 3	Lesson 2 of 4	1.10	Recognizing Proportional Graphs
Week 3	Lesson 3 of 4	1.11	Interpreting Points on Proportional Graphs
Week 3	Lesson 4 of 4	1.12	Comparing Proportional Relationships
Week 3	End of week	Check	Weekly Check 3: Proportional Graphs
Week 4	Lesson 1 of 4	1.13	Solving Proportions
Week 4	Lesson 2 of 4	1.14	Proportional Reasoning in the Real World
Week 4	Lesson 3 of 4	1.15	Multi-Step Ratio Problems
Week 4	Lesson 4 of 4	1.16	Proportional Relationships Problem Solving
Week 4	Before the test	Review	Unit 1 Review
Week 4	End of unit	Test	Unit 1 Test

TEACHING THIS UNIT AT HOME

Have your student compute unit prices on your next grocery trip and decide which size is the better buy.

UNIT VOCABULARY

ratio	A comparison of two quantities.
unit rate	Amount per one unit.
proportional relationship	y ÷ x is always the same.
constant of proportionality	k = y ÷ x.

LESSON 1.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 1: Proportional Relationships

Lesson 1.1: Ratios and Rates

Student goal	I can write ratios and rates and explain what they mean.
Objective	Students write ratios and rates and generate equivalent ratios.
Standard	7.RP.1 (review of 6.RP.1–3) — Review of ratio concepts; prepares for computing unit rates (7.RP.1).
Pacing	Unit 1, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.1, measuring cups (optional)
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	12 × 4; simplify 18/24.
Launch	5	Read Let's Explore.
Teach	15	Build the double number line together. Say: "Scale both quantities by the same factor." Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding the same number to both parts (3:2 → 5:4).

SUPPORT

- Use a double number line or table.

EXTEND

- Find a real ratio on a food label.

STUDENT LESSON 1.1 · WITH ANSWERS

Lesson 1.1: Ratios and Rates

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.1, Part A.
- [ ] Complete the rest of Workbook Lesson 1.1 (Parts B–F).

LET'S EXPLORE

A trail mix recipe uses 3 cups of peanuts for every 2 cups of raisins. If you want a bigger batch, how much of each could you use?

A ratio compares two amounts. The recipe's ratio of peanuts to raisins is 3:2. To make more, multiply both quantities by the same number. That keeps the taste the same.



A rate compares quantities with different units, such as miles and hours or dollars and pounds.

WORKED EXAMPLE · EQUIVALENT RATIOS

Is 9 cups of peanuts to 6 cups of raisins the same mix as 3:2?

1 $3 \times 3 = 9$ and $2 \times 3 = 6$.

2 Both parts were multiplied by 3.

Answer: Yes, 9:6 is equivalent to 3:2.

WORKED EXAMPLE · WRITE A RATE

A car travels 150 miles in 3 hours. Write the rate.

1 Compare miles to hours: 150 miles : 3 hours.

Answer: 150 miles per 3 hours (50 miles per hour)

★ KEY IDEA

Equivalent ratios come from multiplying or dividing both quantities by the same nonzero number.

✎ TRY IT

Answers are shown in color.

1. Write the ratio of 4 dogs to 7 cats three ways.

Answer: 4 to 7, 4:7, $^{4}/_{7}$

2. Is 10:15 equivalent to 2:3?

Answer: Yes (÷ 5)

3. A recipe uses 2 eggs per 3 cups of flour. How many eggs for 12 cups of flour?

Answer: 8 eggs

Workbook Answer Key · Lesson 1.1

#	Question	Answer
1	Write the ratio of 4 dogs to 7 cats three ways.	4 to 7, 4:7, $^{4}/_{7}$
2	Is 10:15 equivalent to 2:3?	Yes (÷ 5)
3	A recipe uses 2 eggs per 3 cups of flour. How many eggs for 12 cups of flour?	8 eggs
4	12 × 4	48
5	Simplify $^{18}/_{24}$	$^{3}/_{4}$
6	2:5	e.g. 4:10, 6:15
7	7:3	e.g. 14:6, 21:9
8	12:16	e.g. 3:4, 6:8
9	1:4	e.g. 2:8, 5:20
10	Lemonade: 4 lemons for every 3 cups of water. How much water for 16 lemons? (figure)	12 cups
11	A team won 12 games and lost 8. What is the ratio of wins to games played?	12:20 = 3:5
12	A printer prints 45 pages in 3 minutes. How many pages in 10 minutes?	150 pages
13	Explain why 6:8 and 9:12 are equivalent.	Both simplify to 3:4 (÷ 2 and ÷ 3).

LESSON 1.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 1: Proportional Relationships

Lesson 1.2: Unit Rates

Student goal	I can find a unit rate and use it to solve problems.
Objective	Students compute and apply unit rates.
Standard	7.RP.1 — Compute unit rates associated with ratios of fractions, including ratios of lengths, areas, and other quantities.
Pacing	Unit 1, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Grocery receipts or ads (optional), workbook Lesson 1.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Division warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work all three examples. Say: Unit rate means per one.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing in the wrong order (12 ÷ 4.20).

SUPPORT

- Label units on every number.

EXTEND

- Compare unit prices on two real products.

STUDENT LESSON 1.2 · WITH ANSWERS

Lesson 1.2: Unit Rates

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.2, Part A.
- [ ] Complete the rest of Workbook Lesson 1.2 (Parts B–E).

LET'S EXPLORE

A 12-pack of water costs $4.20. How much does one bottle cost?

A unit rate tells how much for one. Divide the first quantity by the second.

WORKED EXAMPLE · PRICE PER ITEM

$4.20 for 12 bottles

1 4.20 ÷ 12 = 0.35.

Answer: $0.35 per bottle

WORKED EXAMPLE · SPEED

A cyclist rides 54 miles in 4.5 hours. Find the speed.

1 54 ÷ 4.5 = 12.

Answer: 12 miles per hour

WORKED EXAMPLE · USE A UNIT RATE

At $0.35 per bottle, how much do 30 bottles cost?

1 30 × 0.35 = 10.50.

Answer: $10.50

★ KEY IDEA

Unit rate = first quantity ÷ second quantity. Then multiply the unit rate to scale up.

✎ TRY IT

Answers are shown in color.

1. $18 for 6 notebooks: unit price?

Answer: $3 per notebook

2. 390 miles on 15 gallons: miles per gallon?

Answer: 26 mpg

3. Type 210 words in 5 minutes: words per minute?

Answer: 42 wpm

Workbook Answer Key · Lesson 1.2

#	Question	Answer
1	$18 for 6 notebooks: unit price?	$3 per notebook
2	390 miles on 15 gallons: miles per gallon?	26 mpg
3	Type 210 words in 5 minutes: words per minute?	42 wpm
4	4.20 ÷ 12	0.35
5	54 ÷ 6	9
6	$7.50 for 3 pounds	$2.50 per pound
7	120 km in 2.5 hours	48 km per hour
8	96 heartbeats in 1.5 minutes	64 beats per minute
9	$13.44 for 16 ounces	$0.84 per ounce
10	8 laps in 20 minutes	0.4 lap per minute (or 2.5 minutes per lap)
11	$52 for 8 hours of work	$6.50 per hour
12	Lillie earns $11.50 per hour babysitting. How much for 6 hours?	$69
13	A car uses 3 gallons per 84 miles. How far on 10 gallons?	280 miles
14	A 5-pound bag of rice costs $6.25. What is the price per pound? How much would 12 pounds cost?	$1.25 per pound; $15.00

LESSON 1.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 1: Proportional Relationships

Lesson 1.3: Unit Rates with Fractions

Student goal	I can find unit rates when the quantities are fractions, like $^{1}/_{2}$ mile in $^{1}/_{4}$ hour.
Objective	Students compute unit rates involving fractions.
Standard	7.RP.1 — Compute unit rates associated with ratios of fractions (e.g., 1/2 mile per 1/4 hour).
Pacing	Unit 1, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Fraction division warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Show the table method alongside division.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying instead of dividing.
- Flipping the wrong fraction.

SUPPORT

- Table reasoning before the algorithm.

EXTEND

- Make up a complex-fraction rate problem.

STUDENT LESSON 1.3 · WITH ANSWERS

Lesson 1.3: Unit Rates with Fractions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.3, Part A.
- [ ] Complete the rest of Workbook Lesson 1.3 (Parts B–E).

LET'S EXPLORE

Ellie walks $\frac{1}{2}$ mile in $\frac{1}{6}$ hour. How fast does she walk in miles per hour?

Find the unit rate the same way: divide. $\frac{1}{2} \div \frac{1}{6}$ asks "how many miles in 1 whole hour?"

WORKED EXAMPLE · DIVIDE FRACTIONS

$\frac{1}{2}$ mile in $\frac{1}{6}$ hour

1 $\frac{1}{2} \div \frac{1}{6} = \frac{1}{2} \times \frac{6}{1} = \frac{6}{2} = 3.$

Answer: 3 miles per hour

WORKED EXAMPLE · THINK WITH A TABLE

A painter covers $\frac{2}{3}$ of a wall in $\frac{1}{3}$ hour. How much per hour?

1 In $\frac{1}{3}$ hour: $\frac{2}{3}$ wall. In 1 hour (3 times as long): $3 \times \frac{2}{3} = 2$.

Answer: 2 walls per hour

★ KEY IDEA

To divide by a fraction, multiply by its reciprocal: $a \div \frac{b}{c} = a \times \frac{c}{b}$.

✎ TRY IT

Answers are shown in color.

1. $\frac{3}{4}$ cup of sugar per $\frac{1}{2}$ batch. Cups per batch?

Answer: $1\frac{1}{2}$ cups

2. $\frac{1}{3}$ mile in $\frac{1}{12}$ hour. mph?

Answer: 4 mph

3. \$2.50 for $\frac{1}{2}$ pound. Price per pound?

Answer: $5.00

Workbook Answer Key · Lesson 1.3

#	Question	Answer
1	$\frac{3}{4}$ cup of sugar per $\frac{1}{2}$ batch. Cups per batch?	$1\frac{1}{2}$ cups
2	$\frac{1}{3}$ mile in $\frac{1}{12}$ hour. mph?	4 mph
3	$2.50 for $\frac{1}{2}$ pound. Price per pound?	$5.00
4	$\frac{1}{2} \div \frac{1}{4}$	2
5	$\frac{2}{3} \times 3$	2
6	$\frac{1}{4}$ mile in $\frac{1}{8}$ hour	2 miles per hour
7	$\frac{2}{5}$ pound for $\frac{1}{5}$ dollar	2 pounds per dollar
8	$\frac{3}{4}$ page in $\frac{1}{4}$ minute	3 pages per minute
9	$\frac{5}{6}$ cup for $\frac{2}{3}$ batch	$1\frac{1}{4}$ cups per batch
10	$1\frac{1}{2}$ miles in $\frac{1}{2}$ hour	3 mph
11	$\frac{3}{8}$ gallon for $\frac{3}{4}$ acre	$\frac{1}{2}$ gallon per acre
12	A snail moves $\frac{1}{10}$ meter in $\frac{1}{5}$ minute. How far in 1 minute? In 30 minutes?	$\frac{1}{2}$ meter per minute; 15 meters
13	Lillie mows $\frac{3}{5}$ of a lawn in $\frac{3}{4}$ hour. How much lawn per hour?	$\frac{4}{5}$ lawn per hour

LESSON 1.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 1: Proportional Relationships

Lesson 1.4: Comparing Unit Rates

Student goal	I can use unit rates to decide which deal or option is better.
Objective	Students compare unit rates to make decisions.
Standard	7.RP.1, 7.RP.3 — Use proportional relationships to solve multistep ratio problems.
Pacing	Unit 1, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Store ads or receipts, workbook Lesson 1.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Division warm-up.
Launch	5	Read Let's Explore. Guess first.
Teach	15	Work both examples. Discuss which rate is 'better' in context.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Comparing totals instead of unit rates.

SUPPORT

- Make a 2-column table.

EXTEND

- Compare real unit prices at the grocery store.

STUDENT LESSON 1.4 · WITH ANSWERS

Lesson 1.4: Comparing Unit Rates

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.4, Part A.
- [ ] Complete the rest of Workbook Lesson 1.4 (Parts B–E).

LET'S EXPLORE

A 16-ounce box of cereal costs $3.84. A 24-ounce box costs $5.28. Which is the better buy?

WORKED EXAMPLE · BETTER BUY

16 oz for $3.84 vs 24 oz for $5.28

1 3.84 ÷ 16 = 0.24 per ounce.

2 5.28 ÷ 24 = 0.22 per ounce.

Answer: The 24-ounce box ($0.22/oz)

WORKED EXAMPLE · FASTER RUNNER

Ellie runs 2 miles in 18 minutes. Lillie runs 3 miles in 26 minutes. Who is faster?

1 Ellie: 18 ÷ 2 = 9 minutes per mile.

2 Lillie: 26 ÷ 3 ≈ 8.67 minutes per mile.

Answer: Lillie (fewer minutes per mile)

★ KEY IDEA

Compare like rates: dollars per ounce with dollars per ounce. For time per mile, smaller is faster.

✎ TRY IT

Answers are shown in color.

1. 12 eggs for $3.00 or 18 eggs for $4.14?

Answer: 18 eggs ($0.23 vs $0.25 each)

2. Car A: 300 mi on 12 gal. Car B: 270 mi on 10 gal. Better mileage?

Answer: Car B (27 vs 25 mpg)

3. Who types faster: 150 words in 3 min or 200 words in 5 min?

Answer: 150 in 3 min (50 vs 40 wpm)

Workbook Answer Key · Lesson 1.4

#	Question	Answer
1	12 eggs for $3.00 or 18 eggs for $4.14?	18 eggs ($0.23 vs $0.25 each)
2	Car A: 300 mi on 12 gal. Car B: 270 mi on 10 gal. Better mileage?	Car B (27 vs 25 mpg)
3	Who types faster: 150 words in 3 min or 200 words in 5 min?	150 in 3 min (50 vs 40 wpm)
4	3.84 ÷ 16	0.24
5	5.28 ÷ 24	0.22
6	8 oz for $2.40 or 12 oz for $3.24	12 oz ($0.27 vs $0.30 per oz)
7	3 lb for $4.50 or 5 lb for $7.25	5 lb ($1.45 vs $1.50 per lb)
8	6 pencils for $1.50 or 10 for $2.30	10 for $2.30 ($0.23 vs $0.25)
9	2 L for $1.80 or 3 L for $2.85	2 L ($0.90 vs $0.95 per L)
10	Dad's truck goes 342 miles on 18 gallons. Mom's car goes 406 miles on 14 gallons. Whose vehicle gets better gas mileage, and by how many mpg?	Mom's car: 29 vs 19 mpg, 10 mpg better
11	Explain why a bigger package is not always the better buy.	Compare unit prices; the bigger package can cost more per unit.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.4

About 20 minutes

Weekly Check 1: Ratios and Unit Rates — Answer Key

Answers

#	Question	Answer
1	Write two ratios equivalent to 4:9.	e.g. 8:18, 12:27
2	$11.40 for 6 pounds: unit price	$1.90 per pound
3	$\frac{2}{3}$ mile in $\frac{1}{6}$ hour: mph	4 mph
4	Better buy: 10 oz for $2.50 or 16 oz for $3.84	16 oz ($0.24 vs $0.25)
5	A car goes 252 miles on 9 gallons. mpg?	28 mpg
6	At 28 mpg, how far on 13 gallons?	364 miles

LESSON 1.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 1: Proportional Relationships

Lesson 1.5: What Is a Proportional Relationship?

Student goal	I can tell when two quantities are in a proportional relationship.
Objective	Students decide whether two quantities are proportional.
Standard	7.RP.2a — Decide whether two quantities are in a proportional relationship by testing for equivalent ratios in a table.
Pacing	Unit 1, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	21 ÷ 6.
Launch	5	Read Let's Explore.
Teach	15	Compare the two tables. Say: Divide y by x for every pair.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Checking only one pair.

SUPPORT

- Add a y ÷ x column.

EXTEND

- Find proportional and non-proportional situations at home.

STUDENT LESSON 1.5 · WITH ANSWERS

Lesson 1.5: What Is a Proportional Relationship?

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.5, Part A.
- [ ] Complete the rest of Workbook Lesson 1.5 (Parts B–E).

LET'S EXPLORE

Tickets cost $8 each. Is the total cost proportional to the number of tickets? What if there is also a $5 parking fee?

Two quantities are proportional when one is always the same number times the other. Every ratio in the table is equivalent.

Tickets	Cost
1	$8
2	$16
3	$24
5	$40

Tickets	Cost + parking
1	$13
2	$21
3	$29
5	$45

In the first table, cost ÷ tickets is always 8: proportional. In the second, 13 ÷ 1 = 13 but 21 ÷ 2 = 10.5: not proportional. The $5 fee breaks the pattern.

★ KEY IDEA

Proportional: y ÷ x is the same for every pair, and 0 of one quantity means 0 of the other.

✎ TRY IT

Answers are shown in color.

1. Is distance proportional to time at a steady 60 mph?

Answer: Yes

2. Is a $20 gym fee plus $5 per visit proportional?

Answer: No

3. x: 2, 4, 6 → y: 7, 14, 21. Proportional?

Answer: Yes (y ÷ x = 3.5)

Workbook Answer Key · Lesson 1.5

#	Question	Answer
1	Is distance proportional to time at a steady 60 mph?	Yes
2	Is a $20 gym fee plus $5 per visit proportional?	No
3	x: 2, 4, 6 → y: 7, 14, 21. Proportional?	Yes (y ÷ x = 3.5)
4	21 ÷ 6	3.5
5	Is 4:10 equivalent to 6:15?	Yes
6	x: 1, 2, 3, 4 → y: 5, 10, 15, 20	Yes; y ÷ x = 5
7	x: 1, 2, 3, 4 → y: 3, 5, 7, 9	No; ratios differ
8	x: 2, 5, 8 → y: 3, 7.5, 12	Yes; y ÷ x = 1.5
9	x: 3, 6, 9 → y: 1, 2, 4	No
10	Pay of $12 per hour	proportional
11	A taxi: $3 plus $2 per mile	not proportional
12	Area of a square vs its side length	not proportional
13	Cost of apples at $1.50 per pound	proportional
14	Make a table that is NOT proportional. Explain why.	Answers vary.

LESSON 1.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 1: Proportional Relationships

Lesson 1.6: The Constant of Proportionality

Student goal	I can find the constant of proportionality k from a table.
Objective	Students identify the constant of proportionality in tables.
Standard	7.RP.2b — Identify the constant of proportionality (unit rate) in tables.
Pacing	Unit 1, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Division warm-up.
Launch	5	Read Let's Explore.
Teach	15	Add a y ÷ x column to the table. Name k.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Computing x ÷ y instead of y ÷ x.

SUPPORT

- Label: y per 1 x.

EXTEND

- Find k for x ÷ y too; how do the two relate?

STUDENT LESSON 1.6 · WITH ANSWERS

Lesson 1.6: The Constant of Proportionality

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.6, Part A.
- [ ] Complete the rest of Workbook Lesson 1.6 (Parts B–E).

LET'S EXPLORE

A pancake recipe uses 2.5 cups of milk for every 2 cups of flour. How much milk per 1 cup of flour?

In a proportional relationship, y ÷ x is always the same number. That number is the constant of proportionality, k. It is the same as the unit rate.

Flour (cups)	Milk (cups)	Milk ÷ Flour
2	2.5	1.25
4	5	1.25
6	7.5	1.25
1	1.25	1.25

WORKED EXAMPLE · FIND K

x: 3, 5, 8 → y: 12, 20, 32

1 12 ÷ 3 = 4, 20 ÷ 5 = 4, 32 ÷ 8 = 4.

Answer: k = 4

★ KEY IDEA

k = y ÷ x. It tells how much y for 1 of x.

✎ TRY IT

Answers are shown in color.

1. x: 2, 6, 10 → y: 9, 27, 45. Find k.

Answer: 4.5

2. Hours 3 → $36 pay. k?

Answer: $12 per hour

3. x: 4, 8 → y: 1, 2. k?

Answer: $^1/_4$ (0.25)

Workbook Answer Key · Lesson 1.6

#	Question	Answer
1	x: 2, 6, 10 → y: 9, 27, 45. Find k.	4.5
2	Hours 3 → $36 pay. k?	$12 per hour
3	x: 4, 8 → y: 1, 2. k?	$^1/_4$ (0.25)
4	7.5 ÷ 6	1.25
5	45 ÷ 10	4.5
6	x: 1, 3, 5 → y: 7, 21, 35	7
7	x: 2, 4, 10 → y: 3, 6, 15	1.5
8	x: 5, 10, 20 → y: 2, 4, 8	0.4
9	x: 6, 9, 12 → y: 4, 6, 8	$^2/_3$
10	x: 1, 2, 5, 8 → y: 6, ___, ___, ___	12, 30, 48
11	x: 4, 10, 15 → y: 10, ___, ___	25, 37.5
12	A car travels 165 miles in 3 hours at a steady speed. What is k? What does it mean?	k = 55: 55 miles per hour

LESSON 1.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 1: Proportional Relationships

Lesson 1.7: Equations for Proportional Relationships

Student goal	I can write an equation y = kx for a proportional relationship.
Objective	Students write and use equations y = kx.
Standard	7.RP.2c — Represent proportional relationships by equations.
Pacing	Unit 1, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Connect k from Lesson 1.6 to the equation.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Writing y = x + k.

SUPPORT

- Check the equation against each table pair.

EXTEND

- Write an equation for a real-life rate.

STUDENT LESSON 1.7 · WITH ANSWERS

Lesson 1.7: Equations for Proportional Relationships

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.7, Part A.
- [ ] Complete the rest of Workbook Lesson 1.7 (Parts B–F).

LET'S EXPLORE

Strawberries cost $3.50 per pound. Write a rule for the cost of any number of pounds.

If y is always k times x, the equation is y = kx. For strawberries: cost = 3.50 × pounds, or y = 3.5x.

WORKED EXAMPLE · FROM A TABLE

x: 2, 5, 9 → y: 16, 40, 72

1 k = 16 ÷ 2 = 8.

Answer: y = 8x

WORKED EXAMPLE · USE THE EQUATION

Use y = 3.5x to find the cost of 6 pounds.

1 y = 3.5 × 6 = 21.

Answer: $21

★ KEY IDEA

Proportional equation: y = kx, where k is the unit rate.

✎ TRY IT

Answers are shown in color.

1. A plane flies 480 mph. Equation for distance d in t hours?

Answer: d = 480t

2. x: 3, 6 → y: 2, 4. Equation?

Answer: $y = \frac{2}{3}x$

3. y = 12x. Find y when x = 7.5.

Answer: 90

Workbook Answer Key · Lesson 1.7

#	Question	Answer
1	A plane flies 480 mph. Equation for distance d in t hours?	d = 480t
2	x: 3, 6 → y: 2, 4. Equation?	$y = \frac{2}{3}x$
3	y = 12x. Find y when x = 7.5.	90
4	16 ÷ 2	8
5	3.5 × 6	21
6	k = 9	y = 9x
7	x: 4, 8 → y: 10, 20	y = 2.5x
8	$0.65 per ounce	c = 0.65w
9	15 pages per hour	p = 15h
10	y = 6x; x = 12	72
11	y = 0.8x; x = 25	20
12	y = 4x; y = 52, find x	13
13	$y = \frac{3}{4}x$; x = 20	15
14	Lillie earns $11.50 per hour. Write an equation. How much for 14 hours?	e = 11.5h; $161
15	Explain what k means in y = 45x if x is hours and y is miles.	45 miles per hour

LESSON 1.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 1: Proportional Relationships

Lesson 1.8: Is It Proportional? Tables and Equations

Student goal	I can decide if a table or equation represents a proportional relationship.
Objective	Students identify proportional relationships from tables and equations.
Standard	7.RP.2a — Decide whether two quantities are proportional.
Pacing	Unit 1, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Evaluate at x = 0.
Launch	5	Read Let's Explore.
Teach	15	Sort equation cards into proportional/not.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking any linear equation is proportional.

SUPPORT

- Test x = 0.

EXTEND

- Write 3 non-proportional real situations.

STUDENT LESSON 1.8 · WITH ANSWERS

Lesson 1.8: Is It Proportional? Tables and Equations

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.8, Part A.
- [ ] Complete the rest of Workbook Lesson 1.8 (Parts B–E).

LET'S EXPLORE

Which of these is proportional: y = 4x, y = x + 4, or y = 4?

Only equations of the form y = kx are proportional. Adding a number (like y = x + 4) breaks the pattern, because when x = 0, y would not be 0.

Equation	Proportional?	Why
y = 4x	Yes	form y = kx
y = x + 4	No	adds 4
y = 4	No	y never changes
y = 0.5x	Yes	k = 0.5

WORKED EXAMPLE · TABLE TEST

x: 0, 2, 4, 6 → y: 3, 7, 11, 15

1 When x = 0, y = 3 (not 0).

2 7 ÷ 2 = 3.5 but 11 ÷ 4 = 2.75.

Answer: Not proportional

★ KEY IDEA

Proportional: y = kx, ratios all equal, and (0, 0) fits.

✎ TRY IT

Answers are shown in color.

1. y = 7x

Answer: proportional

2. y = 2x − 1

Answer: not proportional

3. x: 0, 3, 6 → y: 0, 2, 4

Answer: proportional ($k = \frac{2}{3}$)

Workbook Answer Key · Lesson 1.8

#	Question	Answer
1	y = 7x	proportional
2	y = 2x − 1	not proportional
3	x: 0, 3, 6 → y: 0, 2, 4	proportional ($k = \frac{2}{3}$)
4	y = 7x when x = 0	0
5	y = 2x − 1 when x = 0	−1
6	y = 9x	yes
7	y = x + 9	no
8	$y = \frac{1}{3}x$	yes
9	y = 3x + 0	yes
10	y = 10 − x	no
11	y = 2.4x	yes
12	x: 1, 2, 3 → y: 4, 8, 13	no
13	x: 0, 5, 10 → y: 0, 3, 6	yes; k = 0.6
14	A phone plan costs $25 plus $10 per GB. Is the cost proportional to GB? Explain.	No; the $25 fee means 0 GB still costs $25.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.8

About 20 minutes

Weekly Check 2: Proportional Tables and Equations — Answer Key

Answers

#	Question	Answer
1	x: 2, 5, 9 → y: 7, 17.5, 31.5. Proportional? k?	Yes; k = 3.5
2	x: 1, 2, 3 → y: 5, 9, 13. Proportional?	No
3	Write the equation for k = 0.6.	y = 0.6x
4	y = 15x; x = 8	120
5	y = 4x; y = 30, find x	7.5
6	Is y = 3x + 2 proportional? Why?	No; y ≠ 0 when x = 0

LESSON 1.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 1: Proportional Relationships

Lesson 1.9: Graphing Proportional Relationships

Student goal	I can graph a proportional relationship on a coordinate plane.
Objective	Students graph proportional relationships.
Standard	7.RP.2a — Graph proportional relationships; decide whether quantities are proportional by graphing.
Pacing	Unit 1, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Graph paper, ruler, workbook Lesson 1.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Evaluate and name points.
Launch	5	Read Let's Explore.
Teach	15	Build the table, plot points, draw the line with a ruler.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Swapping x and y when plotting.

SUPPORT

- Say: over, then up.

EXTEND

- Graph two jobs' pay on one grid.

STUDENT LESSON 1.9 · WITH ANSWERS

Lesson 1.9: Graphing Proportional Relationships

STUDENT'S ASSIGNMENT

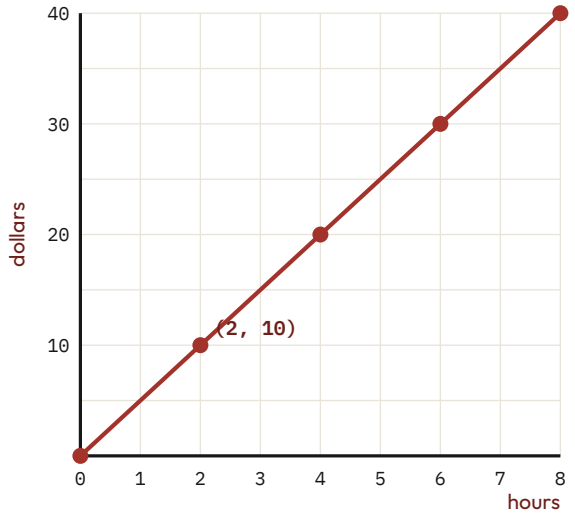
- [ ] Read Lesson 1.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.9, Part A.
- [ ] Complete the rest of Workbook Lesson 1.9 (Parts B–D).

LET'S EXPLORE

Ellie earns $5 per hour. How could a graph show her earnings for 0 to 8 hours?

Make a table, write each pair as a point (x, y), and plot the points.

Hours (x)	Dollars (y)
0	0
2	10
4	20
6	30
8	40



The points lie on a straight line through the origin. That is what every proportional graph looks like.

★ KEY IDEA

A proportional relationship graphs as a straight line through the origin (0, 0).

✎ TRY IT

Answers are shown in color.

1. Plot y = 3x for x = 0, 1, 2, 3. Name the points.

Answer: (0, 0), (1, 3), (2, 6), (3, 9)

2. Does the graph of y = 3x pass through (0, 0)?

Answer: Yes

3. Is (5, 15) on y = 3x?

Answer: Yes

Workbook Answer Key · Lesson 1.9

#	Question	Answer
1	Plot y = 3x for x = 0, 1, 2, 3. Name the points.	(0, 0), (1, 3), (2, 6), (3, 9)
2	Does the graph of y = 3x pass through (0, 0)?	Yes
3	Is (5, 15) on y = 3x?	Yes
4	y = 5x when x = 6	30
5	Name the point 4 right, 20 up	(4, 20)
6	y = 2x for x = 0 to 5 (figure)	(0,0), (1,2), (2,4), (3,6), (4,8), (5,10)
7	y = 0.5x for x = 0 to 8 (figure)	(0,0), (2,1), (4,2), (6,3), (8,4)
8	What do all proportional graphs have in common?	Straight line through the origin.

LESSON 1.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 1: Proportional Relationships

Lesson 1.10: Recognizing Proportional Graphs

Student goal	I can tell from a graph whether a relationship is proportional.
Objective	Students identify proportional relationships from graphs.
Standard	7.RP.2a — Decide whether quantities are proportional by graphing on a coordinate plane and observing a straight line through the origin.
Pacing	Unit 1, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Compare the three graphs. Two tests: straight? through origin?
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Accepting any straight line.

SUPPORT

- Two-question checklist.

EXTEND

- Find a real graph online that is proportional.

STUDENT LESSON 1.10 · WITH ANSWERS

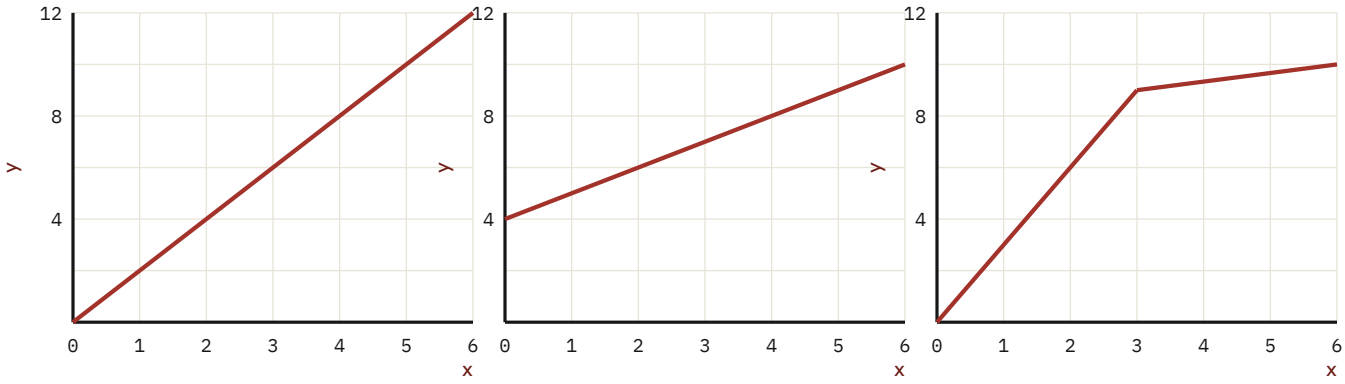
Lesson 1.10: Recognizing Proportional Graphs

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.10, Part A.
- [ ] Complete the rest of Workbook Lesson 1.10 (Parts B–E).

LET'S EXPLORE

Two graphs are straight lines. One starts at (0, 0) and one starts at (0, 4). Are both proportional?



proportional not: does not go through (0, 0) not: not a straight line

★ KEY IDEA

Proportional graph = straight line AND passes through the origin. Both are required.

✎ TRY IT

Answers are shown in color.

1. A straight line through (0, 0) and (3, 12). Proportional? k?

Answer: Yes; k = 4

2. A curved graph through (0, 0). Proportional?

Answer: No

3. A straight line through (0, 2) and (1, 5). Proportional?

Answer: No

Workbook Answer Key · Lesson 1.10

#	Question	Answer
1	A straight line through (0, 0) and (3, 12). Proportional? k?	Yes; k = 4
2	A curved graph through (0, 0). Proportional?	No
3	A straight line through (0, 2) and (1, 5). Proportional?	No
4	12 ÷ 3	4
5	Is (0, 0) the origin?	Yes
6	(figure)	yes
7	(figure)	no
8	A proportional line goes through (4, 10). Name two more points on it.	e.g. (2, 5), (8, 20)
9	Does a line through (2, 6) and (5, 15) show a proportional relationship?	Yes; k = 3 both times
10	Sketch a graph that is a straight line but not proportional.	Line not through origin

LESSON 1.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 1: Proportional Relationships

Lesson 1.11: Interpreting Points on Proportional Graphs

Student goal	I can explain what a point (x, y) means on a proportional graph, including (0, 0) and (1, k).
Objective	Students interpret points on proportional graphs.
Standard	7.RP.2d — Explain what a point (x, y) on the graph of a proportional relationship means in context, with special attention to (0, 0) and (1, r).
Pacing	Unit 1, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	28 × 3.
Launch	5	Read Let's Explore.
Teach	15	Read each labeled point aloud with units.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Ignoring units.

SUPPORT

- Always say units.

EXTEND

- Make a graph of your own unit rate.

STUDENT LESSON 1.11 · WITH ANSWERS

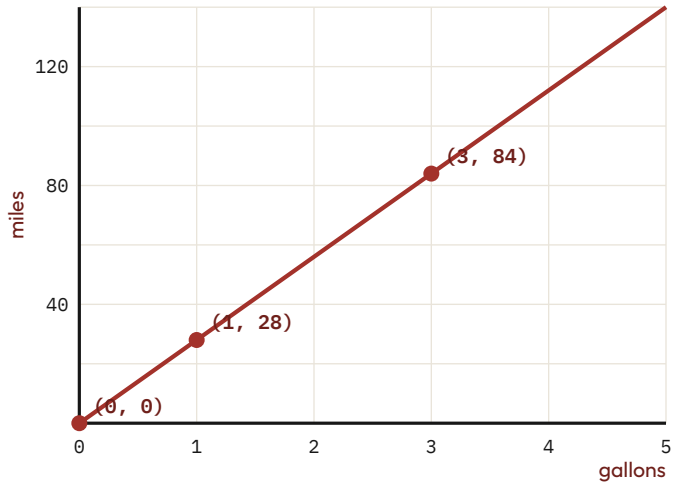
Lesson 1.11: Interpreting Points on Proportional Graphs

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.11, Part A.
- [ ] Complete the rest of Workbook Lesson 1.11 (Parts B–D).

LET'S EXPLORE

On a graph of gallons and miles, what does the point (1, 28) tell you?



- (0, 0): 0 gallons, 0 miles.
- (1, 28): 1 gallon goes 28 miles. The unit rate k = 28 is the y-value when x = 1.
- (3, 84): 3 gallons go 84 miles.

★ KEY IDEA

On a proportional graph, the point (1, k) shows the unit rate.

✎ TRY IT

Answers are shown in color.

1. (1, 7.5) on a graph of hours and dollars means...

Answer: $7.50 per hour

2. If (1, 12) is on a proportional graph, what is y when x = 5?

Answer: 60

3. What does (0, 0) mean for pages read vs minutes?

Answer: 0 minutes, 0 pages

Workbook Answer Key · Lesson 1.11

#	Question	Answer
1	(1, 7.5) on a graph of hours and dollars means...	$7.50 per hour
2	If (1, 12) is on a proportional graph, what is y when x = 5?	60
3	What does (0, 0) mean for pages read vs minutes?	0 minutes, 0 pages
4	28 × 3	84
5	84 ÷ 3	28
6	A graph of cost vs pounds goes through (1, 4.5). What does it mean?	$4.50 per pound
7	A graph of distance vs hours goes through (4, 260). Find k and explain.	k = 65; 65 miles per hour
8	A graph of cups of flour vs batches goes through (2, 5). What is the unit rate?	2.5 cups per batch
9	Write a story for a proportional graph through (1, 15).	e.g. $15 per hour

LESSON 1.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 1: Proportional Relationships

Lesson 1.12: Comparing Proportional Relationships

Student goal	I can compare proportional relationships shown as tables, graphs, equations, and words.
Objective	Students compare proportional relationships across representations.
Standard	7.RP.2 — Recognize and represent proportional relationships between quantities.
Pacing	Unit 1, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Use the table of ways to find k.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Comparing totals instead of k.

SUPPORT

- Find k first, always.

EXTEND

- Compare three phone plans.

STUDENT LESSON 1.12 · WITH ANSWERS

Lesson 1.12: Comparing Proportional Relationships

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.12, Part A.
- [ ] Complete the rest of Workbook Lesson 1.12 (Parts B–D).

LET'S EXPLORE

Ellie earns y = 9x. Lillie's table shows 3 hours → $30. Who earns more per hour?

Find k in each representation, then compare. Ellie: k = 9. Lillie: 30 ÷ 3 = 10. Lillie earns more per hour.

Representation	How to find k
Equation y = kx	the number times x
Table	y ÷ x
Graph	y-value at x = 1
Words	the unit rate (per 1)

★ KEY IDEA

Change every representation to its k (unit rate), then compare.

✎ TRY IT

Answers are shown in color.

1. Car A: y = 32x. Car B: 150 miles on 5 gallons. Better mileage?

Answer: Car A (32 vs 30 mpg)

2. Graph through (1, 2.75) vs y = 2.5x. Greater k?

Answer: the graph (2.75)

3. Store A: $4 per pound. Store B: (5, 18). Cheaper?

Answer: Store B ($3.60/lb)

Workbook Answer Key · Lesson 1.12

#	Question	Answer
1	Car A: y = 32x. Car B: 150 miles on 5 gallons. Better mileage?	Car A (32 vs 30 mpg)
2	Graph through (1, 2.75) vs y = 2.5x. Greater k?	the graph (2.75)
3	Store A: $4 per pound. Store B: (5, 18). Cheaper?	Store B ($3.60/lb)
4	30 ÷ 3	10
5	18 ÷ 5	3.6
6	Pump 1: 12 gallons per minute. Pump 2: y = 14x. Faster?	Pump 2
7	Runner A: table 3 mi in 24 min. Runner B: y = 7.5x (min per mile). Faster?	Runner B (7.5 vs 8 min/mi)
8	Recipe A: 2 cups sugar per 5 cups flour. Recipe B: y = 0.35x (sugar per flour). Sweeter?	Recipe A (0.4 vs 0.35)
9	Plan A: $45 for 3 months. Plan B: graph through (1, 16). Cheaper?	Plan A ($15 vs $16 per month)
10	Explain how to compare a table and a graph that both show proportional relationships.	Find k for each (y ÷ x; y at x = 1) and compare.

WEEKLY CHECK · ANSWER KEY
After Lesson 1.12
About 20 minutes

Weekly Check 3: Proportional Graphs — Answer Key

Answers

#	Question	Answer
1	Proportional? (figure)	yes
2	Proportional? (figure)	no
3	A proportional graph goes through (1, 6.5). What is k?	6.5
4	A proportional graph goes through (4, 18). What is y at x = 10?	45
5	Compare: y = 11x vs table 5 → 57.5. Greater rate?	table (11.5)
6	What does (0, 0) mean on a graph of hours vs pay?	0 hours, $0

LESSON 1.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 1: Proportional Relationships

Lesson 1.13: Solving Proportions

Student goal	I can solve problems with proportions using unit rates, scale factors, or equations.
Objective	Students solve proportions with multiple methods.
Standard	7.RP.3 — Use proportional relationships to solve multistep ratio and percent problems.
Pacing	Unit 1, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Show all three methods. Discuss which is fastest for which numbers.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Setting up the proportion with mismatched units.

SUPPORT

- Label every ratio with units.

EXTEND

- Use a real recipe and scale it.

STUDENT LESSON 1.13 · WITH ANSWERS

Lesson 1.13: Solving Proportions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.13, Part A.
- [ ] Complete the rest of Workbook Lesson 1.13 (Parts B–E).

LET'S EXPLORE

If 5 notebooks cost $8.75, how much do 12 notebooks cost?

WORKED EXAMPLE · UNIT RATE METHOD

5 notebooks for $8.75; 12 notebooks?

1 $8.75 \div 5 = 1.75$ per notebook.

2 $12 \times 1.75 = 21$.

Answer: $21

WORKED EXAMPLE · EQUATION METHOD

$\frac{5}{8.75} = \frac{12}{c}$

1 Equivalent ratios: $c = 8.75 \times 12 \div 5$.

2 $c = 105 \div 5 = 21$.

Answer: $21

WORKED EXAMPLE · SCALE FACTOR

3 cups of flour make 24 cookies. How much flour for 96 cookies?

1 $96 \div 24 = 4$, so multiply by 4.

2 $3 \times 4 = 12$.

Answer: 12 cups

★ KEY IDEA

Use whichever method fits the numbers: unit rate, scale factor, or an equation.

✎ TRY IT

Answers are shown in color.

1. 4 tickets cost $34. Cost of 10 tickets?

Answer: $85

2. Solve $\frac{6}{15} = \frac{x}{25}$

Answer: x = 10

3. A map: 2 cm = 15 km. 7 cm = ?

Answer: 52.5 km

Workbook Answer Key · Lesson 1.13

#	Question	Answer
1	4 tickets cost $34. Cost of 10 tickets?	$85
2	Solve $\frac{6}{15} = \frac{x}{25}$	x = 10
3	A map: 2 cm = 15 km. 7 cm = ?	52.5 km
4	8.75 ÷ 5	1.75
5	96 ÷ 24	4
6	$\frac{3}{8} = \frac{x}{40}$	15
7	$\frac{7}{x} = \frac{21}{33}$	11
8	$\frac{x}{12} = \frac{5}{4}$	15
9	$\frac{2.5}{4} = \frac{x}{10}$	6.25
10	6 bags of mulch cover 72 sq ft. How many bags for 300 sq ft?	25 bags
11	A 20-oz bottle has 240 calories. How many calories in 12 oz?	144 calories
12	Ellie reads 45 pages in 1.5 hours. How long for 120 pages?	4 hours
13	Paint: 2 gallons per 700 sq ft. Gallons for 1,750 sq ft?	5 gallons
14	Solve two ways: 8 lemons make 3 quarts. How many lemons for 12 quarts?	32 lemons

LESSON 1.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 1: Proportional Relationships

Lesson 1.14: Proportional Reasoning in the Real World

Student goal	I can solve real-world problems with proportional relationships, including recipes, maps, and speed.
Objective	Students apply proportional reasoning to real-world problems.
Standard	7.RP.3 — Use proportional relationships to solve multistep ratio problems.
Pacing	Unit 1, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Recipes, a Mississippi map, workbook Lesson 1.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work the three examples. Measure a real distance on a map.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting to convert units.

SUPPORT

- Write the unit rate first.

EXTEND

- Plan a road trip with distances and times.

STUDENT LESSON 1.14 · WITH ANSWERS

Lesson 1.14: Proportional Reasoning in the Real World

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.14, Part A.
- [ ] Complete the rest of Workbook Lesson 1.14 (Parts B–D).

LET'S EXPLORE

A recipe for 6 people uses $1\frac{1}{2}$ cups of rice. How much for 10 people?

WORKED EXAMPLE · RECIPE

$1\frac{1}{2}$ cups for 6 people → 10 people

1 Per person: $\frac{3}{2} \div 6 = \frac{1}{4}$ cup.

2 $10 \times \frac{1}{4} = 2\frac{1}{2}$.

Answer: $2\frac{1}{2}$ cups

WORKED EXAMPLE · MAP

On a map, 1 inch = 25 miles. Jackson to Hattiesburg is 3.6 inches.

1 $3.6 \times 25 = 90$.

Answer: about 90 miles

WORKED EXAMPLE · TIME AND DISTANCE

At 60 mph, how long to drive 210 miles?

1 $210 \div 60 = 3.5$.

Answer: 3.5 hours

★ KEY IDEA

Find the unit rate, then multiply or divide as needed. Check: does the answer make sense?

✎ TRY IT

Answers are shown in color.

1. A car uses 2 gallons per 54 miles. Gallons for 189 miles?

Answer: 7 gallons

2. Map: 1 cm = 40 km. 6.5 cm = ?

Answer: 260 km

3. $3/4$ cup sugar for 12 cookies. For 30 cookies?

Answer: $1\ 7/8$ cups

Workbook Answer Key · Lesson 1.14

#	Question	Answer
1	A car uses 2 gallons per 54 miles. Gallons for 189 miles?	7 gallons
2	Map: 1 cm = 40 km. 6.5 cm = ?	260 km
3	$3/4$ cup sugar for 12 cookies. For 30 cookies?	$1\ 7/8$ cups
4	$3/2 \div 6$	$1/4$
5	3.6×25	90
6	A recipe for 4 people uses 2 lb of chicken. How much for 14 people?	7 lb
7	A cyclist rides 18 miles in 1.5 hours. At that rate, how far in 4 hours?	48 miles
8	A model car is built at a scale of 1 in. : 18 in. The model is 9 in. long. How long is the real car in feet?	162 in. = 13.5 ft
9	Lillie earns $57 for 6 hours of work. How many hours to earn $133?	14 hours
10	Write and solve a recipe-scaling problem for your family.	Answers vary.

LESSON 1.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 1: Proportional Relationships

Lesson 1.15: Multi-Step Ratio Problems

Student goal	I can solve multi-step problems involving ratios and rates.
Objective	Students solve multi-step ratio and rate problems.
Standard	7.RP.3 — Use proportional relationships to solve multistep ratio problems.
Pacing	Unit 1, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Tape diagram for part-to-whole. Then the average-speed example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Averaging the speeds (57.5) instead of using total distance ÷ total time.

SUPPORT

- Tape diagrams.

EXTEND

- Mix paint or juice in a real ratio.

STUDENT LESSON 1.15 · WITH ANSWERS

Lesson 1.15: Multi-Step Ratio Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.15, Part A.
- [ ] Complete the rest of Workbook Lesson 1.15 (Parts B–D).

LET'S EXPLORE

Paint is mixed 3 parts blue to 2 parts white. You need 20 quarts of paint. How much of each color?

WORKED EXAMPLE · PART-TO-WHOLE

3 blue : 2 white, 20 quarts total

1 3 + 2 = 5 parts in the whole.

2 20 ÷ 5 = 4 quarts per part.

3 Blue: 3 × 4 = 12; white: 2 × 4 = 8.

Answer: 12 qt blue, 8 qt white

WORKED EXAMPLE · TWO RATES

Ellie drives 2 hours at 50 mph, then 3 hours at 65 mph. Average speed?

1 Distance: 100 + 195 = 295 miles.

2 Time: 5 hours.

3 295 ÷ 5 = 59.

Answer: 59 mph

★ KEY IDEA

Use the total number of parts for part-to-whole problems. Average speed = total distance ÷ total time.

✎ TRY IT

Answers are shown in color.

1. Boys to girls 4:5; 36 students. How many girls?

Answer: 20

2. Concrete mix 1:2:3 (cement:sand:gravel). 30 buckets total. Gravel?

Answer: 15 buckets

3. Walk 1 hr at 3 mph, then jog 1 hr at 6 mph. Average speed?

Answer: 4.5 mph

Workbook Answer Key · Lesson 1.15

#	Question	Answer
1	Boys to girls 4:5; 36 students. How many girls?	20
2	Concrete mix 1:2:3 (cement:sand:gravel). 30 buckets total. Gravel?	15 buckets
3	Walk 1 hr at 3 mph, then jog 1 hr at 6 mph. Average speed?	4.5 mph
4	20 ÷ 5 × 3	12
5	100 + 195	295
6	A bag has red and green marbles in a ratio of 5:3. There are 48 marbles. How many are red?	30 red
7	A drink mix uses 2 scoops per 3 cups of water. How many scoops for a 9-quart cooler? (1 qt = 4 cups)	24 scoops
8	Lillie saves $3 for every $5 she earns. She earned $85. How much did she save? Spend?	$51 saved, $34 spent
9	A train travels 180 miles in 3 hours, then 120 miles in 1 hour. Average speed for the whole trip?	75 mph
10	A ratio of cats to dogs at a shelter is 2:7. There are 63 animals. How many more dogs than cats?	35 more dogs (49 vs 14)

LESSON 1.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 1: Proportional Relationships

Lesson 1.16: Proportional Relationships Problem Solving

Student goal	I can choose the best representation to solve proportional relationship problems.
Objective	Students solve mixed proportional problems.
Standard	7.RP.1–3 — 7.RP review.
Pacing	Unit 1, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 1.16, Unit 1 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 1.16 and the Unit 1 Review.
Exit Ticket	3	Check. Tomorrow: Unit 1 Test.

COMMON MISCONCEPTIONS

- Mixing up representations.

SUPPORT

- Find k first.

EXTEND

- Research real phone or streaming plans.

STUDENT LESSON 1.16 · WITH ANSWERS

Lesson 1.16: Proportional Relationships Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 1.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 1.16, Part A.
- [ ] Complete the rest of Workbook Lesson 1.16 (Parts B–D).

LET'S EXPLORE

Three phone plans: Plan A is y = 0.12x (dollars per text). Plan B: 50 texts for $5.50. Plan C: a graph through (1, 0.10). Which is cheapest per text?

WORKED EXAMPLE · COMPARE PLANS

Find k for each.

1 A: 0.12.

2 B: 5.50 ÷ 50 = 0.11.

3 C: 0.10.

Answer: Plan C

★ KEY IDEA

Tables, graphs, equations, and words all show the same idea: a constant rate k.

✎ TRY IT

Answers are shown in color.

1. y = 18x: what is k?

Answer: 18

2. Is y = x + 18 proportional?

Answer: No

3. Table: 4 → 14, 10 → 35. Equation?

Answer: y = 3.5x

Workbook Answer Key · Lesson 1.16

#	Question	Answer
1	y = 18x: what is k?	18
2	Is y = x + 18 proportional?	No
3	Table: 4 → 14, 10 → 35. Equation?	y = 3.5x
4	5.50 ÷ 50	0.11
5	0.12 × 100	12
6	$\frac{1}{2}$ mile in $\frac{1}{8}$ hour. mph?	4 mph
7	Better buy: 15 oz for $3.30 or 20 oz for $4.20?	20 oz ($0.21 vs $0.22)
8	x: 2, 3, 7 → y: 5, 7.5, 17.5. Equation?	y = 2.5x
9	A graph passes through (3, 7). Proportional? Need more information?	Need to know it's a straight line through (0, 0).
10	Ratio 3:4, total 56. Larger part?	32
11	$42 for 3.5 hours. Hourly rate?	$12
12	Which representation do you find most useful for proportional relationships? Why?	Answers vary.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 1 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	$9.60 for 12 cans: unit price	$0.80
2	$\frac{3}{4}$ pound for $\frac{1}{2}$ dollar: pounds per dollar	$1\frac{1}{2}$
3	Solve $\frac{9}{12} = \frac{x}{20}$	15
4	Better buy: 6 for $4.50 or 10 for $7.20	10 for $7.20 ($0.72 vs $0.75)
5	x: 3, 6, 12 → y: 2, 4, 8. Equation?	$y = \frac{2}{3}x$
6	Is y = 5x − 5 proportional?	No
7	A proportional graph passes through (1, 3.25). Write the equation.	y = 3.25x
8	Ratio 2:5; total 63. Smaller part?	18
9	A recipe for 8 people uses 3 cups of broth. How much for 20 people?	7.5 cups

UNIT TEST · ANSWER KEY
At the end of the unit
About 45 minutes

Unit 1 Test — Answer Key

Answers

#	Question	Answer
1	$13.75 for 5 pounds: unit price	$2.75 per pound
2	$\frac{1}{2}$ cup per $\frac{2}{3}$ batch: cups per batch	$\frac{3}{4}$ cup
3	Better buy: 24 oz for $4.08 or 32 oz for $5.12	32 oz ($0.16 vs $0.17)
4	Write two ratios equivalent to 6:10.	e.g. 3:5, 12:20
5	x: 4, 6, 10 → y: 10, 15, 25. Proportional? k?	Yes; 2.5
6	x: 2, 4, 6 → y: 3, 7, 11. Proportional?	No
7	Write the equation: $8.25 per hour	y = 8.25x
8	y = 0.4x; x = 35	14
9	Is y = 7 + x proportional?	No
10	A proportional graph passes through (5, 40). k?	8
11	Interpret (1, 12) on a graph of hours vs miles.	12 miles per hour
12	A car travels 196 miles on 7 gallons. Write an equation for miles m and gallons g. How far can it go on a full 15-gallon tank?	m = 28g; 420 miles
13	Sunscreen: 3 oz for $7.20 or 8 oz for $17.60. Which is the better buy? By how much per ounce?	8 oz; $2.20 vs $2.40 per oz, $0.20 less
14	Lemonade is 2 parts juice to 7 parts water. How much juice is in 45 cups of lemonade?	10 cups

Scoring

Score	What it means / next step
17–20 (85%+)	Mastered.
14–16	Proficient; correct missed items.
Below 14	Reteach and retest.

UNIT 2 · TEACHER OVERVIEW

Percents

Percent as a rate per 100, percent equations, discounts, tax, tips, percent change, interest, and percent error.

Percents are rates per 100, and they are everywhere: sales, taxes, grades, interest, and statistics. You will use proportional reasoning and the percent equation to solve real percent problems.

Big ideas

- Understand percent as a rate per 100.
- Convert among percents, fractions, and decimals.
- Find a part, a percent, or a whole.
- Solve discount, markup, tax, tip, and commission problems.
- Find percent change, simple interest, and percent error.

Standards

Code	Standard
7.RP.3	Use proportional relationships to solve multistep ratio and percent problems: simple interest, tax, markups and markdowns, gratuities and commissions, fees, percent increase and decrease, percent error.
7.EE.3	Solve multi-step real-life problems with positive and negative rational numbers; assess reasonableness using estimation.

What students should already know

- Proportional relationships (Unit 1).
- Decimal multiplication and division.

Materials for the unit

- Calculator for checking
- Store ads and receipts
- Hundred grids

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	2.1	What Is a Percent?
Week 1	Lesson 2 of 4	2.2	Percents, Fractions, and Decimals
Week 1	Lesson 3 of 4	2.3	Finding a Percent of a Number
Week 1	Lesson 4 of 4	2.4	Finding What Percent
Week 1	End of week	Check	Weekly Check 4: Percent Basics
Week 2	Lesson 1 of 4	2.5	Finding the Whole
Week 2	Lesson 2 of 4	2.6	The Percent Equation
Week 2	Lesson 3 of 4	2.7	Percents Greater Than 100% and Less Than 1%
Week 2	Lesson 4 of 4	2.8	Estimating with Percents
Week 2	End of week	Check	Weekly Check 5: Percent Equations
Week 3	Lesson 1 of 4	2.9	Discounts and Sale Prices
Week 3	Lesson 2 of 4	2.10	Markups and Sales Tax
Week 3	Lesson 3 of 4	2.11	Percent Increase and Decrease
Week 3	Lesson 4 of 4	2.12	Tips, Commissions, and Fees
Week 3	End of week	Check	Weekly Check 6: Discounts, Tax, and Change
Week 4	Lesson 1 of 4	2.13	Simple Interest
Week 4	Lesson 2 of 4	2.14	Percent Error
Week 4	Lesson 3 of 4	2.15	Multi-Step Percent Problems
Week 4	Lesson 4 of 4	2.16	Percent Problem Solving
Week 4	Before the test	Review	Unit 2 Review
Week 4	End of unit	Test	Unit 2 Test

TEACHING THIS UNIT AT HOME

Let your student calculate the tip and tax at a restaurant or the sale price at a store this unit.

UNIT VOCABULARY

percent	A rate per 100.
discount	Amount off a price.
markup	Amount added to cost.
percent change	Change ÷ original.
simple interest	$I = P \times r \times t$.
percent error	|estimate − actual| ÷ actual.

LESSON 2.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 2: Percents

Lesson 2.1: What Is a Percent?

Student goal	I can explain a percent as a rate per 100.
Objective	Students understand percent as a rate per 100.
Standard	7.RP.3 — Use proportional relationships to solve percent problems.
Pacing	Unit 2, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Hundred grids, workbook Lesson 2.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	25 × 4.
Launch	5	Read Let's Explore.
Teach	15	Shade 35 on a hundred grid; show the double number line. Say: "Percent means per hundred."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Treating 7 out of 20 as 7%.

SUPPORT

- Scale to 100 with a double number line.

EXTEND

- Find percents in the news.

STUDENT LESSON 2.1 · WITH ANSWERS

Lesson 2.1: What Is a Percent?

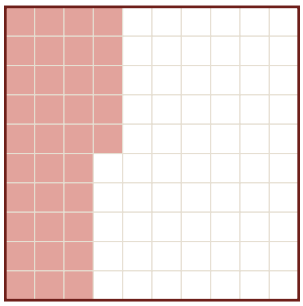
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.1, Part A.
- [ ] Complete the rest of Workbook Lesson 2.1 (Parts B–E).

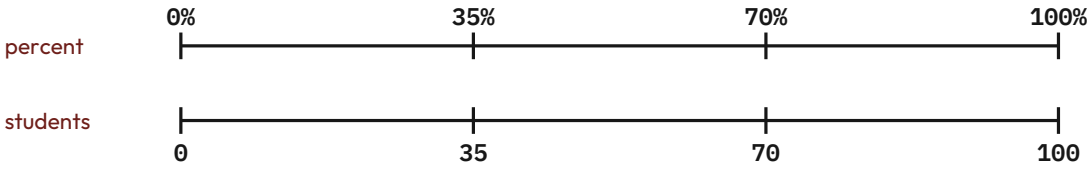
LET'S EXPLORE

A survey says 35% of students walk to school. What does that mean if a school has 100 students? 200 students?

Percent means "per hundred." 35% is 35 per 100. If there are 100 students, 35 walk. If there are 200, twice as many walk: 70.



35 out of 100 squares: 35%



WORKED EXAMPLE · PER 100

In a class, 18 out of 25 students got an A. What percent is that?

1 Scale to 100: 25 × 4 = 100.

2 18 × 4 = 72.

Answer: 72%

★ KEY IDEA

Percent is a rate per 100. Scale the ratio so the whole is 100.

✎ TRY IT

Answers are shown in color.

1. 40 out of 100 = ___%

Answer: 40%

2. 7 out of 20 = ___%

Answer: 35%

3. What percent is 3 out of 4?

Answer: 75%

Workbook Answer Key · Lesson 2.1

#	Question	Answer
1	40 out of 100 = ___%	40%
2	7 out of 20 = ___%	35%
3	What percent is 3 out of 4?	75%
4	25 × 4	100
5	18 × 4	72
6	13 out of 100	13%
7	9 out of 10	90%
8	12 out of 50	24%
9	3 out of 5	60%
10	7 out of 25	28%
11	1 out of 4	25%
12	20 out of 20	100%
13	11 out of 20	55%
14	A basketball player made 17 of 20 free throws. What percent did she make?	85%
15	Lillie answered 46 of 50 questions correctly. Her percent score?	92%
16	Explain why 50% of something is half of it.	50 per 100 = $^{1}/_{2}$

LESSON 2.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 2: Percents

Lesson 2.2: Percents, Fractions, and Decimals

Student goal	I can write percents as fractions and decimals, and fractions and decimals as percents.
Objective	Students convert among percents, fractions, and decimals.
Standard	7.RP.3, 7.NS.2d — Convert a rational number to a decimal using long division; use percents.
Pacing	Unit 2, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	3 ÷ 8.
Launch	5	Read Let's Explore.
Teach	15	Fill in the benchmark table from memory together. Model each conversion.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Writing 0.05 as 50%.

SUPPORT

- Place-value chart.

EXTEND

- Memorize the benchmark table.

STUDENT LESSON 2.2 · WITH ANSWERS

Lesson 2.2: Percents, Fractions, and Decimals

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.2, Part A.
- [ ] Complete the rest of Workbook Lesson 2.2 (Parts B–E).

LET'S EXPLORE

Which is greater: $^{3}/_{8}$, 0.4, or 37%?

Percent	Fraction	Decimal
50%	$^{1}/_{2}$	0.5
25%	$^{1}/_{4}$	0.25
20%	$^{1}/_{5}$	0.2
10%	$^{1}/_{10}$	0.1
75%	$^{3}/_{4}$	0.75
12.5%	$^{1}/_{8}$	0.125

- Percent → decimal: divide by 100 (move the decimal point 2 places left). 37% = 0.37.
- Decimal → percent: multiply by 100. 0.4 = 40%.
- Fraction → percent: divide, then multiply by 100. $^{3}/_{8}$ = 3 ÷ 8 = 0.375 = 37.5%.

WORKED EXAMPLE · COMPARE

$^{3}/_{8}$, 0.4, 37%

1 $^{3}/_{8}$ = 37.5%; 0.4 = 40%; 37%.

Answer: 0.4 is greatest; 37% is least

★ KEY IDEA

Change everything to the same form to compare.

✎ TRY IT

Answers are shown in color.

1. 0.06 as a percent

Answer: 6%

2. 145% as a decimal

Answer: 1.45

3. $^{7}/_{20}$ as a percent

Answer: 35%

Workbook Answer Key · Lesson 2.2

#	Question	Answer
1	0.06 as a percent	6%
2	145% as a decimal	1.45
3	$^{7}/_{20}$ as a percent	35%
4	3 ÷ 8	0.375
5	0.4 × 100	40
6	62% = ___ (decimal)	0.62
7	8% = ___ (decimal)	0.08
8	0.9 = ___%	90%
9	1.25 = ___%	125%
10	$^{2}/_{5}$ = ___%	40%
11	$^{5}/_{8}$ = ___%	62.5%
12	30% = ___ (fraction, simplest)	$^{3}/_{10}$
13	64% = ___ (fraction)	$^{16}/_{25}$
14	$^{2}/_{3}$, 0.65, 64%	64%, 0.65, $^{2}/_{3}$
15	Explain why 0.05 is 5%, not 50%.	0.05 = 5 hundredths = 5 per 100.

LESSON 2.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 2: Percents

Lesson 2.3: Finding a Percent of a Number

Student goal	I can find a percent of a number using models and multiplication.
Objective	Students find a percent of a quantity.
Standard	7.RP.3 — Use proportional relationships to solve percent problems.
Pacing	Unit 2, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	10% of 60.
Launch	5	Read Let's Explore.
Teach	15	Use the percent bar, then multiplication.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying by 30 instead of 0.30.

SUPPORT

- Use 10% and 1% building blocks.

EXTEND

- Compute tips at a restaurant.

STUDENT LESSON 2.3 · WITH ANSWERS

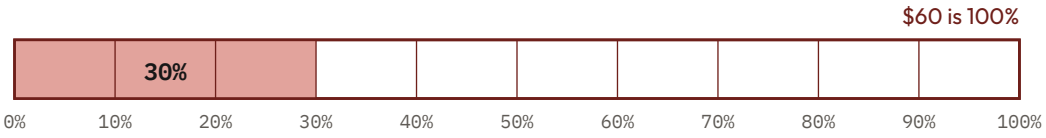
Lesson 2.3: Finding a Percent of a Number

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.3, Part A.
- [ ] Complete the rest of Workbook Lesson 2.3 (Parts B–E).

LET'S EXPLORE

A $60 jacket is 30% off. How many dollars do you save?



10% of 60 is 6, so 30% is 3 × 6 = 18. Or multiply: 0.30 × 60 = 18.

WORKED EXAMPLE · USE 10%

30% of $60

1 10% of 60 = 6.

2 30% = 3 × 6 = 18.

Answer: $18

WORKED EXAMPLE · MULTIPLY BY THE DECIMAL

15% of 240

1 0.15 × 240 = 36.

Answer: 36

★ KEY IDEA

Percent of a number = percent (as a decimal) × number. 10% and 1% are quick building blocks.

✎ TRY IT

Answers are shown in color.

1. 20% of 85

Answer: 17

2. 7% of 300

Answer: 21

3. 120% of 50

Answer: 60

Workbook Answer Key · Lesson 2.3

#	Question	Answer
1	20% of 85	17
2	7% of 300	21
3	120% of 50	60
4	10% of 60	6
5	0.15 × 240	36
6	10% of 90	9
7	25% of 64	16
8	40% of 35	14
9	5% of 180	9
10	75% of 48	36
11	12% of 250	30
12	150% of 30	45
13	1% of 600	6
14	A school has 450 students. 22% play an instrument. How many?	99 students
15	A $25 meal plus an 18% tip. How much is the tip?	$4.50
16	Find 35% of 80 two different ways.	28 (3.5 × 8; 0.35 × 80)

LESSON 2.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 2: Percents

Lesson 2.4: Finding What Percent

Student goal	I can find what percent one number is of another.
Objective	Students find what percent one number is of another.
Standard	7.RP.3 — Solve percent problems.
Pacing	Unit 2, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	42 ÷ 48.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Identify part and whole first.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing whole by part.

SUPPORT

- Label part and whole.

EXTEND

- Track your free throws and compute your percent.

STUDENT LESSON 2.4 · WITH ANSWERS

Lesson 2.4: Finding What Percent

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.4, Part A.
- [ ] Complete the rest of Workbook Lesson 2.4 (Parts B–E).

LET'S EXPLORE

Ellie scored 42 points out of 48 on a test. What percent is that?

Divide the part by the whole, then write the decimal as a percent.

WORKED EXAMPLE · PART ÷ WHOLE

42 out of 48

1 42 ÷ 48 = 0.875.

2 0.875 = 87.5%.

Answer: 87.5%

WORKED EXAMPLE · MORE THAN 100%

This year's garden produced 90 tomatoes. Last year it produced 60. This year is what percent of last year?

1 90 ÷ 60 = 1.5 = 150%.

Answer: 150%

★ KEY IDEA

Percent = part ÷ whole × 100%.

✎ TRY IT

Answers are shown in color.

1. What percent of 80 is 20?

Answer: 25%

2. What percent of 50 is 7?

Answer: 14%

3. What percent of 40 is 52?

Answer: 130%

Workbook Answer Key · Lesson 2.4

#	Question	Answer
1	What percent of 80 is 20?	25%
2	What percent of 50 is 7?	14%
3	What percent of 40 is 52?	130%
4	42 ÷ 48	0.875
5	0.875 = ___%	87.5%
6	9 of 36	25%
7	24 of 30	80%
8	13 of 52	25%
9	45 of 60	75%
10	3 of 12	25%
11	66 of 55	120%
12	A team won 27 of 36 games. What percent did it win?	75%
13	A store sold 340 of its 400 jackets. What percent are left?	15%
14	A shirt costs $12. A sweater costs $30. The shirt's price is what percent of the sweater's?	40%

WEEKLY CHECK · ANSWER KEY
After Lesson 2.4

About 15 minutes

Weekly Check 4: Percent Basics — Answer Key

Answers

#	Question	Answer
1	9 out of 20 = ___%	45%
2	$^{3}/_{8}$ = ___%	37.5%
3	0.07 = ___%	7%
4	30% of 70	21
5	What percent of 60 is 45?	75%
6	A team won 14 of 16 games. Percent won?	87.5%

LESSON 2.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 2: Percents

Lesson 2.5: Finding the Whole

Student goal	I can find the whole when I know a part and its percent.
Objective	Students find the whole given a part and a percent.
Standard	7.RP.3 — Solve percent problems.
Pacing	Unit 2, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Double number line, then equation.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying the part by the percent.

SUPPORT

- Ask: is the whole bigger or smaller than the part?

EXTEND

- Find the original price from a discount amount.

STUDENT LESSON 2.5 · WITH ANSWERS

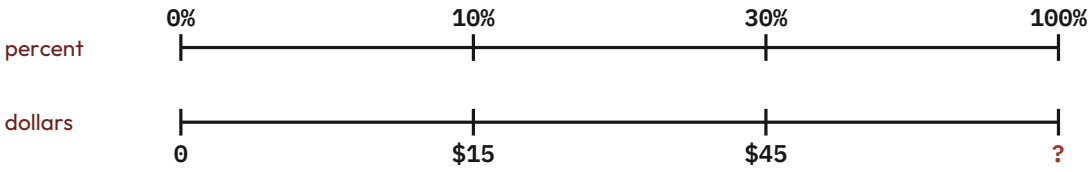
Lesson 2.5: Finding the Whole

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.5, Part A.
- [ ] Complete the rest of Workbook Lesson 2.5 (Parts B–E).

LET'S EXPLORE

Lillie has saved $45, which is 30% of the cost of a bike. How much does the bike cost?



WORKED EXAMPLE · USE 10%

30% is $45

1 10% is 45 ÷ 3 = 15.

2 100% is 15 × 10 = 150.

Answer: $150

WORKED EXAMPLE · DIVIDE BY THE PERCENT

30% of the whole is 45

1 0.30 × w = 45.

2 w = 45 ÷ 0.30 = 150.

Answer: $150

★ KEY IDEA

Whole = part ÷ percent (as a decimal).

✎ TRY IT

Answers are shown in color.

1. 20% of what number is 14?

Answer: 70

2. 75% of what number is 60?

Answer: 80

3. 8 is 40% of what?

Answer: 20

Workbook Answer Key · Lesson 2.5

#	Question	Answer
1	20% of what number is 14?	70
2	75% of what number is 60?	80
3	8 is 40% of what?	20
4	45 ÷ 0.30	150
5	If 10% is 7, 100% is ___	70
6	10% of ___ is 12	120
7	25% of ___ is 9	36
8	50% of ___ is 37	74
9	60% of ___ is 42	70
10	5% of ___ is 3	60
11	125% of ___ is 50	40
12	A player scored 24 points, which was 40% of the team's points. How many points did the team score?	60 points
13	Ellie paid a $6 deposit, which is 15% of the price. What is the price?	$40
14	18 students walk to school. That is 12% of the school. How many students attend?	150 students

LESSON 2.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 2: Percents

Lesson 2.6: The Percent Equation

Student goal	I can use part = percent × whole to solve any percent problem.
Objective	Students use the percent equation.
Standard	7.RP.3, 7.EE.4 — Solve problems using equations; use proportional relationships for percent problems.
Pacing	Unit 2, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Teach part = percent × whole with the three cases.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using the percent as a whole number in the equation.

SUPPORT

- Convert percent to decimal first.

EXTEND

- Solve with a proportion too and compare.

STUDENT LESSON 2.6 · WITH ANSWERS

Lesson 2.6: The Percent Equation

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.6, Part A.
- [ ] Complete the rest of Workbook Lesson 2.6 (Parts B–D).

LET'S EXPLORE

All percent problems have three numbers: a part, a percent, and a whole. If you know two, you can find the third.

Unknown	Equation	Example
part	$p = 0.25 \times 80$	20
percent	$20 = r \times 80$	$r = 0.25 = 25\%$
whole	$20 = 0.25 \times w$	$w = 80$

WORKED EXAMPLE · FIND THE WHOLE

36 is 45% of what number?

1 $36 = 0.45 \times w$.

2 $w = 36 \div 0.45 = 80$.

Answer: 80

★ KEY IDEA

part = percent × whole. Substitute what you know, then solve.

✎ TRY IT

Answers are shown in color.

1. What is 64% of 75?

Answer: 48

2. 27 is what percent of 90?

Answer: 30%

3. 33 is 55% of what?

Answer: 60

Workbook Answer Key · Lesson 2.6

#	Question	Answer
1	What is 64% of 75?	48
2	27 is what percent of 90?	30%
3	33 is 55% of what?	60
4	36 ÷ 0.45	80
5	0.64 × 75	48
6	What is 18% of 350?	63
7	42 is what percent of 168?	25%
8	91 is 70% of what?	130
9	What is 2.5% of 400?	10
10	60 is what percent of 48?	125%
11	15 is 6% of what?	250
12	Write and solve: 14 is 35% of what number?	14 = 0.35w; w = 40

LESSON 2.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 2: Percents

Lesson 2.7: Percents Greater Than 100% and Less Than 1%

Student goal	I can work with percents greater than 100% and less than 1%.
Objective	Students work with very large and very small percents.
Standard	7.RP.3 — Solve percent problems.
Pacing	Unit 2, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work examples; stress the Watch Out.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- 0.5% = 0.5.

SUPPORT

- Use a place-value chart.

EXTEND

- Find a real example of a percent over 100%.

STUDENT LESSON 2.7 · WITH ANSWERS

Lesson 2.7: Percents Greater Than 100% and Less Than 1%

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.7, Part A.
- [ ] Complete the rest of Workbook Lesson 2.7 (Parts B–E).

LET'S EXPLORE

A town's population grew to 250% of what it was. A bank pays 0.5% interest. What do these mean?

250% means 2.5 times the whole. 0.5% means half of 1 percent: 0.005 times the whole.

WORKED EXAMPLE · GREATER THAN 100%

250% of 80

1 2.5 × 80 = 200.

Answer: 200

WORKED EXAMPLE · LESS THAN 1%

0.5% of $2,000

1 0.005 × 2,000 = 10.

Answer: $10

! WATCH OUT

0.5% is not 0.5. As a decimal, 0.5% = 0.005.

★ KEY IDEA

Percent → decimal: divide by 100, even for 250% (2.5) or 0.5% (0.005).

✎ TRY IT

Answers are shown in color.

1. 300% of 15

Answer: 45

2. 0.2% of 5,000

Answer: 10

3. 125% as a decimal

Answer: 1.25

Workbook Answer Key · Lesson 2.7

#	Question	Answer
1	300% of 15	45
2	0.2% of 5,000	10
3	125% as a decimal	1.25
4	2.5 × 80	200
5	0.005 × 2,000	10
6	200% of 34	68
7	110% of 60	66
8	0.1% of 9,000	9
9	0.75% of 400	3
10	175% of 20	35
11	0.4% as a decimal	0.004
12	A puppy's weight is now 300% of its birth weight of 2.5 lb. How much does it weigh?	7.5 lb
13	A savings account earns 0.6% per year on $1,500. How much interest in 1 year?	$9
14	Is 150% of a number more or less than the number? Explain.	More; it is 1.5 times the number.

LESSON 2.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 2: Percents

Lesson 2.8: Estimating with Percents

Student goal	I can estimate percents using benchmarks like 10%, 25%, and 50%.
Objective	Students estimate percents with benchmarks.
Standard	7.RP.3, 7.EE.3 — Use mental computation and estimation strategies to assess reasonableness.
Pacing	Unit 2, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.8, restaurant receipts
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Teach 10%, 5%, 25%, 50% shortcuts.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Moving the decimal the wrong way for 10%.

SUPPORT

- Use money examples.

EXTEND

- Estimate tips on real receipts.

STUDENT LESSON 2.8 · WITH ANSWERS

Lesson 2.8: Estimating with Percents

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.8, Part A.
- [ ] Complete the rest of Workbook Lesson 2.8 (Parts B–D).

LET'S EXPLORE

A $39.95 shirt is 25% off. About how much will it cost?

WORKED EXAMPLE · BENCHMARK

25% off $39.95

1 Round to $40.

2 25% of 40 = 10.

3 40 − 10 = 30.

Answer: about $30

WORKED EXAMPLE · TIP ESTIMATE

A 20% tip on $47.80

1 Round to $48. 10% = $4.80, so 20% ≈ $9.60.

Answer: about $10

★ KEY IDEA

10%: move the decimal point one place left. 5% is half of 10%. 25% is a quarter. 50% is half.

✎ TRY IT

Answers are shown in color.

1. Estimate 10% of $83.20

Answer: about $8.30

2. Estimate 15% of $59

Answer: about $9

3. Estimate 48% of 202

Answer: about 100

Workbook Answer Key · Lesson 2.8

#	Question	Answer
1	Estimate 10% of $83.20	about $8.30
2	Estimate 15% of $59	about $9
3	Estimate 48% of 202	about 100
4	10% of 47.80	4.78
5	Half of 4.80	2.40
6	20% of $31.75	about $6.40
7	25% of 397	about 100
8	5% of $198	about $10
9	33% of 61	about 20
10	Tip of 18% on $52	about $9–10
11	75% of 81	about 60
12	Explain how to find 15% of a restaurant bill in your head.	Find 10%, take half of it for 5%, and add.

WEEKLY CHECK · ANSWER KEY
After Lesson 2.8

About 20 minutes

Weekly Check 5: Percent Equations — Answer Key

Answers

#	Question	Answer
1	40% of what is 26?	65
2	18 is what percent of 24?	75%
3	250% of 18	45
4	0.3% of 2,000	6
5	Estimate 15% of $79.	about $12
6	Write and solve: 63 is 35% of n.	63 = 0.35n; n = 180

LESSON 2.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 2: Percents

Lesson 2.9: Discounts and Sale Prices

Student goal	I can find the sale price after a percent discount.
Objective	Students compute discounts and sale prices.
Standard	7.RP.3 — Use proportional relationships to solve multistep percent problems: discounts.
Pacing	Unit 2, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Store ads, workbook Lesson 2.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Two methods.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding the discount.

SUPPORT

- Write 'you pay ___%'.

EXTEND

- Compare real sale ads.

STUDENT LESSON 2.9 · WITH ANSWERS

Lesson 2.9: Discounts and Sale Prices

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.9, Part A.
- [ ] Complete the rest of Workbook Lesson 2.9 (Parts B–E).

LET'S EXPLORE

A $80 pair of shoes is 25% off. What is the sale price?

WORKED EXAMPLE · SUBTRACT THE DISCOUNT

25% off $80

1 Discount: 0.25 × 80 = 20.

2 Sale price: 80 − 20 = 60.

Answer: $60

WORKED EXAMPLE · MULTIPLY BY WHAT'S LEFT

25% off means you pay 75%

1 0.75 × 80 = 60.

Answer: $60

★ KEY IDEA

Sale price = original × (100% − discount%).

✎ TRY IT

Answers are shown in color.

1. 40% off $45

Answer: $27

2. 15% off $120

Answer: $102

3. A $90 coat is on sale for $63. What percent off?

Answer: 30%

Workbook Answer Key · Lesson 2.9

#	Question	Answer
1	40% off $45	$27
2	15% off $120	$102
3	A $90 coat is on sale for $63. What percent off?	30%
4	0.25 × 80	20
5	100% − 15%	85%
6	20% off $35.00	$28.00
7	30% off $50.00	$35.00
8	10% off $89.90	$80.91
9	35% off $60.00	$39.00
10	50% off $27.80	$13.90
11	12% off $250.00	$220.00
12	A video game costs $59.99. It is 30% off. What is the sale price?	$41.99
13	A $150 bike is on sale for $105. What is the percent discount?	30%
14	Is 20% off and then another 20% off the same as 40% off? Try it with $100.	No: $100 → $80 → $64 (36% off).

LESSON 2.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 2: Percents

Lesson 2.10: Markups and Sales Tax

Student goal	I can find prices after a markup or sales tax.
Objective	Students compute markups and sales tax.
Standard	7.RP.3 — Solve multistep percent problems: tax, markups.
Pacing	Unit 2, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Receipts, workbook Lesson 2.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Subtracting tax.

SUPPORT

- Write 'you pay ___%' (e.g., 107%).

EXTEND

- Check tax on a real receipt.

STUDENT LESSON 2.10 · WITH ANSWERS

Lesson 2.10: Markups and Sales Tax

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.10, Part A.
- [ ] Complete the rest of Workbook Lesson 2.10 (Parts B–E).

LET'S EXPLORE

Mississippi's state sales tax is 7%. What is the total for a $40 purchase?

WORKED EXAMPLE · SALES TAX

7% tax on $40

1 Tax: 0.07 × 40 = 2.80.

2 Total: 40 + 2.80 = 42.80.

Answer: $42.80

WORKED EXAMPLE · MARKUP

A store buys a lamp for $25 and marks it up 60%.

1 Markup: 0.60 × 25 = 15.

2 Price: 25 + 15 = 40. Or 1.60 × 25 = 40.

Answer: $40

★ KEY IDEA

Total with tax or markup = original × (100% + percent).

✎ TRY IT

Answers are shown in color.

1. 7% tax on $15

Answer: $1.05 tax; $16.05 total

2. Markup 40% on $30

Answer: $42

3. Total with 8% tax on $62.50

Answer: $67.50

Workbook Answer Key · Lesson 2.10

#	Question	Answer
1	7% tax on $15	$1.05 tax; $16.05 total
2	Markup 40% on $30	$42
3	Total with 8% tax on $62.50	$67.50
4	0.07 × 40	2.80
5	1.6 × 25	40
6	7% tax on $20.00	$21.40
7	7% tax on $64.00	$68.48
8	5% tax on $38.00	$39.90
9	50% markup on $18.00	$27.00
10	25% markup on $64.00	$80.00
11	120% markup on $15.00	$33.00
12	A bookstore buys a book for $12 and sells it for $18. What is the percent markup?	50%
13	Lillie buys $28.50 of supplies with 7% tax. What does she pay?	$30.50 (30.495 rounds to 30.50)
14	A $50 item has a 20% markup, then a 20% discount. What is the final price?	$48

LESSON 2.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 2: Percents

Lesson 2.11: Percent Increase and Decrease

Student goal	I can find the percent of increase or decrease.
Objective	Students compute percent increase and decrease.
Standard	7.RP.3 — Solve multistep percent problems: percent increase and decrease.
Pacing	Unit 2, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples; stress dividing by the original.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing by the new amount.

SUPPORT

- Label original and new.

EXTEND

- Track a price over months.

STUDENT LESSON 2.11 · WITH ANSWERS

Lesson 2.11: Percent Increase and Decrease

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.11, Part A.
- [ ] Complete the rest of Workbook Lesson 2.11 (Parts B–E).

LET'S EXPLORE

A puppy weighed 8 pounds in March and 14 pounds in June. By what percent did its weight increase?

WORKED EXAMPLE · INCREASE

8 lb → 14 lb

1 Change: 14 − 8 = 6.

2 6 ÷ 8 = 0.75 = 75%.

Answer: 75% increase

WORKED EXAMPLE · DECREASE

A town had 2,500 people and now has 2,200.

1 Change: 300.

2 300 ÷ 2,500 = 0.12 = 12%.

Answer: 12% decrease

! WATCH OUT

Always divide by the original amount, not the new amount.

★ KEY IDEA

Percent change = change ÷ original × 100%.

✎ TRY IT

Answers are shown in color.

1. $40 → $50

Answer: 25% increase

2. 80 → 60

Answer: 25% decrease

3. 150 → 165

Answer: 10% increase

Workbook Answer Key · Lesson 2.11

#	Question	Answer
1	$40 → $50	25% increase
2	80 → 60	25% decrease
3	150 → 165	10% increase
4	6 ÷ 8	0.75
5	300 ÷ 2,500	0.12
6	20 → 25	25% increase
7	50 → 35	30% decrease
8	12 → 18	50% increase
9	200 → 170	15% decrease
10	64 → 80	25% increase
11	90 → 63	30% decrease
12	A gallon of milk went from $3.20 to $3.60. Percent increase?	12.5%
13	A plant was 15 cm tall and is now 24 cm. Percent increase?	60%
14	A price goes from $50 to $100, then back to $50. Are the percent changes the same? Explain.	No: +100%, then −50% (different originals).

LESSON 2.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 2: Percents

Lesson 2.12: Tips, Commissions, and Fees

Student goal	I can solve problems with tips, commissions, and fees.
Objective	Students solve tip, commission, and fee problems.
Standard	7.RP.3 — Solve multistep percent problems: tips, commissions, fees.
Pacing	Unit 2, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting to add the tip to the total when asked.

SUPPORT

- Underline what the question asks.

EXTEND

- Research how commissions work in a job you like.

STUDENT LESSON 2.12 · WITH ANSWERS

Lesson 2.12: Tips, Commissions, and Fees

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.12, Part A.
- [ ] Complete the rest of Workbook Lesson 2.12 (Parts B–D).

LET'S EXPLORE

A salesperson earns 6% commission. She sells a $1,250 sofa. How much does she earn?

WORKED EXAMPLE · COMMISSION

6% of $1,250

1 0.06 × 1,250 = 75.

Answer: $75

WORKED EXAMPLE · TIP

18% tip on a $42 meal

1 0.18 × 42 = 7.56.

2 Total: 42 + 7.56 = 49.56.

Answer: $7.56 tip; $49.56 total

★ KEY IDEA

Tips, commissions, and fees are all “percent of” problems.

✎ TRY IT

Answers are shown in color.

1. 15% tip on $36

Answer: $5.40

2. 4% commission on $8,500

Answer: $340

3. 3% card fee on $120

Answer: $3.60

Workbook Answer Key · Lesson 2.12

#	Question	Answer
1	15% tip on $36	$5.40
2	4% commission on $8,500	$340
3	3% card fee on $120	$3.60
4	0.06 × 1,250	75
5	0.18 × 42	7.56
6	20% tip on a $55 dinner. Total?	$66
7	5% commission on a $24,000 car	$1,200
8	A $2 fee plus 10% on a $30 ticket. Total?	$35
9	Lillie earns $8 per hour plus 10% of her sales. She works 5 hours and sells $300. Pay?	$70
10	15% tip on a $28.40 haircut	$4.26
11	A real estate agent earns 3% on a $180,000 house	$5,400
12	Which earns more: 8% commission on $2,500 or a flat $175?	8% commission ($200)

WEEKLY CHECK · ANSWER KEY
After Lesson 2.12

About 20 minutes

Weekly Check 6: Discounts, Tax, and Change — Answer Key

Answers

#	Question	Answer
1	35% off $80	$52
2	7% tax on $45	$48.15 total
3	$30 → $36 percent change	20% increase
4	$250 → $200 percent change	20% decrease
5	18% tip on $50	$9
6	6% commission on $3,500	$210

LESSON 2.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 2: Percents

Lesson 2.13: Simple Interest

Student goal	I can find simple interest and the total amount in an account or loan.
Objective	Students compute simple interest.
Standard	7.RP.3, 7.EE.3 — Solve multistep percent problems: simple interest.
Pacing	Unit 2, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using r = 3 instead of 0.03.

SUPPORT

- Formula card.

EXTEND

- Compare a real bank's savings rate.

STUDENT LESSON 2.13 · WITH ANSWERS

Lesson 2.13: Simple Interest

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.13, Part A.
- [ ] Complete the rest of Workbook Lesson 2.13 (Parts B–E).

LET'S EXPLORE

Grandpa puts $500 in a savings account that pays 3% simple interest each year. How much interest after 4 years?

WORKED EXAMPLE · SIMPLE INTEREST

P = $500, r = 3%, t = 4 years

1 I = 500 × 0.03 × 4 = 60.

2 Total: 500 + 60 = 560.

Answer: $60 interest; $560 total

WORKED EXAMPLE · A LOAN

Borrow $1,200 at 6% simple interest for 2 years.

1 I = 1,200 × 0.06 × 2 = 144.

2 Total to repay: $1,344.

Answer: $1,344

★ KEY IDEA

I = P × r × t. Use r as a decimal and t in years.

✎ TRY IT

Answers are shown in color.

1. $800 at 5% for 3 years

Answer: $120 interest

2. $2,000 at 4.5% for 2 years. Total?

Answer: $2,180

3. $300 at 2% for 6 months (0.5 year)

Answer: $3 interest

Workbook Answer Key · Lesson 2.13

#	Question	Answer
1	$800 at 5% for 3 years	$120 interest
2	$2,000 at 4.5% for 2 years. Total?	$2,180
3	$300 at 2% for 6 months (0.5 year)	$3 interest
4	500 × 0.03	15
5	15 × 4	60
6	$400.00 at 5% for 2 years	I = $40.00; total $440.00
7	$1,500.00 at 3% for 4 years	I = $180.00; total $1,680.00
8	$250.00 at 8% for 1 years	I = $20.00; total $270.00
9	$6,000.00 at 2.5% for 3 years	I = $450.00; total $6,450.00
10	Which earns more: $1,000 at 4% for 3 years or $1,200 at 3% for 3 years?	$1,000 at 4% ($120 vs $108)
11	Explain why borrowing money costs more than the amount borrowed.	You pay back the principal plus interest.

LESSON 2.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 2: Percents

Lesson 2.14: Percent Error

Student goal	I can find the percent error of an estimate or measurement.
Objective	Students compute percent error.
Standard	7.RP.3 — Solve multistep percent problems, including percent error.
Pacing	Unit 2, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Jar of objects for estimating, workbook Lesson 2.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore. Estimate a jar count.
Teach	15	Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing by the estimate.

SUPPORT

- Absolute value bars mean ‘distance’.

EXTEND

- Estimation contest at home.

STUDENT LESSON 2.14 · WITH ANSWERS

Lesson 2.14: Percent Error

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.14, Part A.
- [ ] Complete the rest of Workbook Lesson 2.14 (Parts B–D).

LET'S EXPLORE

Ellie guessed there were 120 jelly beans in a jar. There were actually 150. How far off was she, as a percent?

WORKED EXAMPLE · PERCENT ERROR

Guess 120, actual 150

1 Error: |120 − 150| = 30.

2 30 ÷ 150 = 0.2 = 20%.

Answer: 20% error

★ KEY IDEA

Percent error = |estimate − actual| ÷ actual × 100%. Always divide by the actual value.

✎ TRY IT

Answers are shown in color.

1. Estimate 45, actual 50

Answer: 10%

2. Measured 8.4 cm, actual 8 cm

Answer: 5%

3. Guessed 300, actual 240

Answer: 25%

Workbook Answer Key · Lesson 2.14

#	Question	Answer
1	Estimate 45, actual 50	10%
2	Measured 8.4 cm, actual 8 cm	5%
3	Guessed 300, actual 240	25%
4	|120 − 150|	30
5	30 ÷ 150	0.2
6	Estimate 90, actual 100	10%
7	Estimate 55, actual 50	10%
8	Estimate 18, actual 24	25%
9	Estimate 420, actual 400	5%
10	Estimate 3.6, actual 4	10%
11	Estimate 130, actual 125	4%
12	Lillie estimated a room was 12 ft long. It was 15 ft. Find the percent error.	20%

LESSON 2.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 2: Percents

Lesson 2.15: Multi-Step Percent Problems

Student goal	I can solve problems that combine discounts, tax, tips, and percent change.
Objective	Students solve multi-step percent problems.
Standard	7.RP.3 — Use proportional relationships to solve multistep ratio and percent problems.
Pacing	Unit 2, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding percents (25% off + 7% tax = 18% off).

SUPPORT

- Step-by-step table.

EXTEND

- Plan a shopping trip with a budget.

STUDENT LESSON 2.15 · WITH ANSWERS

Lesson 2.15: Multi-Step Percent Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.15, Part A.
- [ ] Complete the rest of Workbook Lesson 2.15 (Parts B–D).

LET'S EXPLORE

A $60 game is 25% off, and then 7% tax is added. What is the total?

WORKED EXAMPLE · DISCOUNT, THEN TAX

$60, 25% off, 7% tax

1 Sale: 0.75 × 60 = 45.

2 Total: 1.07 × 45 = 48.15.

Answer: $48.15

WORKED EXAMPLE · TWO CHANGES

A price rises 10%, then falls 10%. Start at $100.

1 100 × 1.10 = 110.

2 110 × 0.90 = 99.

Answer: $99, not $100!

★ KEY IDEA

Apply each percent in order to the current amount.

✎ TRY IT

Answers are shown in color.

1. $40 shirt, 20% off, then 7% tax

Answer: $34.24

2. $25 meal, 8% tax, then 20% tip on the pre-tax amount. Total?

Answer: $32

3. $200 rises 25%, then falls 20%

Answer: $200

Workbook Answer Key · Lesson 2.15

#	Question	Answer
1	$40 shirt, 20% off, then 7% tax	$34.24
2	$25 meal, 8% tax, then 20% tip on the pre-tax amount. Total?	$32
3	$200 rises 25%, then falls 20%	$200
4	0.75 × 60	45
5	1.07 × 45	48.15
6	A $120 jacket is 30% off. Tax is 7%. Total?	$89.88
7	A family's dinner is $64. Tax is 7% and they tip 18% of $64. Total?	$80.00 ($64 + $4.48 + $11.52)
8	A stock worth $50 rises 20% and then falls 15%. What is it worth?	$51
9	A store marks up a $30 item 50%, then puts it on sale for 20% off. Sale price?	$36
10	Does the order of a 10% discount and 7% tax matter for the final price? Test with $100.	No: 100 × 0.9 × 1.07 = 100 × 1.07 × 0.9 = 96.30

LESSON 2.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 2: Percents

Lesson 2.16: Percent Problem Solving

Student goal	I can choose and use the right percent strategy to solve real problems.
Objective	Students solve mixed percent problems.
Standard	7.RP.3 — 7.RP.3 review.
Pacing	Unit 2, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 2.16, Unit 2 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work the plan example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 2.16 and the Unit 2 Review.
Exit Ticket	3	Check. Tomorrow: Unit 2 Test.

COMMON MISCONCEPTIONS

- Using the wrong whole.

SUPPORT

- Identify part, percent, whole.

EXTEND

- Budget a real purchase.

STUDENT LESSON 2.16 · WITH ANSWERS

Lesson 2.16: Percent Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 2.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 2.16, Part A.
- [ ] Complete the rest of Workbook Lesson 2.16 (Parts B–D).

LET'S EXPLORE

Plan a back-to-school shopping trip: a $35 backpack (20% off), $48 shoes (25% off), 7% tax. You have $75. Is it enough?

WORKED EXAMPLE · PLAN

Backpack and shoes

1 Backpack: 0.8 × 35 = 28.

2 Shoes: 0.75 × 48 = 36.

3 Subtotal 64; tax 1.07 × 64 = 68.48.

Answer: Yes, $68.48 ≤ $75, with $6.52 left

★ KEY IDEA

Read carefully, decide what is the whole, and apply each percent in order.

✎ TRY IT

Answers are shown in color.

1. What percent of 64 is 16?

Answer: 25%

2. 12% of what is 9?

Answer: 75

3. $75 → $90 percent change

Answer: 20% increase

Workbook Answer Key · Lesson 2.16

#	Question	Answer
1	What percent of 64 is 16?	25%
2	12% of what is 9?	75
3	$75 → $90 percent change	20% increase
4	0.8 × 35	28
5	1.07 × 64	68.48
6	35% of 260	91
7	54 is what percent of 72?	75%
8	27 is 45% of what?	60
9	Sale price: 15% off $86	$73.10
10	I for $900 at 4% for 5 years	$180
11	Percent error: estimate 38, actual 40	5%
12	Write a multi-step percent problem about something you want to buy. Solve it.	Answers vary.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 2 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	0.45 = ___%	45%
2	$^{7}/_{8}$ = ___%	87.5%
3	24% of 150	36
4	What percent of 80 is 68?	85%
5	15 is 60% of what?	25
6	25% off $64, then 7% tax	$51.36
7	Percent change: 40 → 52	30% increase
8	I = ? for $2,500 at 3% for 4 years	$300
9	Percent error: guess 90, actual 75	20%

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 2 Test — Answer Key

Answers

#	Question	Answer
1	13 out of 25 = ___%	52%
2	1.8 = ___%	180%
3	$^{5}/_{8}$ = ___%	62.5%
4	0.4% as a decimal	0.004
5	16% of 75	12
6	What percent of 120 is 42?	35%
7	51 is 85% of what?	60
8	150% of 46	69
9	Estimate 18% of $41	about $8
10	A $72 jacket is 35% off. Sale price?	$46.80
11	A $36 meal with 7% tax and a 20% tip on $36. Total?	$45.72
12	A town grew from 4,800 to 5,520 people. Percent increase?	15%
13	$1,800 at 3.5% simple interest for 3 years. Interest?	$189
14	Store A sells a $90 bike at 30% off. Store B sells the same bike for $70 with a $5 coupon. Which is cheaper, and by how much?	Store A: $63 (0.70 × 90). Store B: $65. Store A is $2 cheaper.
15	Lillie guessed a pumpkin weighed 12 lb. It weighed 15 lb. Ellie guessed 17 lb. Who had the smaller percent error?	Ellie (13.3%) vs Lillie (20%): Ellie

Scoring

Score	What it means / next step
20–23 (85%+)	Mastered.
17–19	Proficient; correct missed items.
Below 17	Reteach and retest.

UNIT 3 · TEACHER OVERVIEW

Adding and Subtracting Integers

Opposites, absolute value, adding and subtracting integers with chips and number lines, and real-world applications.

Negative numbers let us describe temperatures below zero, elevations below sea level, and money owed. In this unit you will add and subtract integers and explain why the rules work.

Big ideas

- Find opposites and absolute values.
- Add integers with the same and different signs.
- Subtract integers by adding the opposite.
- Find distances between numbers.
- Solve real-world integer problems.

Standards

Code	Standard
7.NS.1	Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers; represent on a number line.
7.NS.3	Solve real-world and mathematical problems involving the four operations with rational numbers.
7.EE.3	Solve multi-step real-life problems with positive and negative numbers.

What students should already know

- Number lines with negatives (Grade 6).
- Operations with whole numbers.

Materials for the unit

- Two-color counters (or paper chips)
- Number lines from −20 to 20
- Thermometer picture

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	3.1	Integers, Opposites, and Absolute Value
Week 1	Lesson 2 of 4	3.2	Adding Integers with the Same Sign
Week 1	Lesson 3 of 4	3.3	Adding Integers with Different Signs
Week 1	Lesson 4 of 4	3.4	Additive Inverses
Week 1	End of week	Check	Weekly Check 7: Adding Integers
Week 2	Lesson 1 of 4	3.5	Adding Integers on a Number Line
Week 2	Lesson 2 of 4	3.6	Subtracting Integers
Week 2	Lesson 3 of 4	3.7	Subtraction as Distance
Week 2	Lesson 4 of 4	3.8	Distance Between Two Numbers
Week 2	End of week	Check	Weekly Check 8: Subtracting Integers
Week 3	Lesson 1 of 4	3.9	Integers: Temperature and Elevation
Week 3	Lesson 2 of 4	3.10	Integers and Money
Week 3	Lesson 3 of 4	3.11	Multi-Step Integer Problems
Week 3	Lesson 4 of 4	3.12	Properties and Mental Math with Integers
Week 3	End of week	Check	Weekly Check 9: Integer Applications
Week 4	Lesson 1 of 4	3.13	Integer Expressions
Week 4	Lesson 2 of 4	3.14	Explaining Integer Rules with Models
Week 4	Lesson 3 of 4	3.15	Finding and Fixing Integer Errors
Week 4	Lesson 4 of 4	3.16	Integer Problem Solving
Week 4	Before the test	Review	Unit 3 Review
Week 4	End of unit	Test	Unit 3 Test

TEACHING THIS UNIT AT HOME

Unit 3 spans winter break. Talk about temperatures and bank balances during the break: “If it's 5° and drops 9°, what is it?”

UNIT VOCABULARY

integer	Whole numbers and their opposites.
opposites	Same distance from 0, opposite sides.
absolute value	Distance from 0.
additive inverse	A number's opposite; sums to 0.

LESSON 3.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.1: Integers, Opposites, and Absolute Value

Student goal	I can use integers to describe real situations and find opposites and absolute values.
Objective	Students represent quantities with integers and find opposites and absolute values.
Standard	7.NS.1 (review of 6.NS.5–7) — Apply understanding of addition and subtraction to rational numbers; represent on a number line.
Pacing	Unit 3, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Thermometer picture, workbook Lesson 3.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Colder temperature.
Launch	5	Read Let's Explore.
Teach	15	Draw a vertical number line like a thermometer. Say: "Opposites are mirror images across 0."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking $-8 > -3$ because $8 > 3$.

SUPPORT

- Vertical number line.

EXTEND

- Find the lowest and highest places in Mississippi (elevation).

STUDENT LESSON 3.1 · WITH ANSWERS

Lesson 3.1: Integers, Opposites, and Absolute Value

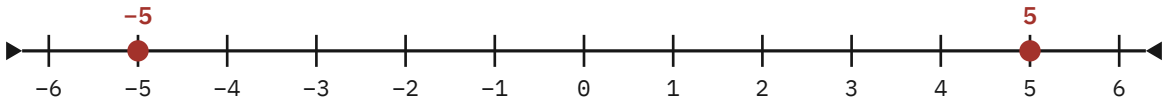
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.1, Part A.
- [ ] Complete the rest of Workbook Lesson 3.1 (Parts B–F).

LET'S EXPLORE

The temperature in Jackson is 5°F. In Fairbanks, Alaska, it is 5 degrees below zero. How would you write both temperatures?

Integers include negative numbers. Negative numbers describe values below zero: temperatures, elevations below sea level, money owed.



5 and −5 are opposites: the same distance from 0 in opposite directions. Both have an absolute value of 5.

WORKED EXAMPLE · ABSOLUTE VALUE

Find |−12| and |9|.

1 −12 is 12 units from 0.

2 9 is 9 units from 0.

Answer: 12 and 9

★ KEY IDEA

Opposites have the same absolute value. Absolute value is a distance, so it is never negative.

✎ TRY IT

Answers are shown in color.

1. Write an integer: a loss of 8 yards

Answer: −8

2. Opposite of −15

Answer: 15

3. |−23|

Answer: 23

Workbook Answer Key · Lesson 3.1

#	Question	Answer
1	Write an integer: a loss of 8 yards	−8
2	Opposite of −15	15
3	|−23|	23
4	Which is colder: −3°F or −8°F?	−8°F
5	Name a number between −2 and 0.	−1
6	a deposit of $40	40
7	20 feet below sea level	−20
8	a temperature drop of 6 degrees	−6
9	a gain of 3 yards	3
10	opposite of 31	−31
11	opposite of −4	4
12	|−18|	18
13	|0|	0
14	−|−6|	−6
15	|14| − |−5|	9
16	3, −7, 0, −2, 5	−7, −2, 0, 3, 5
17	Can absolute value be negative? Explain.	No; it is a distance from 0.

LESSON 3.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.2: Adding Integers with the Same Sign

Student goal	I can add two integers that have the same sign.
Objective	Students add integers with the same sign.
Standard	7.NS.1b — Understand p + q as the number |q| from p in the direction indicated by the sign of q.
Pacing	Unit 3, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Two-color counters (or paper chips), workbook Lesson 3.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Model with chips, then a number line. Say: "Same sign? Add and keep the sign."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Making the sum of two negatives positive.

SUPPORT

- Chips.

EXTEND

- Write a real story for −15 + (−25).

STUDENT LESSON 3.2 · WITH ANSWERS

Lesson 3.2: Adding Integers with the Same Sign

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.2, Part A.
- [ ] Complete the rest of Workbook Lesson 3.2 (Parts B–E).

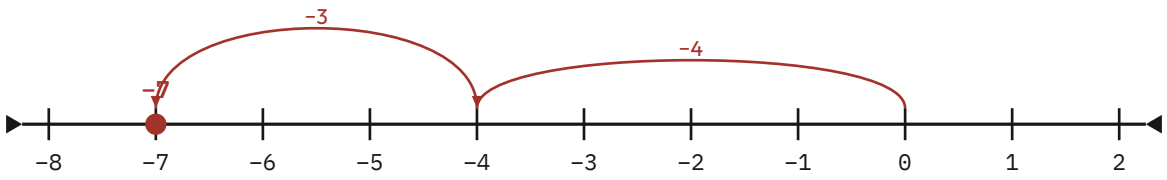
LET'S EXPLORE

You owe your sister $4. Then you borrow $3 more. How much do you owe in all?

Owing $4 is −4. Owing $3 more is −3. Together: −4 + (−3) = −7. You owe $7.



Seven negative chips: −7



WORKED EXAMPLE · SAME SIGN

−9 + (−6)

1 Both negative: add the absolute values, 9 + 6 = 15.

2 Keep the sign: negative.

Answer: −15

★ KEY IDEA

Same signs: add the absolute values and keep the common sign.

✎ TRY IT

Answers are shown in color.

1. −5 + (−8)

Answer: −13

2. 12 + 9

Answer: 21

3. −20 + (−15)

Answer: −35

Workbook Answer Key · Lesson 3.2

#	Question	Answer
1	−5 + (−8)	−13
2	12 + 9	21
3	−20 + (−15)	−35
4	|−9| + |−6|	15
5	4 + 3	7
6	−3 + (−4)	−7
7	−11 + (−2)	−13
8	6 + 8	14
9	−7 + (−7)	−14
10	−15 + (−9)	−24
11	−24 + (−36)	−60
12	13 + 27	40
13	−50 + (−50)	−100
14	A diver is at −12 m and dives 8 m deeper. Where is she now?	−20 m
15	The temperature is −5°F and drops 9 more degrees. New temperature?	−14°F
16	Use chips to show −4 + (−5). What is the sum?	−9

LESSON 3.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.3: Adding Integers with Different Signs

Student goal	I can add two integers with different signs.
Objective	Students add integers with different signs.
Standard	7.NS.1b — Understand p + q; describe situations in which opposite quantities combine to make 0.
Pacing	Unit 3, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Two-color counters, workbook Lesson 3.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	−4 + (−5).
Launch	5	Read Let's Explore.
Teach	15	Chips with zero pairs; then number line. Say: “Cancel, then see what's left.”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding absolute values when signs differ.

SUPPORT

- Chips until confident.

EXTEND

- Why does the bigger absolute value win the sign?

STUDENT LESSON 3.3 · WITH ANSWERS

Lesson 3.3: Adding Integers with Different Signs

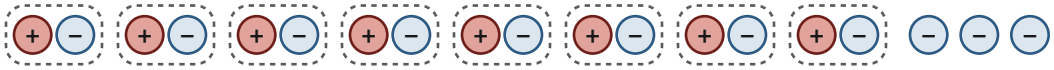
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.3, Part A.
- [ ] Complete the rest of Workbook Lesson 3.3 (Parts B–E).

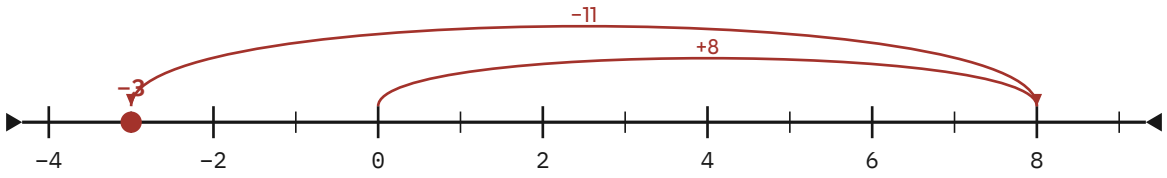
LET'S EXPLORE

You have $8. You spend $11. What is your balance?

Model 8 + (−11) with chips. Each positive cancels a negative to make a zero pair. After 8 zero pairs, 3 negative chips are left: −3.



8 zero pairs cancel; 3 negatives remain.



WORKED EXAMPLE · DIFFERENT SIGNS

−14 + 9

1 Subtract absolute values: 14 − 9 = 5.

2 Use the sign of the number with the larger absolute value: −14.

Answer: −5

★ KEY IDEA

Different signs: subtract the absolute values and use the sign of the number farther from 0.

✎ TRY IT

Answers are shown in color.

1. 7 + (−10)

Answer: −3

2. −6 + 15

Answer: 9

3. −25 + 18

Answer: −7

Workbook Answer Key · Lesson 3.3

#	Question	Answer
1	7 + (−10)	−3
2	−6 + 15	9
3	−25 + 18	−7
4	−4 + (−5)	−9
5	14 − 9	5
6	5 + (−2)	3
7	−5 + 2	−3
8	9 + (−13)	−4
9	−12 + 20	8
10	30 + (−45)	−15
11	−17 + 17	0
12	−8 + 3	−5
13	21 + (−6)	15
14	Your account has $35. You buy a $52 game with an overdraft. Balance?	−$17
15	A football team gains 6 yards, then loses 9. Total?	−3 yards
16	Explain why −7 + 7 = 0 using zero pairs.	7 positives cancel 7 negatives; nothing is left.

LESSON 3.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.4: Additive Inverses

Student goal	I can explain why a number plus its opposite equals 0.
Objective	Students use additive inverses.
Standard	7.NS.1a, 7.NS.1b — Describe situations in which opposite quantities combine to make 0.
Pacing	Unit 3, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Counters, workbook Lesson 3.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work examples; look for opposite pairs.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Missing that −35 needs +35 (not −35).

SUPPORT

- Chips.

EXTEND

- Build a ‘zero’ from 5 different integers.

STUDENT LESSON 3.4 · WITH ANSWERS

Lesson 3.4: Additive Inverses

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.4, Part A.
- [ ] Complete the rest of Workbook Lesson 3.4 (Parts B–E).

LET'S EXPLORE

A hydrogen atom has 1 proton (+1 charge) and 1 electron (−1 charge). What is its total charge?

+1 + (−1) = 0. Opposites always add to 0. The opposite of a number is called its additive inverse.

WORKED EXAMPLE · USE INVERSES

−13 + 8 + 13

1 −13 and 13 are opposites: they make 0.

2 0 + 8 = 8.

Answer: 8

WORKED EXAMPLE · FIND THE MISSING NUMBER

−35 + ___ = 0

1 The additive inverse of −35 is 35.

Answer: 35

★ KEY IDEA

a + (−a) = 0 for every number a. Look for opposites to make 0 and simplify.

✎ TRY IT

Answers are shown in color.

1. −42 + 42

Answer: 0

2. 6 + (−9) + 9

Answer: 6

3. ___ + 17 = 0

Answer: −17

Workbook Answer Key · Lesson 3.4

#	Question	Answer
1	−42 + 42	0
2	6 + (−9) + 9	6
3	___ + 17 = 0	−17
4	−6 + 6	0
5	15 + (−15)	0
6	−7 + 12 + 7	12
7	25 + (−18) + (−25)	−18
8	−3 + 9 + (−9) + 3	0
9	40 + (−11) + (−40)	−11
10	−8 + ___ = 0	8
11	___ + (−26) = 0	26
12	−5 + ___ = 2	7
13	14 + ___ = −1	−15
14	Describe a real situation where two opposite quantities combine to make 0.	e.g. earn $20, spend $20

WEEKLY CHECK · ANSWER KEY
After Lesson 3.4

About 15 minutes

Weekly Check 7: Adding Integers — Answer Key

Answers

#	Question	Answer
1	−9 + (−14)	−23
2	−9 + 14	5
3	21 + (−30)	−9
4	|−17|	17
5	−13 + ___ = 0	13
6	−6 + 11 + 6	11

LESSON 3.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.5: Adding Integers on a Number Line

Student goal	I can use a number line to add integers.
Objective	Students add integers on a number line.
Standard	7.NS.1b — Represent addition on a horizontal or vertical number line diagram.
Pacing	Unit 3, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Floor number line (−10 to 10), workbook Lesson 3.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore. Walk it on a floor number line.
Teach	15	Model hops; say 'positive right, negative left'.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Starting at 0 for the second addend.

SUPPORT

- Floor walking.

EXTEND

- Vertical number line problems (elevation).

STUDENT LESSON 3.5 · WITH ANSWERS

Lesson 3.5: Adding Integers on a Number Line

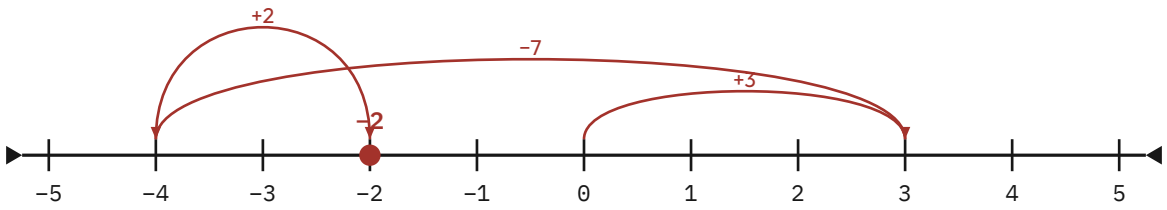
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.5, Part A.
- [ ] Complete the rest of Workbook Lesson 3.5 (Parts B–D).

LET'S EXPLORE

A hiker starts at 3 m above a lake, climbs down 7 m, then up 2 m. Where does she end?

Start at the first number. Move right for a positive addend and left for a negative addend.



WORKED EXAMPLE · NUMBER LINE

3 + (−7) + 2

1 Start at 0, move right 3 to 3.

2 Move left 7 to −4.

3 Move right 2 to −2.

Answer: −2

★ KEY IDEA

Positive → right. Negative → left. The landing point is the sum.

✎ TRY IT

Answers are shown in color.

1. −4 + 6 (use a number line)

Answer: 2

2. 2 + (−5) + (−1)

Answer: −4

3. −3 + (−3) + 8

Answer: 2

Workbook Answer Key · Lesson 3.5

#	Question	Answer
1	−4 + 6 (use a number line)	2
2	2 + (−5) + (−1)	−4
3	−3 + (−3) + 8	2
4	−7 + 4	−3
5	−2 + (−6)	−8
6	−5 + 8 (figure)	3
7	4 + (−9) (figure)	−5
8	−1 + (−4) + 3 (figure)	−2
9	6 + (−2) + (−7) (figure)	−3
10	Draw a number line for −6 + 10 and give the sum.	4

LESSON 3.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.6: Subtracting Integers

Student goal	I can subtract integers by adding the opposite.
Objective	Students subtract integers by adding the additive inverse.
Standard	7.NS.1c — Understand subtraction of rational numbers as adding the additive inverse, $p - q = p + (-q)$.
Pacing	Unit 3, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Two-color counters, workbook Lesson 3.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Opposites.
Launch	5	Read Let's Explore.
Teach	15	Chip model with zero pairs for $3 - (-5)$. Say: "Keep, change, opposite."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Changing the sign of the first number.

SUPPORT

- Write the addition form every time.

EXTEND

- Prove $a - b = a + (-b)$ with chips for 3 cases.

STUDENT LESSON 3.6 · WITH ANSWERS

Lesson 3.6: Subtracting Integers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.6, Part A.
- [ ] Complete the rest of Workbook Lesson 3.6 (Parts B–E).

LET'S EXPLORE

What is 3 − (−5)? How can subtracting make a number bigger?

Subtracting a number is the same as adding its opposite: a − b = a + (−b). So 3 − (−5) = 3 + 5 = 8.

With chips: start with 3 positives. To take away 5 negatives, first add 5 zero pairs (which doesn't change the value). Then remove the 5 negatives. 8 positives are left.



3 positives plus 5 zero pairs; remove the 5 negatives → 8.

WORKED EXAMPLE · ADD THE OPPOSITE

−6 − 4

1 −6 − 4 = −6 + (−4).

2 Same signs: −10.

Answer: −10

WORKED EXAMPLE · SUBTRACT A NEGATIVE

−2 − (−9)

1 −2 − (−9) = −2 + 9 = 7.

Answer: 7

★ KEY IDEA

a − b = a + (−b). Keep the first number, change subtraction to addition, use the opposite of the second number.

✎ TRY IT

Answers are shown in color.

1. 5 − 12

Answer: −7

2. −8 − (−3)

Answer: −5

3. −4 − 10

Answer: −14

Workbook Answer Key · Lesson 3.6

#	Question	Answer
1	5 − 12	−7
2	−8 − (−3)	−5
3	−4 − 10	−14
4	Opposite of −9	9
5	−6 + (−4)	−10
6	7 − 10	−3
7	−3 − 5	−8
8	4 − (−6)	10
9	−9 − (−2)	−7
10	−1 − (−8)	7
11	0 − (−15)	15
12	12 − (−12)	24
13	−20 − 13	−33
14	15 − (−7)	15 + 7 = 22
15	−11 − 4	−11 + (−4) = −15
16	Explain why 6 − (−2) is greater than 6.	Subtracting −2 adds 2: 8.

LESSON 3.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.7: Subtraction as Distance

Student goal	I can use a number line to show subtraction as the change from one number to another.
Objective	Students interpret subtraction as change/distance.
Standard	7.NS.1c — Show that the distance between two rational numbers is the absolute value of their difference.
Pacing	Unit 3, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Number line, workbook Lesson 3.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Model end − start on the number line.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Doing start − end.

SUPPORT

- Label start and end.

EXTEND

- Track a week of temperature changes.

STUDENT LESSON 3.7 · WITH ANSWERS

Lesson 3.7: Subtraction as Distance

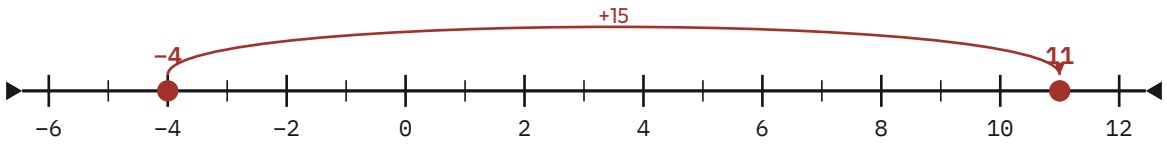
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.7, Part A.
- [ ] Complete the rest of Workbook Lesson 3.7 (Parts B–E).

LET'S EXPLORE

At 6 a.m. it was −4°F. At noon it was 11°F. How much did the temperature rise?

Change = end − start. 11 − (−4) = 11 + 4 = 15. On a number line, from −4 to 11 is 15 units to the right.



WORKED EXAMPLE · ELEVATION

A bird at 25 ft dives to a fish at −8 ft. Change in elevation?

1 −8 − 25 = −33.

Answer: −33 ft (33 ft down)

★ KEY IDEA

End − start tells the change. Positive means up or right; negative means down or left.

✎ TRY IT

Answers are shown in color.

1. From −3 to 9: change?

Answer: 12

2. From 5 to −7: change?

Answer: −12

3. From −10 to −2: change?

Answer: 8

Workbook Answer Key · Lesson 3.7

#	Question	Answer
1	From −3 to 9: change?	12
2	From 5 to −7: change?	−12
3	From −10 to −2: change?	8
4	11 − (−4)	15
5	−8 − 25	−33
6	from −2 to 6	8
7	from 8 to −1	−9
8	from −15 to −5	10
9	from 0 to −9	−9
10	from −7 to −20	−13
11	from 14 to −3	−17
12	A submarine at −120 m rises to −45 m. Change?	75 m up (+75)
13	A company's profit goes from $3 million to −$2 million. What is the change?	−$5 million
14	Explain how a number line shows 4 − (−3) = 7.	The distance from −3 to 4 is 7.

LESSON 3.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.8: Distance Between Two Numbers

Student goal	I can find the distance between two numbers using absolute value.
Objective	Students find distance between rational numbers.
Standard	7.NS.1c — Show that the distance between two rational numbers on the number line is the absolute value of their difference.
Pacing	Unit 3, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 3.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Giving a negative distance.

SUPPORT

- Count units on a number line.

EXTEND

- Find temperature ranges in a weather report.

STUDENT LESSON 3.8 · WITH ANSWERS

Lesson 3.8: Distance Between Two Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.8, Part A.
- [ ] Complete the rest of Workbook Lesson 3.8 (Parts B–E).

LET'S EXPLORE

Death Valley is 282 ft below sea level. Mount Whitney is 14,505 ft above. How far apart are their elevations?

WORKED EXAMPLE · DISTANCE

|14,505 − (−282)|

1 14,505 + 282 = 14,787.

Answer: 14,787 ft

WORKED EXAMPLE · ORDER DOESN'T MATTER

|−5 − 3| and |3 − (−5)|

1 |−8| = 8 and |8| = 8.

Answer: 8 both ways

★ KEY IDEA

Distance between a and b = |a − b|. It is never negative.

✎ TRY IT

Answers are shown in color.

1. Distance between −7 and 2

Answer: 9

2. Distance between −12 and −4

Answer: 8

3. |−6 − 9|

Answer: 15

Workbook Answer Key · Lesson 3.8

#	Question	Answer
1	Distance between −7 and 2	9
2	Distance between −12 and −4	8
3	|−6 − 9|	15
4	|−8|	8
5	3 − (−5)	8
6	−3 and 4	7
7	10 and −10	20
8	−25 and −9	16
9	0 and −14	14
10	−6 and 6	12
11	−100 and 35	135
12	Highest Mississippi point: Woodall Mountain, 807 ft. Lowest: sea level on the Gulf, 0 ft. Difference?	807 ft
13	Low of −11°F and high of 23°F. Temperature range?	34°F
14	Why do we use absolute value for distance?	Distance is never negative; |a − b| works in either order.

WEEKLY CHECK · ANSWER KEY
After Lesson 3.8

About 15 minutes

Weekly Check 8: Subtracting Integers — Answer Key

Answers

#	Question	Answer
1	4 − 11	−7
2	−7 − (−3)	−4
3	−5 − 8	−13
4	10 − (−6)	16
5	Distance between −8 and 5	13
6	From −6°F to 9°F: change?	15°F

LESSON 3.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.9: Integers: Temperature and Elevation

Student goal	I can solve temperature and elevation problems with integers.
Objective	Students solve real-world integer problems.
Standard	7.NS.1, 7.NS.3 — Solve real-world problems involving the four operations with rational numbers.
Pacing	Unit 3, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 3.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples on a vertical number line.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Losing track of the sign of the starting value.

SUPPORT

- Vertical number line.

EXTEND

- Find the elevation of 5 world cities.

STUDENT LESSON 3.9 · WITH ANSWERS

Lesson 3.9: Integers: Temperature and Elevation

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.9, Part A.
- [ ] Complete the rest of Workbook Lesson 3.9 (Parts B–D).

LET'S EXPLORE

New Orleans sits about 6 feet below sea level in places. A levee rises 23 feet above that ground. How high above sea level is the top of the levee?

WORKED EXAMPLE · ELEVATION

Ground at −6 ft, levee 23 ft higher

1 −6 + 23 = 17.

Answer: 17 ft above sea level

WORKED EXAMPLE · TEMPERATURE

It is −3°F at 6 a.m. It rises 4° per hour for 3 hours, then drops 5°.

1 −3 + 12 = 9.

2 9 − 5 = 4.

Answer: 4°F

★ KEY IDEA

Choose a direction for positive (up, warmer) and use negatives for the opposite.

✎ TRY IT

Answers are shown in color.

1. A diver at −30 ft rises 12 ft.

Answer: −18 ft

2. 8°F drops 15°.

Answer: −7°F

3. A balloon at 400 ft descends 650 ft into a canyon. New elevation?

Answer: −250 ft

Workbook Answer Key · Lesson 3.9

#	Question	Answer
1	A diver at −30 ft rises 12 ft.	−18 ft
2	8°F drops 15°.	−7°F
3	A balloon at 400 ft descends 650 ft into a canyon. New elevation?	−250 ft
4	−6 + 23	17
5	9 − 5	4
6	The temperature was −12°F and rose 19°. New temperature?	7°F
7	A plane at 2,500 ft descends 3,100 ft over the Dead Sea basin. New elevation?	−600 ft
8	A miner is 450 ft underground and goes up 120 ft. Elevation?	−330 ft
9	Morning −5°F, afternoon 14°F. How much warmer?	19°F
10	Write a temperature story for −8 + 13 and solve it.	Answers vary; 5.

LESSON 3.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.10: Integers and Money

Student goal	I can use integers to track money: deposits, withdrawals, profits, and debts.
Objective	Students model money situations with integers.
Standard	7.NS.1, 7.NS.3 — Solve real-world problems with rational numbers.
Pacing	Unit 3, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Bank statement example, workbook Lesson 3.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Fill the transaction table.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Treating debt as positive.

SUPPORT

- Running balance table.

EXTEND

- Plan a pretend budget for a month.

STUDENT LESSON 3.10 · WITH ANSWERS

Lesson 3.10: Integers and Money

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.10, Part A.
- [ ] Complete the rest of Workbook Lesson 3.10 (Parts B–E).

LET'S EXPLORE

Lillie's account has $25. She spends $40, then deposits $30. What is her balance after each step?

Transaction	Integer	Balance
start		$25
spend $40	−40	−$15
deposit $30	+30	$15

WORKED EXAMPLE · PROFIT AND LOSS

A lemonade stand earns $18, $−7 (a loss), and $12 on three days.

1 18 + (−7) + 12 = 23.

Answer: $23 profit

★ KEY IDEA

Deposits and profits are positive; withdrawals, purchases, and losses are negative.

✎ TRY IT

Answers are shown in color.

1. $10 balance, buy a $14 item

Answer: −$4

2. −$20 balance, deposit $35

Answer: $15

3. Profits: $50, −$65, $20. Total?

Answer: $5

Workbook Answer Key · Lesson 3.10

#	Question	Answer
1	$10 balance, buy a $14 item	−$4
2	−$20 balance, deposit $35	$15
3	Profits: $50, −$65, $20. Total?	$5
4	25 + (−40)	−15
5	−15 + 30	15
6	Start $0. Earn $45, spend $28, spend $30, earn $20.	$7
7	Start −$12. Deposit $50, withdraw $65.	−$27
8	A business lost $1,200 in January and earned $2,050 in February. Total?	$850 profit
9	You owe $35 and pay back $22. How much do you owe now?	$13 (balance −$13)
10	Make a 4-transaction account record that ends at −$5.	Answers vary.

LESSON 3.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.11: Multi-Step Integer Problems

Student goal	I can solve multi-step problems with integer addition and subtraction.
Objective	Students solve multi-step integer problems.
Standard	7.NS.3, 7.EE.3 — Solve multi-step real-life problems with rational numbers.
Pacing	Unit 3, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 3.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Group gains and losses.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Sign errors in long expressions.

SUPPORT

- Circle each number with its sign.

EXTEND

- Track a real football drive.

STUDENT LESSON 3.11 · WITH ANSWERS

Lesson 3.11: Multi-Step Integer Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.11, Part A.
- [ ] Complete the rest of Workbook Lesson 3.11 (Parts B–E).

LET'S EXPLORE

A football team starts on its own 30-yard line. It loses 7 yards, gains 12, loses 3, and gains 25. Where is it now?

WORKED EXAMPLE · COMBINE

30 − 7 + 12 − 3 + 25

1 Gains: 12 + 25 = 37. Losses: −7 + (−3) = −10.

2 30 + 37 − 10 = 57.

Answer: the 57-yard line (in plain terms, 27 yards net gain)

★ KEY IDEA

Group the positives and the negatives, then combine.

✎ TRY IT

Answers are shown in color.

1. −8 + 15 − 6 + 4

Answer: 5

2. A temperature of 3°F drops 10°, rises 4°, drops 2°.

Answer: −5°F

3. Elevation −20 m: up 35, down 12, up 6

Answer: 9 m

Workbook Answer Key · Lesson 3.11

#	Question	Answer
1	−8 + 15 − 6 + 4	5
2	A temperature of 3°F drops 10°, rises 4°, drops 2°.	−5°F
3	Elevation −20 m: up 35, down 12, up 6	9 m
4	12 + 25	37
5	−7 + (−3)	−10
6	−6 + 9 − 4	−1
7	15 − 22 + 3	−4
8	−10 − 5 + 18	3
9	7 − (−3) − 12	−2
10	−4 + (−8) − (−6)	−6
11	20 − 35 + 10 − 5	−10
12	A hiker starts at −40 ft (in a valley), climbs 125 ft, descends 60 ft, climbs 15 ft. Elevation?	40 ft
13	Write a 3-step temperature story and solve it.	Answers vary.

LESSON 3.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.12: Properties and Mental Math with Integers

Student goal	I can use properties of addition to add integers mentally.
Objective	Students apply properties to integer addition.
Standard	7.NS.1d — Apply properties of operations as strategies to add and subtract rational numbers.
Pacing	Unit 3, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 3.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Reorder to make zero pairs and tens.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Moving a number without its sign.

SUPPORT

- Rewrite subtraction as adding the opposite first.

EXTEND

- Speed challenge with mental integer sums.

STUDENT LESSON 3.12 · WITH ANSWERS

Lesson 3.12: Properties and Mental Math with Integers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.12, Part A.
- [ ] Complete the rest of Workbook Lesson 3.12 (Parts B–D).

LET'S EXPLORE

How would you quickly find −17 + 45 + 17 + (−5)?

WORKED EXAMPLE · REARRANGE

−17 + 45 + 17 + (−5)

1 Reorder: (−17 + 17) + (45 + (−5)).

2 0 + 40 = 40.

Answer: 40

★ KEY IDEA

Rewrite subtraction as addition, then reorder and regroup to make zeros or tens.

✎ TRY IT

Answers are shown in color.

1. −25 + 38 + 25

Answer: 38

2. 14 + (−9) + (−14) + 9

Answer: 0

3. −36 + 50 + (−14)

Answer: 0

Workbook Answer Key · Lesson 3.12

#	Question	Answer
1	−25 + 38 + 25	38
2	14 + (−9) + (−14) + 9	0
3	−36 + 50 + (−14)	0
4	−17 + 17	0
5	45 + (−5)	40
6	−8 + 27 + 8	27
7	45 + (−19) + (−45)	−19
8	−13 + (−7) + 50	30
9	64 − 30 − 34	0
10	−99 + 50 + 99 + (−50)	0
11	125 + (−75) + (−25)	25
12	Show two ways to group −6 + 16 + (−4).	(−6 + −4) + 16 = 6; −6 + (16 + −4) = 6

WEEKLY CHECK · ANSWER KEY
After Lesson 3.12

About 15 minutes

Weekly Check 9: Integer Applications — Answer Key

Answers

#	Question	Answer
1	−15 m diver rises 9 m	−6 m
2	$20 balance, spend $32	−$12
3	−4 + 10 − 9 + 2	−1
4	−29 + 46 + 29 (mentally)	46
5	Temp 6°F, drops 11°, rises 3°	−2°F
6	Profits $40, −$55, $30: total	$15

LESSON 3.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.13: Integer Expressions

Student goal	I can evaluate expressions with integers and variables.
Objective	Students evaluate integer expressions.
Standard	7.NS.1, 7.EE.1 — Apply properties to add and subtract; evaluate expressions.
Pacing	Unit 3, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 3.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Substitute with parentheses.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Writing $x - y$ as $-5 - 3 = -2$.

SUPPORT

- Parentheses always.

EXTEND

- Make a table of $x - 8$ for x from -10 to 10.

STUDENT LESSON 3.13 · WITH ANSWERS

Lesson 3.13: Integer Expressions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.13, Part A.
- [ ] Complete the rest of Workbook Lesson 3.13 (Parts B–D).

LET'S EXPLORE

If $x = -4$, what is $x + 9 - x$?

WORKED EXAMPLE · SUBSTITUTE

$x + 9 - x$ when $x = -4$

1 $(-4) + 9 - (-4) = -4 + 9 + 4 = 9$.

Answer: 9

WORKED EXAMPLE · TWO VARIABLES

$a - b$ when $a = -6$, $b = -11$

1 $-6 - (-11) = -6 + 11 = 5$.

Answer: 5

★ KEY IDEA

Substitute with parentheses around negative values, then add the opposite to subtract.

✎ TRY IT

Answers are shown in color.

1. $x - 7$ when $x = -3$

Answer: -10

2. $-y + 5$ when $y = -2$

Answer: 7

3. $a + b - c$ when $a = 4$, $b = -9$, $c = -2$

Answer: -3

Workbook Answer Key · Lesson 3.13

#	Question	Answer
1	$x - 7$ when $x = -3$	-10
2	$-y + 5$ when $y = -2$	7
3	$a + b - c$ when $a = 4$, $b = -9$, $c = -2$	-3
4	$-6 - (-11)$	5
5	$-(-4)$	4
6	$x + y$	-2
7	$x - y$	-8
8	$y - x$	8
9	$-x$	5
10	$x + x + y$	-7
11	$\|x\| - y$	2
12	Find a value of x so that $x - 8$ is negative and $x + 8$ is positive.	Any x between -8 and 8, e.g. 0

LESSON 3.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.14: Explaining Integer Rules with Models

Student goal	I can explain integer addition and subtraction rules using chips and number lines.
Objective	Students justify integer rules with models.
Standard	7.NS.1 — Understand addition and subtraction of rational numbers with models.
Pacing	Unit 3, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Counters, number lines, workbook Lesson 3.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Model each table row with chips and number lines.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Memorizing rules without meaning.

SUPPORT

- Use chips.

EXTEND

- Teach the rules to a younger sibling.

STUDENT LESSON 3.14 · WITH ANSWERS

Lesson 3.14: Explaining Integer Rules with Models

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.14, Part A.
- [ ] Complete the rest of Workbook Lesson 3.14 (Parts B–D).

LET'S EXPLORE

Why does subtracting a negative make a number bigger? Can you prove it with a model?

Rules are easier to remember when you know why they work. Here are both models side by side.

Problem	Chip model	Number line
−3 + 5	3 zero pairs cancel; 2 positives left	start at −3, move right 5 → 2
4 − (−2)	add 2 zero pairs, remove 2 negatives → 6	the distance from −2 to 4 is 6
−2 − 5	add 5 zero pairs, remove 5 positives → −7	start at −2, move left 5 → −7

★ KEY IDEA

Every rule can be shown with a model. Use a model to check a rule you're unsure of.

✎ TRY IT

Answers are shown in color.

1. Explain −6 + 6 = 0 with chips.

Answer: 6 zero pairs, nothing left

2. Explain 2 − 7 = −5 on a number line.

Answer: Start at 2, move left 7 → −5

3. Explain −1 − (−4) = 3 with chips.

Answer: Start with 1 negative, add 3 zero pairs, remove 4 negatives → 3 positives

Workbook Answer Key · Lesson 3.14

#	Question	Answer
1	Explain −6 + 6 = 0 with chips.	6 zero pairs, nothing left
2	Explain 2 − 7 = −5 on a number line.	Start at 2, move left 7 → −5
3	Explain −1 − (−4) = 3 with chips.	Start with 1 negative, add 3 zero pairs, remove 4 negatives → 3 positives
4	4 − (−2)	6
5	−3 + 5	2
6	−5 + 2	−3; model explanation
7	3 − 8	−5; model explanation
8	−4 − (−6)	2; model explanation
9	0 − (−3)	3; model explanation
10	Which model do you prefer and why?	Answers vary.

LESSON 3.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.15: Finding and Fixing Integer Errors

Student goal	I can find and fix mistakes in integer calculations.
Objective	Students analyze and correct integer errors.
Standard	7.NS.1 — Apply properties of operations to add and subtract rational numbers.
Pacing	Unit 3, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 3.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Go through the table; student explains each fix.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Same errors as listed.

SUPPORT

- Model-check every answer.

EXTEND

- Create error cards for a sibling.

STUDENT LESSON 3.15 · WITH ANSWERS

Lesson 3.15: Finding and Fixing Integer Errors

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.15, Part A.
- [ ] Complete the rest of Workbook Lesson 3.15 (Parts B–D).

LET'S EXPLORE

Sam says $-8 - 3 = -5$. Is he right?

WORKED EXAMPLE · ERROR

$-8 - 3 = -5$?

1 Sam subtracted $8 - 3$ and kept a negative sign.

2 Correct: $-8 - 3 = -8 + (-3) = -11$.

Answer: -11

Common error	Fix
$-8 - 3 = -5$	Rewrite: $-8 + (-3) = -11$
$5 - (-2) = 3$	Subtracting a negative adds: $5 + 2 = 7$
$-4 + 9 = -5$	Different signs: $9 - 4 = 5$, sign of 9: $+5$
$\lvert -6 \rvert = -6$	Absolute value is a distance: 6

★ KEY IDEA

Rewrite subtraction as addition and check signs with a model.

✎ TRY IT

Answers are shown in color.

1. Fix: $-7 + (-2) = -5$

Answer: -9

2. Fix: $3 - 10 = 7$

Answer: -7

3. Fix: −6 − (−6) = −12

Answer: 0

Workbook Answer Key · Lesson 3.15

#	Question	Answer
1	Fix: −7 + (−2) = −5	−9
2	Fix: 3 − 10 = 7	−7
3	Fix: −6 − (−6) = −12	0
4	−8 − 3	−11
5	5 − (−2)	7
6	−12 + 5 = −17	−7
7	9 − (−4) = 5	13
8	−3 − 3 = 0	−6
9	−10 + 15 = −5	5
10	0 − 8 = 8	−8
11	|−2| + |−3| = −5	5
12	What is the most common integer mistake you make? How will you avoid it?	Answers vary.

LESSON 3.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 3: Adding and Subtracting Integers

Lesson 3.16: Integer Problem Solving

Student goal	I can solve problems with integer addition and subtraction from start to finish.
Objective	Students solve mixed integer problems.
Standard	7.NS.1, 7.NS.3 — 7.NS.1 review.
Pacing	Unit 3, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 3.16, Unit 3 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 3.16 and the Unit 3 Review.
Exit Ticket	3	Check. Tomorrow: Unit 3 Test.

COMMON MISCONCEPTIONS

- Sign errors.

SUPPORT

- Models.

EXTEND

- Analyze real weather data.

STUDENT LESSON 3.16 · WITH ANSWERS

Lesson 3.16: Integer Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 3.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 3.16, Part A.
- [ ] Complete the rest of Workbook Lesson 3.16 (Parts B–E).

LET'S EXPLORE

A week of low temperatures: −4, 2, −7, −1, 3, −5, 0. What is the range? What is the sum?

WORKED EXAMPLE · RANGE AND SUM

−4, 2, −7, −1, 3, −5, 0

1 Range: 3 − (−7) = 10.

2 Sum: positives 5, negatives −17 → −12.

Answer: range 10°; sum −12

★ KEY IDEA

Organize the numbers, group signs, and check with a model.

✎ TRY IT

Answers are shown in color.

1. Range of −15 and 22

Answer: 37

2. −9 + 14 − 20

Answer: −15

3. Distance from −6 to −18

Answer: 12

Workbook Answer Key · Lesson 3.16

#	Question	Answer
1	Range of −15 and 22	37
2	−9 + 14 − 20	−15
3	Distance from −6 to −18	12
4	3 − (−7)	10
5	5 + (−17)	−12
6	−23 + 15	−8
7	−23 − 15	−38
8	15 − (−23)	38
9	|−9 − 4|	13
10	−6 + 13 − (−5) − 20	−8
11	x − y when x = −2, y = −9	7
12	A submarine at −250 ft rises 90 ft, dives 145 ft, then rises 60 ft. Final depth?	−245 ft
13	Write and solve a problem that uses both addition and subtraction of negatives.	Answers vary.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 35 minutes

Unit 3 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	−18 + (−7)	−25
2	25 + (−40)	−15
3	−12 − 9	−21
4	6 − (−14)	20
5	−3 − (−10)	7
6	|−5 − 12|	17
7	−8°F rises 21°	13°F
8	Elevation −45 ft to 30 ft: change	75 ft
9	x − y + 3 when x = −1, y = −6	8
10	Fix: −9 − (−4) = −13	−5

UNIT TEST · ANSWER KEY
At the end of the unit

About 40 minutes

Unit 3 Test — Answer Key

Answers

#	Question	Answer
1	Opposite of −21	21
2	|−34|	34
3	Order: 4, −9, 0, −3	−9, −3, 0, 4
4	−17 + ___ = 0	17
5	−15 + (−26)	−41
6	−38 + 22	−16
7	44 + (−19)	25
8	9 − 25	−16
9	−14 − (−30)	16
10	−6 − 18	−24
11	|−11 − 7|	18
12	−8 + 20 − (−5) − 13	4
13	At 5 a.m. it was −9°F. By 2 p.m. it was 16°F. How much did it rise?	25°F
14	A diver at −42 ft rises 15 ft, then dives 28 ft. Where is she?	−55 ft
15	Evaluate a − b + c for a = −3, b = −10, c = −8.	−1
16	Use a chip model or number line to explain why 2 − (−6) = 8.	Add 6 zero pairs to 2 positives, remove 6 negatives → 8 positives; or distance from −6 to 2 is 8.

Scoring

Score	What it means / next step
18–21 (85%+)	Mastered.
15–17	Proficient; correct missed items.
Below 15	Reteach and retest.

UNIT 4 · TEACHER OVERVIEW

Adding and Subtracting Rational Numbers

4

Signed fractions and decimals, converting fractions to decimals, and real-world problems.

The integer rules from Unit 3 work for every rational number. In this unit you will add and subtract signed fractions, mixed numbers, and decimals, and use them to solve problems about money, temperature, and measurement.

Big ideas

- Place and compare signed fractions and decimals.
- Add and subtract signed fractions and mixed numbers.
- Convert fractions to terminating or repeating decimals.
- Add and subtract signed decimals.
- Solve multi-step rational-number problems.

Standards

Code	Standard
7.NS.1	Add and subtract rational numbers; represent on a number line; apply properties.
7.NS.2d	Convert a rational number to a decimal using long division.
7.NS.3	Solve real-world problems with the four operations on rational numbers.
7.EE.3	Solve multi-step real-life problems with rational numbers in any form.

What students should already know

- Integer operations (Unit 3).
- Fraction and decimal operations (Grades 5–6).

Materials for the unit

- Number lines with fractions and decimals
- Two-color counters

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	4.1	Rational Numbers on the Number Line
Week 1	Lesson 2 of 4	4.2	Adding Signed Fractions
Week 1	Lesson 3 of 4	4.3	Subtracting Signed Fractions
Week 1	Lesson 4 of 4	4.4	Adding and Subtracting Signed Mixed Numbers
Week 1	End of week	Check	Weekly Check 10: Signed Fractions
Week 2	Lesson 1 of 4	4.5	Fractions as Decimals
Week 2	Lesson 2 of 4	4.6	Comparing and Ordering Rational Numbers
Week 2	Lesson 3 of 4	4.7	Adding Signed Decimals
Week 2	Lesson 4 of 4	4.8	Subtracting Signed Decimals
Week 2	End of week	Check	Weekly Check 11: Signed Decimals
Week 3	Lesson 1 of 4	4.9	Rational Numbers and Money
Week 3	Lesson 2 of 4	4.10	Rational Numbers in Measurement
Week 3	Lesson 3 of 4	4.11	Properties for Rational Number Sums
Week 3	Lesson 4 of 4	4.12	Multi-Step Rational Number Problems
Week 3	End of week	Check	Weekly Check 12: Rational Number Applications
Week 4	Lesson 1 of 4	4.13	Distance Between Rational Numbers
Week 4	Lesson 2 of 4	4.14	Expressions with Rational Numbers
Week 4	Lesson 3 of 4	4.15	Error Analysis with Rational Numbers
Week 4	Lesson 4 of 4	4.16	Rational Number Problem Solving
Week 4	Before the test	Review	Unit 4 Review
Week 4	End of unit	Test	Unit 4 Test

TEACHING THIS UNIT AT HOME

Look at a bank statement or a weather report together and talk about the negative numbers you see.

UNIT VOCABULARY

rational number	A number that can be written as a fraction of integers.
terminating decimal	A decimal that ends.
repeating decimal	A decimal with a repeating pattern.

LESSON 4.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.1: Rational Numbers on the Number Line

Student goal	I can place positive and negative fractions and decimals on a number line.
Objective	Students locate and compare rational numbers.
Standard	7.NS.1 — Represent addition and subtraction on a number line; understand rational numbers.
Pacing	Unit 4, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Number line −3 to 3 in halves and quarters, workbook Lesson 4.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Compare integers.
Launch	5	Read Let's Explore.
Teach	15	Place fractions and decimals on a number line. Say: "For negatives, farther from 0 is smaller."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking $-1.02 < -1.2$ because it has more digits.

SUPPORT

- Number line.

EXTEND

- Find negative decimals in real data (stocks, temperatures).

STUDENT LESSON 4.1 · WITH ANSWERS

Lesson 4.1: Rational Numbers on the Number Line

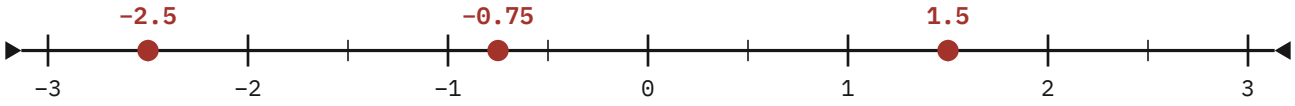
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.1, Part A.
- [ ] Complete the rest of Workbook Lesson 4.1 (Parts B–E).

LET'S EXPLORE

The temperature is −2.5°C. Is that warmer or colder than −2°C?

Rational numbers include integers, fractions, and decimals, positive and negative. −2.5 is halfway between −3 and −2, so it is farther from 0 and colder than −2.



WORKED EXAMPLE · COMPARE

$-\frac{3}{4}$ and $-\frac{2}{3}$

1 $\frac{3}{4} = \frac{9}{12}$ and $\frac{2}{3} = \frac{8}{12}$.

2 $-\frac{9}{12}$ is farther left.

Answer: $-\frac{3}{4} < -\frac{2}{3}$

★ KEY IDEA

For negative numbers, the one farther from 0 is less.

✎ TRY IT

Answers are shown in color.

1. Which is less: −1.2 or −1.02?

Answer: −1.2

2. Opposite of $-3\frac{1}{2}$

Answer: $3\frac{1}{2}$

3. |−4.75|

Answer: 4.75

Workbook Answer Key · Lesson 4.1

#	Question	Answer
1	Which is less: −1.2 or −1.02?	−1.2
2	Opposite of $-3\frac{1}{2}$	$3\frac{1}{2}$
3	|−4.75|	4.75
4	Which is less: −5 or −2?	−5
5	$\frac{3}{4}$ as a decimal	0.75
6	−0.6 ◯ −0.06	<
7	$-\frac{1}{2}$ ◯ −0.5	=
8	$-\frac{2}{5}$ ◯ $-\frac{1}{3}$	<
9	−3.4 ◯ −3.45	>
10	$\frac{5}{8}$ ◯ $-\frac{7}{8}$	>
11	−1.75 ◯ $-1\frac{3}{4}$	=
12	0.5, $-\frac{3}{4}$, −0.7, $\frac{1}{4}$, −1	−1, $-\frac{3}{4}$, −0.7, $\frac{1}{4}$, 0.5
13	Explain why −2.1 is less than −2.	It is farther left (farther from 0 in the negative direction).

LESSON 4.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.2: Adding Signed Fractions

Student goal	I can add positive and negative fractions.
Objective	Students add signed fractions.
Standard	7.NS.1b, 7.NS.1d — Apply properties of operations to add rational numbers.
Pacing	Unit 4, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples. Say: Same rules as integers.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding denominators.
- Losing the negative sign.

SUPPORT

- Write the sign decision first.

EXTEND

- Add three signed fractions.

STUDENT LESSON 4.2 · WITH ANSWERS

Lesson 4.2: Adding Signed Fractions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.2, Part A.
- [ ] Complete the rest of Workbook Lesson 4.2 (Parts B–E).

LET'S EXPLORE

A stock rose $\frac{3}{8}$ of a point on Monday and fell $\frac{5}{8}$ of a point on Tuesday. What was the total change?

The integer rules work for fractions too. Same signs: add absolute values, keep the sign. Different signs: subtract absolute values, use the sign of the one farther from 0.

WORKED EXAMPLE · DIFFERENT SIGNS

$\frac{3}{8} + (-\frac{5}{8})$

1 $|-\frac{5}{8}|$ is larger, so the answer is negative.

2 $\frac{5}{8} - \frac{3}{8} = \frac{2}{8} = \frac{1}{4}$.

Answer: $-\frac{1}{4}$

WORKED EXAMPLE · UNLIKE DENOMINATORS

$-\frac{1}{2} + (-\frac{1}{3})$

1 Common denominator 6: $-\frac{3}{6} + (-\frac{2}{6})$.

2 Same signs: $-\frac{5}{6}$.

Answer: $-\frac{5}{6}$

★ KEY IDEA

Use common denominators, then apply the integer sign rules to the numerators.

✎ TRY IT

Answers are shown in color.

1. $-\frac{2}{5} + \frac{4}{5}$

Answer: $\frac{2}{5}$

2. $\frac{3}{4} + (-\frac{5}{6})$
Answer: $-\frac{1}{12}$
3. $-\frac{1}{6} + (-\frac{2}{3})$
Answer: $-\frac{5}{6}$

Workbook Answer Key · Lesson 4.2

#	Question	Answer
1	$-\frac{2}{5} + \frac{4}{5}$	$\frac{2}{5}$
2	$\frac{3}{4} + (-\frac{5}{6})$	$-\frac{1}{12}$
3	$-\frac{1}{6} + (-\frac{2}{3})$	$-\frac{5}{6}$
4	$-5 + 3$	-2
5	$\frac{1}{2} + \frac{1}{3}$	$\frac{5}{6}$
6	$-\frac{3}{7} + \frac{1}{7}$	$-\frac{2}{7}$
7	$\frac{5}{9} + (-\frac{8}{9})$	$-\frac{1}{3}$
8	$-\frac{1}{4} + (-\frac{3}{8})$	$-\frac{5}{8}$
9	$\frac{2}{3} + (-\frac{1}{6})$	$\frac{1}{2}$
10	$-\frac{7}{10} + \frac{2}{5}$	$-\frac{3}{10}$
11	$-\frac{5}{12} + (-\frac{1}{3})$	$-\frac{3}{4}$
12	$\frac{4}{5} + (-\frac{1}{2})$	$\frac{3}{10}$
13	$-\frac{3}{4} + \frac{3}{4}$	0
14	A water level was $-\frac{3}{4}$ ft (below normal). It rose $\frac{1}{2}$ ft. New level?	$-\frac{1}{4}$ ft
15	$-\frac{5}{6} + \frac{3}{4}$	$-\frac{1}{12}$

LESSON 4.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.3: Subtracting Signed Fractions

Student goal	I can subtract positive and negative fractions by adding the opposite.
Objective	Students subtract signed fractions.
Standard	7.NS.1c — Understand subtraction as adding the additive inverse for rational numbers.
Pacing	Unit 4, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Rewrite each subtraction as addition.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting to change the second sign.

SUPPORT

- Keep-change-opposite.

EXTEND

- Write a real story for −3/4 − 1/2.

STUDENT LESSON 4.3 · WITH ANSWERS

Lesson 4.3: Subtracting Signed Fractions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.3, Part A.
- [ ] Complete the rest of Workbook Lesson 4.3 (Parts B–D).

LET'S EXPLORE

At 6 a.m. a pond's ice was $-\frac{1}{4}$ inch below a mark. At noon it was $\frac{5}{8}$ inch above. How much did it rise?

WORKED EXAMPLE · ADD THE OPPOSITE

$\frac{5}{8} - (-\frac{1}{4})$

$= \frac{5}{8} + \frac{1}{4} = \frac{5}{8} + \frac{2}{8}$.

Answer: $\frac{7}{8}$ inch

WORKED EXAMPLE · NEGATIVE MINUS POSITIVE

$-\frac{2}{3} - \frac{1}{2}$

$= -\frac{2}{3} + (-\frac{1}{2}) = -\frac{4}{6} + (-\frac{3}{6})$.

Answer: $-\frac{7}{6} = -1\frac{1}{6}$

★ KEY IDEA

$a - b = a + (-b)$, for fractions too.

✎ TRY IT

Answers are shown in color.

1. $\frac{1}{3} - \frac{2}{3}$

Answer: $-\frac{1}{3}$

2. $-\frac{3}{4} - (-\frac{1}{8})$

Answer: $-\frac{5}{8}$

3. $-\frac{1}{2} - \frac{3}{10}$

Answer: $-\frac{4}{5}$

Workbook Answer Key · Lesson 4.3

#	Question	Answer
1	$\frac{1}{3} - \frac{2}{3}$	$-\frac{1}{3}$
2	$-\frac{3}{4} - (-\frac{1}{8})$	$-\frac{5}{8}$
3	$-\frac{1}{2} - \frac{3}{10}$	$-\frac{4}{5}$
4	$2 - (-5)$	7
5	Opposite of $-\frac{3}{8}$	$\frac{3}{8}$
6	$\frac{1}{5} - \frac{4}{5}$	$-\frac{3}{5}$
7	$-\frac{2}{9} - \frac{5}{9}$	$-\frac{7}{9}$
8	$\frac{3}{8} - (-\frac{1}{4})$	$\frac{5}{8}$
9	$-\frac{5}{6} - (-\frac{1}{3})$	$-\frac{1}{2}$
10	$-\frac{1}{2} - \frac{1}{4}$	$-\frac{3}{4}$
11	$\frac{2}{3} - \frac{3}{4}$	$-\frac{1}{12}$
12	$-\frac{7}{12} - (-\frac{3}{4})$	$\frac{1}{6}$
13	$0 - (-\frac{5}{8})$	$\frac{5}{8}$
14	Explain why $\frac{1}{2} - (-\frac{1}{2}) = 1$.	Subtracting a negative adds: $\frac{1}{2} + \frac{1}{2}$.

LESSON 4.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.4: Adding and Subtracting Signed Mixed Numbers

Student goal	I can add and subtract signed mixed numbers like $-2\frac{1}{3} + 1\frac{1}{2}$.
Objective	Students add and subtract signed mixed numbers.
Standard	7.NS.1 — Add and subtract rational numbers.
Pacing	Unit 4, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Conversions.
Launch	5	Read Let's Explore.
Teach	15	Work both examples with improper fractions.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Making only the whole part negative: −3½ treated as −3 + ½.

SUPPORT

- Improper fractions.

EXTEND

- Solve with both methods and compare.

STUDENT LESSON 4.4 · WITH ANSWERS

Lesson 4.4: Adding and Subtracting Signed Mixed Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.4, Part A.
- [ ] Complete the rest of Workbook Lesson 4.4 (Parts B–E).

LET'S EXPLORE

A hiker is at $-3\,{}^{1}\!/\!_{2}$ m (in a trench). She climbs $5\,{}^{3}\!/\!_{4}$ m. Where is she now?

Change mixed numbers to improper fractions, or work with whole and fraction parts carefully.

WORKED EXAMPLE · IMPROPER FRACTIONS

$-3\,{}^{1}\!/\!_{2} + 5\,{}^{3}\!/\!_{4}$

$= -{}^{14}\!/\!_{4} + {}^{23}\!/\!_{4} = {}^{9}\!/\!_{4}$.

Answer: $2\,{}^{1}\!/\!_{4}$ m

WORKED EXAMPLE · SUBTRACT

$-1\,{}^{2}\!/\!_{3} - 2\,{}^{1}\!/\!_{6}$

$= -{}^{10}\!/\!_{6} + (-{}^{13}\!/\!_{6}) = -{}^{23}\!/\!_{6}$.

Answer: $-3\,{}^{5}\!/\!_{6}$

★ KEY IDEA

Improper fractions make signed mixed-number problems safer.

✎ TRY IT

Answers are shown in color.

1. $-2\,{}^{1}\!/\!_{4} + 3\,{}^{1}\!/\!_{2}$

Answer: $1\,{}^{1}\!/\!_{4}$

2. $1\,{}^{2}\!/\!_{3} - 3\,{}^{1}\!/\!_{4}$

Answer: $-1\,{}^{7}\!/\!_{12}$

3. $-2\frac{1}{5} + (-1\frac{1}{2})$
Answer: $-3\frac{7}{10}$

Workbook Answer Key · Lesson 4.4

#	Question	Answer
1	$-2\frac{1}{4} + 3\frac{1}{2}$	$1\frac{1}{4}$
2	$1\frac{2}{3} - 3\frac{1}{4}$	$-1\frac{7}{12}$
3	$-2\frac{1}{5} + (-1\frac{1}{2})$	$-3\frac{7}{10}$
4	$2\frac{1}{3}$ as an improper fraction	$\frac{7}{3}$
5	$\frac{17}{6}$ as a mixed number	$2\frac{5}{6}$
6	$-2\frac{1}{2} + 4\frac{1}{4}$	$1\frac{3}{4}$
7	$3\frac{1}{3} + (-3\frac{5}{6})$	$-\frac{1}{2}$
8	$-1\frac{5}{8} + (-\frac{3}{4})$	$-2\frac{3}{8}$
9	$-3\frac{1}{2} - 1\frac{1}{4}$	$-4\frac{3}{4}$
10	$2\frac{1}{4} - (-3\frac{2}{3})$	$5\frac{11}{12}$
11	$-3\frac{1}{5} - (-2\frac{1}{10})$	$-1\frac{1}{10}$
12	A lake was $1\frac{1}{4}$ ft below normal. It dropped $2\frac{2}{3}$ ft more. New level?	$-3\frac{11}{12}$ ft
13	$-2\frac{1}{2} - (-5\frac{3}{4})$	$3\frac{1}{4}$

WEEKLY CHECK · ANSWER KEY
After Lesson 4.4

About 20 minutes

Weekly Check 10: Signed Fractions — Answer Key

Answers

#	Question	Answer
1	$-\frac{3}{5} + \frac{1}{5}$	$-\frac{2}{5}$
2	$-\frac{1}{4} + (-\frac{2}{3})$	$-\frac{11}{12}$
3	$\frac{1}{6} - \frac{5}{6}$	$-\frac{2}{3}$
4	$-\frac{3}{8} - (-\frac{1}{2})$	$\frac{1}{8}$
5	$-3\frac{1}{2} + 2\frac{1}{4}$	$-1\frac{1}{4}$
6	Order: $-\frac{2}{3}$, -0.6, $\frac{1}{2}$	$-\frac{2}{3}$, -0.6, $\frac{1}{2}$

LESSON 4.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.5: Fractions as Decimals

Student goal	I can write fractions as terminating or repeating decimals using long division.
Objective	Students convert fractions to decimals with long division.
Standard	7.NS.2d — Convert a rational number to a decimal using long division; know that the decimal form terminates or eventually repeats.
Pacing	Unit 4, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Division warm-up.
Launch	5	Read Let's Explore.
Teach	15	Long division for 5/6 showing the repeating remainder.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Stopping too early to see the repeat.

SUPPORT

- Watch for repeating remainders.

EXTEND

- Which denominators give terminating decimals? (factors of 2 and 5 only)

STUDENT LESSON 4.5 · WITH ANSWERS

Lesson 4.5: Fractions as Decimals

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.5, Part A.
- [ ] Complete the rest of Workbook Lesson 4.5 (Parts B–D).

LET'S EXPLORE

Is $^{1}/_{3}$ closer to 0.33 or 0.34? Does it ever end?

Divide the numerator by the denominator. $^{3}/_{8} = 3 \div 8 = 0.375$: it terminates. $^{1}/_{3} = 0.3333\ldots$: it repeats forever. We write 0.3 with a bar over the 3.

Fraction	Decimal	Type
$^{1}/_{4}$	0.25	terminating
$^{3}/_{8}$	0.375	terminating
$^{2}/_{3}$	0.666...	repeating
$^{5}/_{11}$	0.4545...	repeating
$^{7}/_{20}$	0.35	terminating

WORKED EXAMPLE · LONG DIVISION

$^{5}/_{6}$

1 $5 \div 6$: $50 \div 6 = 8$ R2; $20 \div 6 = 3$ R2; $20 \div 6 = 3$ R2...

2 The 3 repeats.

Answer: 0.8333... (0.83 with a bar over the 3)

★ KEY IDEA

Every fraction is a terminating or repeating decimal. That's what makes it rational.

✎ TRY IT

Answers are shown in color.

1. $^{5}/_{8}$ as a decimal

Answer: 0.625

2. $^{4}/_{9}$ as a decimal

Answer: 0.444...

3. $-^{7}/_{4}$ as a decimal

Answer: −1.75

Workbook Answer Key · Lesson 4.5

#	Question	Answer
1	$^{5}/_{8}$ as a decimal	0.625
2	$^{4}/_{9}$ as a decimal	0.444...
3	$-^{7}/_{4}$ as a decimal	−1.75
4	3 ÷ 8	0.375
5	1 ÷ 3	0.333...
6	$^{3}/_{5}$	0.6; terminating
7	$^{1}/_{6}$	0.1666...; repeating
8	$^{7}/_{8}$	0.875; terminating
9	$^{2}/_{9}$	0.222...; repeating
10	$^{9}/_{20}$	0.45; terminating
11	$^{4}/_{11}$	0.3636...; repeating
12	Write $-2\,^{1}/_{3}$ as a decimal.	−2.333...

LESSON 4.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.6: Comparing and Ordering Rational Numbers

Student goal	I can compare and order signed fractions and decimals.
Objective	Students compare and order rational numbers.
Standard	7.NS.1, 7.NS.2d — Understand ordering of rational numbers.
Pacing	Unit 4, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Conversions.
Launch	5	Read Let's Explore.
Teach	15	Convert to decimals; place on a number line.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Ordering negatives like positives.

SUPPORT

- Number line.

EXTEND

- Order real golf scores or stock changes.

STUDENT LESSON 4.6 · WITH ANSWERS

Lesson 4.6: Comparing and Ordering Rational Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.6, Part A.
- [ ] Complete the rest of Workbook Lesson 4.6 (Parts B–D).

LET'S EXPLORE

Order these golf scores (relative to par) from best (lowest) to worst: -2.5, $-2\frac{3}{4}$, 1.25, $-\frac{1}{2}$.

WORKED EXAMPLE · ORDER

-2.5, $-2\frac{3}{4}$, 1.25, $-\frac{1}{2}$

1 Decimals: -2.5, -2.75, 1.25, -0.5.

2 Least to greatest: -2.75, -2.5, -0.5, 1.25.

Answer: $-2\frac{3}{4}$, -2.5, $-\frac{1}{2}$, 1.25

★ KEY IDEA

Change all numbers to decimals, then compare on a number line.

✎ TRY IT

Answers are shown in color.

1. $-\frac{5}{8}$ ◯ -0.6

Answer: <

2. Order: -1.3, $-\frac{4}{3}$, -1.25

Answer: $-\frac{4}{3}$, -1.3, -1.25

3. Greatest: -0.09, -0.9, -0.19

Answer: -0.09

Workbook Answer Key · Lesson 4.6

#	Question	Answer
1	$-\frac{5}{8}$ ◯ −0.6	<
2	Order: −1.3, $-\frac{4}{3}$, −1.25	$-\frac{4}{3}$, −1.3, −1.25
3	Greatest: −0.09, −0.9, −0.19	−0.09
4	$\frac{5}{8}$ as a decimal	0.625
5	$\frac{4}{3}$ as a decimal	1.333...
6	−0.4, $-\frac{3}{8}$, 0.3, $-\frac{1}{2}$	$-\frac{1}{2}$, −0.4, $-\frac{3}{8}$, 0.3
7	$1\frac{1}{3}$, −1.3, 1.3, $-1\frac{1}{3}$	$-1\frac{1}{3}$, −1.3, 1.3, $1\frac{1}{3}$
8	−2.05, −2.5, $-2\frac{1}{5}$, −2	−2.5, $-2\frac{1}{5}$, −2.05, −2
9	Which is greater, $-\frac{2}{3}$ or −0.67? Explain.	$-\frac{2}{3}$ (−0.666...) is greater than −0.67.

LESSON 4.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.7: Adding Signed Decimals

Student goal	I can add positive and negative decimals.
Objective	Students add signed decimals.
Standard	7.NS.1 — Add rational numbers.
Pacing	Unit 4, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Misaligned decimals.

SUPPORT

- Write zeros to equal length.

EXTEND

- Track a stock for a week.

STUDENT LESSON 4.7 · WITH ANSWERS

Lesson 4.7: Adding Signed Decimals

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.7, Part A.
- [ ] Complete the rest of Workbook Lesson 4.7 (Parts B–E).

LET'S EXPLORE

A bank account has −$12.50 (overdrawn). A deposit of $40.75 is made. What is the new balance?

WORKED EXAMPLE · DIFFERENT SIGNS

−12.50 + 40.75

1 Subtract absolute values: 40.75 − 12.50 = 28.25.

2 40.75 is farther from 0, so positive.

Answer: $28.25

WORKED EXAMPLE · SAME SIGNS

−3.6 + (−0.85)

1 Add: 3.6 + 0.85 = 4.45. Keep negative.

Answer: −4.45

★ KEY IDEA

Line up decimal points and use the integer sign rules.

✎ TRY IT

Answers are shown in color.

1. −7.2 + 3.5

Answer: −3.7

2. −0.45 + (−1.8)

Answer: −2.25

3. 12.06 + (−12.6)

Answer: −0.54

Workbook Answer Key · Lesson 4.7

#	Question	Answer
1	−7.2 + 3.5	−3.7
2	−0.45 + (−1.8)	−2.25
3	12.06 + (−12.6)	−0.54
4	40.75 − 12.50	28.25
5	3.6 + 0.85	4.45
6	−4.3 + 2.1	−2.2
7	6.75 + (−9.5)	−2.75
8	−0.8 + (−0.15)	−0.95
9	−15.2 + 15.2	0
10	3.05 + (−0.7)	2.35
11	−22.4 + (−7.65)	−30.05
12	A temperature of −3.8°C rises 5.6°C. New temperature?	1.8°C
13	A stock changed −1.35, +0.80, and −0.25 dollars. Total change?	−$0.80
14	−8.06 + 5.9	−2.16

LESSON 4.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.8: Subtracting Signed Decimals

Student goal	I can subtract positive and negative decimals.
Objective	Students subtract signed decimals.
Standard	7.NS.1c — Subtract rational numbers.
Pacing	Unit 4, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Sign errors.

SUPPORT

- Rewrite as addition.

EXTEND

- Find elevation ranges for Gulf states.

STUDENT LESSON 4.8 · WITH ANSWERS

Lesson 4.8: Subtracting Signed Decimals

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.8, Part A.
- [ ] Complete the rest of Workbook Lesson 4.8 (Parts B–E).

LET'S EXPLORE

The lowest point in Louisiana is about −2.4 m (in New Orleans). Driskill Mountain is about 163.1 m. What is the difference in elevation?

WORKED EXAMPLE · SUBTRACT A NEGATIVE

163.1 − (−2.4)

1 = 163.1 + 2.4.

Answer: 165.5 m

WORKED EXAMPLE · NEGATIVE MINUS POSITIVE

−0.75 − 1.6

1 = −0.75 + (−1.6) = −2.35.

Answer: −2.35

★ KEY IDEA

Subtract decimals by adding the opposite.

✎ TRY IT

Answers are shown in color.

1. 4.2 − 6.5

Answer: −2.3

2. −3.8 − (−1.25)

Answer: −2.55

3. −0.9 − 0.45

Answer: −1.35

Workbook Answer Key · Lesson 4.8

#	Question	Answer
1	4.2 − 6.5	−2.3
2	−3.8 − (−1.25)	−2.55
3	−0.9 − 0.45	−1.35
4	163.1 + 2.4	165.5
5	−0.75 + (−1.6)	−2.35
6	2.5 − 7.1	−4.6
7	−6.3 − 2.4	−8.7
8	1.75 − (−0.5)	2.25
9	−8.2 − (−8.2)	0
10	−0.05 − 0.5	−0.55
11	10 − (−3.75)	13.75
12	From −12.5°F to 4.3°F: change?	16.8°F
13	A submarine at −85.4 m dives to −112.9 m. Change?	−27.5 m
14	−5.08 − (−7.3)	2.22

WEEKLY CHECK · ANSWER KEY
After Lesson 4.8

About 20 minutes

Weekly Check 11: Signed Decimals — Answer Key

Answers

#	Question	Answer
1	$^{7}/_{8}$ as a decimal	0.875
2	$^{5}/_{9}$ as a decimal	0.555…
3	−6.4 + 2.75	−3.65
4	−0.35 + (−1.2)	−1.55
5	3.1 − 5.85	−2.75
6	−2.5 − (−7.25)	4.75

LESSON 4.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.9: Rational Numbers and Money

Student goal	I can solve money problems with positive and negative decimals.
Objective	Students solve money problems with signed decimals.
Standard	7.NS.3, 7.EE.3 — Solve real-world problems with rational numbers.
Pacing	Unit 4, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Running balance table.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Treating purchases as positive.

SUPPORT

- Table.

EXTEND

- Track a real allowance budget.

STUDENT LESSON 4.9 · WITH ANSWERS

Lesson 4.9: Rational Numbers and Money

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.9, Part A.
- [ ] Complete the rest of Workbook Lesson 4.9 (Parts B–D).

LET'S EXPLORE

Lillie's account starts at $24.60. She buys a book for $18.95 and a snack for $7.40. Then she deposits $15. What is her balance?

WORKED EXAMPLE · RUNNING BALANCE

24.60 − 18.95 − 7.40 + 15

1 24.60 − 18.95 = 5.65.

2 5.65 − 7.40 = −1.75.

3 −1.75 + 15 = 13.25.

Answer: $13.25

★ KEY IDEA

Write each transaction as a signed number and keep a running balance.

✎ TRY IT

Answers are shown in color.

1. $5.20 − $9.85

Answer: −$4.65

2. −$32.10 + $50

Answer: $17.90

3. Profits: $12.40, −$20.15, $6.80

Answer: −$0.95

Workbook Answer Key · Lesson 4.9

#	Question	Answer
1	$5.20 − $9.85	−$4.65
2	−$32.10 + $50	$17.90
3	Profits: $12.40, −$20.15, $6.80	−$0.95
4	24.6 − 18.95	5.65
5	5.65 − 7.4	−1.75
6	Start at −$8.25. Deposit $30, spend $14.60.	$7.15
7	A lemonade stand: +$18.50, −$6.75 (supplies), +$12.25, −$4.30. Total?	$19.70
8	A stock is at $45.20. It changes −2.85, +1.10, −0.65. New price?	$42.80
9	Ellie owes $12.40 and pays $7.95. How much does she owe now?	$4.45 (−$4.45)
10	Create a 4-transaction account record that ends below $0.	Answers vary.

LESSON 4.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.10: Rational Numbers in Measurement

Student goal	I can solve elevation, temperature, and measurement problems with signed fractions and decimals.
Objective	Students solve measurement problems with rational numbers.
Standard	7.NS.3 — Solve real-world problems with rational numbers.
Pacing	Unit 4, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Sign errors with mixed numbers.

SUPPORT

- Improper fractions.

EXTEND

- Measure and compare real objects.

STUDENT LESSON 4.10 · WITH ANSWERS

Lesson 4.10: Rational Numbers in Measurement

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.10, Part A.
- [ ] Complete the rest of Workbook Lesson 4.10 (Parts B–D).

LET'S EXPLORE

A carpenter needs a board $4\frac{3}{8}$ ft long. Hers is $5\frac{1}{4}$ ft. How much must she cut off?

WORKED EXAMPLE · FRACTIONS

$5\frac{1}{4} - 4\frac{3}{8}$

1 $= \frac{42}{8} - \frac{35}{8} = \frac{7}{8}$.

Answer: $\frac{7}{8}$ ft

WORKED EXAMPLE · TEMPERATURE

−4.5°C at night, 7.2°C at noon

1 $7.2 - (-4.5) = 11.7$.

Answer: 11.7°C warmer

★ KEY IDEA

Choose fractions or decimals, whichever makes the computation easier.

✎ TRY IT

Answers are shown in color.

1. A plant $3\frac{1}{2}$ in. tall shrinks $\frac{3}{4}$ in. New height?

Answer: $2\frac{3}{4}$ in.

2. −1.8°F to 6.5°F: change

Answer: 8.3°F

3. Water level $-\frac{2}{3}$ ft then rises $1\frac{1}{6}$ ft

Answer: $\frac{1}{2}$ ft

Workbook Answer Key · Lesson 4.10

#	Question	Answer
1	A plant $3\frac{1}{2}$ in. tall shrinks $\frac{3}{4}$ in. New height?	$2\frac{3}{4}$ in.
2	−1.8°F to 6.5°F: change	8.3°F
3	Water level $-\frac{2}{3}$ ft then rises $1\frac{1}{6}$ ft	$\frac{1}{2}$ ft
4	$5\frac{1}{4} - 4\frac{3}{8}$	$\frac{7}{8}$
5	7.2 − (−4.5)	11.7
6	A diver at $-10\frac{1}{2}$ ft rises $4\frac{3}{4}$ ft. Depth?	$-5\frac{3}{4}$ ft
7	A runner's times: 12.45 s and 11.9 s. Improvement?	0.55 s
8	−6.25°F drops 3.8°F. New temperature?	−10.05°F
9	Rope $12\frac{1}{3}$ m; cut $4\frac{5}{6}$ m. Left?	$7\frac{1}{2}$ m
10	Write a measurement story for $2\frac{1}{2} - 3\frac{3}{4}$. Solve.	$-1\frac{1}{4}$

LESSON 4.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.11: Properties for Rational Number Sums

Student goal	I can use properties to add and subtract rational numbers efficiently.
Objective	Students use properties to compute with rational numbers.
Standard	7.NS.1d — Apply properties of operations as strategies to add and subtract rational numbers.
Pacing	Unit 4, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Look for opposites and friendly sums.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Moving numbers without their signs.

SUPPORT

- Circle each term with its sign.

EXTEND

- Mental-math race.

STUDENT LESSON 4.11 · WITH ANSWERS

Lesson 4.11: Properties for Rational Number Sums

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.11, Part A.
- [ ] Complete the rest of Workbook Lesson 4.11 (Parts B–D).

LET'S EXPLORE

Find $-\frac{3}{4} + 2.6 + \frac{3}{4} + (-0.6)$ in your head.

WORKED EXAMPLE · REARRANGE

$-\frac{3}{4} + 2.6 + \frac{3}{4} + (-0.6)$

1 $(-\frac{3}{4} + \frac{3}{4}) + (2.6 + (-0.6))$.

2 $0 + 2 = 2$.

Answer: 2

★ KEY IDEA

Group opposites and friendly decimals; rewrite subtraction as addition first.

✎ TRY IT

Answers are shown in color.

1. $-3.7 + 8.25 + 3.7$

Answer: 8.25

2. $\frac{2}{5} - 1.5 - \frac{2}{5} + 0.5$

Answer: −1

3. $-0.25 + 6 + (-0.75)$

Answer: 5

Workbook Answer Key · Lesson 4.11

#	Question	Answer
1	$-3.7 + 8.25 + 3.7$	8.25
2	$2\frac{2}{5} - 1.5 - 2\frac{2}{5} + 0.5$	−1
3	$-0.25 + 6 + (-0.75)$	5
4	$-3\frac{3}{4} + 3\frac{3}{4}$	0
5	$2.6 - 0.6$	2
6	$4.8 + (-2.3) + (-4.8)$	−2.3
7	$-5\frac{5}{6} + 3 + 5\frac{5}{6}$	3
8	$-1.25 + 7 + (-0.75)$	5
9	$2\frac{1}{2} - 4 + 1\frac{1}{2}$	−1
10	$-9.9 + 5 + 9.9 + (-5)$	0
11	$0.75 + (-1\frac{1}{4}) + (-2)$	−1.5
12	Show how grouping helps evaluate $-6.4 + 3.3 + 6.4 - 1.3$.	$(-6.4 + 6.4) + (3.3 - 1.3) = 2$

LESSON 4.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.12: Multi-Step Rational Number Problems

Student goal	I can solve multi-step problems with signed fractions and decimals.
Objective	Students solve multi-step rational problems.
Standard	7.EE.3 — Solve multi-step real-life problems with positive and negative rational numbers.
Pacing	Unit 4, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Convert to one form.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Mixing fractions and decimals carelessly.

SUPPORT

- Convert first.

EXTEND

- Analyze real tide data.

STUDENT LESSON 4.12 · WITH ANSWERS

Lesson 4.12: Multi-Step Rational Number Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.12, Part A.
- [ ] Complete the rest of Workbook Lesson 4.12 (Parts B–D).

LET'S EXPLORE

A tide chart shows changes of $-1\frac{1}{4}$ ft, $+2\frac{1}{2}$ ft, $-\frac{3}{4}$ ft from a starting level of $\frac{1}{2}$ ft. Final level?

WORKED EXAMPLE · TIDE

$\frac{1}{2} - 1\frac{1}{4} + 2\frac{1}{2} - \frac{3}{4}$

1 In fourths: $2 - 5 + 10 - 3 = 4$ fourths.

Answer: 1 ft

★ KEY IDEA

Convert to one form (all fourths, or all decimals) and combine in order.

✎ TRY IT

Answers are shown in color.

1. $-2.4 + 5.1 - 3.25$

Answer: -0.55

2. $\frac{2}{3} - \frac{5}{6} + \left(-\frac{1}{2}\right)$

Answer: $-\frac{2}{3}$

3. A hiker at −15.5 m climbs 40.25 m then descends 8.75 m.

Answer: 16 m

Workbook Answer Key · Lesson 4.12

#	Question	Answer
1	−2.4 + 5.1 − 3.25	−0.55
2	$2/3 - 5/6 + (-1/2)$	$-2/3$
3	A hiker at −15.5 m climbs 40.25 m then descends 8.75 m.	16 m
4	2 − 5 + 10 − 3	4
5	$4/4$	1
6	Temperatures: −2.4°C, then +6.8°C, then −5.1°C. Final?	−0.7°C
7	A stock: start $18.40. Changes: $-3/4$, +$1.20, −$0.55. Final price? (Use 0.75 for $3/4$.)	$18.30
8	Lillie's garden bed is $8\ 1/2$ ft. She uses $2\ 3/4$ ft for tomatoes and $3\ 1/3$ ft for peppers. Space left?	$2\ 5/12$ ft
9	Write and solve a three-step problem with negative decimals.	Answers vary.

WEEKLY CHECK · ANSWER KEY
After Lesson 4.12

About 20 minutes

Weekly Check 12: Rational Number Applications — Answer Key

Answers

#	Question	Answer
1	Start −$6.40, deposit $25, spend $11.85	$6.75
2	−3.2°C to 5.9°C: change	9.1°C
3	Board $6\frac{1}{4}$ ft, cut $2\frac{7}{8}$ ft. Left?	$3\frac{3}{8}$ ft
4	−4.5 + 2.75 + 4.5 (mentally)	2.75
5	−1.6 + 3.25 − 0.9	0.75
6	$\frac{1}{2} - \frac{3}{4} - \frac{1}{8}$	$-\frac{3}{8}$

LESSON 4.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.13: Distance Between Rational Numbers

Student goal	I can find the distance between two rational numbers using absolute value.
Objective	Students find distance between rational numbers.
Standard	7.NS.1c — Show that the distance between two rational numbers is the absolute value of their difference.
Pacing	Unit 4, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Negative distances.

SUPPORT

- Number line.

EXTEND

- Distances between record temperatures.

STUDENT LESSON 4.13 · WITH ANSWERS

Lesson 4.13: Distance Between Rational Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.13, Part A.
- [ ] Complete the rest of Workbook Lesson 4.13 (Parts B–E).

LET'S EXPLORE

How far apart are $-2\frac{1}{2}$ and $1\frac{3}{4}$ on a number line?

WORKED EXAMPLE · DISTANCE

$|1\frac{3}{4} - (-2\frac{1}{2})|$

$= |\frac{7}{4} + \frac{10}{4}| = \frac{17}{4}$.

Answer: $4\frac{1}{4}$

★ KEY IDEA

Distance = $|a - b|$, for fractions and decimals too.

✎ TRY IT

Answers are shown in color.

1. Distance between −3.2 and 1.5

Answer: 4.7

2. Distance between $-\frac{5}{6}$ and $-\frac{1}{3}$

Answer: $\frac{1}{2}$

3. $|-0.8 - 2.45|$

Answer: 3.25

Workbook Answer Key · Lesson 4.13

#	Question	Answer
1	Distance between −3.2 and 1.5	4.7
2	Distance between $-\frac{5}{6}$ and $-\frac{1}{3}$	$\frac{1}{2}$

#	Question	Answer
3	$\lvert -0.8 - 2.45 \rvert$	3.25
4	$\lvert -7.5 \rvert$	7.5
5	$1\frac{3}{4} + 2\frac{1}{2}$	$4\frac{1}{4}$
6	−4.6 and 2.9	7.5
7	$-\frac{2}{3}$ and $\frac{1}{6}$	$\frac{5}{6}$
8	−10.25 and −3.5	6.75
9	$3\frac{1}{2}$ and $-1\frac{1}{4}$	$4\frac{3}{4}$
10	Death Valley: −86 m. Mount McKinley (Denali): 6,190 m. Distance?	6,276 m
11	Explain why the distance from −2.5 to 4 equals the distance from 4 to −2.5.	$\lvert a - b \rvert = \lvert b - a \rvert$

LESSON 4.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.14: Expressions with Rational Numbers

Student goal	I can evaluate expressions with signed fractions and decimals.
Objective	Students evaluate expressions with rational numbers.
Standard	7.NS.1, 7.EE.1 — Evaluate expressions.
Pacing	Unit 4, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Substitute with parentheses.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Sign errors.

SUPPORT

- Parentheses.

EXTEND

- Make a table for an expression.

STUDENT LESSON 4.14 · WITH ANSWERS

Lesson 4.14: Expressions with Rational Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.14, Part A.
- [ ] Complete the rest of Workbook Lesson 4.14 (Parts B–D).

LET'S EXPLORE

Evaluate $x - y + 2.5$ when $x = -1.75$ and $y = -3$.

WORKED EXAMPLE · SUBSTITUTE

$x - y + 2.5$

1 $(-1.75) - (-3) + 2.5 = -1.75 + 3 + 2.5$.

Answer: 3.75

★ KEY IDEA

Substitute with parentheses, rewrite subtraction, then combine.

✎ TRY IT

Answers are shown in color.

1. $a + b$ when $a = -0.6$, $b = 2.35$

Answer: 1.75

2. $x - {}^{1}\!/_{2}$ when $x = -{}^{3}\!/_{4}$

Answer: $-1\,{}^{1}\!/_{4}$

3. $-m + n$ when $m = -4.2$, $n = -1.8$

Answer: 2.4

Workbook Answer Key · Lesson 4.14

#	Question	Answer
1	$a + b$ when $a = -0.6$, $b = 2.35$	1.75
2	$x - {}^{1}\!/_{2}$ when $x = -{}^{3}\!/_{4}$	$-1\,{}^{1}\!/_{4}$

#	Question	Answer
3	$-m + n$ when $m = -4.2$, $n = -1.8$	2.4
4	$-1.75 + 3$	1.25
5	$-(-4.2)$	4.2
6	$p + q$	−1.25
7	$p - q$	−3.75
8	$q - p$	3.75
9	$-p - q$	1.25
10	$p + p + q$	−3.75
11	$\|p\| - \|q\|$	1.25
12	Evaluate $r - s$ when $r = {}^{2}/_{3}$ and $s = -{}^{5}/_{6}$.	$1\,{}^{1}/_{2}$

LESSON 4.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.15: Error Analysis with Rational Numbers

Student goal	I can find and fix errors in rational-number computations.
Objective	Students analyze rational-number errors.
Standard	7.NS.1 — Apply properties to add and subtract rational numbers.
Pacing	Unit 4, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Discuss each error in the table.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Same as listed.

SUPPORT

- Estimate first.

EXTEND

- Write tricky problems for a sibling.

STUDENT LESSON 4.15 · WITH ANSWERS

Lesson 4.15: Error Analysis with Rational Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.15, Part A.
- [ ] Complete the rest of Workbook Lesson 4.15 (Parts B–D).

LET'S EXPLORE

Sam says $-\frac{1}{2} - \frac{1}{3} = -\frac{1}{6}$. Is he right?

WORKED EXAMPLE · FIX IT

$-\frac{1}{2} - \frac{1}{3}$

1 Sam subtracted absolute values. It should be $-\frac{1}{2} + (-\frac{1}{3})$.

2 $= -\frac{3}{6} + (-\frac{2}{6}) = -\frac{5}{6}$.

Answer: $-\frac{5}{6}$

Error	Fix
$-2.5 + 1.3 = -3.8$	Different signs: -1.2
$0.4 - (-0.6) = -0.2$	Add the opposite: 1.0
$-1\frac{1}{2} = -1 + \frac{1}{2}$	$-1\frac{1}{2} = -1 - \frac{1}{2} = -1.5$

★ KEY IDEA

Estimate the sign and size first; then check the computed answer.

✎ TRY IT

Answers are shown in color.

1. Fix: $-3.2 - 1.5 = -1.7$

Answer: -4.7

2. Fix: $\frac{1}{4} - (-\frac{3}{4}) = -\frac{1}{2}$

Answer: 1

3. Fix: $-0.8 + 0.25 = -1.05$

Answer: −0.55

Workbook Answer Key · Lesson 4.15

#	Question	Answer
1	Fix: −3.2 − 1.5 = −1.7	−4.7
2	Fix: $\frac{1}{4} - (-\frac{3}{4}) = -\frac{1}{2}$	1
3	Fix: −0.8 + 0.25 = −1.05	−0.55
4	$-\frac{1}{2} - \frac{1}{3}$	$-\frac{5}{6}$
5	−2.5 + 1.3	−1.2
6	−6.1 + 2.4 = −8.5	−3.7
7	$\frac{2}{3} - \frac{5}{6} = \frac{1}{6}$	$-\frac{1}{6}$
8	−4 − (−1.25) = −5.25	−2.75
9	0.3 − 0.45 = 0.15	−0.15
10	Describe how estimating first can catch a sign error.	Predict positive/negative and size; compare.

LESSON 4.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 4: Adding and Subtracting Rational Numbers

Lesson 4.16: Rational Number Problem Solving

Student goal	I can solve problems with signed fractions and decimals.
Objective	Students solve mixed rational-number problems.
Standard	7.NS.1, 7.NS.3, 7.EE.3 — 7.NS.1–3 review.
Pacing	Unit 4, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 4.16, Unit 4 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work the table example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 4.16 and the Unit 4 Review.
Exit Ticket	3	Check. Tomorrow: Unit 4 Test.

COMMON MISCONCEPTIONS

- Sign errors.

SUPPORT

- Tables.

EXTEND

- Analyze a week of real temperature data.

STUDENT LESSON 4.16 · WITH ANSWERS

Lesson 4.16: Rational Number Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 4.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 4.16, Part A.
- [ ] Complete the rest of Workbook Lesson 4.16 (Parts B–E).

LET'S EXPLORE

A science class recorded temperature changes every hour: −1.5°, −2.25°, +0.75°, +3.5°. Starting at −4°C, what is the final temperature? What was the lowest?

WORKED EXAMPLE · TRACK

Start −4

1 −4 − 1.5 = −5.5.

2 −5.5 − 2.25 = −7.75 (lowest).

3 −7.75 + 0.75 = −7.

4 −7 + 3.5 = −3.5.

Answer: final −3.5°C; lowest −7.75°C

★ KEY IDEA

Organize with a table; check the reasonableness of every step.

✎ TRY IT

Answers are shown in color.

1. $-\frac{3}{8} + \frac{5}{6}$

Answer: $\frac{11}{24}$

2. −7.05 − (−2.6)

Answer: −4.45

3. Distance between $-\frac{3}{4}$ and 2.5

Answer: 3.25

Workbook Answer Key · Lesson 4.16

#	Question	Answer
1	$-\frac{3}{8} + \frac{5}{6}$	$\frac{11}{24}$
2	−7.05 − (−2.6)	−4.45
3	Distance between $-\frac{3}{4}$ and 2.5	3.25
4	−4 − 1.5	−5.5
5	−7.75 + 0.75	−7
6	$-\frac{7}{8} + \frac{1}{4}$	$-\frac{5}{8}$
7	$\frac{2}{5} - (-\frac{1}{2})$	$\frac{9}{10}$
8	−12.4 + 8.75	−3.65
9	0.6 − 2.15	−1.55
10	Order: −0.3, $-\frac{1}{3}$, −0.33	$-\frac{1}{3}$, −0.33, −0.3
11	$\frac{7}{12}$ as a decimal	0.58333…
12	A business had monthly results of −$1,250.50, +$3,400.75, and −$875.25. Total?	$1,275
13	Which is harder for you: signed fractions or signed decimals? Make up and solve one of each.	Answers vary.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 35 minutes

Unit 4 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	$-\frac{5}{6} + \frac{1}{3}$	$-\frac{1}{2}$
2	$-\frac{3}{4} - \frac{5}{8}$	$-1\frac{3}{8}$
3	$-2\frac{3}{4} + (-1\frac{1}{2})$	$-4\frac{1}{4}$
4	−9.6 + 4.35	−5.25
5	−1.2 − (−3.05)	1.85
6	$\frac{3}{11}$ as a decimal	0.2727…
7	Order: −1.5, $-1\frac{2}{3}$, −1.6	$-1\frac{2}{3}$, −1.6, −1.5
8	Distance between −2.75 and 4.1	6.85
9	x − y when x = −0.4, y = 1.35	−1.75
10	A balance of −$15.60 gets a $40 deposit and a $9.25 charge.	$15.15

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 4 Test — Answer Key

Answers

#	Question	Answer
1	$^{9}/_{16}$ as a decimal	0.5625
2	$^{2}/_{3}$ as a decimal	0.666…
3	Order: −0.8, $-^{3}/_{4}$, $-^{5}/_{6}$	$-^{5}/_{6}$, −0.8, $-^{3}/_{4}$
4	|−6.05|	6.05
5	$-^{4}/_{9} + (-^{2}/_{3})$	$-1^{1}/_{9}$
6	$^{7}/_{10} + (-1^{1}/_{5})$	$-^{1}/_{2}$
7	$^{3}/_{8} - (-^{5}/_{12})$	$^{19}/_{24}$
8	$-4^{1}/_{4} - 1^{5}/_{6}$	$-6^{1}/_{12}$
9	−14.25 + 6.8	−7.45
10	0.45 − 2.6	−2.15
11	−8.3 − (−12.05)	3.75
12	−3.75 + 9.2 + 3.75 (mentally)	9.2
13	Temperatures change −2.6°, +4.15°, −0.8° from −1.5°C. Final temperature?	−0.75°C
14	A diver at $-12^{1}/_{2}$ ft rises $7^{3}/_{4}$ ft. Depth now?	$-4^{3}/_{4}$ ft
15	Distance between −18.4 m and 22.75 m	41.15 m
16	Ellie says $-2^{1}/_{4} + 1^{3}/_{4} = -3^{2}/_{4}$. Explain her mistake and give the correct answer.	She added absolute values; signs differ, so subtract: $-^{2}/_{4} = -^{1}/_{2}$.

Scoring

Score	What it means / next step
18–21 (85%+)	Mastered.
15–17	Proficient; correct missed items.
Below 15	Reteach and retest.

Multiplying and Dividing Rational Numbers

Sign rules and why they work, signed fractions and decimals, order of operations, rates, and averages.

You can add and subtract any rational number. Now you will multiply and divide them, discover why the sign rules work, and use them to model repeated change, rates, and averages.

Big ideas

- Multiply and divide integers and explain the sign rules.
- Multiply and divide signed fractions and decimals.
- Use order of operations and the distributive property with negatives.
- Model repeated change and rates with signed numbers.
- Solve multi-step problems with all four operations.

Standards

Code	Standard
7.NS.2	Apply and extend understandings of multiplication and division to rational numbers.
7.NS.3	Solve real-world problems with the four operations on rational numbers.
7.EE.3	Solve multi-step real-life problems with rational numbers.

What students should already know

- Integer and rational addition/subtraction (Units 3–4).
- Fraction multiplication and division (Grade 6).

Materials for the unit

- Two-color counters
- Number lines
- Calculator for checking

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	5.1	Multiplying Integers: Patterns
Week 1	Lesson 2 of 4	5.2	Multiplying Integers: Rules and Reasons
Week 1	Lesson 3 of 4	5.3	Dividing Integers
Week 1	Lesson 4 of 4	5.4	Negative Fractions and Signs of Quotients
Week 1	End of week	Check	Weekly Check 13: Multiplying and Dividing Integers
Week 2	Lesson 1 of 4	5.5	Multiplying Signed Fractions
Week 2	Lesson 2 of 4	5.6	Dividing Signed Fractions
Week 2	Lesson 3 of 4	5.7	Multiplying and Dividing Signed Decimals
Week 2	Lesson 4 of 4	5.8	Rational Number Operations Practice
Week 2	End of week	Check	Weekly Check 14: Rational Products and Quotients
Week 3	Lesson 1 of 4	5.9	Order of Operations with Rational Numbers
Week 3	Lesson 2 of 4	5.10	The Distributive Property with Rational Numbers
Week 3	Lesson 3 of 4	5.11	Repeated Change: Rate × Time
Week 3	Lesson 4 of 4	5.12	Choosing the Operation
Week 3	End of week	Check	Weekly Check 15: Operations and Models
Week 4	Lesson 1 of 4	5.13	Average Change
Week 4	Lesson 2 of 4	5.14	Rate Problems with Negatives
Week 4	Lesson 3 of 4	5.15	Multi-Step Rational Number Problems
Week 4	Lesson 4 of 4	5.16	Rational Number Operations Problem Solving
Week 4	Before the test	Review	Unit 5 Review
Week 4	End of unit	Test	Unit 5 Test

TEACHING THIS UNIT AT HOME

Spring break (Mar 15–19) falls between Weeks 3 and 4 of this unit.

UNIT VOCABULARY

product	Result of multiplication.
quotient	Result of division.
reciprocal	Number whose product with another is 1.
mean	Sum ÷ count.

LESSON 5.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.1: Multiplying Integers: Patterns

Student goal	I can use patterns to discover how to multiply positive and negative integers.
Objective	Students discover integer multiplication rules through patterns.
Standard	7.NS.2a — Understand that multiplication extends from fractions to rational numbers by requiring that operations continue to satisfy the properties.
Pacing	Unit 5, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Repeated addition.
Launch	5	Read Let's Explore.
Teach	15	Build the pattern table together and predict the next rows. Say: "The pattern forces negative × negative to be positive."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Applying addition rules (sign of larger absolute value) to multiplication.

SUPPORT

- Repeated addition for pos × neg.

EXTEND

- Why must $(-1)(-1) = 1$? Use the distributive property: $(-1)(1 + (-1)) = 0$.

STUDENT LESSON 5.1 · WITH ANSWERS

Lesson 5.1: Multiplying Integers: Patterns

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.1, Part A.
- [ ] Complete the rest of Workbook Lesson 5.1 (Parts B–E).

LET'S EXPLORE

A pool drains 3 inches of water each hour. What is the change in water level after 4 hours?

Draining means the level changes by −3 each hour. Four hours: $4 \times (-3) = -3 + (-3) + (-3) + (-3) = -12$. The level drops 12 inches.

Pattern	Product
$3 \times (-2)$	-6
$2 \times (-2)$	-4
$1 \times (-2)$	-2
$0 \times (-2)$	0
$-1 \times (-2)$	2
$-2 \times (-2)$	4

Look at the pattern: each time the first factor goes down by 1, the product goes up by 2. Continue past 0 and the products become positive. So a negative times a negative is positive.

★ KEY IDEA

Positive × negative = negative. Negative × negative = positive.

✎ TRY IT

Answers are shown in color.

1. $5 \times (-6)$

Answer: -30

2. $(-7) \times (-3)$

Answer: 21

3. $(-9) \times 4$

Answer: -36

Workbook Answer Key · Lesson 5.1

#	Question	Answer
1	$5 \times (-6)$	-30
2	$(-7) \times (-3)$	21
3	$(-9) \times 4$	-36
4	$-3 + (-3) + (-3) + (-3)$	-12
5	6×7	42
6	$3 \times (-8)$	-24
7	$(-5) \times 7$	-35
8	$(-6) \times (-6)$	36
9	$(-11) \times 2$	-22
10	$(-4) \times (-9)$	36
11	$12 \times (-3)$	-36
12	$(-1) \times (-15)$	15
13	$0 \times (-23)$	0
14	$3 \times (-5) = -15$, $2 \times (-5) =$ ___, $1 \times (-5) =$ ___, $0 \times (-5) =$ ___, $-1 \times (-5) =$ ___	$-10, -5, 0, 5$
15	Use a pattern to explain why $-3 \times (-4)$ is positive.	Each time the first factor drops by 1, the product rises by 4: $0 \times (-4) = 0$, $-1 \times (-4) = 4$, $-2 \times (-4) = 8$, $-3 \times (-4) = 12$.

LESSON 5.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.2: Multiplying Integers: Rules and Reasons

Student goal	I can multiply integers, including more than two factors, and explain the sign rules.
Objective	Students multiply integers with several factors.
Standard	7.NS.2a, 7.NS.2c — Interpret products of rational numbers; understand (−1)(−1) = 1.
Pacing	Unit 5, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Count negatives; work examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- $(-3)^2 = -9$.

SUPPORT

- Parentheses mean the sign is included.

EXTEND

- Prove $(-a)(-b) = ab$ with the distributive property.

STUDENT LESSON 5.2 · WITH ANSWERS

Lesson 5.2: Multiplying Integers: Rules and Reasons

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.2, Part A.
- [ ] Complete the rest of Workbook Lesson 5.2 (Parts B–E).

LET'S EXPLORE

What is (−2) × (−3) × (−5)? How can you tell the sign before multiplying?

Multiply the absolute values, then decide the sign. Each pair of negatives makes a positive. So an even number of negative factors gives a positive product, and an odd number gives a negative product.

WORKED EXAMPLE · THREE FACTORS

(−2)(−3)(−5)

1 Three negative factors: odd → negative.

2 2 × 3 × 5 = 30.

Answer: −30

WORKED EXAMPLE · USING THE DISTRIBUTIVE PROPERTY

Why is (−1)(−1) = 1?

1 (−1)(1 + (−1)) = (−1)(0) = 0.

2 So (−1)(1) + (−1)(−1) = 0, which means −1 + (−1)(−1) = 0.

3 (−1)(−1) must be 1.

Answer: The rules keep the properties true.

★ KEY IDEA

Count the negative factors: even → positive, odd → negative.

✎ TRY IT

Answers are shown in color.

1. (−4)(5)(−2)

Answer: 40

2. (−1)(−1)(−1)(−1)(−1)

Answer: −1

3. $(-3)^2$

Answer: 9

Workbook Answer Key · Lesson 5.2

#	Question	Answer
1	(−4)(5)(−2)	40
2	(−1)(−1)(−1)(−1)(−1)	−1
3	$(-3)^2$	9
4	(−4) × 5	−20
5	(−20) × (−2)	40
6	(−2)(6)(−3)	36
7	(−5)(−2)(−4)	−40
8	(7)(−1)(3)	−21
9	(−3)(−3)(−3)	−27
10	(−10)(−10)	100
11	(−2)(−2)(−2)(−2)	16
12	Is the product of 9 negative numbers positive or negative?	negative
13	Is (−8)(−7)(−6)(−5) positive or negative?	positive
14	Explain why $(-4)^2$ and -4^2 are different.	$(-4)^2 = 16$; $-4^2 = -(4^2) = -16$.

LESSON 5.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.3: Dividing Integers

Student goal	I can divide integers using the related multiplication fact.
Objective	Students divide integers.
Standard	7.NS.2b — Understand that integers can be divided, provided the divisor is not zero; quotients follow sign rules.
Pacing	Unit 5, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Use fact families to justify each row.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing by 0.

SUPPORT

- Fact families.

EXTEND

- Find the average daily temperature change for a week.

STUDENT LESSON 5.3 · WITH ANSWERS

Lesson 5.3: Dividing Integers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.3, Part A.
- [ ] Complete the rest of Workbook Lesson 5.3 (Parts B–E).

LET'S EXPLORE

A diver descends 60 feet in 5 minutes at a steady rate. What is the change in depth per minute?

$-60 \div 5 = ?$ asks: $5 \times ? = -60$. Since $5 \times (-12) = -60$, the answer is -12 feet per minute.

Division	Because	Quotient
$-60 \div 5$	$5 \times (-12) = -60$	-12
$60 \div (-5)$	$-5 \times (-12) = 60$	-12
$-60 \div (-5)$	$-5 \times 12 = -60$	12

★ KEY IDEA

Division sign rules match multiplication: same signs → positive; different signs → negative. You can never divide by 0.

✎ TRY IT

Answers are shown in color.

1. $-42 \div 6$

Answer: -7

2. $56 \div (-8)$

Answer: -7

3. $-81 \div (-9)$

Answer: 9

Workbook Answer Key · Lesson 5.3

#	Question	Answer
1	−42 ÷ 6	−7
2	56 ÷ (−8)	−7
3	−81 ÷ (−9)	9
4	5 × (−12)	−60
5	(−5) × 12	−60
6	−36 ÷ 4	−9
7	45 ÷ (−9)	−5
8	−72 ÷ (−8)	9
9	100 ÷ (−25)	−4
10	−90 ÷ 10	−9
11	−64 ÷ (−16)	4
12	0 ÷ (−7)	0
13	−150 ÷ 30	−5
14	The temperature dropped 18°F in 6 hours at a steady rate. Change per hour?	−3°F per hour
15	A stock lost $48 over 4 days, the same each day. Daily change?	−$12
16	Why is −5 ÷ 0 undefined?	No number times 0 equals −5.

LESSON 5.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.4: Negative Fractions and Signs of Quotients

Student goal	I can explain why $-(p/q) = -p/q = p/-q$.
Objective	Students interpret signs in fractions.
Standard	7.NS.2b — If p and q are integers, −(p/q) = (−p)/q = p/(−q).
Pacing	Unit 5, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Evaluate each form as division.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking −3/−4 is negative.

SUPPORT

- Treat the fraction bar as division.

EXTEND

- Explain to a sibling with money owed shared among friends.

STUDENT LESSON 5.4 · WITH ANSWERS

Lesson 5.4: Negative Fractions and Signs of Quotients

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.4, Part A.
- [ ] Complete the rest of Workbook Lesson 5.4 (Parts B–E).

LET'S EXPLORE

Are $-\frac{3}{4}$, $\frac{-3}{4}$, and $\frac{3}{-4}$ the same number?

Yes. A fraction is a division: $\frac{-3}{4} = -3 \div 4 = -0.75$ and $\frac{3}{-4} = 3 \div (-4) = -0.75$. Both equal $-\frac{3}{4}$.

WORKED EXAMPLE · WHERE THE SIGN GOES

$\frac{-12}{-3}$

1 $-12 \div (-3) = 4$.

Answer: 4 (two negatives make a positive)

★ KEY IDEA

$-\left(\frac{p}{q}\right) = \frac{-p}{q} = \frac{p}{-q}$. A fraction with one negative sign is negative.

✎ TRY IT

Answers are shown in color.

1. Simplify $\frac{-18}{6}$

Answer: -3

2. Simplify $\frac{20}{-5}$

Answer: -4

3. Simplify $\frac{-7}{-14}$

Answer: $\frac{1}{2}$

Workbook Answer Key · Lesson 5.4

#	Question	Answer
1	Simplify $\frac{-18}{6}$	-3
2	Simplify $\frac{20}{-5}$	-4
3	Simplify $\frac{-7}{-14}$	$\frac{1}{2}$
4	$-18 \div 6$	-3
5	$20 \div (-5)$	-4
6	$\frac{-24}{8}$	-3
7	$\frac{35}{-7}$	-5
8	$\frac{-9}{-3}$	3
9	$\frac{-5}{10}$	$-\frac{1}{2}$
10	$\frac{6}{-8}$	$-\frac{3}{4}$
11	$\frac{-15}{-20}$	$\frac{3}{4}$
12	$-\frac{2}{5}$ and $\frac{-2}{5}$ A. same B. different	same
13	$\frac{-4}{-9}$ and $-\frac{4}{9}$ A. same B. different	different
14	Write $-\frac{5}{6}$ two other ways.	$\frac{-5}{6}$ and $\frac{5}{-6}$

WEEKLY CHECK · ANSWER KEY
After Lesson 5.4

About 15 minutes

Weekly Check 13: Multiplying and Dividing Integers — Answer Key

Answers

#	Question	Answer
1	$(-7) \times 8$	-56
2	$(-6) \times (-9)$	54
3	$(-2)(-3)(-4)$	-24
4	$-63 \div 7$	-9
5	$-48 \div (-6)$	8
6	Simplify $\frac{14}{-21}$	$-\frac{2}{3}$

LESSON 5.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.5: Multiplying Signed Fractions

Student goal	I can multiply positive and negative fractions and mixed numbers.
Objective	Students multiply signed fractions.
Standard	7.NS.2a, 7.NS.2c — Multiply rational numbers.
Pacing	Unit 5, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Sign first, then multiply and simplify.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using common denominators for multiplication.

SUPPORT

- Cross-simplify before multiplying.

EXTEND

- Scale a recipe by −? (Why doesn't that make sense?)

STUDENT LESSON 5.5 · WITH ANSWERS

Lesson 5.5: Multiplying Signed Fractions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.5, Part A.
- [ ] Complete the rest of Workbook Lesson 5.5 (Parts B–E).

LET'S EXPLORE

A recipe needs $\frac{3}{4}$ cup of sugar, but you're making $\frac{2}{3}$ of the recipe. A water tank loses $\frac{1}{2}$ gallon per minute for $\frac{3}{4}$ minute. What are the products?

WORKED EXAMPLE · FRACTIONS

$\frac{2}{3} \times \frac{3}{4}$

1 Multiply numerators and denominators: $\frac{6}{12} = \frac{1}{2}$.

Answer: $\frac{1}{2}$ cup

WORKED EXAMPLE · WITH A NEGATIVE

$(-\frac{1}{2}) \times \frac{3}{4}$

1 Different signs → negative.

2 $\frac{1}{2} \times \frac{3}{4} = \frac{3}{8}$.

Answer: $-\frac{3}{8}$ gallon

WORKED EXAMPLE · MIXED NUMBERS

$(-1\frac{1}{2}) \times (-2\frac{2}{3})$

1 $= \frac{3}{2} \times \frac{8}{3}$ (two negatives → positive).

2 $= \frac{24}{6} = 4$.

Answer: 4

★ KEY IDEA

Decide the sign, multiply the absolute values, and simplify. Change mixed numbers to improper fractions.

✎ TRY IT

Answers are shown in color.

1. $(-\frac{2}{5}) \times \frac{3}{4}$

Answer: $-\frac{3}{10}$

2. $(-\frac{5}{6}) \times (-\frac{3}{10})$

Answer: $\frac{1}{4}$

3. $(-2\frac{1}{4}) \times \frac{2}{3}$

Answer: $-1\frac{1}{2}$

Workbook Answer Key · Lesson 5.5

#	Question	Answer
1	$(-\frac{2}{5}) \times \frac{3}{4}$	$-\frac{3}{10}$
2	$(-\frac{5}{6}) \times (-\frac{3}{10})$	$\frac{1}{4}$
3	$(-2\frac{1}{4}) \times \frac{2}{3}$	$-1\frac{1}{2}$
4	$\frac{2}{3} \times \frac{3}{4}$	$\frac{1}{2}$
5	$(-2) \times 3$	-6
6	$(-\frac{1}{3}) \times \frac{6}{7}$	$-\frac{2}{7}$
7	$\frac{4}{5} \times (-\frac{5}{8})$	$-\frac{1}{2}$
8	$(-\frac{3}{4}) \times (-\frac{2}{9})$	$\frac{1}{6}$
9	$(-3\frac{1}{2}) \times \frac{4}{7}$	-2
10	$(-2\frac{1}{2}) \times (-2\frac{1}{2})$	$6\frac{1}{4}$
11	$3\frac{1}{3} \times (-\frac{9}{20})$	$-1\frac{1}{2}$
12	A diver descends at $2\frac{1}{4}$ ft per second for $\frac{4}{3}$ seconds. Change in depth?	-3 ft
13	$(-3\frac{3}{4}) \times (-1\frac{3}{5})$	6

LESSON 5.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.6: Dividing Signed Fractions

Student goal	I can divide positive and negative fractions.
Objective	Students divide signed fractions.
Standard	7.NS.2b, 7.NS.2c — Divide rational numbers.
Pacing	Unit 5, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Multiply by reciprocal; sign rules.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Flipping the dividend.

SUPPORT

- Keep-change-flip with signs.

EXTEND

- Write a story for −2 ÷ 1/4.

STUDENT LESSON 5.6 · WITH ANSWERS

Lesson 5.6: Dividing Signed Fractions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.6, Part A.
- [ ] Complete the rest of Workbook Lesson 5.6 (Parts B–E).

LET'S EXPLORE

A tank's level fell $3/4$ ft over $3/8$ hour. What is the change per hour?

WORKED EXAMPLE · DIVIDE

$(-3/4) \div 3/8$

1 Different signs → negative.

2 $3/4 \times 8/3 = 24/12 = 2$.

Answer: −2 ft per hour

WORKED EXAMPLE · TWO NEGATIVES

$(-5/6) \div (-5/12)$

1 Positive. $5/6 \times 12/5 = 2$.

Answer: 2

★ KEY IDEA

To divide, multiply by the reciprocal of the divisor. Sign rules are the same.

✎ TRY IT

Answers are shown in color.

1. $(-2/3) \div 4/9$

Answer: $-1\,1/2$

2. $7/8 \div (-7/16)$

Answer: −2

3. $(-2\,1/2) \div (-3/4)$

Answer: $3\,1/3$

Workbook Answer Key · Lesson 5.6

#	Question	Answer
1	$(-\frac{2}{3}) \div \frac{4}{9}$	$-1\frac{1}{2}$
2	$\frac{7}{8} \div (-\frac{7}{16})$	-2
3	$(-2\frac{1}{2}) \div (-\frac{3}{4})$	$3\frac{1}{3}$
4	Reciprocal of $-\frac{3}{8}$	$-\frac{8}{3}$
5	$\frac{3}{4} \times \frac{8}{3}$	2
6	$(-\frac{1}{2}) \div \frac{1}{4}$	-2
7	$\frac{3}{5} \div (-\frac{9}{10})$	$-\frac{2}{3}$
8	$(-\frac{4}{7}) \div (-\frac{2}{7})$	2
9	$(-6) \div \frac{3}{4}$	-8
10	$1\frac{1}{4} \div (-2\frac{1}{2})$	$-\frac{1}{2}$
11	$(-4\frac{1}{2}) \div (-\frac{3}{8})$	12
12	A hiker descends $4\frac{1}{2}$ miles in $\frac{3}{4}$ hour. Change in elevation per hour (in miles)?	-6 miles per hour
13	$(-3\frac{1}{3}) \div \frac{5}{6}$	-4

LESSON 5.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.7: Multiplying and Dividing Signed Decimals

Student goal	I can multiply and divide positive and negative decimals.
Objective	Students multiply and divide signed decimals.
Standard	7.NS.2 — Multiply and divide rational numbers.
Pacing	Unit 5, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Decimal placement errors.

SUPPORT

- Estimate first.

EXTEND

- Compute real stock changes.

STUDENT LESSON 5.7 · WITH ANSWERS

Lesson 5.7: Multiplying and Dividing Signed Decimals

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.7, Part A.
- [ ] Complete the rest of Workbook Lesson 5.7 (Parts B–E).

LET'S EXPLORE

A stock fell $1.25 per share. Ellie owns 8 shares. What is the total change? Over 4 days, what was the average daily change?

WORKED EXAMPLE · MULTIPLY

8 × (−1.25)

1 8 × 1.25 = 10. Different signs → negative.

Answer: −$10

WORKED EXAMPLE · DIVIDE

−10 ÷ 4

1 10 ÷ 4 = 2.5. Negative.

Answer: −$2.50 per day

★ KEY IDEA

Multiply or divide the decimals, then apply the sign rules.

✎ TRY IT

Answers are shown in color.

1. (−0.6) × 4.5

Answer: −2.7

2. (−7.2) ÷ (−0.9)

Answer: 8

3. (−2.5) × (−0.4)

Answer: 1

Workbook Answer Key · Lesson 5.7

#	Question	Answer
1	(−0.6) × 4.5	−2.7
2	(−7.2) ÷ (−0.9)	8
3	(−2.5) × (−0.4)	1
4	8 × 1.25	10
5	10 ÷ 4	2.5
6	(−3.2) × 5	−16
7	(−0.25) × (−12)	3
8	1.5 × (−0.6)	−0.9
9	(−8.4) ÷ 4	−2.1
10	6.3 ÷ (−0.7)	−9
11	(−0.48) ÷ (−0.06)	8
12	A submarine changes depth by −12.5 m per minute. Change in 3.2 minutes?	−40 m
13	A temperature fell 7.2°F in 1.5 hours. Change per hour?	−4.8°F
14	(−13.5) ÷ 0.45	−30

LESSON 5.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.8: Rational Number Operations Practice

Student goal	I can multiply and divide any rational numbers fluently.
Objective	Students compute fluently with rational numbers.
Standard	7.NS.2 — Apply properties of operations to multiply and divide rational numbers.
Pacing	Unit 5, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Convert; compute.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Sign errors in multi-step products.

SUPPORT

- Sign first.

EXTEND

- Rational-number bingo.

STUDENT LESSON 5.8 · WITH ANSWERS

Lesson 5.8: Rational Number Operations Practice

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.8, Part A.
- [ ] Complete the rest of Workbook Lesson 5.8 (Parts B–D).

LET'S EXPLORE

Which is greater: $(-\frac{3}{4})(-0.8)$ or $(-1.2) \div (-\frac{3}{2})$?

WORKED EXAMPLE · COMPARE

Both products/quotients

1 $(-\frac{3}{4})(-0.8) = 0.75 \times 0.8 = 0.6$.

2 $(-1.2) \div (-1.5) = 0.8$.

Answer: the second ($0.8 > 0.6$)

★ KEY IDEA

Convert to one form (all fractions or all decimals) when operations mix.

✎ TRY IT

Answers are shown in color.

1. $(-0.5) \times \frac{4}{5}$

Answer: -0.4

2. $\frac{9}{10} \div (-0.3)$

Answer: -3

3. $(-1\frac{1}{4})(-0.8)$

Answer: 1

Workbook Answer Key · Lesson 5.8

#	Question	Answer
1	$(-0.5) \times {}^{4}/_{5}$	−0.4
2	${}^{9}/_{10} \div (-0.3)$	−3
3	$(-1\,{}^{1}/_{4})(-0.8)$	1
4	0.75×0.8	0.6
5	$1.2 \div 1.5$	0.8
6	$(-0.25)(-{}^{4}/_{5})$	0.2
7	$-{}^{3}/_{8} \div 0.75$	−0.5
8	$(-2.4)({}^{5}/_{6})$	−2
9	$-2\,{}^{1}/_{2} \div (-0.5)$	5
10	$(-1.5)(-1.5)$	2.25
11	${}^{1}/_{3} \div (-{}^{1}/_{9})$	−3
12	Evaluate $(-{}^{2}/_{5})(1.5) \div (-0.3)$.	2

WEEKLY CHECK · ANSWER KEY
After Lesson 5.8

About 20 minutes

Weekly Check 14: Rational Products and Quotients — Answer Key

Answers

#	Question	Answer
1	$(-\frac{3}{8}) \times \frac{4}{9}$	$-\frac{1}{6}$
2	$(-2\frac{1}{2}) \times (-1\frac{1}{5})$	3
3	$(-\frac{4}{5}) \div \frac{2}{15}$	−6
4	$2\frac{1}{3} \div (-1\frac{5}{9})$	$-1\frac{1}{2}$
5	(−1.6) × 0.5	−0.8
6	(−2.7) ÷ (−0.09)	30

LESSON 5.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.9: Order of Operations with Rational Numbers

Student goal	I can use the order of operations with positive and negative numbers.
Objective	Students apply order of operations with rational numbers.
Standard	7.NS.3, 7.EE.3 — Solve problems with the four operations on rational numbers.
Pacing	Unit 5, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples step by step.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- $-4^2 = 16$.

SUPPORT

- Write each step on a new line.

EXTEND

- Create an expression equal to −10 using 4 operations.

STUDENT LESSON 5.9 · WITH ANSWERS

Lesson 5.9: Order of Operations with Rational Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.9, Part A.
- [ ] Complete the rest of Workbook Lesson 5.9 (Parts B–D).

LET'S EXPLORE

Evaluate $-3 + 4 \times (-2)^2 \div 8$.

WORKED EXAMPLE · ORDER

$-3 + 4 \times (-2)^2 \div 8$

1 Exponent: $(-2)^2 = 4$.

2 Multiply/divide left to right: $4 \times 4 = 16$; $16 \div 8 = 2$.

3 Add: $-3 + 2 = -1$.

Answer: -1

WORKED EXAMPLE · PARENTHESES WITH NEGATIVES

$(-6 - 4) \div (-5) + 3$

1 $(-10) \div (-5) = 2$.

2 $2 + 3 = 5$.

Answer: 5

★ KEY IDEA

Same order of operations as always; keep track of every sign.

✎ TRY IT

Answers are shown in color.

1. $-8 + 3 \times (-2)$

Answer: -14

2. $(-4)^2 - 10$

Answer: 6

3. $12 \div (-3) \times (-2)$

Answer: 8

Workbook Answer Key · Lesson 5.9

#	Question	Answer
1	$-8 + 3 \times (-2)$	-14
2	$(-4)^2 - 10$	6
3	$12 \div (-3) \times (-2)$	8
4	$(-2)^2$	4
5	$-3 + 2$	-1
6	$5 - 2 \times (-4)$	13
7	$-18 \div 3 + (-2)$	-8
8	$(-3)^2 - (-3)$	12
9	$-2(5 - 9)$	8
10	$(1 - 7)(-2) \div 4$	3
11	$-4^2 + (-4)^2$	0
12	Evaluate $-0.5(6 - 10) + 1.5^2$.	4.25

LESSON 5.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.10: The Distributive Property with Rational Numbers

Student goal	I can use the distributive property with negative numbers.
Objective	Students apply the distributive property with negatives.
Standard	7.NS.2a, 7.EE.1 — Apply properties of operations as strategies to multiply rational numbers and to expand linear expressions.
Pacing	Unit 5, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- $-3(x - 5) = -3x - 15$.

SUPPORT

- Arrows from the factor to each term.

EXTEND

- Mental math with distribution.

STUDENT LESSON 5.10 · WITH ANSWERS

Lesson 5.10: The Distributive Property with Rational Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.10, Part A.
- [ ] Complete the rest of Workbook Lesson 5.10 (Parts B–D).

LET'S EXPLORE

How could you compute -4×2.75 in your head?

WORKED EXAMPLE · MENTAL MATH

-4×2.75

1 $2.75 = 3 - 0.25$.

2 $-4(3 - 0.25) = -12 + 1 = -11$.

Answer: -11

WORKED EXAMPLE · DISTRIBUTE A NEGATIVE

$-3(x - 5)$

1 $-3 \cdot x + (-3)(-5) = -3x + 15$.

Answer: $-3x + 15$

★ KEY IDEA

Distribute the sign too: $-a(b - c) = -ab + ac$.

✎ TRY IT

Answers are shown in color.

1. $-5(2 + 0.4)$

Answer: -12

2. $-2(y - 7)$

Answer: $-2y + 14$

3. $-\frac{1}{2}(8 - 6)$

Answer: −1

Workbook Answer Key · Lesson 5.10

#	Question	Answer
1	−5(2 + 0.4)	−12
2	−2(y − 7)	−2y + 14
3	$-\frac{1}{2}(8 - 6)$	−1
4	(−4) × 3	−12
5	(−4)(−0.25)	1
6	−6 × 1.5 (think 6 × (1 + 0.5))	−9
7	−8(10 − 0.5)	−76
8	−3(4x + 2)	−12x − 6
9	−(a − 9)	−a + 9
10	$\frac{2}{3}(9 - 12)$	−2
11	−0.5(4n − 10)	−2n + 5
12	Show two ways to evaluate −7(5 − 2).	−7(3) = −21; −35 + 14 = −21

LESSON 5.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.11: Repeated Change: Rate × Time

Student goal	I can model repeated change with multiplication of signed numbers.
Objective	Students model repeated change with signed multiplication.
Standard	7.NS.2a, 7.NS.3 — Interpret products of rational numbers by describing real-world contexts.
Pacing	Unit 5, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Timeline with past (negative) and future (positive).
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Confusing direction.

SUPPORT

- Timeline diagram.

EXTEND

- Make a ‘time machine’ story for (−5)(−2).

STUDENT LESSON 5.11 · WITH ANSWERS

Lesson 5.11: Repeated Change: Rate × Time

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.11, Part A.
- [ ] Complete the rest of Workbook Lesson 5.11 (Parts B–D).

LET'S EXPLORE

A balloon descends 4.5 meters per minute. Where was it 3 minutes ago, compared to now? Where will it be 3 minutes from now?

Use (rate) × (time). Descending is a negative rate: −4.5 m/min. Future time is positive; past time is negative.

WORKED EXAMPLE · FUTURE

(−4.5)(3)

1 Different signs → negative.

Answer: −13.5 m (13.5 m lower)

WORKED EXAMPLE · PAST

(−4.5)(−3)

1 Same signs → positive.

Answer: +13.5 m (it was 13.5 m higher)

★ KEY IDEA

Negative × negative = positive makes sense: going back in time while descending means you were higher.

✎ TRY IT

Answers are shown in color.

1. Rate −2°F/hour; 5 hours later

Answer: −10°F

2. Rate −2°F/hour; 5 hours ago

Answer: +10°F

3. Rate +3 cm/day; 4 days ago

Answer: −12 cm

Workbook Answer Key · Lesson 5.11

#	Question	Answer
1	Rate −2°F/hour; 5 hours later	−10°F
2	Rate −2°F/hour; 5 hours ago	+10°F
3	Rate +3 cm/day; 4 days ago	−12 cm
4	(−4.5) × 3	−13.5
5	(−4.5) × (−3)	13.5
6	A tank drains 1.5 gallons per minute. Change after 8 minutes?	−12 gallons
7	Same tank: change compared to 8 minutes ago?	It had 12 gallons more
8	Temperature rises 1.2° per hour. Compared to now, what was it 6 hours ago?	7.2° lower (−7.2)
9	A diver ascends 0.75 m/s. Change in 20 s?	+15 m
10	Explain (−3)(−4) = 12 using a temperature that drops 3° per hour.	4 hours ago it was 12° warmer.

LESSON 5.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.12: Choosing the Operation

Student goal	I can decide which operation fits a situation with rational numbers.
Objective	Students choose operations for rational-number situations.
Standard	7.NS.3 — Solve real-world problems with the four operations on rational numbers.
Pacing	Unit 5, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Sort situation cards by operation.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Key-word guessing.

SUPPORT

- Act out or draw.

EXTEND

- Write a 4-operation story.

STUDENT LESSON 5.12 · WITH ANSWERS

Lesson 5.12: Choosing the Operation

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.12, Part A.
- [ ] Complete the rest of Workbook Lesson 5.12 (Parts B–D).

LET'S EXPLORE

A hiker's elevation changed by −1,240 ft over 4 hours. Is the average change per hour found by multiplying or dividing?

Situation	Operation
combine changes	add
difference / change between	subtract
repeated equal change	multiply
equal share / average rate	divide

WORKED EXAMPLE · AVERAGE CHANGE

−1,240 ft over 4 hours

1 Equal share per hour → divide: −1,240 ÷ 4.

Answer: −310 ft per hour

★ KEY IDEA

Think about what is happening, not just key words.

✎ TRY IT

Answers are shown in color.

1. Lost 3.5 lb a week for 6 weeks: total change?

Answer: −21 lb (multiply)

2. From −8°F to 5°F: change?

Answer: 13°F (subtract)

3. −$84 over 7 days: average daily change?

Answer: −$12 (divide)

Workbook Answer Key · Lesson 5.12

#	Question	Answer
1	Lost 3.5 lb a week for 6 weeks: total change?	−21 lb (multiply)
2	From −8°F to 5°F: change?	13°F (subtract)
3	−$84 over 7 days: average daily change?	−$12 (divide)
4	−1,240 ÷ 4	−310
5	(−3.5) × 6	−21
6	A balance of −$45 gets a $60 deposit.	add; $15
7	A plane descends 300 ft/min for 12 min.	multiply; −3,600 ft
8	Temperatures −6° and 9°: how far apart?	subtract; 15°
9	A stock drops $36.60 over 6 days. Average?	divide; −$6.10/day
10	Write a situation for each operation using negative numbers.	Answers vary.

WEEKLY CHECK · ANSWER KEY
After Lesson 5.12
About 20 minutes

Weekly Check 15: Operations and Models — Answer Key

Answers

#	Question	Answer
1	$-4 + 6 \times (-2)$	-16
2	$(-5)^2 - 3(-4)$	37
3	$-3(2x - 7)$	$-6x + 21$
4	Rate −1.5°/hr; 4 hours ago?	6° warmer (+6)
5	A $45 loss over 5 days: average daily change	−$9
6	From 30 to −10 in 8 hours: rate	−5 per hour

LESSON 5.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.13: Average Change

Student goal	I can find the mean of positive and negative numbers.
Objective	Students compute means of signed data.
Standard	7.NS.3 — Solve real-world problems with rational numbers.
Pacing	Unit 5, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Compute mean.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Ignoring signs when summing.

SUPPORT

- Group positives and negatives.

EXTEND

- Mean of a week's real temperature changes.

STUDENT LESSON 5.13 · WITH ANSWERS

Lesson 5.13: Average Change

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.13, Part A.
- [ ] Complete the rest of Workbook Lesson 5.13 (Parts B–E).

LET'S EXPLORE

Daily temperature changes: −4°, +2°, −7°, +1°, −2°. What was the average daily change?

WORKED EXAMPLE · MEAN

−4, 2, −7, 1, −2

1 Sum: −4 + 2 − 7 + 1 − 2 = −10.

2 Divide by 5: −10 ÷ 5 = −2.

Answer: −2° per day

★ KEY IDEA

Mean = sum ÷ count. A negative mean means a typical decrease.

✎ TRY IT

Answers are shown in color.

1. Mean of −3, 5, −8, 2

Answer: −1

2. Mean of −1.5, −2.5, 1

Answer: −1

3. Golf scores −2, 0, −3, +1: mean

Answer: −1

Workbook Answer Key · Lesson 5.13

#	Question	Answer
1	Mean of −3, 5, −8, 2	−1
2	Mean of −1.5, −2.5, 1	−1
3	Golf scores −2, 0, −3, +1: mean	−1
4	−4 + 2 − 7 + 1 − 2	−10
5	−10 ÷ 5	−2
6	−6, 4, −2, 0	−1
7	−10, −15, −5	−10
8	3, −9, 6, −12, 2	−2
9	−1, −1, −1, 7	1
10	A stock changed −$1.20, +$0.60, −$0.90, +$0.30 over 4 days. Mean daily change?	−$0.30
11	Find 4 numbers with a mean of −3.	e.g. −3, −3, −3, −3 or −10, 2, −5, 1

LESSON 5.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.14: Rate Problems with Negatives

Student goal	I can solve rate problems that involve decreases.
Objective	Students solve rate problems with negative rates.
Standard	7.NS.3, 7.RP.1 — Compute unit rates; solve real-world problems with rational numbers.
Pacing	Unit 5, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Start − end instead of end − start.

SUPPORT

- Label start/end.

EXTEND

- Track a real cold front.

STUDENT LESSON 5.14 · WITH ANSWERS

Lesson 5.14: Rate Problems with Negatives

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.14, Part A.
- [ ] Complete the rest of Workbook Lesson 5.14 (Parts B–D).

LET'S EXPLORE

A cold front drops the temperature from 18°F to −6°F in 8 hours. What is the rate of change per hour?

WORKED EXAMPLE · RATE OF CHANGE

18 → −6 in 8 hours

1 Change: −6 − 18 = −24.

2 Rate: −24 ÷ 8 = −3.

Answer: −3°F per hour

WORKED EXAMPLE · PREDICT

At −3°F per hour, when will it reach −15°F from −6°F?

1 Change needed: −15 − (−6) = −9.

2 −9 ÷ (−3) = 3 hours.

Answer: 3 hours later

★ KEY IDEA

Rate of change = change ÷ time. Change = end − start.

✎ TRY IT

Answers are shown in color.

1. A pool level drops from 60 in. to 48 in. in 4 days. Rate?

Answer: −3 in. per day

2. A balance goes from $50 to −$25 in 5 weeks. Rate?

Answer: −$15 per week

3. Elevation 2,400 ft to 600 ft in 3 hours. Rate?

Answer: −600 ft per hour

Workbook Answer Key · Lesson 5.14

#	Question	Answer
1	A pool level drops from 60 in. to 48 in. in 4 days. Rate?	−3 in. per day
2	A balance goes from $50 to −$25 in 5 weeks. Rate?	−$15 per week
3	Elevation 2,400 ft to 600 ft in 3 hours. Rate?	−600 ft per hour
4	−6 − 18	−24
5	−24 ÷ 8	−3
6	A submarine goes from −40 m to −220 m in 12 minutes. Rate?	−15 m per minute
7	At that rate, how long from −220 m to −400 m?	12 minutes
8	A plant's height changes from 14.5 cm to 11.3 cm in 4 days (drought). Rate?	−0.8 cm per day
9	−$2.40 per day. After how many days is the change −$16.80?	7 days
10	Temperature drops 2.5° per hour from 10°. What is it after 6 hours?	−5°

LESSON 5.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.15: Multi-Step Rational Number Problems

Student goal	I can solve multi-step problems using all four operations with rational numbers.
Objective	Students solve multi-step rational problems.
Standard	7.EE.3, 7.NS.3 — Solve multi-step real-life problems with rational numbers in any form; assess reasonableness.
Pacing	Unit 5, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work the example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Skipping steps.

SUPPORT

- Table of steps.

EXTEND

- Plan a budget with gains and losses.

STUDENT LESSON 5.15 · WITH ANSWERS

Lesson 5.15: Multi-Step Rational Number Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.15, Part A.
- [ ] Complete the rest of Workbook Lesson 5.15 (Parts B–D).

LET'S EXPLORE

A business lost $1,250 per month for 3 months, then earned $2,400 per month for 2 months. What is the average monthly result?

WORKED EXAMPLE · ALL OPERATIONS

3 months at −$1,250, 2 months at +$2,400

1 Total: 3(−1,250) + 2(2,400) = −3,750 + 4,800 = 1,050.

2 Average: 1,050 ÷ 5 = 210.

Answer: +$210 per month

★ KEY IDEA

Organize the steps; estimate to check the sign and size.

✎ TRY IT

Answers are shown in color.

1. 4(−2.5) + 3(1.5)

Answer: −5.5

2. $(-\frac{3}{4} + \frac{1}{2}) \times 8$

Answer: −2

3. (−12 − 6) ÷ (−3) + 2

Answer: 8

Workbook Answer Key · Lesson 5.15

#	Question	Answer
1	4(−2.5) + 3(1.5)	−5.5
2	$(-\frac{3}{4} + \frac{1}{2}) \times 8$	−2
3	(−12 − 6) ÷ (−3) + 2	8
4	3 × (−1,250)	−3,750
5	2 × 2,400	4,800
6	A diver descends 2.4 m every 10 seconds for 50 seconds, then rises 3 m. Final change?	−9 m
7	Ellie's savings change by −$12.50 a week for 4 weeks, then +$20 a week for 3 weeks. Net change?	+$10
8	The temperature was 4°C. It fell 1.5°C per hour for 6 hours, then rose 2°C. Final temperature?	−3°C
9	Write a problem whose answer is −8 using all four operations.	Answers vary.

LESSON 5.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 5: Multiplying and Dividing Rational Numbers

Lesson 5.16: Rational Number Operations Problem Solving

Student goal	I can use all four operations with rational numbers to solve problems.
Objective	Students solve mixed rational-number problems.
Standard	7.NS.1–3 — 7.NS review.
Pacing	Unit 5, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 5.16, Unit 5 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work backward example.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 5.16 and the Unit 5 Review.
Exit Ticket	3	Check. Tomorrow: Unit 5 Test.

COMMON MISCONCEPTIONS

- Sign errors.

SUPPORT

- Sign first.

EXTEND

- Design your own challenge.

STUDENT LESSON 5.16 · WITH ANSWERS

Lesson 5.16: Rational Number Operations Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 5.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 5.16, Part A.
- [ ] Complete the rest of Workbook Lesson 5.16 (Parts B–E).

LET'S EXPLORE

Design a 'temperature challenge': start at 5°F, apply four changes so that the final temperature is −7°F and the average change is −3°F.

WORKED EXAMPLE · WORK BACKWARD

Average −3 over 4 changes

1 Total change = 4 × (−3) = −12.

2 5 + (−12) = −7 ✓. Any 4 changes summing to −12 work, e.g., −5, +2, −6, −3.

Answer: −5, +2, −6, −3

★ KEY IDEA

Use multiplication and division to plan totals and averages.

✎ TRY IT

Answers are shown in color.

1. (−2.5)(−4) ÷ (−5)

Answer: −2

2. $-\frac{2}{3} \div \frac{4}{9}$

Answer: $-1\frac{1}{2}$

3. Mean of −4.2, 1.8, −3

Answer: −1.8

Workbook Answer Key · Lesson 5.16

#	Question	Answer
1	$(-2.5)(-4) \div (-5)$	-2
2	$-\frac{2}{3} \div \frac{4}{9}$	$-1\frac{1}{2}$
3	Mean of −4.2, 1.8, −3	−1.8
4	$4 \times (-3)$	−12
5	$5 + (-12)$	−7
6	$(-8) \times 7$	−56
7	$-96 \div (-12)$	8
8	$(-\frac{5}{6}) \times \frac{9}{10}$	$-\frac{3}{4}$
9	$(-1\frac{3}{4}) \div 3\frac{1}{2}$	$-\frac{1}{2}$
10	$(-0.4) \times (-2.75)$	1.1
11	$-3 + (-2)^3$	−11
12	A stock's price changes −$0.85 per day for 6 days, then +$1.20 per day for 4 days. Net change? Average per day?	−$0.30 net; −$0.03 per day
13	Explain how you decide the sign of a product of five numbers.	Count negatives; odd → negative.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 35 minutes

Unit 5 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	$(-12) \times (-5)$	60
2	$84 \div (-7)$	-12
3	$(-\frac{2}{3}) \times (-\frac{9}{14})$	$\frac{3}{7}$
4	$(-\frac{5}{8}) \div \frac{15}{16}$	$-\frac{2}{3}$
5	$3.5 \times (-0.6)$	-2.1
6	$(-4.5) \div 1.5$	-3
7	$(-1)^6$	1
8	$10 - 3(-2)^2$	-2
9	Mean of $-8, 3, -5, -2$	-3
10	Temp falls from 12° to −9° in 7 hours. Rate?	−3° per hour
11	$-0.5(8x - 6)$	$-4x + 3$
12	A pool drains 2.5 in./hour. Change after 3.5 hours?	−8.75 in.

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 5 Test — Answer Key

Answers

#	Question	Answer
1	$(-9) \times 6$	-54
2	$(-11) \times (-4)$	44
3	$(-1)(-2)(-3)(-4)$	24
4	$-108 \div 9$	-12
5	$-75 \div (-15)$	5
6	Simplify $\frac{-16}{24}$	$-\frac{2}{3}$
7	$(-\frac{4}{7}) \times 2\frac{5}{8}$	$-1\frac{1}{2}$
8	$(-\frac{3}{5}) \div (-\frac{9}{20})$	$1\frac{1}{3}$
9	$(-2.4) \times 1.25$	-3
10	$6.4 \div (-0.8)$	-8
11	$-8 + 12 \div (-4)$	-11
12	$-4(3 - 2.5)$	-2
13	The temperature fell from 15°F to −9°F in 6 hours. Find the rate of change.	−4°F per hour
14	Mean of −3.5, 2, −4.5, 1, −5	−2
15	A submarine changes depth by −8.5 m per minute. Where was it, relative to now, 4 minutes ago?	34 m higher (+34 m)
16	Use a pattern or the distributive property to explain why $(-2)(-5) = 10$.	Pattern: $2(-5) = -10$, $1(-5) = -5$, $0(-5) = 0$, $-1(-5) = 5$, $-2(-5) = 10$; or $-2(5 + (-5)) = 0 \Rightarrow -10 + (-2)(-5) = 0$.

Scoring

Score	What it means / next step
18–21 (85%+)	Mastered.
15–17	Proficient; correct missed items.
Below 15	Reteach and retest.

Expressions, Equations, and Inequalities

Writing and simplifying expressions, solving two-step equations, and solving and graphing inequalities.

Algebra lets you describe real situations with symbols and solve for unknowns. You will write and simplify expressions, solve equations like $3x + 5 = 20$ and $2(x - 4) = 10$, and solve and graph inequalities.

Big ideas

- Write and evaluate expressions with rational numbers.
- Combine like terms, expand, and factor.
- Solve equations of the forms $px + q = r$ and $p(x + q) = r$.
- Write, solve, and graph inequalities.
- Interpret solutions in real situations.

Standards

Code	Standard
7.EE.1	Apply properties of operations to add, subtract, factor, and expand linear expressions with rational coefficients.
7.EE.2	Understand that rewriting an expression can shed light on a problem.
7.EE.3	Solve multi-step real-life problems; assess reasonableness.
7.EE.4	Use variables to construct and solve equations and inequalities to solve problems.

What students should already know

- Rational-number operations (Units 3–5).
- One-step equations (Grade 6).

Materials for the unit

- Algebra tiles (optional)
- Balance-scale picture

- Number lines

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	6.1	Writing Expressions
Week 1	Lesson 2 of 4	6.2	Evaluating Expressions
Week 1	Lesson 3 of 4	6.3	Combining Like Terms
Week 1	Lesson 4 of 4	6.4	Expressions in Context
Week 1	End of week	Check	Weekly Check 16: Expressions
Week 2	Lesson 1 of 4	6.5	The Distributive Property: Expanding
Week 2	Lesson 2 of 4	6.6	Factoring Expressions
Week 2	Lesson 3 of 4	6.7	Simplifying with Rational Coefficients
Week 2	Lesson 4 of 4	6.8	Equivalent Expressions
Week 2	End of week	Check	Weekly Check 17: Equivalent Expressions
Week 3	Lesson 1 of 4	6.9	One-Step Equations
Week 3	Lesson 2 of 4	6.10	Two-Step Equations
Week 3	Lesson 3 of 4	6.11	Equations with Parentheses
Week 3	Lesson 4 of 4	6.12	Equation Word Problems
Week 3	End of week	Check	Weekly Check 18: Equations
Week 4	Lesson 1 of 4	6.13	Writing and Graphing Inequalities
Week 4	Lesson 2 of 4	6.14	Solving One-Step Inequalities
Week 4	Lesson 3 of 4	6.15	Two-Step Inequalities
Week 4	Lesson 4 of 4	6.16	Inequality Word Problems
Week 4	Before the test	Review	Unit 6 Review
Week 4	End of unit	Test	Unit 6 Test

TEACHING THIS UNIT AT HOME

When a cost has a fixed fee plus a rate (gym, taxi, phone), ask your student to write and solve an equation for it.

UNIT VOCABULARY

variable	A letter for a number.
expression	No equals sign.
equation	Two expressions are equal.
inequality	Comparison with $<$, $>$, $\leq$, $\geq$.
like terms	Same variable part.

LESSON 6.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.1: Writing Expressions

Student goal	I can write algebraic expressions from words and real situations.
Objective	Students write algebraic expressions.
Standard	7.EE.2, 7.EE.4 — Rewrite expressions in different forms; write expressions for real situations.
Pacing	Unit 6, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Arithmetic warm-up.
Launch	5	Read Let's Explore.
Teach	15	Translate the table phrases together. Say: "Order matters for subtraction and division."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Writing 3 − x for "3 less than x."

SUPPORT

- Test with a number: 3 less than 10 is 7.

EXTEND

- Write expressions for family costs.

STUDENT LESSON 6.1 · WITH ANSWERS

Lesson 6.1: Writing Expressions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.1, Part A.
- [ ] Complete the rest of Workbook Lesson 6.1 (Parts B–E).

LET'S EXPLORE

A movie ticket costs $9 and popcorn costs $6. Write an expression for the cost of t tickets and one popcorn.

An expression uses a variable for an unknown or changing number. The cost is 9 for each ticket, so 9t, plus 6: 9t + 6.

Words	Expression
5 more than a number n	n + 5
3 less than twice a number	2n − 3
the product of −4 and x	−4x
a number divided by 6	n ÷ 6
half of the sum of a number and 8	$\frac{1}{2}(n + 8)$

! WATCH OUT

“3 less than x” is x − 3, not 3 − x. The order matters for subtraction and division.

★ KEY IDEA

Read the words carefully. Use a variable for the unknown and write operations in the correct order.

✎ TRY IT

Answers are shown in color.

1. 7 less than a number m

Answer: m − 7

2. the quotient of a number and −3

Answer: n ÷ (−3)

3. A $25 bill split among k friends

Answer: 25 ÷ k

Workbook Answer Key · Lesson 6.1

#	Question	Answer
1	7 less than a number m	m − 7
2	the quotient of a number and −3	n ÷ (−3)
3	A $25 bill split among k friends	25 ÷ k
4	9 × 4 + 6	42
5	Name the coefficient in −5y	−5
6	a number increased by 12	n + 12
7	8 times a number, minus 4	8n − 4
8	the difference of a number and −2	n − (−2) or n + 2
9	one-third of a number	n/3
10	6 less than 5 times a number	5n − 6
11	twice the sum of a number and 3	2(n + 3)
12	A phone plan costs $20 plus $0.10 per text t.	20 + 0.10t
13	Lillie has d dollars and spends $7.50.	d − 7.50
14	Write a real situation for the expression 12h + 15.	Answers vary, e.g. $12 per hour plus a $15 fee

LESSON 6.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.2: Evaluating Expressions

Student goal	I can evaluate expressions for given values, including negative and fractional values.
Objective	Students evaluate expressions with rational values.
Standard	7.EE.3 — Solve problems using expressions with rational numbers.
Pacing	Unit 6, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Substitute with parentheses.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- x^2 with $x = -3$ written as -9.

SUPPORT

- Parentheses first.

EXTEND

- Convert this week's temperatures.

STUDENT LESSON 6.2 · WITH ANSWERS

Lesson 6.2: Evaluating Expressions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.2, Part A.
- [ ] Complete the rest of Workbook Lesson 6.2 (Parts B–E).

LET'S EXPLORE

The expression 1.8C + 32 converts Celsius to Fahrenheit. What is −10°C in Fahrenheit?

WORKED EXAMPLE · SUBSTITUTE

1.8C + 32 when C = −10

1 1.8(−10) + 32 = −18 + 32.

Answer: 14°F

WORKED EXAMPLE · TWO VARIABLES

3a − 2b when a = −4, b = 1.5

1 3(−4) − 2(1.5) = −12 − 3.

Answer: −15

★ KEY IDEA

Substitute with parentheses, then follow the order of operations.

✎ TRY IT

Answers are shown in color.

1. 4x − 7 when x = −2

Answer: −15

2. $\frac{1}{2}n + 3$ when n = −8

Answer: −1

3. $a^2 - b$ when a = −3, b = 4

Answer: 5

Workbook Answer Key · Lesson 6.2

#	Question	Answer
1	$4x - 7$ when $x = -2$	−15
2	$\frac{1}{2}n + 3$ when $n = -8$	−1
3	$a^2 - b$ when $a = -3$, $b = 4$	5
4	$1.8 \times (-10)$	−18
5	$-18 + 32$	14
6	$2x + y$	−2
7	$y - 5x$	19
8	$10z - x$	8
9	xy	−12
10	$x^2 + y^2$	25
11	$(y - x) \div z$	14
12	A taxi costs $2.5m + 3$ dollars for m miles. Cost for 12 miles?	$33
13	Convert 25°C to Fahrenheit with $1.8C + 32$.	77°F
14	Evaluate $-2(x - 6)$ when $x = -1$.	14

LESSON 6.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.3: Combining Like Terms

Student goal	I can simplify expressions by combining like terms, including rational coefficients.
Objective	Students combine like terms.
Standard	7.EE.1 — Apply properties to add, subtract, factor, and expand linear expressions with rational coefficients.
Pacing	Unit 6, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Algebra tiles (optional), workbook Lesson 6.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Color like terms the same color.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Combining x and constants.

SUPPORT

- Highlight like terms.

EXTEND

- Write 3 expressions that simplify to 2x − 1.

STUDENT LESSON 6.3 · WITH ANSWERS

Lesson 6.3: Combining Like Terms

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.3, Part A.
- [ ] Complete the rest of Workbook Lesson 6.3 (Parts B–D).

LET'S EXPLORE

Simplify $4x + 7 - 6x + 2$.

Like terms have the same variable part. Add their coefficients. Constants combine with constants.

WORKED EXAMPLE · COMBINE

$4x + 7 - 6x + 2$

1 x-terms: $4x - 6x = -2x$.

2 Constants: $7 + 2 = 9$.

Answer: $-2x + 9$

WORKED EXAMPLE · RATIONAL COEFFICIENTS

$\frac{1}{2}a - 0.75a + 3$

1 $\frac{1}{2} - 0.75 = 0.5 - 0.75 = -0.25$.

Answer: $-0.25a + 3$

★ KEY IDEA

Only like terms combine. $3x + 2$ is already simplified.

✎ TRY IT

Answers are shown in color.

1. $5y - 8y + 3$

Answer: $-3y + 3$

2. $-2m + 6 + 7m - 10$

Answer: $5m - 4$

3. $1.5x - 2 + 0.5x$

Answer: $2x - 2$

Workbook Answer Key · Lesson 6.3

#	Question	Answer
1	$5y - 8y + 3$	$-3y + 3$
2	$-2m + 6 + 7m - 10$	$5m - 4$
3	$1.5x - 2 + 0.5x$	$2x - 2$
4	$4 - 6$	-2
5	$0.5 - 0.75$	-0.25
6	$9x + 4x$	$13x$
7	$3a - 8 + 5a$	$8a - 8$
8	$-6n + 2 - n + 9$	$-7n + 11$
9	$2.4p - 1.4p + 6$	$p + 6$
10	$\frac{2}{3}k + \frac{1}{3}k - 5$	$k - 5$
11	$4x + 3y - x - 5y$	$3x - 2y$
12	Is $5x + 3$ equal to $8x$? Explain.	No. They are not like terms. Test $x = 2$: $5(2) + 3 = 13$, but $8(2) = 16$.

LESSON 6.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 6: Expressions, Equations, and Inequalities

Lesson 6.4: Expressions in Context

Student goal	I can rewrite an expression to show what it means in a situation.
Objective	Students interpret equivalent expressions in context.
Standard	7.EE.2 — Understand that rewriting an expression in different forms can shed light on the problem.
Pacing	Unit 6, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Writing p + 0.05 instead of p + 0.05p.

SUPPORT

- Plug in p = 100.

EXTEND

- Rewrite a real sale ad as an expression.

STUDENT LESSON 6.4 · WITH ANSWERS

Lesson 6.4: Expressions in Context

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.4, Part A.
- [ ] Complete the rest of Workbook Lesson 6.4 (Parts B–D).

LET'S EXPLORE

A store raises a price p by 5%. Two students write p + 0.05p and 1.05p. Are both right?

Both are right! p + 0.05p shows "the price plus 5% of the price." 1.05p shows "105% of the price." Rewriting an expression can reveal a new way of seeing the problem.

WORKED EXAMPLE · DISCOUNT

A 20% discount on price p

1 p − 0.2p = 0.8p.

2 0.8p means you pay 80% of the price.

Answer: 0.8p

WORKED EXAMPLE · PERIMETER

A rectangle has length L and width 3. Perimeter?

1 L + 3 + L + 3 = 2L + 6 = 2(L + 3).

Answer: 2(L + 3): twice the sum of length and width

★ KEY IDEA

Equivalent expressions can show different meanings of the same quantity.

✎ TRY IT

Answers are shown in color.

1. A 15% tip on bill b, two ways

Answer: b + 0.15b; 1.15b

2. A 30% markdown on price m

Answer: 0.7m

3. Equal sides: s + s + s

Answer: 3s

Workbook Answer Key · Lesson 6.4

#	Question	Answer
1	A 15% tip on bill b, two ways	b + 0.15b; 1.15b
2	A 30% markdown on price m	0.7m
3	Equal sides: s + s + s	3s
4	1 + 0.05	1.05
5	1 − 0.2	0.8
6	p − 0.35p	0.65p: pay 65%
7	c + 0.07c	1.07c: price with 7% tax
8	2w + 2(w + 4)	4w + 8: perimeter of a rectangle w by w + 4
9	x + x + 0.5x	2.5x
10	Write two expressions for the price p after a 10% increase. Explain each.	p + 0.1p (price plus 10%); 1.1p (110% of price)

WEEKLY CHECK · ANSWER KEY
After Lesson 6.4

About 15 minutes

Weekly Check 16: Expressions — Answer Key

Answers

#	Question	Answer
1	Write: 4 less than 3 times a number	$3n - 4$
2	Evaluate $5x - 2y$ when $x = -2$, $y = 3$	-16
3	Simplify $7a - 3 - 9a + 8$	$-2a + 5$
4	Simplify $0.6k + 1.4k - 2$	$2k - 2$
5	Rewrite $p - 0.25p$	$0.75p$
6	Evaluate $x^2 - x$ when $x = -4$	20

LESSON 6.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.5: The Distributive Property: Expanding

Student goal	I can expand expressions using the distributive property.
Objective	Students expand expressions.
Standard	7.EE.1 — Apply properties to expand linear expressions with rational coefficients.
Pacing	Unit 6, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Draw arrows from the factor to each term.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying only the first term.

SUPPORT

- Arrows.

EXTEND

- Use area models to show 3(x + 4).

STUDENT LESSON 6.5 · WITH ANSWERS

Lesson 6.5: The Distributive Property: Expanding

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.5, Part A.
- [ ] Complete the rest of Workbook Lesson 6.5 (Parts B–E).

LET'S EXPLORE

Expand $-2(3x - 5)$.

WORKED EXAMPLE · EXPAND

$-2(3x - 5)$

1 $-2 \cdot 3x = -6x$.

2 $-2 \cdot (-5) = 10$.

Answer: $-6x + 10$

WORKED EXAMPLE · FRACTIONS

$\frac{3}{4}(8y + 12)$

1 $\frac{3}{4} \cdot 8y = 6y$; $\frac{3}{4} \cdot 12 = 9$.

Answer: $6y + 9$

WORKED EXAMPLE · EXPAND AND COMBINE

$4(x + 2) - 3(x - 1)$

1 $4x + 8 - 3x + 3$.

Answer: $x + 11$

★ KEY IDEA

Multiply the outside factor (with its sign) by every term inside.

✎ TRY IT

Answers are shown in color.

1. $5(2a - 3)$

Answer: $10a - 15$

2. $-(4m + 1)$

Answer: $-4m - 1$

3. $-\frac{1}{2}(6n - 10)$

Answer: $-3n + 5$

Workbook Answer Key · Lesson 6.5

#	Question	Answer
1	$5(2a - 3)$	$10a - 15$
2	$-(4m + 1)$	$-4m - 1$
3	$-\frac{1}{2}(6n - 10)$	$-3n + 5$
4	$-2 \cdot (-5)$	10
5	$\frac{3}{4} \cdot 12$	9
6	$3(x + 7)$	$3x + 21$
7	$-4(2y - 1)$	$-8y + 4$
8	$0.5(6k + 8)$	$3k + 4$
9	$-(-3 + p)$	$3 - p$
10	$\frac{2}{3}(9x - 3)$	$6x - 2$
11	$-1.5(4 - 2w)$	$-6 + 3w$
12	$2(x - 4) + 3x$	$5x - 8$
13	$6 - 2(n + 5)$	$-2n - 4$
14	$3(a + 1) - (a - 4)$	$2a + 7$
15	$-5(2 - y) + 4y$	$9y - 10$
16	Expand and simplify: $-3(2x + 4) + 7x$	$x - 12$

LESSON 6.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.6: Factoring Expressions

Student goal	I can factor an expression by finding a common factor.
Objective	Students factor linear expressions.
Standard	7.EE.1 — Apply properties to factor linear expressions with rational coefficients.
Pacing	Unit 6, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Area model in reverse.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Not factoring from every term.

SUPPORT

- Check by expanding.

EXTEND

- Factor with fractional GCFs.

STUDENT LESSON 6.6 · WITH ANSWERS

Lesson 6.6: Factoring Expressions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.6, Part A.
- [ ] Complete the rest of Workbook Lesson 6.6 (Parts B–D).

LET'S EXPLORE

A rectangle has area 6x + 9. One side is 3. What is the other side?

WORKED EXAMPLE · FACTOR OUT THE GCF

6x + 9

1 GCF of 6 and 9 is 3.

2 6x = 3 · 2x; 9 = 3 · 3.

Answer: 3(2x + 3)

WORKED EXAMPLE · NEGATIVE FACTOR

−4y + 12

1 Factor −4: −4(y − 3). Check: −4y + 12 ✓.

Answer: −4(y − 3)

★ KEY IDEA

Factoring undoes the distributive property. Always check by expanding.

✎ TRY IT

Answers are shown in color.

1. 10a + 15

Answer: 5(2a + 3)

2. 8m − 12

Answer: 4(2m − 3)

3. −7n − 21

Answer: −7(n + 3)

Workbook Answer Key · Lesson 6.6

#	Question	Answer
1	10a + 15	5(2a + 3)
2	8m − 12	4(2m − 3)
3	−7n − 21	−7(n + 3)
4	GCF of 6 and 9	3
5	Expand 3(2x + 3)	6x + 9
6	4x + 20	4(x + 5)
7	18y − 24	6(3y − 4)
8	9k + 9	9(k + 1)
9	−5p + 30	−5(p − 6)
10	12 − 16w	4(3 − 4w)
11	2.5x + 5	2.5(x + 2)
12	A rectangle's area is 8x + 32 and one side is 8. Find the other side.	x + 4

LESSON 6.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.7: Simplifying with Rational Coefficients

Student goal	I can simplify expressions with fraction and decimal coefficients.
Objective	Students simplify expressions with rational coefficients.
Standard	7.EE.1 — Add, subtract, factor, and expand linear expressions with rational coefficients.
Pacing	Unit 6, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Distribute then combine.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Distributing a negative incorrectly.

SUPPORT

- Rewrite subtraction of a group as + (−).

EXTEND

- Write a challenge expression.

STUDENT LESSON 6.7 · WITH ANSWERS

Lesson 6.7: Simplifying with Rational Coefficients

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.7, Part A.
- [ ] Complete the rest of Workbook Lesson 6.7 (Parts B–D).

LET'S EXPLORE

Simplify $\frac{1}{2}(4x - 6) - \frac{1}{3}(3x + 9)$.

WORKED EXAMPLE · EXPAND THEN COMBINE

$\frac{1}{2}(4x - 6) - \frac{1}{3}(3x + 9)$

1 $2x - 3 - x - 3$.

Answer: $x - 6$

WORKED EXAMPLE · DECIMALS

$0.4(5n + 10) - 1.5n$

1 $2n + 4 - 1.5n$.

Answer: $0.5n + 4$

★ KEY IDEA

Distribute carefully, then combine like terms.

✎ TRY IT

Answers are shown in color.

1. $\frac{1}{4}(8a + 4) + a$

Answer: $3a + 1$

2. $0.2(10y - 5) - y$

Answer: $y - 1$

3. $\frac{2}{5}(10 - 5m) + 3m$

Answer: $4 + m$

Workbook Answer Key · Lesson 6.7

#	Question	Answer
1	$\frac{1}{4}(8a + 4) + a$	$3a + 1$
2	$0.2(10y - 5) - y$	$y - 1$
3	$\frac{2}{5}(10 - 5m) + 3m$	$4 + m$
4	$\frac{1}{2} \cdot 4x$	$2x$
5	$0.4 \cdot 5n$	$2n$
6	$\frac{1}{3}(6x + 3) - x$	$x + 1$
7	$0.5(4k - 8) + 3k$	$5k - 4$
8	$-\frac{3}{4}(4p - 8) + p$	$-2p + 6$
9	$1.2(5 + 10w) - 2w$	$6 + 10w$
10	$\frac{1}{2}x + \frac{1}{4}x - \frac{3}{4}x$	0
11	$\frac{2}{3}(3n - 6) - \frac{1}{2}(2n + 4)$	$n - 6$
12	Simplify $\frac{3}{5}(10x - 5) - 0.5(2x + 4)$	$5x - 5$

LESSON 6.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.8: Equivalent Expressions

Student goal	I can tell whether two expressions are equivalent and justify my answer.
Objective	Students determine and justify equivalence.
Standard	7.EE.1, 7.EE.2 — Understand equivalent expressions.
Pacing	Unit 6, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Simplify vs counterexample.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Testing only one value to prove equivalence.

SUPPORT

- Simplify fully.

EXTEND

- Find a value where $3x + 2$ and $3(x + 2)$ agree? (None.)

STUDENT LESSON 6.8 · WITH ANSWERS

Lesson 6.8: Equivalent Expressions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.8, Part A.
- [ ] Complete the rest of Workbook Lesson 6.8 (Parts B–D).

LET'S EXPLORE

Are 2(x + 3) + x and 3x + 6 equivalent? What about 3(x + 2) and 3x + 2?

WORKED EXAMPLE · SIMPLIFY BOTH

2(x + 3) + x vs 3x + 6

1 2x + 6 + x = 3x + 6.

2 Same.

Answer: Equivalent

WORKED EXAMPLE · TEST A VALUE

3(x + 2) vs 3x + 2

1 x = 1: 9 vs 5.

2 Different for x = 1.

Answer: Not equivalent

★ KEY IDEA

Simplify both to prove equivalence. One counterexample proves they are not equivalent.

✎ TRY IT

Answers are shown in color.

1. 4(x − 1) and 4x − 4

Answer: equivalent

2. 5 − 2x and 2x − 5

Answer: not equivalent

3. x + x + x + 2 and 3(x + 1) − 1

Answer: equivalent

Workbook Answer Key · Lesson 6.8

#	Question	Answer
1	4(x − 1) and 4x − 4	equivalent
2	5 − 2x and 2x − 5	not equivalent
3	x + x + x + 2 and 3(x + 1) − 1	equivalent
4	Expand 2(x + 3)	2x + 6
5	3(1) + 2	5
6	6x − 9 and 3(2x − 3)	Yes (expand)
7	−(x − 4) and −x − 4	No (x = 0: 4 vs −4)
8	0.5(4n + 2) and 2n + 1	Yes
9	3(a + 2) − a and 2a + 2	No (= 2a + 6)
10	Write three expressions equivalent to 4x + 8.	e.g. 4(x + 2), 2(2x + 4), 5x + 8 − x

WEEKLY CHECK · ANSWER KEY
After Lesson 6.8
About 15 minutes

Weekly Check 17: Equivalent Expressions — Answer Key

Answers

#	Question	Answer
1	Expand $-3(4x - 2)$	$-12x + 6$
2	Factor $12y + 18$	$6(2y + 3)$
3	Simplify $\frac{1}{2}(6n + 4) - n$	$2n + 2$
4	Simplify $5(x - 2) - 2(x + 1)$	$3x - 12$
5	Equivalent? $2(x + 5)$ and $2x + 5$	No
6	Factor $-8m + 24$	$-8(m - 3)$

LESSON 6.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.9: One-Step Equations

Student goal	I can solve one-step equations with rational numbers.
Objective	Students solve one-step equations.
Standard	7.EE.4a — Solve word problems leading to equations of the form px + q = r.
Pacing	Unit 6, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Balance picture, workbook Lesson 6.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Balance model; inverse operations.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Doing an operation to only one side.

SUPPORT

- Balance scale picture.

EXTEND

- Write equations for real puzzles.

STUDENT LESSON 6.9 · WITH ANSWERS

Lesson 6.9: One-Step Equations

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.9, Part A.
- [ ] Complete the rest of Workbook Lesson 6.9 (Parts B–D).

LET'S EXPLORE

$x - 4.5 = -2$. What number is x?

Use inverse operations to get the variable alone. Do the same thing to both sides to keep the equation balanced.

WORKED EXAMPLE · ADD

$x - 4.5 = -2$

1 Add 4.5 to both sides: $x = -2 + 4.5$.

Answer: $x = 2.5$

WORKED EXAMPLE · DIVIDE

$-3y = 18$

1 Divide both sides by −3.

Answer: $y = -6$

WORKED EXAMPLE · MULTIPLY

$\frac{n}{4} = -5$

1 Multiply both sides by 4.

Answer: $n = -20$

★ KEY IDEA

Undo the operation on the variable. Check by substituting.

✎ TRY IT

Answers are shown in color.

1. $x + 7 = -3$

Answer: $x = -10$

2. $-5m = -35$

Answer: $m = 7$

3. $\frac{2}{3}k = 8$

Answer: $k = 12$

Workbook Answer Key · Lesson 6.9

#	Question	Answer
1	$x + 7 = -3$	$x = -10$
2	$-5m = -35$	$m = 7$
3	$\frac{2}{3}k = 8$	$k = 12$
4	$-2 + 4.5$	2.5
5	$18 \div (-3)$	-6
6	$a + 12 = 5$	$a = -7$
7	$b - 3.2 = -1.8$	$b = 1.4$
8	$6c = -45$	$c = -7.5$
9	$-d = 9$	$d = -9$
10	$p \div (-2) = 11$	$p = -22$
11	$\frac{3}{4}w = -9$	$w = -12$
12	Solve and check: $-\frac{1}{2}x = 7$.	$x = -14$; $-\frac{1}{2} \cdot (-14) = 7$

LESSON 6.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 6: Expressions, Equations, and Inequalities

Lesson 6.10: Two-Step Equations

Student goal	I can solve equations of the form px + q = r.
Objective	Students solve two-step equations.
Standard	7.EE.4a — Solve equations of the form px + q = r fluently.
Pacing	Unit 6, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Tape diagram, then algebra.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing before subtracting.

SUPPORT

- Undo in reverse order.

EXTEND

- Write a two-step equation for a real cost.

STUDENT LESSON 6.10 · WITH ANSWERS

Lesson 6.10: Two-Step Equations

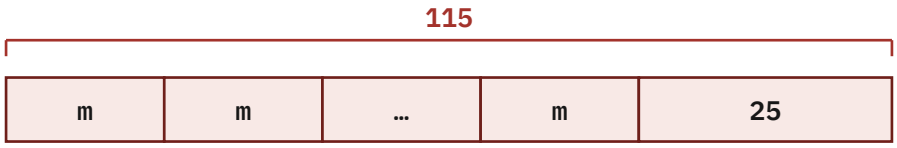
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.10, Part A.
- [ ] Complete the rest of Workbook Lesson 6.10 (Parts B–E).

LET'S EXPLORE

A gym charges a $25 sign-up fee plus $15 per month. Ellie paid $115. How many months?

15m + 25 = 115. Undo the operations in reverse order: subtract 25 first, then divide by 15.



WORKED EXAMPLE · SOLVE

15m + 25 = 115

1 Subtract 25: 15m = 90.

2 Divide by 15: m = 6.

3 Check: 15(6) + 25 = 115 ✓.

Answer: 6 months

WORKED EXAMPLE · NEGATIVE COEFFICIENT

−4x + 7 = 31

1 −4x = 24.

2 x = −6.

Answer: x = −6

★ KEY IDEA

Undo addition/subtraction first, then multiplication/division.

✎ TRY IT

Answers are shown in color.

1. $3x + 5 = 20$

Answer: $x = 5$

2. $-2x + 9 = -7$

Answer: $x = 8$

3. $5x - 12 = 13$

Answer: $x = 5$

Workbook Answer Key · Lesson 6.10

#	Question	Answer
1	$3x + 5 = 20$	$x = 5$
2	$-2x + 9 = -7$	$x = 8$
3	$5x - 12 = 13$	$x = 5$
4	$115 - 25$	90
5	$90 \div 15$	6
6	$2x + 7 = 19$	$x = 6$
7	$6x - 4 = 32$	$x = 6$
8	$-3x + 10 = 1$	$x = 3$
9	$4x + 15 = 3$	$x = -3$
10	$-7x - 2 = 26$	$x = -4$
11	$8x + 5 = 9$	$x = 1/2$
12	$1/2x + 4 = 10$	$x = 12$
13	$0.25x - 3 = 2$	$x = 20$
14	$2/3x - 5 = -1$	$x = 6$
15	$1.5x + 2.5 = -5.5$	$x = -5\ 1/3$
16	Write and solve: a taxi charges \$3 plus \$2.50 per mile. The fare was \$20.50. How many miles?	$2.5m + 3 = 20.5$; $m = 7$

LESSON 6.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 6: Expressions, Equations, and Inequalities

Lesson 6.11: Equations with Parentheses

Student goal	I can solve equations of the form p(x + q) = r.
Objective	Students solve p(x + q) = r equations.
Standard	7.EE.4a — Solve equations of the form p(x + q) = r fluently.
Pacing	Unit 6, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Show both methods.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Distributing to only the first term.

SUPPORT

- Divide first when it's easy.

EXTEND

- Which method is faster for 0.5(w + 8) = −1?

STUDENT LESSON 6.11 · WITH ANSWERS

Lesson 6.11: Equations with Parentheses

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.11, Part A.
- [ ] Complete the rest of Workbook Lesson 6.11 (Parts B–D).

LET'S EXPLORE

Three friends each buy a $6 ticket plus the same snack. They spend $33 total. How much was each snack?

3(s + 6) = 33. You can divide first or distribute first.

WORKED EXAMPLE · DIVIDE FIRST

3(s + 6) = 33

1 Divide by 3: s + 6 = 11.

2 Subtract 6: s = 5.

Answer: $5

WORKED EXAMPLE · DISTRIBUTE FIRST

3(s + 6) = 33

1 3s + 18 = 33.

2 3s = 15; s = 5.

Answer: $5

★ KEY IDEA

For p(x + q) = r: divide by p first, or distribute and solve. Both work.

✎ TRY IT

Answers are shown in color.

1. 2(x − 4) = 10

Answer: x = 9

2. −5(y + 1) = 20

Answer: y = −5

3. $\frac{1}{2}(n + 6) = -3$

Answer: n = −12

Workbook Answer Key · Lesson 6.11

#	Question	Answer
1	2(x − 4) = 10	x = 9
2	−5(y + 1) = 20	y = −5
3	$\frac{1}{2}(n + 6) = -3$	n = −12
4	33 ÷ 3	11
5	11 − 6	5
6	4(x + 3) = 28	x = 4
7	−2(y − 7) = 6	y = 4
8	6(k − 1.5) = 21	k = 5
9	−3(m + 4) = −3	m = −3
10	$\frac{2}{3}(p - 9) = 4$	p = 15
11	0.5(w + 8) = −1	w = −10
12	The perimeter of a rectangle is 2(L + 7) = 46. Find L.	L = 16

LESSON 6.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 6: Expressions, Equations, and Inequalities

Lesson 6.12: Equation Word Problems

Student goal	I can write and solve equations for word problems and compare to an arithmetic solution.
Objective	Students solve equation word problems.
Standard	7.EE.4a — Solve word problems leading to px + q = r and p(x + q) = r; compare algebraic and arithmetic solutions.
Pacing	Unit 6, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Compare the two solution paths.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Setting up equations with the wrong operation.

SUPPORT

- Define the variable in words first.

EXTEND

- Write equations for real family situations.

STUDENT LESSON 6.12 · WITH ANSWERS

Lesson 6.12: Equation Word Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.12, Part A.
- [ ] Complete the rest of Workbook Lesson 6.12 (Parts B–D).

LET'S EXPLORE

The perimeter of a rectangle is 54 cm. Its length is 6 cm. What is its width?

WORKED EXAMPLE · ALGEBRAIC

2(6) + 2w = 54

1 12 + 2w = 54.

2 2w = 42; w = 21.

Answer: 21 cm

WORKED EXAMPLE · ARITHMETIC

Same problem without a variable

1 54 − 12 = 42 for both widths.

2 42 ÷ 2 = 21.

Answer: 21 cm (same steps!)

★ KEY IDEA

Equation steps match arithmetic steps. Writing the equation helps organize harder problems.

✎ TRY IT

Answers are shown in color.

1. A plumber charges $60 plus $45 per hour. The bill is $240. Hours?

Answer: 4 hours

2. Lillie has $18 and saves $7.50 a week. When will she have $63?

Answer: 6 weeks

3. The sum of three consecutive integers is 57. The smallest?

Answer: 18

Workbook Answer Key · Lesson 6.12

#	Question	Answer
1	A plumber charges $60 plus $45 per hour. The bill is $240. Hours?	4 hours
2	Lillie has $18 and saves $7.50 a week. When will she have $63?	6 weeks
3	The sum of three consecutive integers is 57. The smallest?	18
4	2x + 12 = 54	x = 21
5	54 − 12	42
6	A family pays $28 for 4 movie tickets and a $4 snack. Price per ticket?	4t + 4 = 28; $6
7	A water tank has 120 gallons and drains 7.5 gallons per minute. When will it have 30 gallons left?	120 − 7.5m = 30; 12 minutes
8	Ellie's age times 3, minus 4, is 32. How old is she?	3a − 4 = 32; 12
9	A temperature of −8°F rises 2.5° per hour. When will it reach 7°F?	−8 + 2.5h = 7; 6 hours
10	Write your own equation word problem with answer x = −3.	Answers vary.

WEEKLY CHECK · ANSWER KEY
After Lesson 6.12

About 20 minutes

Weekly Check 18: Equations — Answer Key

Answers

#	Question	Answer
1	$x - 6.5 = -2$	$x = 4.5$
2	$-7y = 56$	$y = -8$
3	$3x - 8 = 13$	$x = 7$
4	$-5x + 4 = 29$	$x = -5$
5	$4(n - 2) = -20$	$n = -3$
6	A $15 fee plus $9 per hour; total $78. Hours?	7 hours

LESSON 6.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.13: Writing and Graphing Inequalities

Student goal	I can write inequalities for situations and graph them on a number line.
Objective	Students write and graph inequalities.
Standard	7.EE.4b — Write and graph inequalities; interpret solutions in context.
Pacing	Unit 6, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Inequality truth checks.
Launch	5	Read Let's Explore.
Teach	15	Teach the word/symbol table and circle rules.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Open vs closed circle confusion.

SUPPORT

- 'Or equal' fills the circle.

EXTEND

- Find inequalities in signs: speed limits, height limits.

STUDENT LESSON 6.13 · WITH ANSWERS

Lesson 6.13: Writing and Graphing Inequalities

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.13, Part A.
- [ ] Complete the rest of Workbook Lesson 6.13 (Parts B–E).

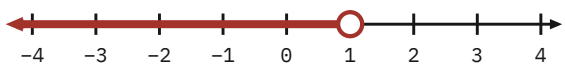
LET'S EXPLORE

You must be at least 48 inches tall to ride a roller coaster. Write and graph an inequality.

"At least 48" means 48 or more: $h \geq 48$. A closed circle means the number is included; an open circle means it is not.



$h \geq 48$ (closed circle)



$x < 1$ (open circle)

Words	Symbol
at least, no less than	≥
at most, no no more than	≤
more than, greater than	>
fewer than, less than	<

★ KEY IDEA

≤ and ≥ use closed circles; < and > use open circles. The arrow points toward the solutions.

✎ TRY IT

Answers are shown in color.

1. No more than 30 students: write it

Answer: $s \leq 30$

2. Temperature below −5

Answer: $t < -5$

3. Graph $x \geq -2$: open or closed circle?

Answer: closed

Workbook Answer Key · Lesson 6.13

#	Question	Answer
1	No more than 30 students: write it	$s \leq 30$
2	Temperature below −5	$t < -5$
3	Graph $x \geq -2$: open or closed circle?	closed
4	Is $48 \geq 48$ true?	Yes
5	Is $48 > 48$ true?	No
6	a speed limit of 65 mph	$s \leq 65$
7	more than 12 players	$p > 12$
8	at least $20	$d \geq 20$
9	colder than −10°	$t < -10$
10	(figure)	$x > 2$
11	(figure)	$x \leq -1$
12	(figure)	$x \geq -4$
13	(figure)	$x < 0$
14	Graph $t \leq 3$ on a number line. (figure)	closed circle at 3, arrow left

LESSON 6.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4

Unit 6: Expressions, Equations, and Inequalities

Lesson 6.14: Solving One-Step Inequalities

Student goal	I can solve one-step inequalities, including multiplying or dividing by a negative.
Objective	Students solve one-step inequalities.
Standard	7.EE.4b — Solve inequalities; graph the solution set.
Pacing	Unit 6, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Show why the symbol flips with a number line.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting to flip.

SUPPORT

- Test a value from the answer.

EXTEND

- Explain the flip rule to a sibling.

STUDENT LESSON 6.14 · WITH ANSWERS

Lesson 6.14: Solving One-Step Inequalities

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.14, Part A.
- [ ] Complete the rest of Workbook Lesson 6.14 (Parts B–D).

LET'S EXPLORE

Solve $-3x < 12$. Is $x = -5$ a solution? Is $x = 0$?

Solve inequalities like equations, with one big difference: when you multiply or divide both sides by a negative number, reverse the inequality symbol.

WORKED EXAMPLE · WHY REVERSE?

$2 < 5$. Multiply both sides by -1.

1 -2 and -5: now $-2 > -5$.

2 Multiplying by a negative flips the order on the number line.

Answer: the symbol must flip

WORKED EXAMPLE · SOLVE

$-3x < 12$

1 Divide by -3 and flip: $x > -4$.

2 Check $x = 0$: $-3(0) = 0 < 12$ ✓. Check $x = -5$: $15 < 12$ ✗.

Answer: $x > -4$

★ KEY IDEA

Flip the inequality symbol when multiplying or dividing by a negative number.

✎ TRY IT

Answers are shown in color.

1. $x + 6 \geq 2$

Answer: $x \geq -4$

2. $-5y \leq 30$

Answer: $y \geq -6$

3. $\frac{n}{-2} > 3$

Answer: $n < -6$

Workbook Answer Key · Lesson 6.14

#	Question	Answer
1	$x + 6 \geq 2$	$x \geq -4$
2	$-5y \leq 30$	$y \geq -6$
3	$\frac{n}{-2} > 3$	$n < -6$
4	Is $-2 > -5$?	Yes
5	$12 \div (-3)$	-4
6	$x - 4 < 7$	$x < 11$
7	$4k \geq -20$	$k \geq -5$
8	$-6m > 18$	$m < -3$
9	$-p \leq 2.5$	$p \geq -2.5$
10	$\frac{1}{3}w < -2$	$w < -6$
11	$-0.5z \geq -4$	$z \leq 8$
12	Solve $-2x \geq 10$ and graph it. (figure)	$x \leq -5$

LESSON 6.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 6: Expressions, Equations, and Inequalities

Lesson 6.15: Two-Step Inequalities

Student goal	I can solve inequalities of the form px + q > r and graph the solutions.
Objective	Students solve two-step inequalities.
Standard	7.EE.4b — Solve word problems leading to inequalities px + q > r or px + q < r; graph and interpret.
Pacing	Unit 6, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work both examples.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Flipping when subtracting a negative constant.

SUPPORT

- Flip only for × or ÷ by a negative.

EXTEND

- Budget inequality for a purchase.

STUDENT LESSON 6.15 · WITH ANSWERS

Lesson 6.15: Two-Step Inequalities

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.15, Part A.
- [ ] Complete the rest of Workbook Lesson 6.15 (Parts B–D).

LET'S EXPLORE

Lillie has $20 and earns $8 per hour. How many hours must she work to have at least $100?

WORKED EXAMPLE · SOLVE

$8h + 20 \geq 100$

1 Subtract 20: $8h \geq 80$.

2 Divide by 8: $h \geq 10$.

Answer: at least 10 hours

WORKED EXAMPLE · NEGATIVE COEFFICIENT

$-2x + 5 > 11$

1 $-2x > 6$.

2 Divide by -2 and flip: $x < -3$.

Answer: $x < -3$

★ KEY IDEA

Use the same steps as two-step equations; flip when dividing by a negative.

✎ TRY IT

Answers are shown in color.

1. $3x - 4 \leq 11$

Answer: $x \leq 5$

2. $-4y + 2 > 14$

Answer: $y < -3$

3. $\frac{1}{2}n + 3 \geq -1$

Answer: $n \geq -8$

Workbook Answer Key · Lesson 6.15

#	Question	Answer
1	$3x - 4 \leq 11$	$x \leq 5$
2	$-4y + 2 > 14$	$y < -3$
3	$\frac{1}{2}n + 3 \geq -1$	$n \geq -8$
4	$100 - 20$	80
5	$80 \div 8$	10
6	$2x + 7 > 15$	$x > 4$
7	$5y - 3 \leq -18$	$y \leq -3$
8	$-3k + 1 \geq 10$	$k \leq -3$
9	$9 - 2m < 1$	$m > 4$
10	$0.5p + 4 > 1$	$p > -6$
11	$-x - 6 \leq -2$	$x \geq -4$
12	Solve $6 - 3x \leq 12$ and graph it. (figure)	$x \geq -2$

LESSON 6.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 6: Expressions, Equations, and Inequalities

Lesson 6.16: Inequality Word Problems

Student goal	I can write, solve, and interpret inequalities for real situations.
Objective	Students solve inequality word problems.
Standard	7.EE.4b — Solve word problems leading to inequalities; interpret solutions in context.
Pacing	Unit 6, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 6.16, Unit 6 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Work the example; discuss interpretation.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 6.16 and the Unit 6 Review.
Exit Ticket	3	Check. Tomorrow: Unit 6 Test.

COMMON MISCONCEPTIONS

- Answering 11.2 pizzas.

SUPPORT

- Ask: does a decimal make sense?

EXTEND

- Plan a party budget.

STUDENT LESSON 6.16 · WITH ANSWERS

Lesson 6.16: Inequality Word Problems

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 6.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 6.16, Part A.
- [ ] Complete the rest of Workbook Lesson 6.16 (Parts B–D).

LET'S EXPLORE

A school club has $150 for a pizza party. Each pizza costs $12.50 and delivery is $10. How many pizzas can they buy?

WORKED EXAMPLE · WRITE, SOLVE, INTERPRET

$12.5p + 10 \le 150$

1 $12.5p \le 140$.

2 $p \le 11.2$.

Answer: at most 11 pizzas (you can't buy part of a pizza)

★ KEY IDEA

Interpret the solution in context: whole numbers only? Must be positive?

✎ TRY IT

Answers are shown in color.

1. A phone plan costs $30 + $5 per GB. Budget $60. Max GB?

Answer: 6 GB

2. Ellie needs at least 85 points. She has 62. Each task is worth 4 points. How many tasks?

Answer: at least 6 tasks

3. An elevator holds at most 2,000 lb. A 180-lb operator rides with boxes of 65 lb. Max boxes?

Answer: 28 boxes

Workbook Answer Key · Lesson 6.16

#	Question	Answer
1	A phone plan costs $30 + $5 per GB. Budget $60. Max GB?	6 GB
2	Ellie needs at least 85 points. She has 62. Each task is worth 4 points. How many tasks?	at least 6 tasks
3	An elevator holds at most 2,000 lb. A 180-lb operator rides with boxes of 65 lb. Max boxes?	28 boxes
4	150 − 10	140
5	140 ÷ 12.5	11.2
6	A taxi charges $3 plus $2 per mile. You have $25. How far can you ride?	$3 + 2m \leq 25$; at most 11 miles
7	To keep a B, Lillie needs an average of at least 80 on 4 tests. Her first three scores are 76, 84, 78. What score does she need on test 4?	$(238 + s)/4 \geq 80 \rightarrow s \geq 82$
8	A freezer is at 5°F and cools 3° per hour. When will it be below −10°F?	$5 - 3h < -10$; after more than 5 hours
9	Write an inequality problem where the answer must be a whole number. Solve it.	Answers vary.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 6 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	Evaluate $-2a + b^2$ when $a = 3$, $b = -4$	10
2	Simplify $3(2x - 1) - (x + 5)$	$5x - 8$
3	Factor $15k - 25$	$5(3k - 5)$
4	Rewrite 1.08c in words	price plus 8% tax
5	$4x + 9 = -11$	$x = -5$
6	$\frac{2}{5}(x + 10) = 6$	$x = 5$
7	$-3x + 4 > 19$	$x < -5$
8	$7 + 2y \leq 1$	$y \leq -3$
9	A pool has 900 gal and drains 25 gal/min. When is it below 400 gal?	after more than 20 minutes

UNIT TEST · ANSWER KEY
At the end of the unit
About 45 minutes

Unit 6 Test — Answer Key

Answers

#	Question	Answer
1	Write: twice the difference of a number and 5	$2(n - 5)$
2	Evaluate $3x - y$ when $x = -2$, $y = -7$	1
3	Simplify $-4m + 9 + 6m - 12$	$2m - 3$
4	Expand $-5(2 - 3k)$	$-10 + 15k$
5	Factor $21a + 14$	$7(3a + 2)$
6	Simplify $\frac{3}{4}(8x - 4) - 2x$	$4x - 3$
7	$y + 8.25 = 3$	$y = -5.25$
8	$-6x + 0 = 39$	$x = -6\frac{1}{2}$
9	$2x + 11 = -5$	$x = -8$
10	$-3(p + 4) = 9$	$p = -7$
11	$\frac{1}{3}n - 2 = 5$	$n = 21$
12	$x - 3 \geq -8$	$x \geq -5$
13	$-4y < 28$	$y > -7$
14	$5 - 2k \leq 17$	$k \geq -6$
15	Graph the answer to the previous problem. (figure)	closed circle at −6, arrow right
16	A plumber charges $55 plus $40 per hour. The bill was $235. Write and solve an equation to find the hours.	$55 + 40h = 235$; $h = 4.5$ hours
17	Ellie has $35 and wants to buy shirts at $8.50 each with a $3 shipping fee. Write and solve an inequality for the number of shirts. Interpret.	$8.5s + 3 \leq 35$; $s \leq 3.76...$, so at most 3 shirts

Scoring

Score	What it means / next step
18–21 (85%+)	Mastered.
15–17	Proficient; correct missed items.

Score	What it means / next step
Below 15	Reteach and retest.

Statistics and Probability

Sampling, comparing populations, probability models, and compound events.

How can a survey of 100 people tell us about a whole city? How likely is it to roll doubles? In this unit you will learn to sample fairly, compare groups with data, and measure chance with probability.

Big ideas

- Use random samples to make inferences about populations.
- Compare two data sets using center and variability.
- Find theoretical and experimental probabilities.
- Build probability models.
- Find probabilities of compound events with lists, tables, trees, and simulations.

Standards

Code	Standard
7.SP.1–2	Use random sampling to draw inferences about a population.
7.SP.3–4	Draw informal comparative inferences about two populations.
7.SP.5–7	Investigate chance processes and develop probability models.
7.SP.8	Find probabilities of compound events.

What students should already know

- Mean, median, and dot plots (Grade 6).
- Fractions, decimals, and percents (Units 2–5).

Materials for the unit

- Coins, number cubes, paper spinners
- Bag of beads or candies

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	7.1	Populations and Samples
Week 1	Lesson 2 of 4	7.2	Random Samples and Bias
Week 1	Lesson 3 of 4	7.3	Making Inferences from Samples
Week 1	Lesson 4 of 4	7.4	Variability in Samples
Week 1	End of week	Check	Weekly Check 19: Sampling
Week 2	Lesson 1 of 4	7.5	Measures of Center and Variability
Week 2	Lesson 2 of 4	7.6	Comparing Dot Plots
Week 2	Lesson 3 of 4	7.7	Comparing Box Plots
Week 2	Lesson 4 of 4	7.8	Comparing Populations
Week 2	End of week	Check	Weekly Check 20: Comparing Data
Week 3	Lesson 1 of 4	7.9	Probability and Likelihood
Week 3	Lesson 2 of 4	7.10	Theoretical Probability
Week 3	Lesson 3 of 4	7.11	Experimental Probability
Week 3	Lesson 4 of 4	7.12	Probability Models
Week 3	End of week	Check	Weekly Check 21: Probability
Week 4	Lesson 1 of 4	7.13	Sample Spaces for Compound Events
Week 4	Lesson 2 of 4	7.14	Tree Diagrams
Week 4	Lesson 3 of 4	7.15	Probability of Compound Events
Week 4	Lesson 4 of 4	7.16	Simulations
Week 4	Before the test	Review	Unit 7 Review
Week 4	End of unit	Test	Unit 7 Test

TEACHING THIS UNIT AT HOME

Play board games together and ask your student to estimate the chance of a roll or draw before it happens.

UNIT VOCABULARY

population	The whole group.
sample	Part of the population.
MAD	Average distance from the mean.
probability	A number 0–1 for likelihood.
sample space	All possible outcomes.

LESSON 7.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 7: Statistics and Probability

Lesson 7.1: Populations and Samples

Student goal	I can tell the difference between a population and a sample.
Objective	Students distinguish populations and samples.
Standard	7.SP.1 — Understand that statistics can be used to gain information about a population by examining a sample.
Pacing	Unit 7, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Percent warm-up.
Launch	5	Discuss Let's Explore: could you ask everyone?
Teach	15	Define population, sample, census. Say: "A spoonful of soup is a sample of the pot."
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Calling the sample the population.

SUPPORT

- Ask: Who do we want to know about? Who did we actually ask?

EXTEND

- Find a survey in the news and name its population and sample.

STUDENT LESSON 7.1 · WITH ANSWERS

Lesson 7.1: Populations and Samples

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.1, Part A.
- [ ] Complete the rest of Workbook Lesson 7.1 (Parts B–D).

LET'S EXPLORE

You want to know the favorite lunch of every 7th grader in your state. Could you ask all of them? What could you do instead?

A population is the whole group you are curious about. Studying everyone is usually too slow or expensive, so statisticians study a sample, a smaller part of the population, and use it to learn about the whole.

WORKED EXAMPLE · IDENTIFY

A library asks 80 of its 2,000 members how often they visit.

1 Population: all 2,000 members.

2 Sample: the 80 members surveyed.

Answer: Population: 2,000 members; sample: 80 members

★ KEY IDEA

A sample is useful only when it represents the population well.

✎ TRY IT

Answers are shown in color.

1. A chef tastes one spoonful of soup. Population? Sample?

Answer: Population: the whole pot; sample: the spoonful

2. A factory tests 50 of 10,000 light bulbs. Population? Sample?

Answer: Population: 10,000 bulbs; sample: 50 bulbs

3. When is a census a better choice than a sample?

Answer: When the population is small, e.g. one class of 20

Workbook Answer Key · Lesson 7.1

#	Question	Answer
1	A chef tastes one spoonful of soup. Population? Sample?	Population: the whole pot; sample: the spoonful
2	A factory tests 50 of 10,000 light bulbs. Population? Sample?	Population: 10,000 bulbs; sample: 50 bulbs
3	When is a census a better choice than a sample?	When the population is small, e.g. one class of 20
4	What fraction of 2,000 is 80?	$\frac{1}{25}$
5	What percent of 2,000 is 80?	4%
6	A city surveys 400 residents about a new park.	Population: all city residents; sample: 400 surveyed
7	A coach times 6 of her 30 swimmers.	Population: 30 swimmers; sample: 6 swimmers
8	A scientist tags 25 frogs in a pond.	Population: all frogs in the pond; sample: 25 frogs
9	A teacher checks every student's homework.	Census: the whole class is studied
10	Describe a question you could answer with a sample. Name the population and the sample.	Answers vary.

LESSON 7.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 7: Statistics and Probability

Lesson 7.2: Random Samples and Bias

Student goal	I can tell whether a sample is random and representative, or biased.
Objective	Students identify random and biased samples.
Standard	7.SP.1 — Random sampling tends to produce representative samples and support valid inferences.
Pacing	Unit 7, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Names in a hat (optional), workbook Lesson 7.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Discuss Ellie's soccer survey.
Teach	15	Go through the table. Ask: Who is left out?
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking a big sample is always fair.

SUPPORT

- Ask: Does everyone have the same chance?

EXTEND

- Design a fair survey for the family.

STUDENT LESSON 7.2 · WITH ANSWERS

Lesson 7.2: Random Samples and Bias

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.2, Part A.
- [ ] Complete the rest of Workbook Lesson 7.2 (Parts B–D).

LET'S EXPLORE

To find the town's favorite sport, Ellie surveys people leaving a soccer game. Is that fair?

Ellie's sample is biased: people at a soccer game probably like soccer more than the town as a whole. A random sample, such as drawing names from the whole town's list, gives everyone the same chance and tends to represent the population.

Sampling method	Random?	Why
Draw 50 names from a hat with every student's name	Yes	Everyone has an equal chance
Ask the first 50 students who arrive	No	Early arrivers may differ
Ask volunteers to answer online	No	Only people who care respond
Number students; use a random-number generator	Yes	Equal chance

! WATCH OUT

A large sample is not automatically fair. 1,000 biased answers are still biased.

★ KEY IDEA

Random samples tend to be representative; samples that leave out groups are biased.

✎ TRY IT

Answers are shown in color.

1. Survey the basketball team about favorite sports.

Answer: Biased

2. Roll a die to pick one student from each table group of 6.

Answer: Random

3. Ask your friends whom they will vote for.

Answer: Biased

Workbook Answer Key · Lesson 7.2

#	Question	Answer
1	Survey the basketball team about favorite sports.	Biased
2	Roll a die to pick one student from each table group of 6.	Random
3	Ask your friends whom they will vote for.	Biased
4	Name a population for “favorite sport in town.”	All town residents
5	Name a sample of that population.	e.g. 100 residents chosen at random
6	A school picks 40 student ID numbers by computer to survey about lunch.	Random: every student has an equal chance
7	A pet store asks its customers whether people in the city prefer cats or dogs.	Biased: only pet store customers
8	A magazine asks readers to mail in opinions about the magazine.	Biased: only motivated readers reply
9	A farmer tests corn from 10 spots chosen at random across the field.	Random
10	Describe a random way to choose 10 students from a school of 300.	e.g. number students 1–300 and use a random-number generator

LESSON 7.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 7: Statistics and Probability

Lesson 7.3: Making Inferences from Samples

Student goal	I can use data from a random sample to estimate something about a population.
Objective	Students estimate population values from samples.
Standard	7.SP.2 — Use data from a random sample to draw inferences about a population.
Pacing	Unit 7, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Percent warm-up.
Launch	5	Read Let's Explore.
Teach	15	Proportion method, then percent method.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using the sample count as the population count.

SUPPORT

- Write the rate as a percent first.

EXTEND

- Sample the family's pantry and estimate.

STUDENT LESSON 7.3 · WITH ANSWERS

Lesson 7.3: Making Inferences from Samples

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.3, Part A.
- [ ] Complete the rest of Workbook Lesson 7.3 (Parts B–E).

LET'S EXPLORE

In a random sample of 50 students, 18 prefer pizza for lunch. The school has 600 students. About how many prefer pizza?

WORKED EXAMPLE · USE A PROPORTION

18 of 50 prefer pizza; 600 students in all

1 Sample rate: $^{18}/_{50}$ = 36%.

2 36% of 600 = 0.36 × 600 = 216.

Answer: about 216 students

WORKED EXAMPLE · DEFECTS

4 of 200 sampled phones are defective. Estimate for 15,000 phones.

1 $^{4}/_{200}$ = 2%.

2 0.02 × 15,000 = 300.

Answer: about 300 phones

★ KEY IDEA

Sample rate × population size = estimate. Estimates are approximate, not exact.

✎ TRY IT

Answers are shown in color.

1. 12 of 40 sampled students walk to school. School: 500. Estimate walkers.

Answer: about 150

2. 3 of 60 sampled apples are bruised. Shipment: 2,400. Estimate bruised.

Answer: about 120

3. 45 of 75 sampled voters favor a plan. Percent?

Answer: 60%

Workbook Answer Key · Lesson 7.3

#	Question	Answer
1	12 of 40 sampled students walk to school. School: 500. Estimate walkers.	about 150
2	3 of 60 sampled apples are bruised. Shipment: 2,400. Estimate bruised.	about 120
3	45 of 75 sampled voters favor a plan. Percent?	60%
4	18 ÷ 50 as a percent	36%
5	0.36 × 600	216
6	9 of 30 in the sample; population 450	about 135
7	14 of 70 in the sample; population 1,500	about 300
8	6 of 25 in the sample; population 800	about 192
9	33 of 120 in the sample; population 2,400	about 660
10	A random sample of 80 households found 52 own a pet. The town has 6,000 households. Estimate the number with pets.	about 3,900
11	A survey of 25 random students found 7 play an instrument. Estimate for 350 students.	about 98
12	Why should the sample be random for this estimate to be trusted?	A biased sample would not match the population rate.

LESSON 7.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 7: Statistics and Probability

Lesson 7.4: Variability in Samples

Student goal	I can use several samples to judge how much an estimate might vary.
Objective	Students explore sampling variability.
Standard	7.SP.2 — Generate multiple samples to gauge variation in estimates.
Pacing	Unit 7, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Bag of mixed candies or beads (optional), workbook Lesson 7.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Do Let's Explore with real beads if available.
Teach	15	Record several samples; compute the mean.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Expecting every sample to match.

SUPPORT

- Physically draw samples.

EXTEND

- Compare samples of size 5 vs 20.

STUDENT LESSON 7.4 · WITH ANSWERS

Lesson 7.4: Variability in Samples

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.4, Part A.
- [ ] Complete the rest of Workbook Lesson 7.4 (Parts B–D).

LET'S EXPLORE

Five students each scoop 20 candies from the same jar and count the red ones. Will they all get the same number?

No! Different random samples give different results. That is sampling variability. Looking at several samples helps you see how much an estimate might change.

Sample	1	2	3	4	5
Red (of 20)	6	8	5	7	9
Percent red	30%	40%	25%	35%	45%

WORKED EXAMPLE · COMBINE SAMPLES

Estimate the percent of red candies in the jar.

1 Mean red per sample: (6 + 8 + 5 + 7 + 9) ÷ 5 = 7.

2 7 ÷ 20 = 35%.

Answer: about 35%

WORKED EXAMPLE · USE IT

The jar holds 500 candies. About how many are red?

1 35% of 500.

Answer: about 175

★ KEY IDEA

Estimates from samples vary. Larger samples and more samples give more reliable estimates.

✎ TRY IT

Answers are shown in color.

1. What is the range of the five percents above?

Answer: 25% to 45%

2. Which gives a more reliable estimate: one sample of 20 or five samples of 20?

Answer: five samples of 20

3. A sixth sample has 7 red. Does it change the estimate much?

Answer: No; 7 of 20 is 35%, the same as the estimate

Workbook Answer Key · Lesson 7.4

#	Question	Answer
1	What is the range of the five percents above?	25% to 45%
2	Which gives a more reliable estimate: one sample of 20 or five samples of 20?	five samples of 20
3	A sixth sample has 7 red. Does it change the estimate much?	No; 7 of 20 is 35%, the same as the estimate
4	Mean of 6, 8, 5, 7, 9	7
5	7 out of 20 as a percent	35%
6	Four random samples of 50 voters show 28, 31, 25, and 30 supporting a plan. Find the mean percent.	57%
7	Use your answer to estimate supporters among 8,000 voters.	about 4,560
8	Lillie's sample of 10 says 70% like a book. Ellie's sample of 100 says 52%. Whose estimate is more reliable? Why?	Ellie's; the larger sample varies less
9	Why don't all random samples give the same result?	Each sample includes different members by chance.

WEEKLY CHECK · ANSWER KEY
After Lesson 7.4

About 15 minutes

Weekly Check 19: Sampling — Answer Key

Answers

#	Question	Answer
1	A school surveys 60 of its 900 students. Name the population.	all 900 students
2	Surveying only the chess club about favorite games: random or biased?	Biased
3	15 of 60 sampled students bike. Estimate bikers in 900.	225
4	Samples show 40%, 46%, 43%, 39%. Mean percent?	42%
5	Use that mean to estimate for 2,000 people.	about 840
6	Which is more reliable: a sample of 30 or of 300?	300

LESSON 7.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 7: Statistics and Probability

Lesson 7.5: Measures of Center and Variability

Student goal	I can find the mean, median, mean absolute deviation, range, and interquartile range of a data set.
Objective	Students compute measures of center and variability.
Standard	7.SP.4 — Use measures of center and variability to describe and compare data.
Pacing	Unit 7, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Ordering warm-up.
Launch	5	Read Let's Explore.
Teach	15	Review mean and median. Compute MAD with a distance table. Review quartiles and IQR.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting to order data before finding the median.
- Using signed differences for MAD.

SUPPORT

- Number line distances.

EXTEND

- Compute the MAD of family heights.

STUDENT LESSON 7.5 · WITH ANSWERS

Lesson 7.5: Measures of Center and Variability

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.5, Part A.
- [ ] Complete the rest of Workbook Lesson 7.5 (Parts B–E).

LET'S EXPLORE

Two quizzes have scores 2, 4, 6, 8, 10 and 12, 15, 15, 18, 20. Which set is more spread out?

Center (mean or median) describes a typical value. Variability (MAD, range, or IQR) describes how spread out the values are.

WORKED EXAMPLE · MEAN AND MAD

2, 4, 6, 8, 10

1 Mean: 30 ÷ 5 = 6.

2 Distances from 6: 4, 2, 0, 2, 4.

3 MAD: 12 ÷ 5 = 2.4.

Answer: mean 6, MAD 2.4

WORKED EXAMPLE · MEDIAN AND IQR

52, 54, 55, 56, 58, 59, 60, 61, 62, 64, 66

1 Median (6th value): 59.

2 Q1 = median of lower half = 55; Q3 = 62.

3 IQR = 62 − 55 = 7.

Answer: median 59, IQR 7

★ KEY IDEA

Use mean and MAD for symmetric data; median and IQR when there are outliers or skew.

✎ TRY IT

Answers are shown in color.

1. Mean and MAD of 12, 15, 15, 18, 20

Answer: mean 16, MAD 2.4

2. Median of 7, 3, 9, 4, 10, 6

Answer: 6.5

3. IQR of 1, 3, 4, 6, 7, 9, 12, 15

Answer: 7

Workbook Answer Key · Lesson 7.5

#	Question	Answer
1	Mean and MAD of 12, 15, 15, 18, 20	mean 16, MAD 2.4
2	Median of 7, 3, 9, 4, 10, 6	6.5
3	IQR of 1, 3, 4, 6, 7, 9, 12, 15	7
4	Order: 9, 2, 7, 4, 5	2, 4, 5, 7, 9
5	Median of 2, 4, 5, 7, 9	5
6	4, 6, 8, 10, 12	mean 8, MAD 2.4
7	5, 5, 7, 9, 9	mean 7, MAD 1.6
8	10, 12, 14, 16, 18, 20	mean 15, MAD 3
9	1, 4, 4, 5, 6	mean 4, MAD 1.2
10	2, 3, 5, 6, 8, 9, 11	median 6, IQR 6
11	10, 12, 13, 15, 18, 20, 21, 25	median 16.5, IQR 8
12	Which set has more variability: 2, 4, 6, 8, 10 or 12, 15, 15, 18, 20? Use MAD.	2, 4, 6, 8, 10 (MAD 2.4 vs 2.4)

LESSON 7.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 7: Statistics and Probability

Lesson 7.6: Comparing Dot Plots

Student goal	I can compare two data sets using their centers and variability.
Objective	Students compare dot plots.
Standard	7.SP.3 — Informally assess the degree of visual overlap of two distributions; express the difference in centers as a multiple of variability.
Pacing	Unit 7, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Means warm-up.
Launch	5	Look at the two dot plots.
Teach	15	Compute difference ÷ MAD.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Comparing only the highest values.

SUPPORT

- Shade overlap.

EXTEND

- Compare two family members' reaction times.

STUDENT LESSON 7.6 · WITH ANSWERS

Lesson 7.6: Comparing Dot Plots

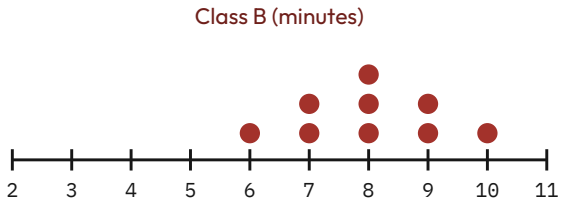
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.6, Part A.
- [ ] Complete the rest of Workbook Lesson 7.6 (Parts B–D).

LET'S EXPLORE

Two classes record minutes of homework. Class B seems to do more. How much more, compared to the spread?





WORKED EXAMPLE · COMPARE

Class A vs Class B

1 Means: A = 5, B = 8; difference 3.

2 MAD of each set: ≈ 0.89.

3 $3 \div {}^{8}\!/_{9} \approx 3.38$.

Answer: The centers differ by about 3.4 MADs: a big difference

★ KEY IDEA

Divide the difference in centers by the variability (MAD). A result above about 2 means the groups are clearly different.

✎ TRY IT

Answers are shown in color.

1. Do the dot plots overlap?

Answer: Yes, from 6 to 7

2. Which class has more variability?

Answer: Neither; same MAD

3. Is a typical B student doing more homework?

Answer: Yes, about 3 more minutes

Workbook Answer Key · Lesson 7.6

#	Question	Answer
1	Do the dot plots overlap?	Yes, from 6 to 7
2	Which class has more variability?	Neither; same MAD
3	Is a typical B student doing more homework?	Yes, about 3 more minutes
4	Mean of 3, 4, 4, 5, 5, 5, 6, 6, 7	5
5	Mean of 6, 7, 7, 8, 8, 8, 9, 9, 10	8
6	Team X jumps (ft): 10, 11, 12, 12, 13, 14. Team Y: 12, 13, 14, 14, 15, 16. Difference of means?	2
7	Find the MAD of either team.	1
8	How many MADs apart are the means?	2
9	Is the difference large or small? Explain.	Yes: about 2 MADs apart, a noticeable difference
10	Two data sets have means 20 and 22 and MADs of 5. Are they very different? Why?	No; 2 ÷ 5 = 0.4 MADs, lots of overlap

LESSON 7.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 7: Statistics and Probability

Lesson 7.7: Comparing Box Plots

Student goal	I can compare two data sets displayed in box plots.
Objective	Students compare box plots.
Standard	7.SP.3, 7.SP.4 — Use measures of center and variability to compare two populations.
Pacing	Unit 7, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Read both box plots.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking a longer whisker means more data points.

SUPPORT

- Each section holds about 25% of the data.

EXTEND

- Box plot of family ages.

STUDENT LESSON 7.7 · WITH ANSWERS

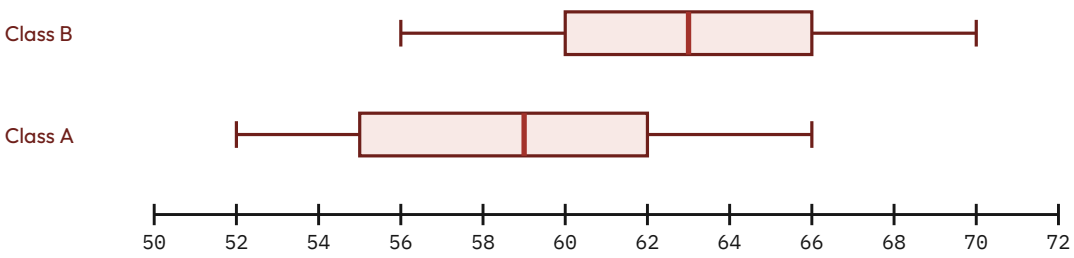
Lesson 7.7: Comparing Box Plots

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.7, Part A.
- [ ] Complete the rest of Workbook Lesson 7.7 (Parts B–D).

LET'S EXPLORE

Two 7th-grade classes measured heights in inches. Which class tends to be taller?



WORKED EXAMPLE · READ THE PLOTS

Class A vs Class B

1 Medians: A = 59, B = 63.

2 IQRs: A = 7, B = 6.

Answer: Class B's median is 4 in. greater; the spreads are similar

★ KEY IDEA

Compare medians for center and IQRs for spread.

✎ TRY IT

Answers are shown in color.

1. Range of Class A

Answer: 14

2. Range of Class B

Answer: 14

3. About what percent of Class B is taller than Class A's median?

Answer: more than 75% (B's Q1 = 60)

Workbook Answer Key · Lesson 7.7

#	Question	Answer
1	Range of Class A	14
2	Range of Class B	14
3	About what percent of Class B is taller than Class A's median?	more than 75% (B's Q1 = 60)
4	What does the line inside a box show?	the median
5	What does the box's length show?	the IQR
6	Data: 3, 5, 6, 8, 9, 11, 14. Five-number summary?	3, 5, 8, 11, 14
7	Data: 20, 22, 25, 26, 28, 30, 31, 35. Five-number summary?	20, 23.5, 27, 30.5, 35
8	Plot Set A {2, 4, 5, 7, 9} and Set B {5, 7, 8, 10, 12}. Which has the greater median? (figure)	Set B (8 vs 5)
9	When might you compare medians instead of means?	When the data are skewed or have outliers

LESSON 7.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 7: Statistics and Probability

Lesson 7.8: Comparing Populations

Student goal	I can use random samples to compare two populations.
Objective	Students compare populations from samples.
Standard	7.SP.3, 7.SP.4 — Use measures of center and variability of random samples to draw informal comparative inferences.
Pacing	Unit 7, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Compute difference ÷ MAD.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Concluding a difference from tiny gaps.

SUPPORT

- Use the 2-MAD rule of thumb.

EXTEND

- Design a two-group study at home.

STUDENT LESSON 7.8 · WITH ANSWERS

Lesson 7.8: Comparing Populations

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.8, Part A.
- [ ] Complete the rest of Workbook Lesson 7.8 (Parts B–D).

LET'S EXPLORE

Random samples of 7th and 4th graders report weekly reading hours. Do 7th graders read more?

Group	Sample data (hours)	Mean	MAD
Grade 7	4, 5, 6, 6, 7, 8	6	1
Grade 4	2, 3, 3, 4, 4, 5	3.5	≈ 0.83

WORKED EXAMPLE · COMPARE

Difference in means vs MAD

1 Difference: 6 − 3.5 = 2.5.

2 Divide by the larger MAD: 2.5 ÷ 1 = 2.5.

Answer: about 2.5 MADs: 7th graders likely read more

★ KEY IDEA

If the samples are random and the difference is more than about 2 MADs, the populations probably differ.

✎ TRY IT

Answers are shown in color.

1. If the difference were 0.5 MAD, what would you conclude?

Answer: Not enough evidence of a difference

2. Why do the samples need to be random?

Answer: So they represent their populations

3. Name one other way to compare these groups.

Answer: Compare medians and IQRs

Workbook Answer Key · Lesson 7.8

#	Question	Answer
1	If the difference were 0.5 MAD, what would you conclude?	Not enough evidence of a difference
2	Why do the samples need to be random?	So they represent their populations
3	Name one other way to compare these groups.	Compare medians and IQRs
4	Mean of 4, 5, 6, 6, 7, 8	6
5	Mean of 2, 3, 3, 4, 4, 5	3.5
6	Random samples of tomato plants: Fertilizer A heights (cm) have mean 42, MAD 3. Fertilizer B: mean 36, MAD 3. How many MADs apart?	2 MADs
7	Is there a meaningful difference? Explain.	Yes; about 2 MADs, so A plants tend to be taller
8	Samples of two dog breeds: mean weights 61 lb and 58 lb, each MAD 6 lb. Meaningful difference?	No; 3 ÷ 6 = 0.5 MAD
9	Write a question comparing two populations you could answer with random samples.	Answers vary.

WEEKLY CHECK · ANSWER KEY
After Lesson 7.8

About 15 minutes

Weekly Check 20: Comparing Data — Answer Key

Answers

#	Question	Answer
1	Mean of 6, 8, 10, 12, 14	10
2	MAD of 6, 8, 10, 12, 14	2.4
3	Median of 3, 7, 8, 12, 15, 20	10
4	IQR of 2, 4, 5, 7, 8, 10, 13, 14	7
5	Means 30 and 36, MAD 3. How many MADs apart?	2
6	Is that a meaningful difference?	Yes (about 2 MADs)

LESSON 7.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 7: Statistics and Probability

Lesson 7.9: Probability and Likelihood

Student goal	I can describe the likelihood of an event with a number from 0 to 1.
Objective	Students describe probability on a 0–1 scale.
Standard	7.SP.5 — Understand probability as a number from 0 to 1 expressing likelihood.
Pacing	Unit 7, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Fraction warm-up.
Launch	5	Discuss Let's Explore.
Teach	15	Build the likelihood line.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking probability can exceed 1.

SUPPORT

- Place events on a number line.

EXTEND

- Ask family members to estimate probabilities of daily events.

STUDENT LESSON 7.9 · WITH ANSWERS

Lesson 7.9: Probability and Likelihood

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.9, Part A.
- [ ] Complete the rest of Workbook Lesson 7.9 (Parts B–D).

LET'S EXPLORE

Is it likely, unlikely, certain, or impossible that the sun rises tomorrow? That you roll a 7 on a number cube?

Probability measures likelihood on a scale from 0 (impossible) to 1 (certain). A probability of $^1/_2$ means equally likely to happen or not.

0	near 0	$^1/_2$	near 1	1
impossible	unlikely	equally likely	likely	certain

★ KEY IDEA

Probabilities near 0 are unlikely; near 1 are likely.

✎ TRY IT

Answers are shown in color.

1. Rolling a 7 on a standard number cube

Answer: impossible (0)

2. A coin landing heads

Answer: equally likely ($^1/_2$)

3. Drawing a red marble from a bag of 9 red, 1 blue

Answer: likely

Workbook Answer Key · Lesson 7.9

#	Question	Answer
1	Rolling a 7 on a standard number cube	impossible (0)
2	A coin landing heads	equally likely ($\frac{1}{2}$)
3	Drawing a red marble from a bag of 9 red, 1 blue	likely
4	Is $\frac{3}{4}$ closer to 0 or 1?	1
5	Write 0.2 as a fraction	$\frac{1}{5}$
6	Probability 0.95	likely
7	Probability $\frac{1}{8}$	unlikely
8	Probability 0	impossible
9	Probability 0.5	equally likely
10	Snow in July in Texas	very unlikely
11	Choosing a vowel from A, E, I	certain
12	Describe an event that is likely but not certain. Estimate its probability.	Answers vary.

LESSON 7.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 7: Statistics and Probability

Lesson 7.10: Theoretical Probability

Student goal	I can find the theoretical probability of an event.
Objective	Students find theoretical probabilities.
Standard	7.SP.7a — Develop a uniform probability model and use it to find probabilities.
Pacing	Unit 7, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Spinner or number cube, workbook Lesson 7.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Favorable ÷ total. Complement.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using favorable ÷ unfavorable.

SUPPORT

- List the sample space.

EXTEND

- Probabilities with a deck of cards.

STUDENT LESSON 7.10 · WITH ANSWERS

Lesson 7.10: Theoretical Probability

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.10, Part A.
- [ ] Complete the rest of Workbook Lesson 7.10 (Parts B–E).

LET'S EXPLORE

A bag holds 3 red, 5 blue, and 2 green marbles. What is the probability of drawing blue?

When all outcomes are equally likely, P(event) = favorable outcomes ÷ total outcomes.

WORKED EXAMPLE · MARBLES

3 red, 5 blue, 2 green

1 Total: 10. Blue: 5.

Answer: P(blue) = $^{1}/_{2}$

WORKED EXAMPLE · NUMBER CUBE

P(number greater than 4)

1 Favorable: 5, 6. Total: 6.

Answer: P = $^{1}/_{3}$

WORKED EXAMPLE · COMPLEMENT

P(not green)

1 P(green) = $^{1}/_{5}$, so P(not green) = 1 − $^{1}/_{5}$.

Answer: $^{4}/_{5}$

★ KEY IDEA

P(not A) = 1 − P(A).

✎ TRY IT

Answers are shown in color.

1. P(red)

Answer: $\frac{3}{10}$

2. P(even) on a number cube

Answer: $\frac{1}{2}$

3. P(not 6) on a number cube

Answer: $\frac{5}{6}$

Workbook Answer Key · Lesson 7.10

#	Question	Answer
1	P(red)	$\frac{3}{10}$
2	P(even) on a number cube	$\frac{1}{2}$
3	P(not 6) on a number cube	$\frac{5}{6}$
4	Simplify $\frac{5}{10}$	$\frac{1}{2}$
5	1 − 0.3	0.7
6	P(3)	$\frac{1}{8}$
7	P(even)	$\frac{1}{2}$
8	P(less than 4)	$\frac{3}{8}$
9	P(9)	0
10	P(not 8)	$\frac{7}{8}$
11	P(multiple of 3)	$\frac{1}{4}$
12	A class has 12 boys and 16 girls. A name is drawn at random. P(girl)?	$\frac{4}{7}$
13	The probability of rain is 0.35. What is the probability of no rain?	0.65
14	Letters of MISSISSIPPI are on cards. P(S)?	$\frac{4}{11}$

LESSON 7.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 7: Statistics and Probability

Lesson 7.11: Experimental Probability

Student goal	I can find experimental probability and compare it to theoretical probability.
Objective	Students compute experimental probability.
Standard	7.SP.6, 7.SP.7b — Approximate probability by collecting data; predict relative frequency.
Pacing	Unit 7, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Number cube or coin, workbook Lesson 7.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Roll a cube 30 times and record.
Teach	15	Compare experimental and theoretical.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Expecting exact matches with few trials.

SUPPORT

- Pool data from several experiments.

EXTEND

- Graph running proportion of heads.

STUDENT LESSON 7.11 · WITH ANSWERS

Lesson 7.11: Experimental Probability

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.11, Part A.
- [ ] Complete the rest of Workbook Lesson 7.11 (Parts B–E).

LET'S EXPLORE

Ellie rolled a number cube 60 times. She got a 6 fifteen times. Is the cube fair?

Result	1	2	3	4	5	6
Frequency	7	9	11	8	10	15

WORKED EXAMPLE · EXPERIMENTAL

P(6) from the data

1 6 occurred 15 times in 60 rolls.

Answer: $^{1}/_{4}$ = 25%

WORKED EXAMPLE · COMPARE

Theoretical P(6)

1 $^{1}/_{6} \approx 17\%$.

2 25% is higher, but 60 rolls is a small number of trials.

Answer: Roll many more times before deciding the cube is unfair

★ KEY IDEA

As the number of trials grows, experimental probability tends to get closer to theoretical probability.

✎ TRY IT

Answers are shown in color.

1. Experimental P(3)

Answer: $\frac{11}{60}$

2. Experimental P(odd)

Answer: $\frac{7}{15}$

3. Predict how many 6s in 600 rolls of a fair cube

Answer: about 100

Workbook Answer Key · Lesson 7.11

#	Question	Answer
1	Experimental P(3)	$\frac{11}{60}$
2	Experimental P(odd)	$\frac{7}{15}$
3	Predict how many 6s in 600 rolls of a fair cube	about 100
4	15 ÷ 60 as a percent	25%
5	$\frac{1}{6}$ of 600	100
6	Experimental P(heads)	$\frac{29}{50}$
7	Theoretical P(heads)	$\frac{1}{2}$
8	Experimental P(tails)	$\frac{21}{50}$
9	Predict heads in 1,000 flips of a fair coin	about 500
10	A player made 18 of 24 free throws. Predict makes in 40 attempts.	about 30
11	A spinner landed on blue 12 times in 80 spins. Predict blues in 200 spins.	about 30
12	Why might experimental and theoretical probability differ?	Chance; small number of trials

LESSON 7.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 7: Statistics and Probability

Lesson 7.12: Probability Models

Student goal	I can build uniform and non-uniform probability models.
Objective	Students develop probability models.
Standard	7.SP.7a, 7.SP.7b — Develop uniform and non-uniform probability models.
Pacing	Unit 7, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Paper spinner, workbook Lesson 7.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Discuss Let's Explore.
Teach	15	Build models; check the sum is 1.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Assuming every outcome is equally likely.

SUPPORT

- Count sections.

EXTEND

- Design a fair game.

STUDENT LESSON 7.12 · WITH ANSWERS

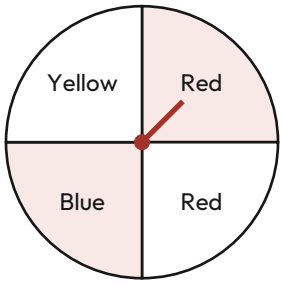
Lesson 7.12: Probability Models

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.12, Part A.
- [ ] Complete the rest of Workbook Lesson 7.12 (Parts B–D).

LET'S EXPLORE

A spinner has 4 equal sections: red, red, blue, yellow. Are all colors equally likely?



WORKED EXAMPLE · NON-UNIFORM MODEL

Spinner above

1 P(red) = $^{2}/_{4}$ = $^{1}/_{2}$, P(blue) = $^{1}/_{4}$, P(yellow) = $^{1}/_{4}$.

2 Sum: $^{1}/_{2}$ + $^{1}/_{4}$ + $^{1}/_{4}$ = 1 ✓.

Answer: Not uniform: red is twice as likely

WORKED EXAMPLE · MODEL FROM DATA

A survey of 40: 18 dogs, 14 cats, 8 fish (favorite pet)

1 P(dog) = $^{9}/_{20}$, P(cat) = $^{7}/_{20}$, P(fish) = $^{1}/_{5}$.

Answer: probabilities add to 1

★ KEY IDEA

In every probability model, the probabilities add up to 1.

✎ TRY IT

Answers are shown in color.

1. Uniform model for a coin

Answer: H: $\frac{1}{2}$, T: $\frac{1}{2}$

2. Spinner P(not red)

Answer: $\frac{1}{2}$

3. A model lists 0.3, 0.5, and ? Find ?

Answer: 0.2

Workbook Answer Key · Lesson 7.12

#	Question	Answer
1	Uniform model for a coin	H: $\frac{1}{2}$, T: $\frac{1}{2}$
2	Spinner P(not red)	$\frac{1}{2}$
3	A model lists 0.3, 0.5, and ? Find ?	0.2
4	$\frac{1}{2} + \frac{1}{4} + \frac{1}{4}$	1
5	1 − 0.3 − 0.5	0.2
6	Spinner with sections 1, 2, 2, 3, 3, 3. Give P(1), P(2), P(3). (figure)	$\frac{1}{6}$, $\frac{1}{3}$, $\frac{1}{2}$
7	Is the model P(A) = 0.4, P(B) = 0.4, P(C) = 0.3 valid? Why?	No; the sum is 1.1
8	Of 50 cars, 20 were white, 15 black, 10 silver, and 5 red. Write a model.	white $\frac{2}{5}$, black $\frac{3}{10}$, silver $\frac{1}{5}$, red $\frac{1}{10}$
9	Design a spinner where P(green) = 0.5 and P(red) = 0.25.	e.g. 4 sections: green, green, red, any other color

WEEKLY CHECK · ANSWER KEY
After Lesson 7.12

About 15 minutes

Weekly Check 21: Probability — Answer Key

Answers

#	Question	Answer
1	Bag: 4 red, 6 blue. P(red)	$\frac{2}{5}$
2	P(not 2) on a number cube	$\frac{5}{6}$
3	Probability 0.9 is…	likely
4	32 heads in 50 flips: experimental P(heads)	$\frac{16}{25}$
5	Model: 0.25, 0.45, ? Find ?	0.3
6	Coin landed heads 12 of 40 times. Predict heads in 200 flips with that rate.	60

LESSON 7.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 7: Statistics and Probability

Lesson 7.13: Sample Spaces for Compound Events

Student goal	I can list the sample space of a compound event using lists and tables.
Objective	Students list sample spaces.
Standard	7.SP.8b — Represent sample spaces for compound events using organized lists and tables.
Pacing	Unit 7, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Two number cubes, workbook Lesson 7.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Discuss Let's Explore.
Teach	15	Build the sum table together.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Counting 3+4 and 4+3 as one outcome.

SUPPORT

- Color-code the table.

EXTEND

- Play a sum-guessing game.

STUDENT LESSON 7.13 · WITH ANSWERS

Lesson 7.13: Sample Spaces for Compound Events

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.13, Part A.
- [ ] Complete the rest of Workbook Lesson 7.13 (Parts B–E).

LET'S EXPLORE

You roll two number cubes. How many outcomes are possible?

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

WORKED EXAMPLE · COUNT OUTCOMES

Two number cubes

1 $6 \times 6 = 36$ outcomes.

Answer: 36

WORKED EXAMPLE · USE THE TABLE

P(sum is 7)

1 Count 7s in the table: 6.

Answer: $\frac{1}{6}$

★ KEY IDEA

Organized lists and tables make sure you count every outcome once.

✎ TRY IT

Answers are shown in color.

1. P(sum is 2)

Answer: $\frac{1}{36}$

2. P(sum is 10)

Answer: $\frac{1}{12}$

3. P(doubles)

Answer: $\frac{1}{6}$

Workbook Answer Key · Lesson 7.13

#	Question	Answer
1	P(sum is 2)	$\frac{1}{36}$
2	P(sum is 10)	$\frac{1}{12}$
3	P(doubles)	$\frac{1}{6}$
4	6 × 6	36
5	Simplify $\frac{6}{36}$	$\frac{1}{6}$
6	P(sum is 8)	$\frac{5}{36}$
7	P(sum greater than 9)	$\frac{1}{6}$
8	P(sum is even)	$\frac{1}{2}$
9	P(sum less than 5)	$\frac{1}{6}$
10	P(sum is 12)	$\frac{1}{36}$
11	P(sum is 1)	0
12	Flip a coin and spin a 3-color spinner (R, B, Y). List all outcomes.	HR, HB, HY, TR, TB, TY (6)
13	Which sum of two number cubes is most likely? Why?	7; it occurs 6 ways, more than any other

LESSON 7.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 7: Statistics and Probability

Lesson 7.14: Tree Diagrams

Student goal	I can use a tree diagram to find the sample space of a compound event.
Objective	Students use tree diagrams.
Standard	7.SP.8b — Represent sample spaces for compound events with tree diagrams.
Pacing	Unit 7, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Draw the outfit tree.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Missing branches.

SUPPORT

- Draw every branch at each stage.

EXTEND

- Plan outfits for the week with a tree.

STUDENT LESSON 7.14 · WITH ANSWERS

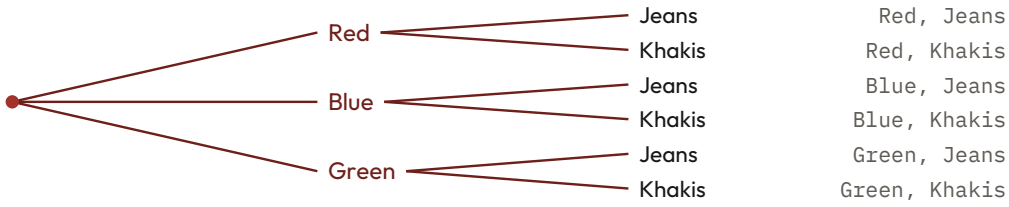
Lesson 7.14: Tree Diagrams

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.14, Part A.
- [ ] Complete the rest of Workbook Lesson 7.14 (Parts B–E).

LET'S EXPLORE

Lillie has a red, blue, or green shirt and jeans or khakis. How many outfits can she make?



WORKED EXAMPLE · COUNT

3 shirts, 2 pants

1 Each of 3 shirts branches to 2 pants.

Answer: 6 outfits

WORKED EXAMPLE · PROBABILITY

Outfit chosen at random: P(blue shirt with jeans)

1 1 favorable outcome of 6.

Answer: $\frac{1}{6}$

★ KEY IDEA

Multiply the number of choices at each stage to count outcomes (the counting principle).

✎ TRY IT

Answers are shown in color.

1. P(outfit with khakis)

Answer: $\frac{1}{2}$

2. Add a yellow shirt. How many outfits?

Answer: 8

3. Coin flipped 3 times: how many outcomes?

Answer: 8

Workbook Answer Key · Lesson 7.14

#	Question	Answer
1	P(outfit with khakis)	$\frac{1}{2}$
2	Add a yellow shirt. How many outfits?	8
3	Coin flipped 3 times: how many outcomes?	8
4	3 × 2	6
5	2 × 2 × 2	8
6	Draw a tree: flip a coin twice. List outcomes.	HH, HT, TH, TT
7	P(exactly one head)	$\frac{1}{2}$
8	P(at least one tail)	$\frac{3}{4}$
9	4 sandwiches, 3 drinks: combos	12
10	2 sizes, 5 flavors: cones	10
11	Coin and number cube: outcomes	12
12	3 coins: P(all heads)	$\frac{1}{8}$
13	A café offers 3 breads, 4 fillings, and 2 sauces. How many sandwiches?	24

LESSON 7.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 7: Statistics and Probability

Lesson 7.15: Probability of Compound Events

Student goal	I can find the probability of a compound event using the sample space.
Objective	Students compute compound probabilities.
Standard	7.SP.8a, 7.SP.8c — Find probabilities of compound events using lists, tables, tree diagrams, and simulation.
Pacing	Unit 7, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 7.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Count from the tree; then multiply.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Adding probabilities instead of multiplying for “and.”

SUPPORT

- Use the sample space.

EXTEND

- Find P(at least one 6) in two rolls: 11/36.

STUDENT LESSON 7.15 · WITH ANSWERS

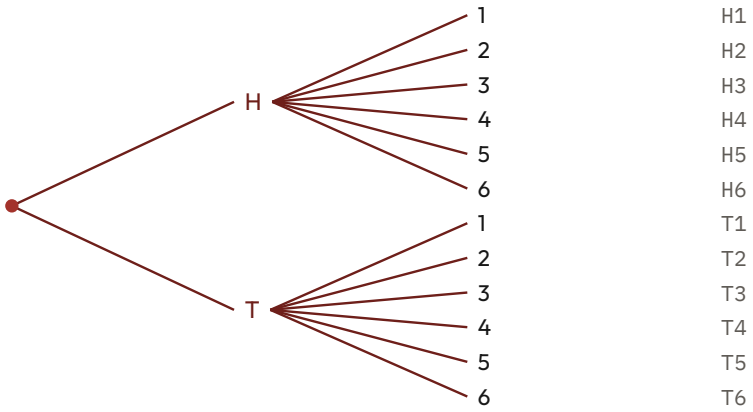
Lesson 7.15: Probability of Compound Events

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.15, Part A.
- [ ] Complete the rest of Workbook Lesson 7.15 (Parts B–E).

LET'S EXPLORE

Flip a coin and roll a number cube. What is P(heads and an even number)?



WORKED EXAMPLE · COUNT

P(heads and even)

1 Sample space: 2 × 6 = 12.

2 Favorable: H2, H4, H6 = 3.

Answer: $^{1}/_{4}$

WORKED EXAMPLE · MULTIPLY (INDEPENDENT EVENTS)

Same event

1 P(H) × P(even) = $^{1}/_{2} \times {}^{1}/_{2}$.

Answer: $^{1}/_{4}$

★ KEY IDEA

Probability of a compound event = favorable outcomes ÷ sample space size. For independent events you can also multiply.

✎ TRY IT

Answers are shown in color.

1. P(tails and 5)

Answer: $\frac{1}{12}$

2. P(heads and greater than 2)

Answer: $\frac{1}{3}$

3. P(tails or 6)

Answer: $\frac{7}{12}$

Workbook Answer Key · Lesson 7.15

#	Question	Answer
1	P(tails and 5)	$\frac{1}{12}$
2	P(heads and greater than 2)	$\frac{1}{3}$
3	P(tails or 6)	$\frac{7}{12}$
4	$\frac{1}{2} \times \frac{1}{3}$	$\frac{1}{6}$
5	2 × 6	12
6	Two coins: P(HH)	$\frac{1}{4}$
7	Two number cubes: P(both 6)	$\frac{1}{36}$
8	Two cubes: P(both even)	$\frac{1}{4}$
9	Coin + cube: P(T and 1 or 2)	$\frac{1}{6}$
10	Spinner 1–4 twice: P(sum 5)	$\frac{1}{4}$
11	Three coins: P(exactly 2 heads)	$\frac{3}{8}$
12	Ellie guesses on 2 true/false questions. P(both correct)?	$\frac{1}{4}$
13	A family has 3 children. Assuming boy and girl are equally likely, P(at least one girl)?	$\frac{7}{8}$
14	Roll two number cubes. P(sum is 4)?	$\frac{1}{12}$

LESSON 7.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 7: Statistics and Probability

Lesson 7.16: Simulations

Student goal	I can design and use a simulation to estimate the probability of a compound event.
Objective	Students design and run simulations.
Standard	7.SP.8c — Design and use a simulation to generate frequencies for compound events.
Pacing	Unit 7, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Coins, number cubes, spinners, workbook Lesson 7.16, Unit 7 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Run the cereal-prize simulation 5 times.
Teach	15	Pool results; compute the mean.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 7.16 and the Unit 7 Review.
Exit Ticket	3	Check. Tomorrow: Unit 7 Test.

COMMON MISCONCEPTIONS

- Choosing a tool with the wrong probability.

SUPPORT

- Match probabilities first.

EXTEND

- Run 50 trials and compare.

STUDENT LESSON 7.16 · WITH ANSWERS

Lesson 7.16: Simulations

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 7.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 7.16, Part A.
- [ ] Complete the rest of Workbook Lesson 7.16 (Parts B–D).

LET'S EXPLORE

A cereal box holds one of 4 equally likely prizes. About how many boxes must you buy to get all 4?

Instead of buying cereal, simulate: roll a 4-sided die (or spin a 4-part spinner) where each number stands for a prize. Count rolls until all four appear. Repeat many times.

Trial	Rolls	Boxes needed
1	2, 1, 2, 4, 1, 3	6
2	3, 3, 1, 4, 2	5
3	1, 4, 4, 2, 4, 1, 3	7
4	4, 2, 2, 1, 3	5
5	2, 2, 3, 2, 1, 1, 4	7

WORKED EXAMPLE · ESTIMATE

Mean of the five trials

1 $(6 + 5 + 7 + 5 + 7) \div 5 = 6$.

Answer: about 6 boxes (more trials improve the estimate)

★ KEY IDEA

A simulation matches each real outcome to a random-tool outcome with the same probability.

✎ TRY IT

Answers are shown in color.

1. Experimental P(needing more than 6 boxes) from the table

Answer: $^{2}/_{5}$

2. Which tool simulates a 50% event?

Answer: a coin

3. Which tool simulates a 1-in-6 event?

Answer: a number cube

Workbook Answer Key · Lesson 7.16

#	Question	Answer
1	Experimental P(needing more than 6 boxes) from the table	$^{2}/_{5}$
2	Which tool simulates a 50% event?	a coin
3	Which tool simulates a 1-in-6 event?	a number cube
4	Mean of 6, 5, 7, 5, 7	6
5	What tool has 4 equal outcomes?	4-section spinner
6	A player makes 50% of shots. Design a simulation for P(making 3 in a row).	Flip 3 coins; heads = make; repeat many times
7	Rain is forecast at 30% each day. Design a simulation for P(rain on at least 1 of 2 days).	Use digits 0–9; 0–2 = rain; draw 2 digits per trial
8	In 20 trials of the shot simulation, 3 had three heads. Experimental probability?	$^{3}/_{20}$
9	What is the theoretical probability of three heads?	$^{1}/_{8}$
10	Why do more trials give a better estimate?	Results settle closer to the true probability

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 7 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	Random or biased: surveying shoppers at a bookstore about reading habits	Biased
2	24 of 80 sampled like a show. Estimate for 5,000.	1,500
3	Mean and MAD of 5, 7, 9, 11, 13	mean 9, MAD 2.4
4	Five-number summary of 1, 4, 6, 7, 9, 12, 14	1, 4, 7, 12, 14
5	Spinner 1–10: P(prime)	$\frac{2}{5}$
6	Two coins: P(no heads)	$\frac{1}{4}$
7	Two cubes: P(sum is 6)	$\frac{5}{36}$
8	3 shirts, 4 pants, 2 shoes: outfits	24
9	Design a simulation for a 25% event.	e.g. 4-section spinner; 1 section = success

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 7 Test — Answer Key

Answers

#	Question	Answer
1	A factory tests 100 of 20,000 parts. Name the sample.	the 100 parts tested
2	Asking the first 20 people in line for a roller coaster about favorite rides: random or biased?	Biased
3	9 of 45 sampled students have a summer job. Estimate for 1,200 students.	240
4	Samples: 12, 15, 9, 14 successes out of 30 each. Mean count?	12.5
5	Mean of 8, 10, 11, 13, 18	12
6	MAD of 8, 10, 11, 13, 18	2.8
7	IQR of 5, 6, 9, 10, 12, 15, 16, 20	8
8	Means 52 and 60, MADs 4. MADs apart?	2
9	Bag: 2 red, 3 green, 5 yellow. P(green)	$\frac{3}{10}$
10	P(not yellow)	$\frac{1}{2}$
11	Coin + cube: P(H and 3)	$\frac{1}{12}$
12	Two cubes: P(sum is 9)	$\frac{1}{9}$
13	Three coins: P(at least one head)	$\frac{7}{8}$
14	18 of 30 spins landed on red. Experimental P(red)?	$\frac{3}{5}$
15	Random samples of two orchards: Orchard A apples mean 150 g, MAD 10 g; Orchard B mean 175 g, MAD 10 g. Compare the orchards. Is the difference meaningful? Explain.	Difference 25 g = 2.5 MADs; yes, B's apples tend to be heavier
16	Design a simulation to estimate the probability that a family with 4 children has at least 3 girls. Describe the tool, one trial, and what you record.	Flip 4 coins (H = girl) per trial; record whether ≥ 3 heads; repeat many times; theoretical $\frac{5}{16}$

Scoring

Score	What it means / next step
17–20 (85%+)	Mastered.
14–16	Proficient; correct missed items.

Score	What it means / next step
Below 14	Reteach and retest.

Scale Drawings and Geometric Figures

Scaled copies, scale drawings and maps, drawing triangles from conditions, and cross sections.

Architects, mapmakers, and game designers all work with scale. In this unit you will make and use scale drawings and maps, discover which measurements determine a triangle, and imagine slicing solids to see what shapes appear.

Big ideas

- Find and use scale factors, including their effect on area.
- Use scales to find actual lengths, areas, and distances.
- Redraw scale drawings at a new scale.
- Decide when conditions give one, many, or no triangles.
- Describe cross sections of prisms, pyramids, cylinders, and cones.

Standards

Code	Standard
7.G.1	Solve problems involving scale drawings, including computing actual lengths and areas and reproducing a drawing at a different scale.
7.G.2	Draw geometric shapes with given conditions; focus on triangles from three measures of angles or sides.
7.G.3	Describe the two-dimensional figures that result from slicing three-dimensional figures.

What students should already know

- Ratios and proportional relationships (Unit 1).
- Angles and the protractor (Grade 4 and 5).

Materials for the unit

- Ruler, protractor, grid paper
- Straws, clay, plastic knife

- Tape measure

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	8.1	Scaled Copies
Week 1	Lesson 2 of 4	8.2	Scale Factors
Week 1	Lesson 3 of 4	8.3	Scale Factors and Area
Week 1	Lesson 4 of 4	8.4	Scale Drawings
Week 1	End of week	Check	Weekly Check 22: Scale Factors
Week 2	Lesson 1 of 4	8.5	Finding Drawing Lengths
Week 2	Lesson 2 of 4	8.6	Area in Scale Drawings
Week 2	Lesson 3 of 4	8.7	Maps and Scale
Week 2	Lesson 4 of 4	8.8	Changing the Scale
Week 2	End of week	Check	Weekly Check 23: Scale Drawings and Maps
Week 3	Lesson 1 of 4	8.9	Triangle Side Lengths
Week 3	Lesson 2 of 4	8.10	Triangle Angles
Week 3	Lesson 3 of 4	8.11	Drawing Triangles from Conditions
Week 3	Lesson 4 of 4	8.12	Drawing Quadrilaterals with Conditions
Week 3	End of week	Check	Weekly Check 24: Triangles and Quadrilaterals
Week 4	Lesson 1 of 4	8.13	Cross Sections of Prisms
Week 4	Lesson 2 of 4	8.14	Cross Sections of Pyramids
Week 4	Lesson 3 of 4	8.15	Cross Sections of Cylinders and Cones
Week 4	Lesson 4 of 4	8.16	Scale Drawing Project
Week 4	Before the test	Review	Unit 8 Review
Week 4	End of unit	Test	Unit 8 Test

TEACHING THIS UNIT AT HOME

Look at a road map or a floor plan together and ask your student to find a real distance using the scale.

UNIT VOCABULARY

scale factor	Multiplier for every length.
scale	Drawing length : actual length.
triangle inequality	Two sides sum to more than the third.
cross section	The shape of a slice.

LESSON 8.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.1: Scaled Copies

Student goal	I can tell whether one figure is a scaled copy of another.
Objective	Students identify scaled copies.
Standard	7.G.1 — Solve problems involving scale drawings of geometric figures.
Pacing	Unit 8, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Grid paper, workbook Lesson 8.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Division warm-up.
Launch	5	Discuss Let's Explore.
Teach	15	Compare the L-shapes. Say: “Multiply, don't add.”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking adding the same amount makes a scaled copy.

SUPPORT

- Compute side ratios.

EXTEND

- Enlarge a favorite drawing on grid paper.

STUDENT LESSON 8.1 · WITH ANSWERS

Lesson 8.1: Scaled Copies

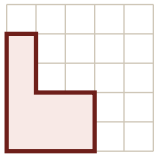
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.1, Part A.
- [ ] Complete the rest of Workbook Lesson 8.1 (Parts B–E).

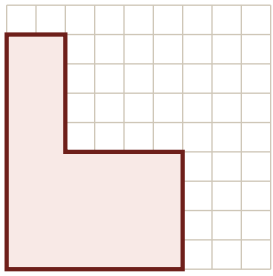
LET'S EXPLORE

Ellie draws a 3-by-4 rectangle, then a 6-by-8 rectangle and a 5-by-6 rectangle. Which one is a scaled copy?

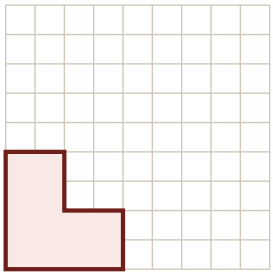
In a scaled copy, every length is multiplied by the same number and every angle stays the same. Adding the same amount to each side does not make a scaled copy.



Original



Copy A: every length × 2



Copy B: not scaled

WORKED EXAMPLE · CHECK THE RECTANGLES

3×4 vs 6×8 and 5×6

1 $6 \div 3 = 2$ and $8 \div 4 = 2$: same factor.

2 $5 \div 3 = {}^{5}/_{3}$ but $6 \div 4 = {}^{3}/_{2}$: different.

Answer: 6×8 is a scaled copy; 5×6 is not

★ KEY IDEA

Scaled copy: all lengths multiplied by the same factor, all angles unchanged.

✎ TRY IT

Answers are shown in color.

1. Is a 4 × 10 rectangle a scaled copy of 2 × 5?

Answer: Yes (× 2)

2. Is a 5 × 7 rectangle a scaled copy of 3 × 5?

Answer: No

3. Is any square a scaled copy of any other square?

Answer: Yes

Workbook Answer Key · Lesson 8.1

#	Question	Answer
1	Is a 4 × 10 rectangle a scaled copy of 2 × 5?	Yes (× 2)
2	Is a 5 × 7 rectangle a scaled copy of 3 × 5?	No
3	Is any square a scaled copy of any other square?	Yes
4	6 ÷ 3	2
5	8 ÷ 4	2
6	2 × 3 and 6 × 9	Yes
7	4 × 6 and 6 × 8	No
8	5 × 10 and 1 × 2	Yes
9	3 × 7 and 9 × 14	No
10	8 × 12 and 2 × 3	Yes
11	1 × 4 and 3 × 7	No
12	(figure)	Sides 9, 6, 6, 6, 3, 12 (each original side × 3)
13	Lillie adds 2 to every side of a triangle. Is the new triangle a scaled copy? Explain.	No; adding changes the ratios, so lengths are not multiplied by one factor.

LESSON 8.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.2: Scale Factors

Student goal	I can find and use a scale factor.
Objective	Students find and apply scale factors.
Standard	7.G.1 — Compute lengths in scale drawings.
Pacing	Unit 8, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 8.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Find the factor, then use it. Enlargement vs reduction.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing original by copy.

SUPPORT

- Say: copy ÷ original.

EXTEND

- Resize a photo on a computer and record the factor.

STUDENT LESSON 8.2 · WITH ANSWERS

Lesson 8.2: Scale Factors

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.2, Part A.
- [ ] Complete the rest of Workbook Lesson 8.2 (Parts B–E).

LET'S EXPLORE

A photo 4 in. by 6 in. is enlarged so the short side is 10 in. How long is the long side?

WORKED EXAMPLE · FIND THE FACTOR

4 in. → 10 in.

1 Scale factor = 10 ÷ 4 = 2.5.

Answer: 2.5

WORKED EXAMPLE · USE THE FACTOR

Long side: 6 × 2.5

1 6 × 2.5 = 15.

Answer: 15 in.

A scale factor greater than 1 makes an enlargement. A scale factor between 0 and 1 makes a reduction. A scale factor of 1 makes a same-size copy.

★ KEY IDEA

Scale factor = copy length ÷ original length.

✎ TRY IT

Answers are shown in color.

1. Original 5 cm, copy 15 cm. Scale factor?

Answer: 3

2. Original 12 ft, copy 3 ft. Scale factor?

Answer: $\frac{1}{4}$

3. Side 8 m, scale factor 0.75. Copy length?

Answer: 6 m

Workbook Answer Key · Lesson 8.2

#	Question	Answer
1	Original 5 cm, copy 15 cm. Scale factor?	3
2	Original 12 ft, copy 3 ft. Scale factor?	$\frac{1}{4}$
3	Side 8 m, scale factor 0.75. Copy length?	6 m
4	10 ÷ 4	2.5
5	6 × 2.5	15
6	3 → 12	4
7	10 → 4	0.4
8	6 → 9	1.5
9	20 → 5	0.25
10	8 → 8	1
11	4 → 14	3.5
12	Original 6 cm, scale factor 3	18 cm
13	Original 12 cm, scale factor $\frac{1}{3}$	4 cm
14	Original 9 cm, scale factor $1\frac{2}{3}$	15 cm
15	Original 14 cm, scale factor 0.5	7 cm
16	Original 10 cm, scale factor 1.5	15 cm
17	Original 16 cm, scale factor 0.75	12 cm
18	A triangle has sides 6, 8, 10. A copy's shortest side is 9. Find the other two sides.	12 and 15 (factor 1.5)

LESSON 8.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.3: Scale Factors and Area

Student goal	I can find how area changes when a figure is scaled.
Objective	Students relate scale factor to area.
Standard	7.G.1 — Compute actual areas from a scale drawing.
Pacing	Unit 8, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	Grid paper, workbook Lesson 8.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Square warm-up.
Launch	5	Count squares in both rectangles.
Teach	15	Show k × k squares fit in each original square.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying area by k instead of k^2.

SUPPORT

- Draw it and count.

EXTEND

- Volume scales by k^3: try with cubes.

STUDENT LESSON 8.3 · WITH ANSWERS

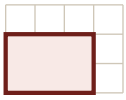
Lesson 8.3: Scale Factors and Area

STUDENT'S ASSIGNMENT

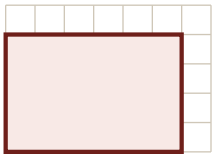
- [ ] Read Lesson 8.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.3, Part A.
- [ ] Complete the rest of Workbook Lesson 8.3 (Parts B–E).

LET'S EXPLORE

A 2-by-3 rectangle is scaled by a factor of 2. Does its area double?



Area 6



Area 24

No! The original area is 6 square units; the copy is 4 × 6 = 24 square units, which is 4 times as much. Length and width both doubled, so area is multiplied by 2 × 2.

WORKED EXAMPLE · FACTOR 3

Area 10 cm², scale factor 3

1 Area factor: $3^2 = 9$.

2 $10 \times 9 = 90$.

Answer: 90 cm²

WORKED EXAMPLE · REDUCTION

Area 48 in², scale factor $\frac{1}{2}$

1 Area factor: $(\frac{1}{2})^2 = \frac{1}{4}$.

2 $48 \times \frac{1}{4} = 12$.

Answer: 12 in²

★ KEY IDEA

If lengths are multiplied by k, area is multiplied by k^2.

✎ TRY IT

Answers are shown in color.

1. Area 5, scale factor 4. New area?

Answer: 80

2. Area 36, scale factor $\frac{1}{3}$

Answer: 4

3. Area changed from 7 to 63. Scale factor?

Answer: 3

Workbook Answer Key · Lesson 8.3

#	Question	Answer
1	Area 5, scale factor 4. New area?	80
2	Area 36, scale factor $\frac{1}{3}$	4
3	Area changed from 7 to 63. Scale factor?	3
4	2^2	4
5	$(\frac{1}{2})^2$	$\frac{1}{4}$
6	Area 8, scale factor 2	32
7	Area 6, scale factor 5	150
8	Area 50, scale factor 0.2	2
9	Area 12, scale factor 1.5	27
10	Area 20, scale factor 0.5	5
11	Area 3, scale factor 10	300
12	A garden 4 m by 5 m is redesigned with every length × 1.5. Find the new area.	45 m²
13	A poster's area is 9 times a flyer's area. What is the scale factor?	3
14	Lillie says scaling by 3 triples the area. Explain her mistake.	Area is multiplied by $3^2 = 9$, not 3.

LESSON 8.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.4: Scale Drawings

Student goal	I can use a scale to find actual lengths from a scale drawing.
Objective	Students use scales to find actual lengths.
Standard	7.G.1 — Solve problems involving scale drawings.
Pacing	Unit 8, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Ruler, workbook Lesson 8.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Double number line for the scale.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing instead of multiplying.

SUPPORT

- Double number line.

EXTEND

- Find the scale on a real map or floor plan.

STUDENT LESSON 8.4 · WITH ANSWERS

Lesson 8.4: Scale Drawings

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.4, Part A.
- [ ] Complete the rest of Workbook Lesson 8.4 (Parts B–D).

LET'S EXPLORE

A floor plan uses the scale 1 in. : 8 ft. A bedroom measures 1.5 in. by 1.75 in. on the plan. How big is the real room?

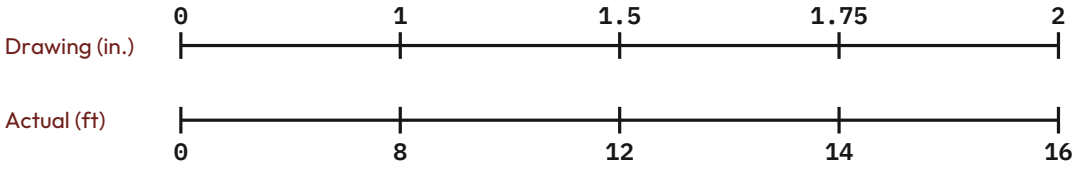
WORKED EXAMPLE · USE THE SCALE

1.5 in. by 1.75 in.; 1 in. : 8 ft

1 1.5 × 8 = 12 ft.

2 1.75 × 8 = 14 ft.

Answer: 12 ft by 14 ft



★ KEY IDEA

Multiply drawing lengths by the scale's actual-per-unit value.

✎ TRY IT

Answers are shown in color.

1. Scale 1 cm : 5 m. Drawing 7 cm. Actual?

Answer: 35 m

2. Scale 1 in. : 20 mi. Drawing 3.5 in. Actual?

Answer: 70 mi

3. Scale 2 cm : 1 m. Drawing 9 cm. Actual?

Answer: 4.5 m

Workbook Answer Key · Lesson 8.4

#	Question	Answer
1	Scale 1 cm : 5 m. Drawing 7 cm. Actual?	35 m
2	Scale 1 in. : 20 mi. Drawing 3.5 in. Actual?	70 mi
3	Scale 2 cm : 1 m. Drawing 9 cm. Actual?	4.5 m
4	1.5 × 8	12
5	1.75 × 8	14
6	Scale 1 in. : 6 ft; drawing 4 in.	24 ft
7	Scale 1 cm : 3 m; drawing 8.5 cm	25.5 m
8	Scale 1 in. : 50 mi; drawing 2.25 in.	112.5 mi
9	Scale 1 cm : 10 km; drawing 6.3 cm	63 km
10	Scale 2 in. : 5 ft; drawing 6 in.	15 ft
11	Scale $1/4$ in. : 1 ft; drawing 3 in.	12 ft
12	A model car uses 1 in. : 2 ft. The model is 7 in. long. How long is the real car?	14 ft

WEEKLY CHECK · ANSWER KEY
After Lesson 8.4

About 15 minutes

Weekly Check 22: Scale Factors — Answer Key

Answers

#	Question	Answer
1	Is 6 × 15 a scaled copy of 2 × 5?	Yes
2	Original 8 cm, copy 20 cm. Scale factor?	2.5
3	Side 12 in., scale factor $^{2}/_{3}$	8 in.
4	Area 7, scale factor 3. New area?	63
5	Scale 1 in. : 5 ft; drawing 6.4 in. Actual?	32 ft
6	Scale 1 cm : 4 m; drawing 3.5 cm. Actual?	14 m

LESSON 8.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.5: Finding Drawing Lengths

Student goal	I can find the length on a scale drawing for a given actual length.
Objective	Students find drawing lengths from actual lengths.
Standard	7.G.1 — Solve problems involving scale drawings.
Pacing	Unit 8, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Grid paper, ruler, workbook Lesson 8.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Divide by the scale value.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying instead of dividing.

SUPPORT

- Ask: should the drawing be smaller?

EXTEND

- Draw your room to scale.

STUDENT LESSON 8.5 · WITH ANSWERS

Lesson 8.5: Finding Drawing Lengths

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.5, Part A.
- [ ] Complete the rest of Workbook Lesson 8.5 (Parts B–E).

LET'S EXPLORE

A soccer field is 100 m long. Using 1 cm : 5 m, how long is it on the drawing?

WORKED EXAMPLE · DIVIDE BY THE SCALE

100 m with 1 cm : 5 m

1 $100 \div 5 = 20$.

Answer: 20 cm

WORKED EXAMPLE · SCALE WITH A FRACTION

A 30-ft wall with $\frac{1}{2}$ in. : 3 ft

1 $30 \div 3 = 10$ groups of 3 ft.

2 $10 \times \frac{1}{2}$ in. = 5 in.

Answer: 5 in.

★ KEY IDEA

To go from actual to drawing, divide by how much 1 drawing unit represents.

✎ TRY IT

Answers are shown in color.

1. Scale 1 in. : 4 ft. Actual 22 ft. Drawing?

Answer: 5.5 in.

2. Scale 1 cm : 20 km. Actual 130 km. Drawing?

Answer: 6.5 cm

3. Scale 2 cm : 3 m. Actual 12 m. Drawing?

Answer: 8 cm

Workbook Answer Key · Lesson 8.5

#	Question	Answer
1	Scale 1 in. : 4 ft. Actual 22 ft. Drawing?	5.5 in.
2	Scale 1 cm : 20 km. Actual 130 km. Drawing?	6.5 cm
3	Scale 2 cm : 3 m. Actual 12 m. Drawing?	8 cm
4	100 ÷ 5	20
5	$10 \times {}^{1}\!/_{2}$	5
6	Scale 1 in. : 10 ft; actual 45 ft	4.5 in.
7	Scale 1 cm : 2 m; actual 15 m	7.5 cm
8	Scale 1 in. : 25 mi; actual 175 mi	7 in.
9	Scale 1 cm : 50 km; actual 325 km	6.5 cm
10	Scale 3 cm : 1 m; actual 4 m	12 cm
11	Scale ${}^{1}\!/_{4}$ in. : 1 ft; actual 18 ft	4.5 in.
12	Lillie's room is 12 ft by 10 ft. Draw it using the scale 1 square : 2 ft. What are the drawing dimensions? (figure)	6 squares by 5 squares
13	A 60-ft tree is drawn with 1 in. : 15 ft. How tall is the drawing?	4 in.

LESSON 8.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.6: Area in Scale Drawings

Student goal	I can find actual areas from scale drawings.
Objective	Students compute actual areas from scale drawings.
Standard	7.G.1 — Compute actual lengths and areas from a scale drawing.
Pacing	Unit 8, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 8.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Two methods.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Multiplying area by the scale once.

SUPPORT

- Convert lengths first.

EXTEND

- Estimate the cost of flooring for a room at home.

STUDENT LESSON 8.6 · WITH ANSWERS

Lesson 8.6: Area in Scale Drawings

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.6, Part A.
- [ ] Complete the rest of Workbook Lesson 8.6 (Parts B–E).

LET'S EXPLORE

A park drawing is 4 cm by 6 cm with 1 cm : 50 m. What is the park's actual area?

WORKED EXAMPLE · CONVERT LENGTHS FIRST

4 cm by 6 cm; 1 cm : 50 m

1 Actual: 4 × 50 = 200 m and 6 × 50 = 300 m.

2 Area: 200 × 300 = 60,000 m².

Answer: 60,000 m²

WORKED EXAMPLE · AREA FACTOR

Same park

1 Drawing area: 24 cm².

2 Each cm² represents 50 × 50 = 2,500 m².

3 24 × 2,500 = 60,000.

Answer: 60,000 m²

! WATCH OUT

Do not multiply the drawing area by 50. Each square centimeter stands for 50 m × 50 m.

★ KEY IDEA

Convert each length first, then compute the area. Or multiply drawing area by (scale)².

✎ TRY IT

Answers are shown in color.

1. Drawing 3 in. × 5 in.; 1 in. : 4 ft. Actual area?

Answer: 240 ft²

2. Drawing 2 cm × 2 cm; 1 cm : 10 m. Actual area?

Answer: 400 m²

3. Drawing area 6 in²; 1 in. : 3 ft. Actual area?

Answer: 54 ft²

Workbook Answer Key · Lesson 8.6

#	Question	Answer
1	Drawing 3 in. × 5 in.; 1 in. : 4 ft. Actual area?	240 ft²
2	Drawing 2 cm × 2 cm; 1 cm : 10 m. Actual area?	400 m²
3	Drawing area 6 in²; 1 in. : 3 ft. Actual area?	54 ft²
4	200 × 300	60,000
5	50 × 50	2,500
6	Drawing 2 × 3 in.; 1 in. : 5 ft	150 ft²
7	Drawing 5 × 8 cm; 1 cm : 2 m	160 m²
8	Drawing 3/2 × 4 in.; 1 in. : 6 ft	216 ft²
9	Drawing 5/2 × 7/2 cm; 1 cm : 10 m	875 m²
10	A rectangular rug on a plan (1 in. : 2 ft) measures 2.5 in. by 4 in. Carpet costs $3 per ft². Cost of the rug?	$120
11	A plan uses 1 cm : 3 m. A room is 4 cm by 5 cm on the plan. Find the actual area.	180 m²

LESSON 8.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.7: Maps and Scale

Student goal	I can use a map scale to find real distances and travel times.
Objective	Students use map scales.
Standard	7.G.1, 7.RP.2 — Solve problems involving scale drawings; use proportional reasoning.
Pacing	Unit 8, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	A real road map (optional), workbook Lesson 8.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Distance, then time. Unitless scales.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting to convert cm to km.

SUPPORT

- Write the unit at every step.

EXTEND

- Plan a trip using a real map.

STUDENT LESSON 8.7 · WITH ANSWERS

Lesson 8.7: Maps and Scale

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.7, Part A.
- [ ] Complete the rest of Workbook Lesson 8.7 (Parts B–E).

LET'S EXPLORE

On a map with scale 1 cm : 25 km, two cities are 7.2 cm apart. How far apart are they? How long would a 90 km/h drive take?

WORKED EXAMPLE · DISTANCE

7.2 cm; 1 cm : 25 km

1 7.2 × 25 = 180.

Answer: 180 km

WORKED EXAMPLE · TIME

180 km at 90 km/h

1 180 ÷ 90 = 2.

Answer: 2 hours

WORKED EXAMPLE · SCALE WITHOUT UNITS

Scale 1 : 50,000. Map distance 6 cm.

1 Actual = 6 × 50,000 = 300,000 cm.

2 300,000 cm = 3,000 m = 3 km.

Answer: 3 km

★ KEY IDEA

A scale like 1 : 50,000 has no units: 1 of any unit on the map is 50,000 of the same unit in real life.

✎ TRY IT

Answers are shown in color.

1. Map 1 in. : 30 mi. Distance 4.5 in.?

Answer: 135 mi

2. Scale 1 : 100,000. Map 4 cm. Actual km?

Answer: 4 km

3. 150 mi at 60 mph: hours?

Answer: 2.5 hours

Workbook Answer Key · Lesson 8.7

#	Question	Answer
1	Map 1 in. : 30 mi. Distance 4.5 in.?	135 mi
2	Scale 1 : 100,000. Map 4 cm. Actual km?	4 km
3	150 mi at 60 mph: hours?	2.5 hours
4	7.2 × 25	180
5	300,000 cm in km	3 km
6	Scale 1 cm : 40 km; map 3.5 cm	140 km
7	Scale 1 in. : 15 mi; map 6 in.	90 mi
8	Scale 1 : 25,000; map 8 cm	2 km
9	Scale 1 : 200,000; map 5 cm	10 km
10	On a map (1 in. : 50 mi), Ellie's trip is 4.8 in. She drives 60 mph. How long is the drive?	240 mi; 4 hours
11	A hiking map is 1 : 24,000. A trail is 15 cm on the map. How many kilometers is it?	3.6 km
12	Two towns are 75 km apart. With 1 cm : 15 km, how far apart on the map?	5 cm

LESSON 8.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.8: Changing the Scale

Student goal	I can reproduce a scale drawing at a different scale.
Objective	Students redraw at a different scale.
Standard	7.G.1 — Reproduce a scale drawing at a different scale.
Pacing	Unit 8, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Grid paper, workbook Lesson 8.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Actual first, then redraw.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Scaling the wrong direction.

SUPPORT

- Ask: does each cm represent more or less now?

EXTEND

- Redraw a floor plan for a smaller sheet of paper.

STUDENT LESSON 8.8 · WITH ANSWERS

Lesson 8.8: Changing the Scale

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.8, Part A.
- [ ] Complete the rest of Workbook Lesson 8.8 (Parts B–D).

LET'S EXPLORE

A garden plan uses 1 cm : 2 m and is 10 cm by 6 cm. Redraw it at 1 cm : 4 m. What size is the new drawing?

WORKED EXAMPLE · FIND ACTUAL, THEN REDRAW

10 cm × 6 cm at 1 cm : 2 m

1 Actual: 20 m by 12 m.

2 New drawing at 1 cm : 4 m: 20 ÷ 4 = 5 cm; 12 ÷ 4 = 3 cm.

Answer: 5 cm by 3 cm

WORKED EXAMPLE · COMPARE SCALES

1 cm : 2 m vs 1 cm : 4 m

1 Each cm now stands for twice as much.

2 So the new drawing is $^{1}/_{2}$ the size.

Answer: drawing lengths × $^{1}/_{2}$

★ KEY IDEA

Find actual lengths first, then convert to the new scale.

✎ TRY IT

Answers are shown in color.

1. Plan 8 in. at 1 in. : 5 ft; redraw at 1 in. : 10 ft

Answer: 4 in.

2. Plan 3 cm at 1 cm : 6 m; redraw at 1 cm : 2 m

Answer: 9 cm

3. Plan 12 cm at 1 cm : 1 m; redraw at 1 cm : 3 m

Answer: 4 cm

Workbook Answer Key · Lesson 8.8

#	Question	Answer
1	Plan 8 in. at 1 in. : 5 ft; redraw at 1 in. : 10 ft	4 in.
2	Plan 3 cm at 1 cm : 6 m; redraw at 1 cm : 2 m	9 cm
3	Plan 12 cm at 1 cm : 1 m; redraw at 1 cm : 3 m	4 cm
4	10 × 2	20
5	20 ÷ 4	5
6	6 in. at 1 in. : 4 ft; new scale 1 in. : 8 ft	3 in.
7	5 cm at 1 cm : 10 m; new scale 1 cm : 5 m	10 cm
8	9 in. at 1 in. : 2 ft; new scale 1 in. : 3 ft	6 in.
9	4 cm at 1 cm : 3 m; new scale 1 cm : 2 m	6 cm
10	A rectangle 8 cm × 4 cm uses 1 cm : 3 m. Redraw it at 1 cm : 6 m on the grid (1 square = 1 cm). (figure)	4 squares by 2 squares

WEEKLY CHECK · ANSWER KEY
After Lesson 8.8
About 15 minutes

Weekly Check 23: Scale Drawings and Maps — Answer Key

Answers

#	Question	Answer
1	Scale 1 in. : 8 ft; actual 36 ft. Drawing?	4.5 in.
2	Drawing 3 cm × 4 cm; 1 cm : 5 m. Actual area?	300 m²
3	Map 1 cm : 30 km; 4.5 cm. Actual?	135 km
4	Scale 1 : 50,000; map 4 cm. Actual km?	2 km
5	Drawing 10 cm at 1 cm : 2 m. Redraw at 1 cm : 5 m.	4 cm
6	180 mi at 45 mph: hours?	4 hours

LESSON 8.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.9: Triangle Side Lengths

Student goal	I can decide whether three side lengths can form a triangle.
Objective	Students apply the triangle inequality.
Standard	7.G.2 — Draw geometric shapes with given conditions; notice when conditions determine a unique triangle, more than one, or none.
Pacing	Unit 8, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Straws or strips of paper, ruler, scissors, workbook Lesson 8.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Build triangles from straws (cut to 2, 3, 6 and 3, 4, 5 in.).
Teach	15	State the rule.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Checking only one pair of sides.

SUPPORT

- Sort the lengths first.

EXTEND

- Find all whole-number triangles with perimeter 12.

STUDENT LESSON 8.9 · WITH ANSWERS

Lesson 8.9: Triangle Side Lengths

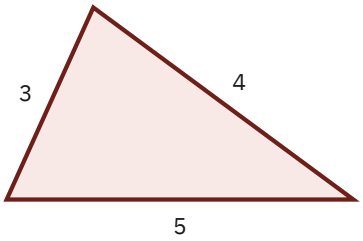
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.9, Part A.
- [ ] Complete the rest of Workbook Lesson 8.9 (Parts B–E).

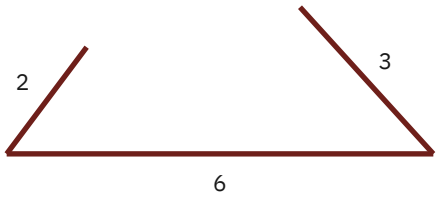
LET'S EXPLORE

Can you make a triangle with straws 2 in., 3 in., and 6 in. long?

No! The two short straws together reach only 5 in., which is shorter than 6 in. They can't meet.



3, 4, 5: forms a triangle



2, 3, 6: the sides can't meet

WORKED EXAMPLE · TEST

4, 7, 9

1 Shortest two: 4 + 7 = 11.

2 11 > 9 ✓.

Answer: Yes, a triangle

WORKED EXAMPLE · TEST

5, 5, 10

1 5 + 5 = 10, not greater than 10.

Answer: No triangle (it lies flat)

★ KEY IDEA

Check only the two shortest sides: their sum must be greater than the longest side.

✎ TRY IT

Answers are shown in color.

1. 3, 3, 5

Answer: Yes

2. 1, 2, 4

Answer: No

3. 8, 8, 8

Answer: Yes

Workbook Answer Key · Lesson 8.9

#	Question	Answer
1	3, 3, 5	Yes
2	1, 2, 4	No
3	8, 8, 8	Yes
4	2 + 3	5
5	Is 5 > 6?	No
6	5, 6, 7	Yes
7	2, 2, 5	No
8	10, 4, 7	Yes
9	3, 9, 6	No
10	12, 5, 13	Yes
11	1.5, 2.5, 3.5	Yes
12	Two sides are 5 cm and 8 cm. What lengths could the third side be?	between 3 cm and 13 cm
13	Two sides are 10 in. and 4 in. Can the third side be 6 in.? 7 in.?	6: no; 7: yes
14	Explain why 4, 5, 10 cannot form a triangle.	4 + 5 = 9, which is less than 10.

LESSON 8.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.10: Triangle Angles

Student goal	I can use the angle sum of a triangle to decide whether angles can form a triangle.
Objective	Students use the triangle angle sum.
Standard	7.G.2 — Notice when conditions determine one triangle, more than one, or none.
Pacing	Unit 8, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Paper triangles, scissors, workbook Lesson 8.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Tear corners off a triangle.
Teach	15	Missing-angle examples; AAA gives many triangles.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting the right angle is 90°.

SUPPORT

- Label all three angles.

EXTEND

- Can a triangle have two obtuse angles? Why not?

STUDENT LESSON 8.10 · WITH ANSWERS

Lesson 8.10: Triangle Angles

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.10, Part A.
- [ ] Complete the rest of Workbook Lesson 8.10 (Parts B–E).

LET'S EXPLORE

Can a triangle have angles of 90°, 60°, and 40°?

Tear the three corners off any paper triangle and line them up: they always form a straight angle, 180°.

WORKED EXAMPLE · CHECK

90°, 60°, 40°

1 $90 + 60 + 40 = 190 \neq 180$.

Answer: No triangle

WORKED EXAMPLE · MISSING ANGLE

Two angles are 48° and 75°

1 $180 - 48 - 75 = 57$.

Answer: 57°

WORKED EXAMPLE · HOW MANY TRIANGLES?

Angles 50°, 60°, 70°

1 They sum to 180°.

2 Triangles of many sizes have these angles (scaled copies).

Answer: Many triangles

★ KEY IDEA

Angles alone can't fix a triangle's size: three angles that sum to 180° give infinitely many triangles.

✎ TRY IT

Answers are shown in color.

1. 30°, 60°, 90°: triangle?

Answer: Yes

2. Two angles 110° and 80°: triangle?

Answer: No (190° already)

3. Missing angle: 35°, 35°, ?

Answer: 110°

Workbook Answer Key · Lesson 8.10

#	Question	Answer
1	30°, 60°, 90°: triangle?	Yes
2	Two angles 110° and 80°: triangle?	No (190° already)
3	Missing angle: 35°, 35°, ?	110°
4	180 − 48 − 75	57
5	90 + 60 + 40	190
6	60° and 60°	60°
7	90° and 25°	65°
8	120° and 35°	25°
9	45° and 45°	90°
10	100° and 47°	33°
11	15° and 82°	83°
12	70°, 70°, 40°	Yes
13	90°, 90°, 10°	No
14	100°, 40°, 50°	No
15	33°, 57°, 90°	Yes
16	A right triangle has one 38° angle. Find the third angle.	52°

LESSON 8.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.11: Drawing Triangles from Conditions

Student goal	I can draw triangles with a ruler and protractor and tell whether the conditions give one, many, or no triangles.
Objective	Students draw triangles from conditions.
Standard	7.G.2 — Draw triangles from three measures of angles or sides; notice when conditions determine a unique triangle.
Pacing	Unit 8, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Ruler, protractor, workbook Lesson 8.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Draw the SAS triangle; compare.
Teach	15	Teach the table. Model drawing ASA.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Measuring the angle from the wrong scale on the protractor.

SUPPORT

- Start from 0 on the baseline.

EXTEND

- Explore SSA: two sides and a non-included angle can give two triangles!

STUDENT LESSON 8.11 · WITH ANSWERS

Lesson 8.11: Drawing Triangles from Conditions

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.11, Part A.
- [ ] Complete the rest of Workbook Lesson 8.11 (Parts B–E).

LET'S EXPLORE

Draw a triangle with sides 5 cm and 7 cm and a 40° angle between them. Will your triangle match a classmate's?

Yes. When two sides and the angle between them are given, there is only one possible third side, so the triangle is unique.

Given	Example	How many triangles?
3 sides (that pass the inequality)	5, 6, 7 cm	one
2 sides + angle between them	5 cm, 40°, 7 cm	one
2 angles + side between them	50°, 6 cm, 60°	one
3 angles that sum to 180°	50°, 60°, 70°	many
Conditions that break a rule	2, 3, 6 cm; or 100°, 90°	none

WORKED EXAMPLE · STEPS TO DRAW SAS

5 cm, 40°, 7 cm

1 Draw a 7 cm segment.
2 At one end, measure 40° with a protractor and draw a ray.
3 Mark 5 cm on the ray.
4 Connect the endpoints.

Answer: one triangle

★ KEY IDEA

SSS, SAS, and ASA give one triangle. AAA gives many. Broken rules give none.

✎ TRY IT

Answers are shown in color.

1. Sides 4, 4, 4 cm

Answer: one

2. Angles 45°, 45°, 90°

Answer: many

3. Angles 100°, 100°, ?

Answer: none

Workbook Answer Key · Lesson 8.11

#	Question	Answer
1	Sides 4, 4, 4 cm	one
2	Angles 45°, 45°, 90°	many
3	Angles 100°, 100°, ?	none
4	Is 5 + 6 > 7?	Yes
5	50 + 60 + 70	180
6	sides 3, 4, 8 cm	none
7	sides 6, 8, 10 cm	one
8	angles 30°, 60°, 90°	many
9	sides 5 and 9 cm with a 70° angle between	one
10	angles 40° and 80° with a 6 cm side between	one
11	angles 95°, 95°, ?	none
12	Use a ruler and protractor to draw a triangle with sides 6 cm and 4 cm and a 60° angle between them. Measure the third side.	about 5.3 cm
13	Why do three angles not give a unique triangle?	Scaled copies have the same angles but different sizes.

LESSON 8.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.12: Drawing Quadrilaterals with Conditions

Student goal	I can draw quadrilaterals with given conditions and decide whether they are unique.
Objective	Students draw quadrilaterals with conditions.
Standard	7.G.2 — Draw geometric shapes with given conditions.
Pacing	Unit 8, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Ruler, protractor, straws and pipe cleaners (optional), workbook Lesson 8.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Push a straw square into a rhombus.
Teach	15	Work examples.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking 4 equal sides means a square.

SUPPORT

- Build with straws.

EXTEND

- Which conditions make a kite unique?

STUDENT LESSON 8.12 · WITH ANSWERS

Lesson 8.12: Drawing Quadrilaterals with Conditions

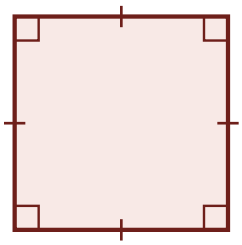
STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.12, Part A.
- [ ] Complete the rest of Workbook Lesson 8.12 (Parts B–E).

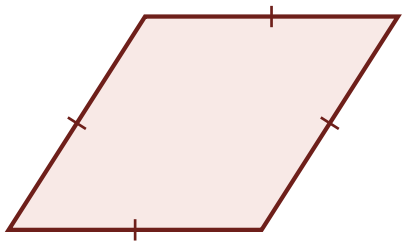
LET'S EXPLORE

Draw a quadrilateral with all sides 4 cm. Is there only one?

No: a square works, but so does any rhombus with 4 cm sides. Push a square's corner and it leans into a rhombus. Quadrilaterals need more conditions than triangles to be unique.



square



rhombus

WORKED EXAMPLE · UNIQUE?

A rectangle 5 cm by 3 cm

1 Right angles fix the shape; lengths fix the size.

Answer: one rectangle

★ KEY IDEA

Side lengths alone don't fix a quadrilateral. Angles help.

✎ TRY IT

Answers are shown in color.

1. Rectangle with sides 6 and 2: how many?

Answer: one

2. Parallelogram with sides 6 and 2: how many?

Answer: many

3. Square with sides 3 cm: how many?

Answer: one

Workbook Answer Key · Lesson 8.12

#	Question	Answer
1	Rectangle with sides 6 and 2: how many?	one
2	Parallelogram with sides 6 and 2: how many?	many
3	Square with sides 3 cm: how many?	one
4	Angles in a rectangle	four 90°
5	Angle sum of a quadrilateral	360°
6	square, side 5 cm	one
7	rhombus, side 5 cm	many
8	rectangle, 4 cm by 7 cm	one
9	quadrilateral with sides 3, 4, 5, 6 cm	many
10	Draw a parallelogram with sides 5 cm and 3 cm and a 60° angle. Is it unique?	Yes; the angle fixes the shape
11	A quadrilateral has angles 90°, 90°, 90°, and ? Find the fourth angle.	90°

WEEKLY CHECK · ANSWER KEY
After Lesson 8.12
About 15 minutes

Weekly Check 24: Triangles and Quadrilaterals — Answer Key

Answers

#	Question	Answer
1	Triangle? 4, 6, 11	No
2	Triangle? 7, 8, 12	Yes
3	Missing angle: 64° and 71°	45°
4	Angles 50°, 60°, 70°: one, many, or none?	many
5	Sides 5 cm, 7 cm with 45° between: one, many, or none?	one
6	Rhombus with side 4 cm: one or many?	many

LESSON 8.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.13: Cross Sections of Prisms

Student goal	I can describe the cross sections made by slicing a rectangular prism.
Objective	Students describe prism cross sections.
Standard	7.G.3 — Describe the two-dimensional figures that result from slicing three-dimensional figures.
Pacing	Unit 8, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Clay or a block of cheese and a plastic knife (optional), workbook Lesson 8.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Slice a clay block.
Teach	15	Parallel vs perpendicular slices.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Naming a 3-D shape instead of the 2-D face.

SUPPORT

- Look at the cut face only.

EXTEND

- Find a hexagonal slice of a cube.

STUDENT LESSON 8.13 · WITH ANSWERS

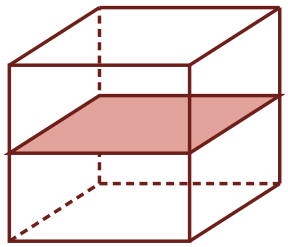
Lesson 8.13: Cross Sections of Prisms

STUDENT'S ASSIGNMENT

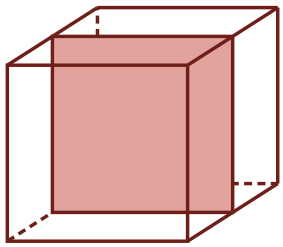
- [ ] Read Lesson 8.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.13, Part A.
- [ ] Complete the rest of Workbook Lesson 8.13 (Parts B–D).

LET'S EXPLORE

Slice a block of cheese straight across, parallel to the bottom. What shape is the cut face?



Parallel to the base: a rectangle the same size as the base



Perpendicular to the base: a rectangle

WORKED EXAMPLE · SLICE A CUBE

Cube, slice parallel to a face

1 The cut matches the face.

Answer: a square

WORKED EXAMPLE · SLANTED SLICE

Cube, slice through one corner

1 Cutting off one corner makes three cut edges.

Answer: a triangle

★ KEY IDEA

Slices of a rectangular prism parallel or perpendicular to the base are rectangles. Slanted slices can make triangles and other polygons.

✎ TRY IT

Answers are shown in color.

1. Slice of a 4 × 6 × 10 box parallel to the 4 × 6 base

Answer: 4 × 6 rectangle

2. Vertical slice of a cube

Answer: square

3. Slice cutting off one corner of a cube

Answer: triangle

Workbook Answer Key · Lesson 8.13

#	Question	Answer
1	Slice of a 4 × 6 × 10 box parallel to the 4 × 6 base	4 × 6 rectangle
2	Vertical slice of a cube	square
3	Slice cutting off one corner of a cube	triangle
4	How many faces does a cube have?	6
5	Shape of each face of a cube	square
6	A box 3 × 5 × 8, sliced parallel to the 3 × 5 face	3 × 5 rectangle
7	A cube sliced diagonally from one top edge to the opposite bottom edge	rectangle
8	A loaf of bread sliced like usual	rectangle (the loaf's end shape)
9	A triangular prism sliced parallel to its triangle base	triangle
10	Can a slice of a cube be a pentagon? Try to sketch one.	Yes, a slanted plane through 5 faces

LESSON 8.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.14: Cross Sections of Pyramids

Student goal	I can describe the cross sections made by slicing a pyramid.
Objective	Students describe pyramid cross sections.
Standard	7.G.3 — Describe plane sections of right rectangular pyramids.
Pacing	Unit 8, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Clay pyramid (optional), workbook Lesson 8.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Slice a clay pyramid.
Teach	15	Horizontal vs vertical.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking the slice is the same size as the base.

SUPPORT

- Compare with the base.

EXTEND

- Sketch slices at 1/4 and 3/4 height.

STUDENT LESSON 8.14 · WITH ANSWERS

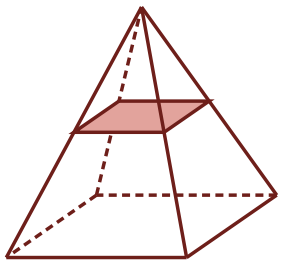
Lesson 8.14: Cross Sections of Pyramids

STUDENT'S ASSIGNMENT

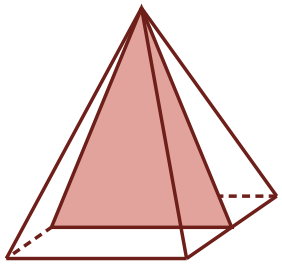
- [ ] Read Lesson 8.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.14, Part A.
- [ ] Complete the rest of Workbook Lesson 8.14 (Parts B–D).

LET'S EXPLORE

Slice a square pyramid parallel to its base. What shape is the cut? What if you slice straight down through the top?



Parallel to the base: a smaller square



Perpendicular through the top: a triangle

WORKED EXAMPLE · SIZE CHANGES

Slice parallel to the base, halfway up

1 Each side of the cut is half the base side.

Answer: a square half as wide

★ KEY IDEA

Parallel slices of a pyramid have the base's shape but get smaller toward the top. Vertical slices through the top are triangles.

✎ TRY IT

Answers are shown in color.

1. Square pyramid, horizontal slice

Answer: square

2. Triangular pyramid, slice parallel to base

Answer: triangle

3. Square pyramid, vertical slice not through the top

Answer: trapezoid

Workbook Answer Key · Lesson 8.14

#	Question	Answer
1	Square pyramid, horizontal slice	square
2	Triangular pyramid, slice parallel to base	triangle
3	Square pyramid, vertical slice not through the top	trapezoid
4	Shape of a square pyramid's base	square
5	Shape of its side faces	triangles
6	Square pyramid, slice parallel to the base near the top	small square
7	Square pyramid, vertical slice through the apex	triangle
8	Rectangular pyramid, horizontal slice	rectangle
9	Triangular pyramid, horizontal slice	triangle
10	A square pyramid's base is 10 in. on a side. A slice halfway up and parallel to the base: what shape and size?	a 5 in. by 5 in. square

LESSON 8.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4

Unit 8: Scale Drawings and Geometric Figures

Lesson 8.15: Cross Sections of Cylinders and Cones

Student goal	I can describe the cross sections of cylinders and cones.
Objective	Students describe cross sections of round solids.
Standard	7.G.3 — Describe the two-dimensional figures that result from slicing three-dimensional figures.
Pacing	Unit 8, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Clay, a can, an ice-cream cone (optional), workbook Lesson 8.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Slice clay cylinders and cones.
Teach	15	Name each cut.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Calling a circle a sphere.

SUPPORT

- The cut face is flat.

EXTEND

- What slice of a cylinder makes an oval?

STUDENT LESSON 8.15 · WITH ANSWERS

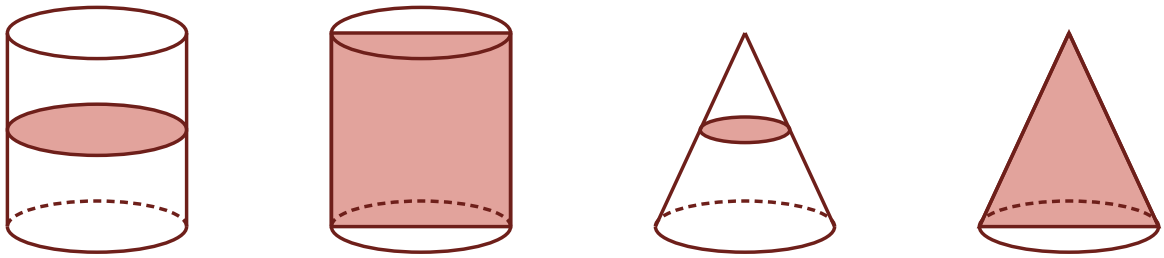
Lesson 8.15: Cross Sections of Cylinders and Cones

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.15, Part A.
- [ ] Complete the rest of Workbook Lesson 8.15 (Parts B–D).

LET'S EXPLORE

What shape do you see when you cut a can straight across? Straight down the middle?



circle rectangle smaller circle triangle

★ KEY IDEA

Horizontal slices of cylinders and cones are circles. Vertical slices through the center are rectangles (cylinder) or triangles (cone).

✎ TRY IT

Answers are shown in color.

1. Slice of a cylinder parallel to the base

Answer: circle

2. Vertical slice of a cone through its tip

Answer: triangle

3. Vertical slice of a cylinder through its center

Answer: rectangle

Workbook Answer Key · Lesson 8.15

#	Question	Answer
1	Slice of a cylinder parallel to the base	circle
2	Vertical slice of a cone through its tip	triangle
3	Vertical slice of a cylinder through its center	rectangle
4	Shape of a cylinder's bases	circles
5	Shape of a cone's base	circle
6	Cylinder, horizontal	circle
7	Cylinder, vertical through center	rectangle
8	Cone, horizontal	circle
9	Cone, vertical through tip	triangle
10	Cylinder (r = 3, h = 10), vertical through center: dimensions	6 × 10 rectangle
11	Sphere, any slice	circle
12	A cylinder has diameter 4 in. and height 9 in. Describe its vertical cross section through the center.	a rectangle 4 in. by 9 in.

LESSON 8.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 8: Scale Drawings and Geometric Figures

Lesson 8.16: Scale Drawing Project

Student goal	I can create a scale drawing of a real space and use it to solve problems.
Objective	Students create a scale drawing of a real room.
Standard	7.G.1 — Solve problems involving scale drawings; reproduce a drawing at a scale.
Pacing	Unit 8, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Tape measure, ruler, grid paper, workbook Lesson 8.16, Unit 8 Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Measure the room together.
Teach	15	Choose a scale; draw.
Guided Practice	10	Project parts 1–4.
Independent	12	Workbook Lesson 8.16 and the Unit 8 Review.
Exit Ticket	3	Present the drawing. Tomorrow: Unit 8 Test.

COMMON MISCONCEPTIONS

- Choosing a scale that doesn't fit.

SUPPORT

- Check the largest length first.

EXTEND

- Redesign the room layout with cut-out furniture.

STUDENT LESSON 8.16 · WITH ANSWERS

Lesson 8.16: Scale Drawing Project

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 8.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 8.16, Part A.
- [ ] Complete the rest of Workbook Lesson 8.16 (Parts B–D).

LET'S EXPLORE

Design a scale floor plan of your bedroom or kitchen.

Measure the room and its furniture. Choose a scale that fits your paper (for example, 1 in. : 2 ft or 1 cm : 50 cm). Draw the room and furniture to scale, and label each length.

WORKED EXAMPLE · SAMPLE

A room 14 ft by 12 ft on 8.5 × 11 in. paper

1 1 in. : 2 ft gives 7 in. by 6 in.: fits!

2 A 6 ft bed becomes 3 in.

Answer: Scale 1 in. : 2 ft

Object	Actual size	Drawing size (1 in. : 2 ft)
Room	14 ft × 12 ft	7 in. × 6 in.
Bed	6 ft × 3.5 ft	3 in. × 1.75 in.
Desk	4 ft × 2 ft	2 in. × 1 in.

★ KEY IDEA

Choose a scale so the biggest length fits your paper.

✎ TRY IT

Answers are shown in color.

1. Rug 5 ft × 8 ft at 1 in. : 2 ft

Answer: 2.5 in. × 4 in.

2. Window 3 ft wide at 1 in. : 2 ft

Answer: 1.5 in.

3. Area of the room above

Answer: 168 ft²

Workbook Answer Key · Lesson 8.16

#	Question	Answer
1	Rug 5 ft × 8 ft at 1 in. : 2 ft	2.5 in. × 4 in.
2	Window 3 ft wide at 1 in. : 2 ft	1.5 in.
3	Area of the room above	168 ft²
4	14 ÷ 2	7
5	14 × 12	168
6	Measure your room. Length and width:	Answers vary.
7	Choose a scale:	Answers vary.
8	Draw your room and at least three objects on the grid. (figure)	Drawing
9	Find the floor area of your room.	Answers vary.
10	Why did you choose your scale?	Answers vary: the room fits the paper.

UNIT REVIEW · ANSWER KEY
Study before the unit test

About 40 minutes

Unit 8 Review — Answer Key

Homework or a study session before the test.

Answers

#	Question	Answer
1	Scale factor from 15 to 6	0.4
2	Area 20, scale factor 1.5	45
3	Plan 1 in. : 3 ft; room 5 in. × 4.5 in. Actual size?	15 ft × 13.5 ft
4	Same room: actual area?	202.5 ft²
5	Map 1 : 100,000; 7.5 cm. Actual km?	7.5 km
6	Redraw 9 cm at 1 cm : 4 m using 1 cm : 6 m	6 cm
7	Two sides 6 and 9: possible third sides	between 3 and 15
8	Missing angle: 90° and 27°	63°
9	ASA: one, many, or none?	one
10	Cross section of a cylinder, horizontal	circle
11	Cross section of a square pyramid through the apex, vertical	triangle
12	Cross section of a 2 × 3 × 7 box parallel to the 2 × 3 face	2 × 3 rectangle

UNIT TEST · ANSWER KEY
At the end of the unit

About 45 minutes

Unit 8 Test — Answer Key

Answers

#	Question	Answer
1	Is 9 × 12 a scaled copy of 3 × 4?	Yes
2	Scale factor from 10 to 25	2.5
3	Side 18 cm, scale factor $^1/_3$	6 cm
4	Area 12, scale factor 2	48
5	Scale 1 in. : 6 ft; drawing 4.5 in. Actual?	27 ft
6	Scale 1 cm : 25 km; actual 200 km. Drawing?	8 cm
7	Triangle? 3, 5, 9	No
8	Missing angle: 38° and 106°	36°
9	Sides 6, 8, 10: one, many, or none?	one
10	Angles 45°, 45°, 90°: one, many, or none?	many
11	Cone, horizontal slice	circle
12	Cube, slice parallel to a face	square
13	A floor plan uses 1 in. : 4 ft. A living room is 5 in. by 3.5 in. on the plan. Find the actual dimensions and area. Carpet costs $2.50 per square foot: how much to carpet the room?	20 ft × 14 ft; 280 ft²; $700
14	A map uses 1 cm : 20 km. Two towns are 9 cm apart. Ellie drives 60 km/h. How long is the drive? Show your work.	180 km; 3 hours
15	Use a ruler and protractor to draw a triangle with a 6 cm side between angles of 50° and 70°. Find the third angle. Is the triangle unique?	60°; yes (ASA)

Scoring

Score	What it means / next step
18–21 (85%+)	Mastered.
15–17	Proficient; correct missed items.
Below 15	Reteach and retest.

Angles, Circles, Area, and Volume

Angle relationships, circles, composite area, surface area and volume of prisms, and a review of the whole year.

Geometry is everywhere: wheels, windows, pools, and boxes. In this final unit you will use angle facts to find unknown angles, discover π, find areas of composite figures, and measure the surface area and volume of prisms. The last week reviews the whole year.

Big ideas

- Use complementary, supplementary, vertical, and adjacent angles to find unknowns.
- Find the circumference and area of circles.
- Find areas of composite figures.
- Find surface area and volume of prisms.
- Review and connect every unit from the year.

Standards

Code	Standard
7.G.4	Know and use the formulas for the area and circumference of a circle.
7.G.5	Use facts about supplementary, complementary, vertical, and adjacent angles.
7.G.6	Solve problems involving area, volume, and surface area of two- and three-dimensional objects.

What students should already know

- Equations (Unit 6).
- Scale and cross sections (Unit 8).
- Area of rectangles and triangles (Grade 6).

Materials for the unit

- String, ruler, protractor, round objects
- Paper circle and scissors

- Empty boxes, unit cubes

Pacing guide

Teach the lessons in order, then give the week's check. Set the days to fit your own calendar.

Week	Order	Lesson	Title
Week 1	Lesson 1 of 4	9.1	Angle Relationships
Week 1	Lesson 2 of 4	9.2	Solving for Unknown Angles
Week 1	Lesson 3 of 4	9.3	Circles and Circumference
Week 1	Lesson 4 of 4	9.4	Area of a Circle
Week 1	End of week	Check	Weekly Check 25: Angles and Circles
Week 2	Lesson 1 of 4	9.5	Problems with Circles
Week 2	Lesson 2 of 4	9.6	Area of Composite Figures
Week 2	Lesson 3 of 4	9.7	Surface Area of Prisms
Week 2	Lesson 4 of 4	9.8	Volume of Prisms
Week 2	End of week	Check	Weekly Check 26: Area, Surface Area, and Volume
Week 3	Lesson 1 of 4	9.9	Volume and Surface Area in Context
Week 3	Lesson 2 of 4	9.10	Composite Solids
Week 3	Lesson 3 of 4	9.11	Comparing Volume and Surface Area
Week 3	Lesson 4 of 4	9.12	Geometry Problem Solving
Week 3	End of week	Check	Weekly Check 27: Geometry Problem Solving
Week 4	Lesson 1 of 4	9.13	Year Review: Ratios, Proportions, and Percents
Week 4	Lesson 2 of 4	9.14	Year Review: Rational Numbers
Week 4	Lesson 3 of 4	9.15	Year Review: Expressions, Equations, and Inequalities
Week 4	Lesson 4 of 4	9.16	Year Review: Statistics, Probability, and Geometry
Week 4	Before the test	Review	Year-End Review
Week 4	End of unit	Test	Grade 7 Year-End Test

TEACHING THIS UNIT AT HOME

This is the final unit of Grade 7! Ask your student to estimate how much paint a room needs or how much water a container holds.

UNIT VOCABULARY

supplementary	Sum to 180°.
complementary	Sum to 90°.
circumference	$C = \pi d$.
area of a circle	$A = \pi r^2$.
volume of a prism	$V = Bh$.

LESSON 9.1 · TEACHER PLAN
Week 1 · Lesson 1 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.1: Angle Relationships

Student goal	I can identify complementary, supplementary, vertical, and adjacent angles.
Objective	Students identify angle relationships.
Standard	7.G.5 — Use facts about supplementary, complementary, vertical, and adjacent angles.
Pacing	Unit 9, week 1, lesson 1 of 4
Time	About 50 minutes
Materials	Protractor, two pencils, workbook Lesson 9.1
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Subtraction warm-up.
Launch	5	Cross two pencils and measure the four angles.
Teach	15	Define each pair. Say: “Vertical angles are across from each other.”
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Mixing up complementary (90) and supplementary (180).

SUPPORT

- C comes before S, and 90 before 180.

EXTEND

- Find angle pairs in the house: scissors, doors, roads on a map.

STUDENT LESSON 9.1 · WITH ANSWERS

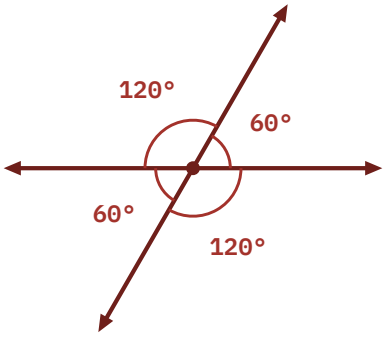
Lesson 9.1: Angle Relationships

STUDENT'S ASSIGNMENT

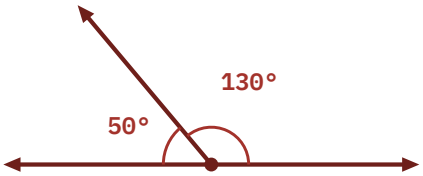
- [ ] Read Lesson 9.1 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.1, Part A.
- [ ] Complete the rest of Workbook Lesson 9.1 (Parts B–E).

LET'S EXPLORE

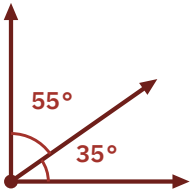
Two straight roads cross. One angle at the corner measures 60°. What are the other three angles?



Vertical angles are equal; neighbors are supplementary.



Supplementary: 130° + 50° = 180°



Complementary: 35° + 55° = 90°

WORKED EXAMPLE · CROSSING ROADS

One angle is 60°

1 The vertical angle is also 60°.

2 Each neighbor is 180° − 60° = 120°.

Answer: 60°, 120°, 60°, 120°

★ KEY IDEA

Complementary → 90°. Supplementary → 180°. Vertical angles are equal.

✎ TRY IT

Answers are shown in color.

1. Complement of 28°

Answer: 62°

2. Supplement of 28°

Answer: 152°

3. Two lines cross; one angle is 75°. Its vertical angle?

Answer: 75°

Workbook Answer Key · Lesson 9.1

#	Question	Answer
1	Complement of 28°	62°
2	Supplement of 28°	152°
3	Two lines cross; one angle is 75°. Its vertical angle?	75°
4	90 − 35	55
5	180 − 130	50
6	Complement of 40°	50°
7	Supplement of 40°	140°
8	Complement of 72°	18°
9	Supplement of 115°	65°
10	Vertical angle to 33°	33°
11	Neighbor (adjacent, on a line) of 33°	147°
12	(figure)	vertical angles (∠1 and ∠3); ∠1 and ∠2 are supplementary
13	Can two angles be both complementary and vertical? Explain.	Yes, if each is 45°
14	Two angles are supplementary. One is 4 times the other. Find both.	36° and 144°

LESSON 9.2 · TEACHER PLAN
Week 1 · Lesson 2 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.2: Solving for Unknown Angles

Student goal	I can write and solve equations to find unknown angles.
Objective	Students solve for unknown angles.
Standard	7.G.5, 7.EE.4a — Write and solve simple equations for an unknown angle in a figure.
Pacing	Unit 9, week 1, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.2
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Equation warm-up.
Launch	5	Read Let's Explore.
Teach	15	Name the relationship first, then write the equation.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Stopping at x instead of finding the angle.

SUPPORT

- Underline what the question asks for.

EXTEND

- Write your own angle puzzle.

STUDENT LESSON 9.2 · WITH ANSWERS

Lesson 9.2: Solving for Unknown Angles

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.2 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.2, Part A.
- [ ] Complete the rest of Workbook Lesson 9.2 (Parts B–D).

LET'S EXPLORE

Two adjacent angles form a straight line. One is (2x + 10)° and the other is 70°. Find x.

WORKED EXAMPLE · SUPPLEMENTARY

(2x + 10) + 70 = 180

1 2x + 80 = 180.

2 2x = 100, so x = 50.

3 The angle is 2(50) + 10 = 110°.

Answer: x = 50

WORKED EXAMPLE · VERTICAL

Vertical angles 3x° and 105°

1 3x = 105.

Answer: x = 35

WORKED EXAMPLE · COMPLEMENTARY

Angles x° and 2x° are complementary

1 x + 2x = 90; 3x = 90.

Answer: x = 30; angles 30° and 60°

★ KEY IDEA

Write an equation from the relationship (sum to 90°, sum to 180°, or equal), then solve.

✎ TRY IT

Answers are shown in color.

1. x + 48 = 90

Answer: x = 42

2. Vertical angles (4x − 20)° and 60°

Answer: x = 20

3. Supplementary: 5x and 4x

Answer: x = 20

Workbook Answer Key · Lesson 9.2

#	Question	Answer
1	x + 48 = 90	x = 42
2	Vertical angles (4x − 20)° and 60°	x = 20
3	Supplementary: 5x and 4x	x = 20
4	2x + 80 = 180	x = 50
5	3x = 105	x = 35
6	Complementary: x° and 64°	x + 64 = 90; x = 26
7	Supplementary: (x + 30)° and 2x°	3x + 30 = 180; x = 50
8	Vertical: (5x − 10)° and 80°	5x − 10 = 80; x = 18
9	Complementary: (3x)° and (x + 10)°	4x + 10 = 90; x = 20
10	Three angles on a line: 40°, x°, and 2x°	3x + 40 = 180; x = 46 $^{2}/_{3}$
11	Angles around a point: 90°, 150°, and x°	x = 120
12	Two vertical angles measure (6x + 4)° and (4x + 20)°. Find x and the angle.	x = 8; 52°

LESSON 9.3 · TEACHER PLAN
Week 1 · Lesson 3 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.3: Circles and Circumference

Student goal	I can find the circumference of a circle from its radius or diameter.
Objective	Students find circumference.
Standard	7.G.4 — Know the formula for the circumference of a circle and use it to solve problems.
Pacing	Unit 9, week 1, lesson 3 of 4
Time	About 50 minutes
Materials	String, ruler, 3 round objects, workbook Lesson 9.3
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Measure C and d of three circles; divide.
Teach	15	Discover π; derive C = πd.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using the radius in C = πd.

SUPPORT

- Write d = 2r first.

EXTEND

- Measure the circumference of a car tire; how many turns per mile?

STUDENT LESSON 9.3 · WITH ANSWERS

Lesson 9.3: Circles and Circumference

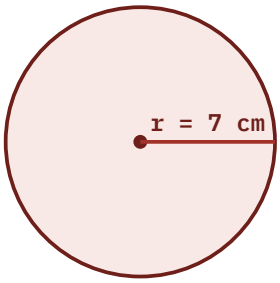
STUDENT'S ASSIGNMENT

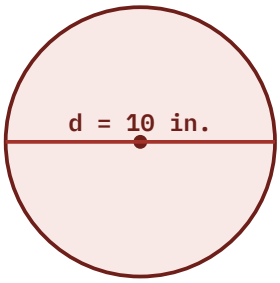
- [ ] Read Lesson 9.3 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.3, Part A.
- [ ] Complete the rest of Workbook Lesson 9.3 (Parts B–E).

LET'S EXPLORE

Measure around a can and across its top. Divide. Try another circle. What do you notice?

For every circle, circumference ÷ diameter is the same number, π (about 3.14). So C = πd, or C = 2πr. In this book use 3.14 for π unless told otherwise.





WORKED EXAMPLE · DIAMETER

d = 10 in.

1 C = 3.14 × 10 = 31.4.

Answer: 31.4 in.

WORKED EXAMPLE · RADIUS

r = 7 cm

1 d = 14; C = 3.14 × 14 = 43.96.

Answer: 43.96 cm

WORKED EXAMPLE · WORK BACKWARD

C = 62.8 m

1 d = 62.8 ÷ 3.14 = 20.

Answer: d = 20 m

★ KEY IDEA

C = πd = 2πr. Use π ≈ 3.14.

✎ TRY IT

Answers are shown in color.

1. C when d = 6 ft

Answer: 18.84 ft

2. C when r = 3 m

Answer: 18.84 m

3. d when C = 31.4 cm

Answer: 10 cm

Workbook Answer Key · Lesson 9.3

#	Question	Answer
1	C when d = 6 ft	18.84 ft
2	C when r = 3 m	18.84 m
3	d when C = 31.4 cm	10 cm
4	3.14 × 10	31.4
5	2 × 7	14
6	d = 4 cm	12.56 cm
7	r = 5 in.	31.4 in.
8	d = 20 ft	62.8 ft
9	r = 12 m	75.36 m
10	d = 5/2 cm	7.85 cm
11	r = 50 mm	314 mm
12	A bike wheel has diameter 26 in. How far does it roll in one turn? In 100 turns?	81.64 in.; 8,164 in.
13	A circular track is 314 m around. What is its diameter?	100 m
14	A circle has radius 9 cm. Find its circumference.	56.52 cm

LESSON 9.4 · TEACHER PLAN
Week 1 · Lesson 4 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.4: Area of a Circle

Student goal	I can find the area of a circle.
Objective	Students find the area of a circle.
Standard	7.G.4 — Know the formula for the area of a circle; give an informal derivation of the relationship between circumference and area.
Pacing	Unit 9, week 1, lesson 4 of 4
Time	About 50 minutes
Materials	Paper circle, scissors, workbook Lesson 9.4
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Cut and rearrange the wedges.
Teach	15	Derive $A = \pi r^2$.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using the diameter in $A = \pi r^2$.
- Computing $(\pi r)^2$.

SUPPORT

- Circle the r before you start.

EXTEND

- Compare pizza deals at a real restaurant.

STUDENT LESSON 9.4 · WITH ANSWERS

Lesson 9.4: Area of a Circle

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.4 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.4, Part A.
- [ ] Complete the rest of Workbook Lesson 9.4 (Parts B–E).

LET'S EXPLORE

Cut a paper circle into 16 wedges and rearrange them. What shape do they almost make?

The wedges form a shape that is almost a parallelogram. Its base is half the circumference (πr) and its height is the radius (r). So its area is $\pi r \times r = \pi r^2$.

WORKED EXAMPLE · RADIUS

$r = 5$ cm

1 $A = 3.14 \times 5^2 = 3.14 \times 25 = 78.5$.

Answer: 78.5 cm²

WORKED EXAMPLE · DIAMETER

$d = 8$ in.

1 $r = 4$; $A = 3.14 \times 16 = 50.24$.

Answer: 50.24 in²

! WATCH OUT

Square the radius before multiplying by π. And use the radius, not the diameter.

★ KEY IDEA

$A = \pi r^2$.

✎ TRY IT

Answers are shown in color.

1. A when $r = 10$ m

Answer: 314 m²

2. A when d = 6 ft

Answer: 28.26 ft²

3. A when r = 1.5 cm

Answer: 7.065 cm²

Workbook Answer Key · Lesson 9.4

#	Question	Answer
1	A when r = 10 m	314 m²
2	A when d = 6 ft	28.26 ft²
3	A when r = 1.5 cm	7.065 cm²
4	5^2	25
5	3.14 × 25	78.5
6	r = 2 cm	12.56 cm²
7	r = 7 in.	153.86 in.²
8	d = 10 ft	78.5 ft²
9	d = 12 m	113.04 m²
10	r = 20 yd	1,256 yd²
11	d = 3 cm	7.065 cm²
12	A sprinkler waters a circle of radius 15 ft. What area does it water?	706.5 ft²
13	A 12-in. pizza (diameter) costs $12. A 16-in. pizza costs $18. Which is the better deal per square inch?	16-in. pizza: 200.96 in² for $18 (about 9¢ per in²) beats 113.04 in² for $12 (about 11¢ per in²)
14	Find the area of a circle with diameter 14 cm.	153.86 cm²

WEEKLY CHECK · ANSWER KEY
After Lesson 9.4

About 15 minutes

Weekly Check 25: Angles and Circles — Answer Key

Answers

#	Question	Answer
1	Complement of 37°	53°
2	Vertical angles (3x + 6)° and 51°: x?	x = 15
3	Supplementary: x° and (2x + 30)°	x = 50
4	Circumference, r = 5 in.	31.4 in.
5	Area, d = 10 cm	78.5 cm²
6	Diameter if C = 47.1 m	15 m

LESSON 9.5 · TEACHER PLAN
Week 2 · Lesson 1 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.5: Problems with Circles

Student goal	I can find areas and perimeters of figures made from parts of circles.
Objective	Students solve problems with parts of circles.
Standard	7.G.4 — Solve real-world problems involving area and circumference of circles.
Pacing	Unit 9, week 2, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.5
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Area: add parts. Perimeter: only outside edges.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Including the dashed inside segment in the perimeter.

SUPPORT

- Trace the outside with a finger.

EXTEND

- Design a garden with curved edges and find its area.

STUDENT LESSON 9.5 · WITH ANSWERS

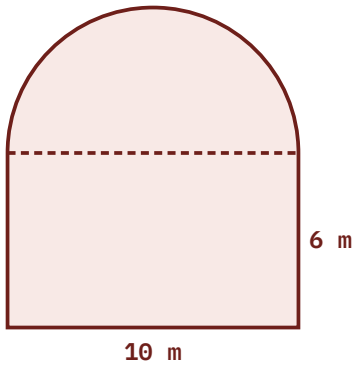
Lesson 9.5: Problems with Circles

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.5 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.5, Part A.
- [ ] Complete the rest of Workbook Lesson 9.5 (Parts B–E).

LET'S EXPLORE

A window is a 10 m by 6 m rectangle topped by a semicircle. How much glass does it need? How much trim goes around it?



WORKED EXAMPLE · AREA

Rectangle + half circle (r = 5)

1 Rectangle: 10 × 6 = 60.

2 Half circle: 3.14 × 25 ÷ 2 = 39.25.

3 Total: 60 + 39.25.

Answer: 99.25 m²

WORKED EXAMPLE · PERIMETER

Three sides of the rectangle + half the circumference

1 10 + 6 + 6 = 22.

2 Half of 3.14 × 10 = 15.7.

3 22 + 15.7.

Answer: 37.7 m

★ KEY IDEA

Break the figure into rectangles and circle parts. Include only the edges on the outside for perimeter.

✎ TRY IT

Answers are shown in color.

1. Area of a semicircle, r = 4 cm

Answer: 25.12 cm²

2. Area of a quarter circle, r = 10 ft

Answer: 78.5 ft²

3. Perimeter of a semicircle, d = 10 in. (curve + diameter)

Answer: 25.7 in.

Workbook Answer Key · Lesson 9.5

#	Question	Answer
1	Area of a semicircle, r = 4 cm	25.12 cm²
2	Area of a quarter circle, r = 10 ft	78.5 ft²
3	Perimeter of a semicircle, d = 10 in. (curve + diameter)	25.7 in.
4	3.14 × 25 ÷ 2	39.25
5	3.14 × 10 ÷ 2	15.7
6	Semicircle, r = 6 m	56.52 m²
7	Quarter circle, r = 2 cm	3.14 cm²
8	Square 8 × 8 with a circle (d = 8) cut out	13.76 square units
9	Rectangle 12 × 4 with semicircles on both short ends	60.56 square units
10	A running track is a 100 m by 60 m rectangle with semicircles on both 60 m ends. Find the distance around the track.	388.4 m
11	Find the area of the window above if the rectangle is 4 ft by 3 ft (semicircle on the 4 ft side).	18.28 ft²

LESSON 9.6 · TEACHER PLAN
Week 2 · Lesson 2 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.6: Area of Composite Figures

Student goal	I can find the area of composite figures made of triangles, rectangles, and trapezoids.
Objective	Students find areas of composite figures.
Standard	7.G.6 — Solve real-world problems involving area of two-dimensional objects composed of triangles, quadrilaterals, and polygons.
Pacing	Unit 9, week 2, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.6
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Discuss Let's Explore.
Teach	15	Decompose the L two ways; same answer.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Double-counting overlap.

SUPPORT

- Shade each part a different color.

EXTEND

- Find the area of a room with an alcove.

STUDENT LESSON 9.6 · WITH ANSWERS

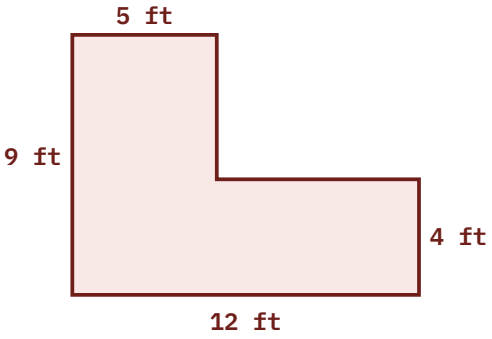
Lesson 9.6: Area of Composite Figures

STUDENT'S ASSIGNMENT

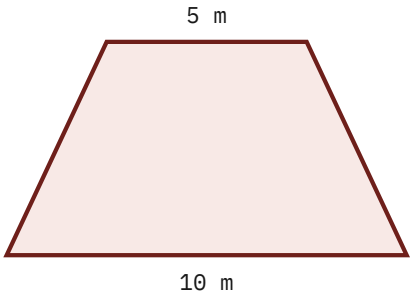
- [ ] Read Lesson 9.6 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.6, Part A.
- [ ] Complete the rest of Workbook Lesson 9.6 (Parts B–E).

LET'S EXPLORE

A garden is shaped like an L. How can you find its area without a formula for L-shapes?



Split into two rectangles.



Trapezoid, height 4 m

WORKED EXAMPLE · L-SHAPE

Split into rectangles

1 Bottom: 12 × 4 = 48.

2 Top: 5 × 5 = 25.

Answer: 73 ft²

WORKED EXAMPLE · TRAPEZOID

Bases 10 m and 5 m, height 4 m

1 A = (10 + 5) ÷ 2 × 4 = 30.

Answer: 30 m²

Shape	Area formula
rectangle	$A = lw$
triangle	$A = \frac{1}{2}bh$
parallelogram	$A = bh$
trapezoid	$A = \frac{1}{2}(b_1 + b_2)h$

★ KEY IDEA

Decompose into shapes you know, find each area, and add (or subtract cut-out parts).

✎ TRY IT

Answers are shown in color.

1. Triangle, b = 9, h = 6

Answer: 27

2. Trapezoid, bases 7 and 11, h = 5

Answer: 45

3. Rectangle 10 × 8 with a 3 × 3 square cut out

Answer: 71

Workbook Answer Key · Lesson 9.6

#	Question	Answer
1	Triangle, b = 9, h = 6	27
2	Trapezoid, bases 7 and 11, h = 5	45
3	Rectangle 10 × 8 with a 3 × 3 square cut out	71
4	12 × 4 + 5 × 5	73
5	(10 + 5) ÷ 2 × 4	30
6	L-shape: full width 10, full height 8, top width 4, bottom height 3 (figure)	50
7	Trapezoid: bases 6 and 14, height 5	50
8	A rectangle 8 × 5 with a triangle (base 8, height 3) on top	52
9	Parallelogram b = 12, h = 7	84
10	A deck is a 16 ft by 12 ft rectangle with a 4 ft by 4 ft square cut out for a tree. Stain covers 50 ft² per can. How many cans?	176 ft²; 4 cans

#	Question	Answer
11	Find the area: a 10 × 6 rectangle with a right triangle (legs 6 and 4) attached to the right side.	72

LESSON 9.7 · TEACHER PLAN
Week 2 · Lesson 3 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.7: Surface Area of Prisms

Student goal	I can find the surface area of rectangular and triangular prisms.
Objective	Students compute surface area of prisms.
Standard	7.G.6 — Solve real-world problems involving surface area of objects composed of triangles, quadrilaterals, and prisms.
Pacing	Unit 9, week 2, lesson 3 of 4
Time	About 50 minutes
Materials	Empty cereal box (to cut into a net), workbook Lesson 9.7
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Unfold a cereal box into a net.
Teach	15	Pair the faces. Triangular prism: two triangles + three rectangles.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Counting only three faces.

SUPPORT

- Use the net; check off each face.

EXTEND

- Find the SA of a real box at home.

STUDENT LESSON 9.7 · WITH ANSWERS

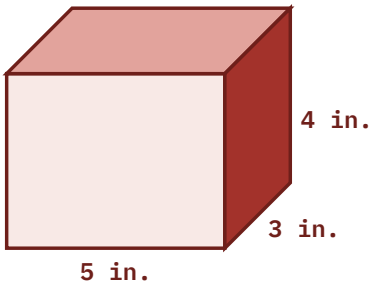
Lesson 9.7: Surface Area of Prisms

STUDENT'S ASSIGNMENT

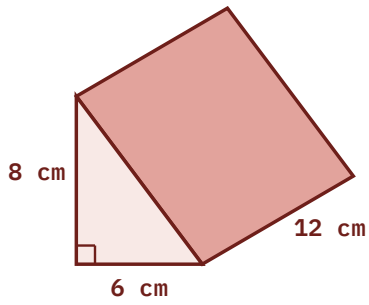
- [ ] Read Lesson 9.7 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.7, Part A.
- [ ] Complete the rest of Workbook Lesson 9.7 (Parts B–E).

LET'S EXPLORE

How much cardboard does a 5 in. by 3 in. by 4 in. box need?



Rectangular prism



Triangular prism: legs 6 and 8, hypotenuse 10, length 12

WORKED EXAMPLE · RECTANGULAR PRISM

$5 \times 3 \times 4$

1 Top and bottom: $2 \times (5 \times 3) = 30$.

2 Front and back: $2 \times (5 \times 4) = 40$.

3 Sides: $2 \times (3 \times 4) = 24$.

Answer: 94 in²

WORKED EXAMPLE · TRIANGULAR PRISM

Legs 6 and 8, hypotenuse 10, length 12

1 Two triangles: $2 \times \frac{1}{2} \times 6 \times 8 = 48$.

2 Three rectangles: $(6 + 8 + 10) \times 12 = 288$.

Answer: 336 cm²

★ KEY IDEA

Surface area = the sum of the areas of all faces. A net helps you see every face.

✎ TRY IT

Answers are shown in color.

1. Cube with edge 3 cm

Answer: 54 cm²

2. Box 10 × 2 × 4

Answer: 136

3. Box 6 × 6 × 1

Answer: 96

Workbook Answer Key · Lesson 9.7

#	Question	Answer
1	Cube with edge 3 cm	54 cm²
2	Box 10 × 2 × 4	136
3	Box 6 × 6 × 1	96
4	2 × (15 + 20 + 12)	94
5	$\frac{1}{2}$ × 6 × 8	24
6	Box 4 × 4 × 4	96
7	Box 8 × 3 × 2	92
8	Box 10 × 5 × 6	280
9	Box 3/2 × 2 × 4	34
10	Triangular prism: legs 3 and 4, hypotenuse 5, length 10	132
11	Triangular prism: legs 5 and 12, hypotenuse 13, length 8	300
12	Lillie wraps a gift box 12 in. × 8 in. × 3 in. How much wrapping paper covers it exactly?	312 in²
13	Find the surface area of a box 7 cm × 2 cm × 5 cm.	118 cm²

LESSON 9.8 · TEACHER PLAN
Week 2 · Lesson 4 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.8: Volume of Prisms

Student goal	I can find the volume of any right prism using V = Bh.
Objective	Students find volumes of prisms.
Standard	7.G.6 — Solve real-world problems involving volume of right prisms.
Pacing	Unit 9, week 2, lesson 4 of 4
Time	About 50 minutes
Materials	Unit cubes (optional), workbook Lesson 9.8
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Build the box with layers of cubes.
Teach	15	V = Bh for any prism.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Using a side length instead of the base area for B.

SUPPORT

- Shade the base first.

EXTEND

- Compare the volumes of two cereal boxes.

STUDENT LESSON 9.8 · WITH ANSWERS

Lesson 9.8: Volume of Prisms

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.8 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.8, Part A.
- [ ] Complete the rest of Workbook Lesson 9.8 (Parts B–E).

LET'S EXPLORE

How many 1-inch cubes fill a 5 in. by 3 in. by 4 in. box?

The base holds 5 × 3 = 15 cubes per layer, and there are 4 layers: 60 cubes. For any prism, V = Bh, where B is the area of the base and h is the prism's height (the distance between the bases).

WORKED EXAMPLE · RECTANGULAR PRISM

5 × 3 × 4

1 B = 15; h = 4.

Answer: 60 in³

WORKED EXAMPLE · TRIANGULAR PRISM

Base triangle legs 6 and 8; length 12

1 $B = \frac{1}{2} \times 6 \times 8 = 24$.

2 V = 24 × 12.

Answer: 288 cm³

WORKED EXAMPLE · TRAPEZOID BASE

Bases 4 and 6, height 3; prism length 10

1 B = 15.

2 V = 15 × 10.

Answer: 150 cubic units

★ KEY IDEA

V = Bh for every right prism. Volume is measured in cubic units.

✎ TRY IT

Answers are shown in color.

1. Box 6 × 2 × 5

Answer: 60

2. Cube edge 4

Answer: 64

3. Prism with B = 18 cm², h = 7 cm

Answer: 126 cm³

Workbook Answer Key · Lesson 9.8

#	Question	Answer
1	Box 6 × 2 × 5	60
2	Cube edge 4	64
3	Prism with B = 18 cm², h = 7 cm	126 cm³
4	5 × 3 × 4	60
5	24 × 12	288
6	Box 9 × 2 × 3	54
7	Box 5/2 × 4 × 6	60
8	Box 10 × 10 × 1/2	50
9	Triangular prism: legs 3 and 4, length 10	60
10	Trapezoid base (bases 5, 9; height 2), length 6	84
11	B = 32.5 in², h = 4 in.	130 in³
12	A fish tank is 30 in. × 12 in. × 16 in. One gallon is about 231 in³. About how many gallons does it hold?	5,760 in³; about 25 gallons
13	A prism's base is a triangle with base 10 cm and height 6 cm. The prism is 9 cm long. Find its volume.	270 cm³

WEEKLY CHECK · ANSWER KEY
After Lesson 9.8

About 15 minutes

Weekly Check 26: Area, Surface Area, and Volume — Answer Key

Answers

#	Question	Answer
1	Semicircle area, r = 10	157
2	Trapezoid: bases 8 and 12, height 5	50
3	Surface area, box 4 × 3 × 2	52
4	Volume, box 4 × 3 × 2	24
5	Volume, triangular prism: legs 5, 6; length 10	150
6	SA, cube edge 5	150

LESSON 9.9 · TEACHER PLAN
Week 3 · Lesson 1 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.9: Volume and Surface Area in Context

Student goal	I can decide whether a problem needs volume or surface area and solve it.
Objective	Students choose volume or surface area.
Standard	7.G.6 — Solve real-world problems involving area, volume, and surface area.
Pacing	Unit 9, week 3, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.9
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Discuss the sandbox.
Teach	15	Sort situations: fill vs cover.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Including the top when the box is open.

SUPPORT

- Sketch which faces count.

EXTEND

- Plan a raised garden bed: soil and wood.

STUDENT LESSON 9.9 · WITH ANSWERS

Lesson 9.9: Volume and Surface Area in Context

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.9 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.9, Part A.
- [ ] Complete the rest of Workbook Lesson 9.9 (Parts B–D).

LET'S EXPLORE

A sandbox is 6 ft by 4 ft by 1 ft. You want to fill it with sand and paint its outside walls. Which needs volume? Which needs area?

WORKED EXAMPLE · FILL → VOLUME

$6 \times 4 \times 1$

1 $V = 24$.

Answer: 24 ft³ of sand

WORKED EXAMPLE · PAINT → AREA

Four outside walls (no top or bottom)

1 $2 \times (6 \times 1) + 2 \times (4 \times 1) = 12 + 8$.

Answer: 20 ft² of paint

Volume (cubic units)	Surface area (square units)
fill, hold, pour, pack inside	wrap, cover, paint, label outside

★ KEY IDEA

Filling uses volume; covering uses surface area. Decide which faces are included.

✎ TRY IT

Answers are shown in color.

1. Water to fill a pool

Answer: volume

2. Tiles to line the pool's walls and floor

Answer: surface area

3. Label around a soup can

Answer: surface area (lateral)

Workbook Answer Key · Lesson 9.9

#	Question	Answer
1	Water to fill a pool	volume
2	Tiles to line the pool's walls and floor	surface area
3	Label around a soup can	surface area (lateral)
4	6 × 4 × 1	24
5	2 × 6 + 2 × 4	20
6	How much soil fills a planter 3 ft × 2 ft × 1.5 ft?	volume; 9 ft³
7	How much fabric covers a cube-shaped ottoman, 2 ft edges, all faces except the bottom?	surface area; 20 ft²
8	An open-top box (no lid) is 10 cm × 8 cm × 5 cm. How much cardboard?	surface area; 260 cm²
9	Ice cubes are 2 cm on each edge. How many fit in a 10 cm × 8 cm × 6 cm tray (exactly packed)?	volume; 60 cubes
10	Paint covers 350 ft² per gallon. Walls of a room total 1,050 ft². How many gallons?	3 gallons

LESSON 9.10 · TEACHER PLAN
Week 3 · Lesson 2 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.10: Composite Solids

Student goal	I can find the volume and surface area of solids made from prisms.
Objective	Students find volumes of composite solids.
Standard	7.G.6 — Solve real-world problems involving volume of composite objects.
Pacing	Unit 9, week 3, lesson 2 of 4
Time	About 50 minutes
Materials	Unit cubes, workbook Lesson 9.10
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Build the steps with cubes.
Teach	15	Add or subtract parts.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting to subtract a hole.

SUPPORT

- Sketch each part.

EXTEND

- Find the volume of a room with a closet.

STUDENT LESSON 9.10 · WITH ANSWERS

Lesson 9.10: Composite Solids

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.10 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.10, Part A.
- [ ] Complete the rest of Workbook Lesson 9.10 (Parts B–D).

LET'S EXPLORE

A set of steps is made of two boxes: a bottom box 6 ft × 3 ft × 1 ft and a top box 3 ft × 3 ft × 1 ft on one end. How much concrete is needed?

WORKED EXAMPLE · VOLUME

Add the boxes

1 Bottom: 18. Top: 9.

Answer: 27 ft³

WORKED EXAMPLE · ANOTHER SOLID

A 10 × 10 × 10 cube with a 2 × 2 × 10 hole drilled through

1 Cube: 1,000. Hole: 40.

Answer: 960 cubic units

★ KEY IDEA

Volume of a composite solid: add the parts, or subtract the parts removed.

✎ TRY IT

Answers are shown in color.

1. Two cubes, edges 3 and 2, stacked

Answer: 35

2. Box 8 × 5 × 4 with a 2 × 2 × 4 hole

Answer: 144

3. Box 5 × 5 × 5 plus a 5 × 5 × 2 box on top

Answer: 175

Workbook Answer Key · Lesson 9.10

#	Question	Answer
1	Two cubes, edges 3 and 2, stacked	35
2	Box 8 × 5 × 4 with a 2 × 2 × 4 hole	144
3	Box 5 × 5 × 5 plus a 5 × 5 × 2 box on top	175
4	6 × 3 × 1 + 3 × 3 × 1	27
5	1,000 − 40	960
6	L-shaped block: 10 × 4 × 2 plus 4 × 4 × 2	112
7	Box 12 × 6 × 6 minus a cube of edge 3	405
8	Triangular prism (legs 4, 6; length 10) on a box 6 × 10 × 2	240
9	Three stacked boxes: 6 × 6 × 1, 4 × 4 × 1, 2 × 2 × 1	56
10	A planter is a 4 ft × 2 ft × 2 ft box with a 4 ft × 1 ft × 1 ft box attached on the side. How much soil does it hold?	20 ft³

LESSON 9.11 · TEACHER PLAN
Week 3 · Lesson 3 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.11: Comparing Volume and Surface Area

Student goal	I can compare solids with the same volume but different surface areas.
Objective	Students compare volume and surface area.
Standard	7.G.6 — Solve real-world and mathematical problems involving volume and surface area.
Pacing	Unit 9, week 3, lesson 3 of 4
Time	About 50 minutes
Materials	Unit cubes, workbook Lesson 9.11
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Build all boxes from 8 cubes.
Teach	15	Table of V and SA.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Thinking same volume means same SA.

SUPPORT

- Build them.

EXTEND

- Why are many animals in cold climates round and compact?

STUDENT LESSON 9.11 · WITH ANSWERS

Lesson 9.11: Comparing Volume and Surface Area

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.11 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.11, Part A.
- [ ] Complete the rest of Workbook Lesson 9.11 (Parts B–E).

LET'S EXPLORE

Boxes 8 × 1 × 1, 4 × 2 × 1, and 2 × 2 × 2 all hold 8 cubes. Which uses the least cardboard?

Box	Volume	Surface area
8 × 1 × 1	8	34
4 × 2 × 1	8	28
2 × 2 × 2	8	24

All three hold the same amount, but the cube uses the least material. The closer a box is to a cube, the less surface area it needs for its volume.

★ KEY IDEA

Same volume can have different surface areas. Cube-like shapes use the least surface area.

✎ TRY IT

Answers are shown in color.

1. SA of 12 × 1 × 1

Answer: 50

2. SA of 3 × 2 × 2

Answer: 32

3. Both have volume 12. Which uses less material?

Answer: 3 × 2 × 2

Workbook Answer Key · Lesson 9.11

#	Question	Answer
1	SA of 12 × 1 × 1	50
2	SA of 3 × 2 × 2	32
3	Both have volume 12. Which uses less material?	3 × 2 × 2
4	8 × 1 × 1	8
5	2 × 2 × 2	8
6	24 × 1 × 1	98
7	12 × 2 × 1	76
8	6 × 4 × 1	68
9	4 × 3 × 2	52
10	A company wants a box that holds 24 cubes and uses the least cardboard. Which box above should it choose? Why?	4 × 3 × 2; it is the most cube-like, with the least surface area (52)
11	Can two boxes have the same surface area but different volumes? Give an example.	Yes: 1 × 1 × 5 (SA 22, V 5) and 2 × 1 × 3 (SA 22, V 6)

LESSON 9.12 · TEACHER PLAN
Week 3 · Lesson 4 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.12: Geometry Problem Solving

Student goal	I can choose and use the right geometry formulas to solve multi-step problems.
Objective	Students solve mixed geometry problems.
Standard	7.G.4, 7.G.5, 7.G.6 — Solve real-world and mathematical problems involving angle measure, area, surface area, and volume.
Pacing	Unit 9, week 3, lesson 4 of 4
Time	About 50 minutes
Materials	Formula card, workbook Lesson 9.12
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Read Let's Explore.
Teach	15	Model the plan: draw, label, choose, compute, check.
Guided Practice	10	Try It 1–3. Weekly Check tomorrow.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Picking the wrong formula.

SUPPORT

- Use the formula table.

EXTEND

- Measure and solve a real geometry problem at home.

STUDENT LESSON 9.12 · WITH ANSWERS

Lesson 9.12: Geometry Problem Solving

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.12 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.12, Part A.
- [ ] Complete the rest of Workbook Lesson 9.12 (Parts B–D).

LET'S EXPLORE

A circular pool (diameter 20 ft) is surrounded by a 3 ft wide concrete path. What is the area of the path?

WORKED EXAMPLE · RING AREA

Outer radius 13 ft, inner radius 10 ft

1 Outer: 3.14 × 169 = 530.66.

2 Inner: 3.14 × 100 = 314.

3 Subtract.

Answer: 216.66 ft²

Measure	Formula
Circumference	$C = \pi d = 2\pi r$
Circle area	$A = \pi r^2$
Triangle area	$A = {}^{1}\!/_{2}bh$
Trapezoid area	$A = {}^{1}\!/_{2}(b_1 + b_2)h$
Prism volume	$V = Bh$
Surface area	sum of all face areas

★ KEY IDEA

Plan: draw, label, choose the formula, compute, and check the units.

✎ TRY IT

Answers are shown in color.

1. Circumference of the pool

Answer: 62.8 ft

2. Outer edge of the path

Answer: 81.64 ft

3. Volume of a 1 ft deep layer of water in the pool

Answer: 314 ft³

Workbook Answer Key · Lesson 9.12

#	Question	Answer
1	Circumference of the pool	62.8 ft
2	Outer edge of the path	81.64 ft
3	Volume of a 1 ft deep layer of water in the pool	314 ft³
4	13^2	169
5	169 − 100	69
6	A circular table has diameter 4 ft. A cloth hangs 1 ft over the edge all around. What is the cloth's area?	r = 3; 28.26 ft²
7	Two angles are supplementary. One is 24° more than the other. Find both.	78° and 102°
8	A tent is a triangular prism: triangle base 6 ft, height 4 ft, length 8 ft. Find the space inside.	96 ft³
9	A cylindrical-looking hay bale is wrapped with a band around its 5 ft diameter. How long is the band?	15.7 ft
10	Write your own two-step geometry problem and solve it.	Answers vary.

WEEKLY CHECK · ANSWER KEY
After Lesson 9.12

About 15 minutes

Weekly Check 27: Geometry Problem Solving — Answer Key

Answers

#	Question	Answer
1	Filling a tank: volume or SA?	volume
2	Open box 5 × 4 × 3: cardboard	74
3	Box 6 × 6 × 6 minus a 2 × 2 × 6 hole	192
4	Which has less SA: 6 × 1 × 1 or 3 × 2 × 1?	3 × 2 × 1
5	Ring: outer r = 5, inner r = 3	50.24
6	Two complementary angles; one is twice the other	30° and 60°

LESSON 9.13 · TEACHER PLAN
Week 4 · Lesson 1 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.13: Year Review: Ratios, Proportions, and Percents

Student goal	I can solve problems with unit rates, proportional relationships, and percents.
Objective	Review Units 1–2.
Standard	7.RP.1–3 — Analyze proportional relationships and use them to solve problems.
Pacing	Unit 9, week 4, lesson 1 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.13
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Better buy.
Teach	15	Review unit rates, k, percent change.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Dividing the wrong way for unit rate.

SUPPORT

- Label units.

EXTEND

- Compare real grocery prices.

STUDENT LESSON 9.13 · WITH ANSWERS

Lesson 9.13: Year Review: Ratios, Proportions, and Percents

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.13 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.13, Part A.
- [ ] Complete the rest of Workbook Lesson 9.13 (Parts B–D).

LET'S EXPLORE

Which is the better buy: 3 lb of apples for $4.47 or 5 lb for $7.25?

WORKED EXAMPLE · UNIT RATES

$4.47 ÷ 3 vs $7.25 ÷ 5

1 $1.49 per lb vs $1.45 per lb.

Answer: 5 lb for $7.25

WORKED EXAMPLE · PERCENT CHANGE

Price rises from $80 to $100

1 Change: 20.

2 20 ÷ 80 = 0.25.

Answer: 25% increase

★ KEY IDEA

Unit rates compare; proportions scale; percent = part ÷ whole × 100.

✎ TRY IT

Answers are shown in color.

1. Unit rate: $\frac{3}{4}$ mi in $\frac{1}{4}$ hr

Answer: 3 mph

2. y is proportional to x; y = 10 when x = 4. Find k.

Answer: 2.5

3. 20% off $65

Answer: $52

Workbook Answer Key · Lesson 9.13

#	Question	Answer
1	Unit rate: $^{3}/_{4}$ mi in $^{1}/_{4}$ hr	3 mph
2	y is proportional to x; y = 10 when x = 4. Find k.	2.5
3	20% off $65	$52
4	4.47 ÷ 3	1.49
5	7.25 ÷ 5	1.45
6	Ellie reads 45 pages in 1.5 hours. Pages per hour?	30
7	Solve: $^{6}/_{15} = {}^{x}/_{40}$	x = 16
8	Is y = 4x + 1 proportional?	No
9	15% tip on $48	$7.20
10	Price $120 → $90: percent change	25% decrease
11	Simple interest: $500 at 4% for 3 years	$60
12	6% tax on $35	$2.10
13	Map: 2 cm : 15 km. 7 cm?	52.5 km
14	A recipe uses 3 cups flour for 2 dozen cookies. How much for 5 dozen?	7.5 cups

LESSON 9.14 · TEACHER PLAN
Week 4 · Lesson 2 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.14: Year Review: Rational Numbers

Student goal	I can add, subtract, multiply, and divide rational numbers.
Objective	Review Units 3–5.
Standard	7.NS.1–3 — Apply operations with rational numbers.
Pacing	Unit 9, week 4, lesson 2 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.14
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Temperature story.
Teach	15	Review sign rules.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Sign errors in subtraction.

SUPPORT

- Number line.

EXTEND

- Make a sign-rule poster.

STUDENT LESSON 9.14 · WITH ANSWERS

Lesson 9.14: Year Review: Rational Numbers

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.14 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.14, Part A.
- [ ] Complete the rest of Workbook Lesson 9.14 (Parts B–D).

LET'S EXPLORE

The temperature was −6°F at dawn and rose 14°F by noon, then fell 9°F by night. What was the night temperature?

WORKED EXAMPLE · INTEGER STEPS

−6 + 14 − 9

1 −6 + 14 = 8.

2 8 − 9 = −1.

Answer: −1°F

WORKED EXAMPLE · FRACTIONS

$-\frac{3}{4} \div \frac{1}{2}$

1 $-\frac{3}{4} \times 2 = -\frac{6}{4}$.

Answer: $-1\frac{1}{2}$

★ KEY IDEA

Same signs multiply/divide to positive; different signs to negative. Subtracting means adding the opposite.

✎ TRY IT

Answers are shown in color.

1. −7 + 12

Answer: 5

2. −3 − (−8)

Answer: 5

3. -2.5×4

Answer: -10

Workbook Answer Key · Lesson 9.14

#	Question	Answer
1	$-7 + 12$	5
2	$-3 - (-8)$	5
3	-2.5×4	-10
4	$-6 + 14$	8
5	$8 - 9$	-1
6	$-15 + (-9)$	-24
7	$4 - 11$	-7
8	$-1.2 + 3.5$	2.3
9	$(-6)(-7)$	42
10	$-48 \div 6$	-8
11	$\frac{2}{3} \times (-\frac{9}{10})$	$-\frac{3}{5}$
12	Write $\frac{5}{8}$ as a decimal	0.625
13	Write $\frac{2}{3}$ as a decimal	0.6 repeating
14	$(-2)^3$	-8
15	$\lvert -9 \rvert - \lvert 4 \rvert$	5
16	A diver at -40 ft rises 15 ft, then descends 28 ft. Where is she?	-53 ft

LESSON 9.15 · TEACHER PLAN
Week 4 · Lesson 3 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.15: Year Review: Expressions, Equations, and Inequalities

Student goal	I can simplify expressions and solve equations and inequalities.
Objective	Review Unit 6.
Standard	7.EE.1–4 — Solve equations and inequalities.
Pacing	Unit 9, week 4, lesson 3 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.15
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Gym problem.
Teach	15	Review distributing, factoring, solving.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook lesson pages: Warm-Up through Think Deeper.
Exit Ticket	3	Student completes the Exit Ticket alone. Check it before the next lesson.

COMMON MISCONCEPTIONS

- Forgetting to flip the inequality.

SUPPORT

- Test a value.

EXTEND

- Write equations for family budgets.

STUDENT LESSON 9.15 · WITH ANSWERS

Lesson 9.15: Year Review: Expressions, Equations, and Inequalities

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.15 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.15, Part A.
- [ ] Complete the rest of Workbook Lesson 9.15 (Parts B–D).

LET'S EXPLORE

A gym charges $20 to join and $15 per month. Lillie has $110. How many months can she afford?

WORKED EXAMPLE · INEQUALITY

$20 + 15m \le 110$

1 $15m \le 90$.

2 $m \le 6$.

Answer: at most 6 months

WORKED EXAMPLE · SIMPLIFY

$3(x - 4) + 2x$

1 $3x - 12 + 2x$.

Answer: $5x - 12$

★ KEY IDEA

Undo operations in reverse order; flip the inequality when multiplying or dividing by a negative.

✎ TRY IT

Answers are shown in color.

1. Factor $18y + 12$

Answer: $6(3y + 2)$

2. $4x - 7 = 21$

Answer: $x = 7$

3. $-3x + 5 > 17$

Answer: $x < -4$

Workbook Answer Key · Lesson 9.15

#	Question	Answer
1	Factor $18y + 12$	$6(3y + 2)$
2	$4x - 7 = 21$	$x = 7$
3	$-3x + 5 > 17$	$x < -4$
4	$110 - 20$	90
5	$90 \div 15$	6
6	Simplify $5a - 2(a + 3)$	$3a - 6$
7	Simplify $\frac{1}{2}(8x - 6) + x$	$5x - 3$
8	Evaluate $2x^2 - x$ when $x = -3$	21
9	$-2(x + 3) = 10$	$x = -8$
10	$\frac{1}{2}x - 2 = 4$	$x = 12$
11	$0.4n + 1.2 = 3.6$	$n = 6$
12	$-4y \geq 20$	$y \leq -5$
13	$2x - 9 < 7$	$x < 8$
14	Write and solve: three times a number, minus 8, is 25.	$3n - 8 = 25$; $n = 11$

LESSON 9.16 · TEACHER PLAN
Week 4 · Lesson 4 of 4
Unit 9: Angles, Circles, Area, and Volume

Lesson 9.16: Year Review: Statistics, Probability, and Geometry

Student goal	I can solve problems using statistics, probability, and geometry.
Objective	Review Units 7–9.
Standard	7.SP, 7.G — Use sampling, probability, and geometry to solve problems.
Pacing	Unit 9, week 4, lesson 4 of 4
Time	About 50 minutes
Materials	Workbook Lesson 9.16, Year-End Review
Get ready	Read the student lesson and work the Try It problems.

Lesson Flow

Part	Min	What to do and say
Warm-Up	5	Warm-up.
Launch	5	Sampling example.
Teach	15	Review probability and geometry formulas.
Guided Practice	10	Try It 1–3.
Independent	12	Workbook Lesson 9.16 and the Year-End Review.
Exit Ticket	3	Check. Tomorrow: Year-End Test.

COMMON MISCONCEPTIONS

- Mixing up area and circumference formulas.

SUPPORT

- Use the formula table in Lesson 9.12.

EXTEND

- Celebrate the year!

STUDENT LESSON 9.16 · WITH ANSWERS

Lesson 9.16: Year Review: Statistics, Probability, and Geometry

STUDENT'S ASSIGNMENT

- [ ] Read Lesson 9.16 in this book with your teacher.
- [ ] Answer the Try It questions in Workbook Lesson 9.16, Part A.
- [ ] Complete the rest of Workbook Lesson 9.16 (Parts B–D).

LET'S EXPLORE

In a random sample of 40 students, 14 walk to school. About how many of the 600 students walk?

WORKED EXAMPLE · INFERENCE

14 of 40; 600 students

1 $^{14}/_{40}$ = 35%; 0.35 × 600.

Answer: about 210

WORKED EXAMPLE · PROBABILITY

Two number cubes: P(sum is 8)

1 (2,6), (3,5), (4,4), (5,3), (6,2): 5 outcomes of 36.

Answer: $^{5}/_{36}$

★ KEY IDEA

Use the right tool: proportions for samples, sample spaces for probability, formulas for geometry.

✎ TRY IT

Answers are shown in color.

1. P(heads, then heads)

Answer: $^{1}/_{4}$

2. Area of a circle, r = 6

Answer: 113.04

3. Missing angle in a triangle: 50°, 75°

Answer: 55°

Workbook Answer Key · Lesson 9.16

#	Question	Answer
1	P(heads, then heads)	$\frac{1}{4}$
2	Area of a circle, r = 6	113.04
3	Missing angle in a triangle: 50°, 75°	55°
4	0.35 × 600	210
5	6 × 6	36
6	Mean and MAD of 4, 6, 8, 10, 12	mean 8, MAD 2.4
7	Bag: 3 red, 5 blue, 2 green. P(not blue)	$\frac{1}{2}$
8	Outfits: 4 shirts, 3 pants	12
9	Scale 1 in. : 6 ft. Drawing 3.5 in.?	21 ft
10	Triangle? 3, 4, 8	No
11	Supplement of 47°	133°
12	Circumference, d = 15 cm	47.1 cm
13	Volume, box 6 × 4 × 2.5	60
14	Surface area, box 6 × 4 × 2	88
15	Cross section of a cylinder, horizontal	circle
16	Which topic from this year do you feel strongest in? Which one will you review before the test?	Answers vary.

YEAR-END REVIEW · ANSWER KEY
Study before the unit test

About 50 minutes

Year-End Review — Answer Key

Spread over the week as homework, or use as a study session before the Year-End Test.

Answers

#	Question	Answer
1	Unit price: 6 for $8.70	$1.45 each
2	y = kx; y = 21 when x = 3. Find y when x = 8.	56
3	35% of 240	84
4	Price $60 → $75: percent change	25% increase
5	−8 − (−3) + 12	7
6	$-\frac{5}{6} \times \frac{3}{10}$	$-\frac{1}{4}$
7	−3.6 ÷ (−0.4)	9
8	$\frac{7}{20}$ as a decimal	0.35
9	Simplify 4(2x − 1) − 3x	5x − 4
10	−5x + 8 = −17	x = 5
11	3(n − 2) = 15	n = 7
12	6 − 2x ≥ 14	x ≤ −4
13	18 of 60 sampled like a show; estimate for 2,500	750
14	Coin + cube: P(T and even)	$\frac{1}{4}$
15	Area, circle r = 4	50.24
16	Volume, triangular prism: legs 4 and 9, length 5	90
17	Scale 1 cm : 8 m; 6.5 cm	52 m
18	Missing angle: 90° and 34° in a triangle	56°

YEAR-END TEST · ANSWER KEY
At the end of the unit
About 60 minutes

Grade 7 Year-End Test — Answer Key

Answers

#	Question	Answer
1	Unit rate: 150 miles in 2.5 hours	60 mph
2	y = kx passes through (5, 35). Find k.	7
3	30% of 90	27
4	A $40 shirt is 15% off. Sale price?	$34
5	Price $50 → $65: percent change	30% increase
6	−14 + 9	−5
7	6 − (−11)	17
8	(−4)(−2.5)	10
9	$-\frac{2}{3} \div \frac{4}{9}$	$-1\frac{1}{2}$
10	Write $\frac{3}{8}$ as a decimal	0.375
11	Simplify 2(3x − 5) + 4x	10x − 10
12	Factor 24n − 16	8(3n − 2)
13	−6x + 4 = 40	x = −6
14	5(y + 2) = −15	y = −5
15	−2k + 7 ≤ 15	k ≥ −4
16	A random sample of 50 has 12 left-handed people. Estimate for 1,500.	360
17	Mean of 3, 7, 8, 10, 12	8
18	Bag: 4 red, 6 blue. P(blue)	$\frac{3}{5}$
19	Two coins: P(at least one head)	$\frac{3}{4}$
20	Two number cubes: P(sum is 10)	$\frac{1}{12}$
21	Scale 1 in. : 12 ft; 3.5 in. Actual?	42 ft
22	Supplement of 68°	112°
23	Circumference, d = 12 m	37.68 m

#	Question	Answer
24	Area, r = 9 cm	254.34 cm²
25	Volume, box 7 × 3 × 4	84
26	Surface area, box 7 × 3 × 4	122
27	A phone plan costs $25 per month plus $0.10 per text. Ellie can spend at most $40 per month. Write and solve an inequality for the number of texts. Interpret.	25 + 0.10t ≤ 40; t ≤ 150; at most 150 texts
28	A circular garden has diameter 14 ft. Find its area. Edging costs $2 per foot: how much to edge it? Show your work.	153.86 ft²; C = 43.96 ft; cost $87.92

Scoring

Score	What it means / next step
28–32 (85%+)	Mastered: ready for Grade 8 math.
24–27	Proficient: review missed strands over the summer.
Below 24	Review missed strands, then retest those parts.

Grade 7 Mathematics

Teacher Edition

A full year of Grade 7 mathematics, taught one clear idea at a time. Every lesson opens with a question worth thinking about, walks through worked examples step by step, and builds from models and pictures to fluent, confident problem solving.

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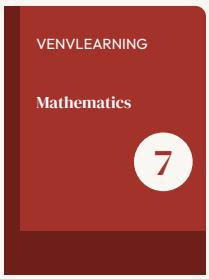
INSIDE THIS YEAR

01 Proportional Relationships
02 Percents
03 Adding and Subtracting Integers
04 Adding and Subtracting Rational Numbers
05 Multiplying and Dividing Rational Numbers
06 Expressions, Equations, and Inequalities
07 Statistics and Probability
08 Scale Drawings and Geometric Figures
09 Angles, Circles, Area, and Volume

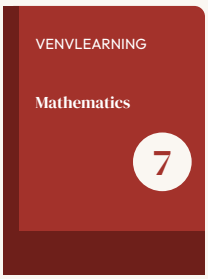
IN THIS BOOK

- Week-by-week pacing guide
- Guided lessons with timing and teaching moves
- Misconceptions, supports, and extensions
- Answer keys and scoring guides for every page

THE COMPLETE SET



Textbook



Workbook



Teacher Edition

You are holding this book

Each lesson has the same number in all three books, so the textbook, workbook, and teacher pages always match.